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(54) **SMART CONTRACT MANAGEMENT OF
LICENSING AND APPORTIONMENT USING
A DISTRIBUTED LEDGER**

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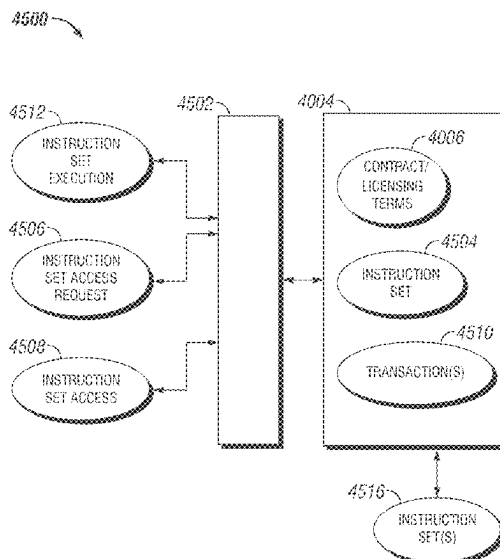
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(57) **ABSTRACT**

Transaction-enabled methods for providing provable access to a distributed ledger with a tokenized instruction set for polymer production processes are described. A method may include accessing a distributed ledger comprising an instruction set for a polymer production process and tokenizing the instruction set. The method may further include interpreting an instruction set access request and providing provable access to the instruction set. The method may further include providing commands to a production tool of the polymer production process and recording the transaction on the distributed ledger.

20 Claims, 72 Drawing Sheets



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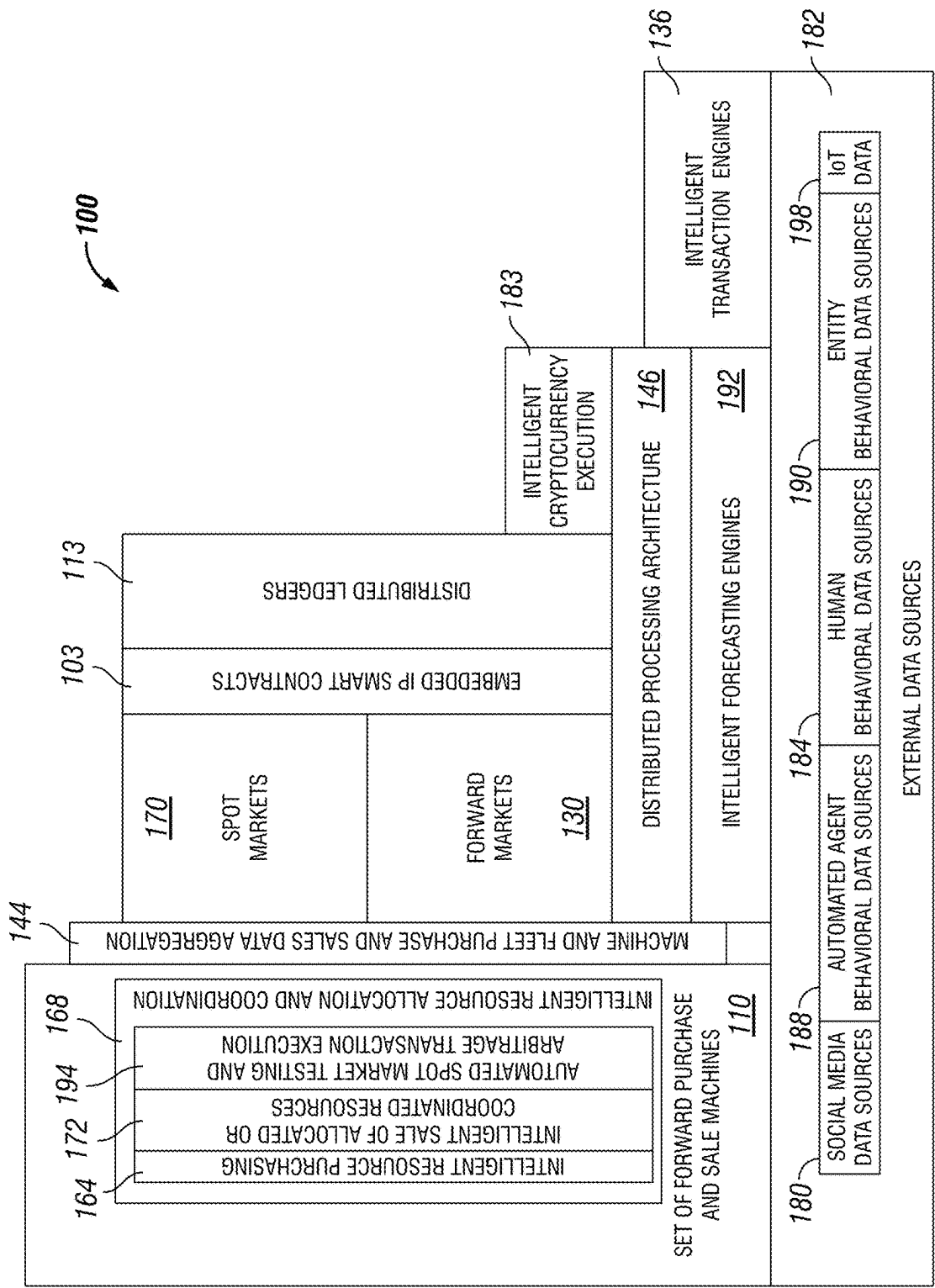
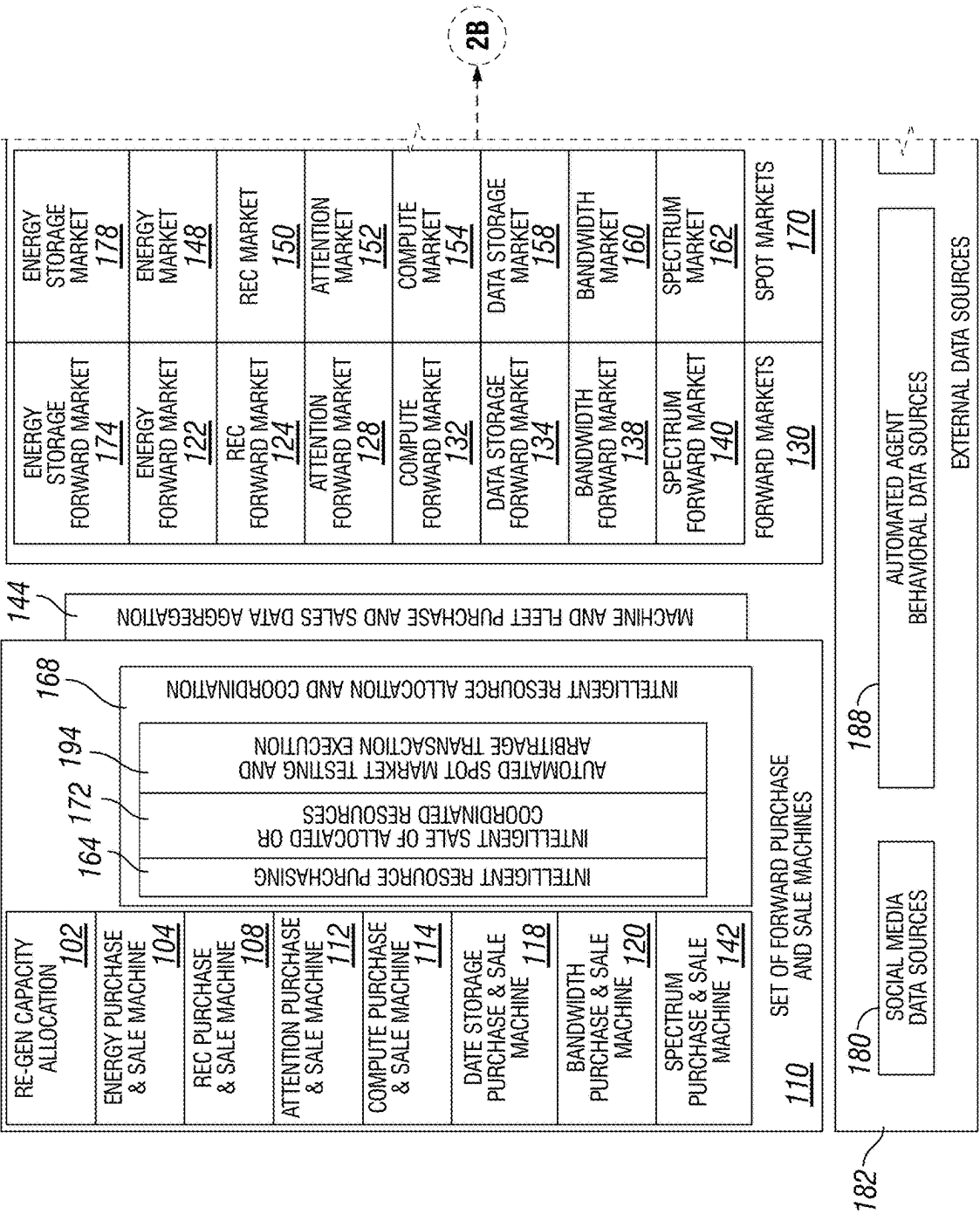
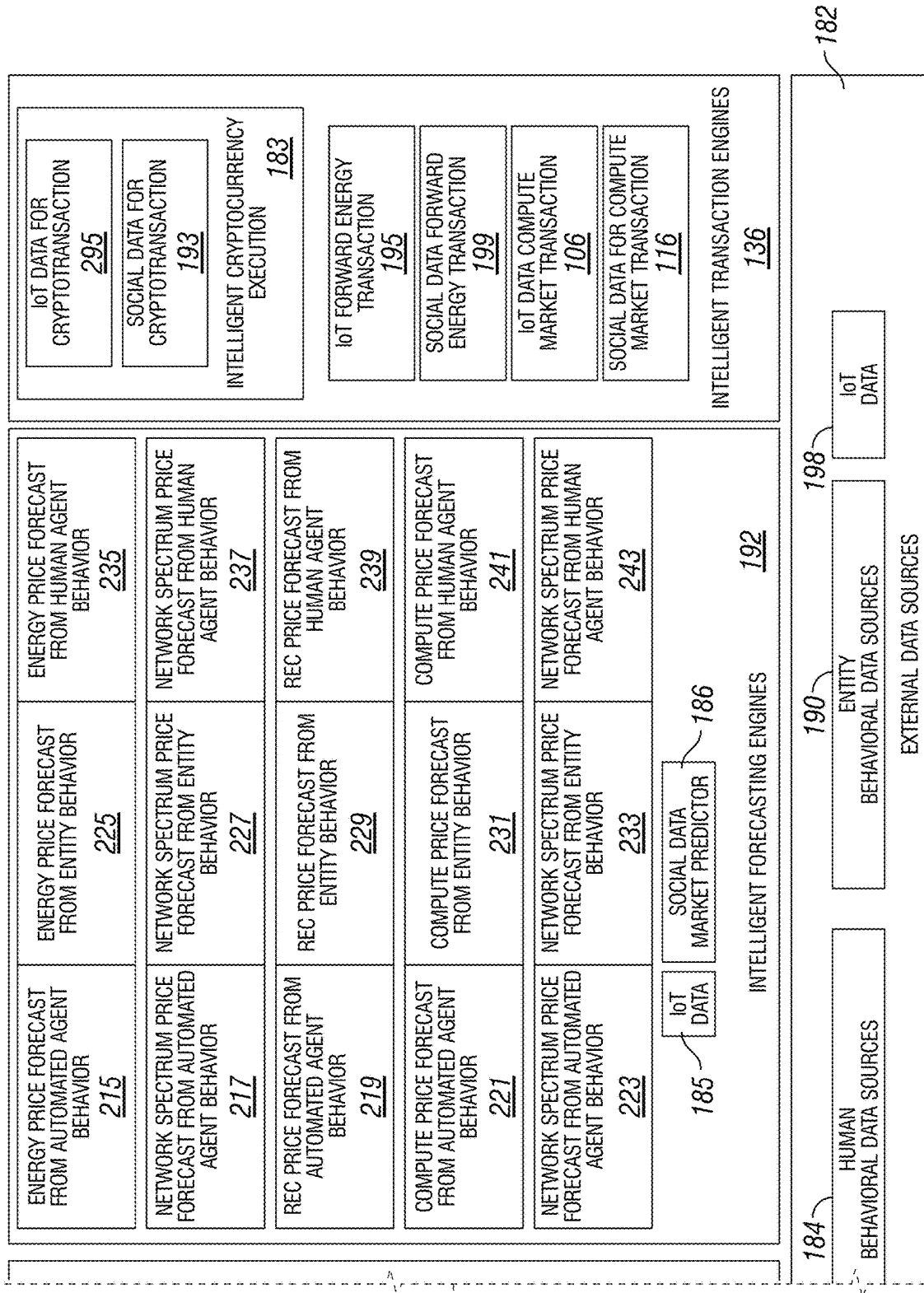


FIG. 1





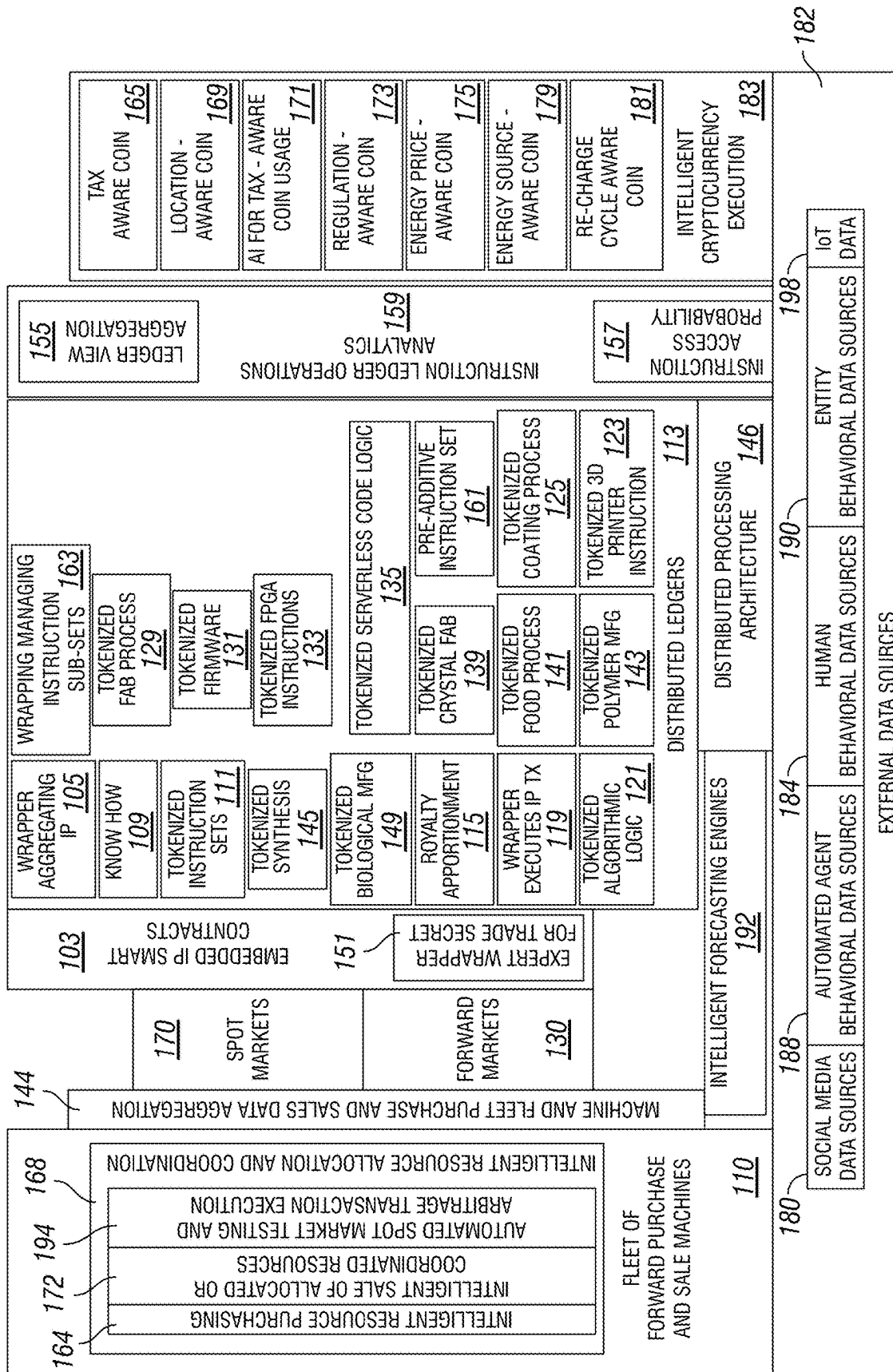


FIG. 3

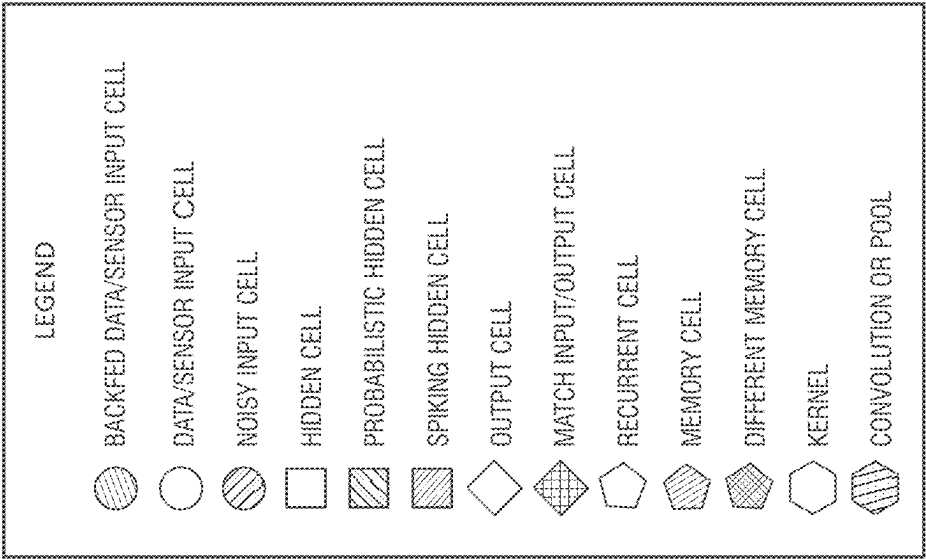


FIG. 4

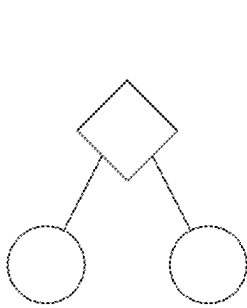


FIG. 5

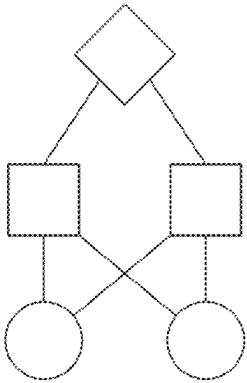


FIG. 6

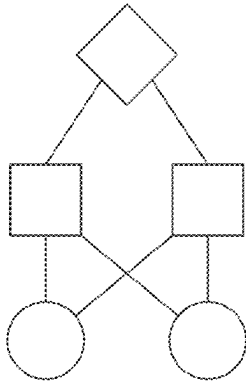


FIG. 7

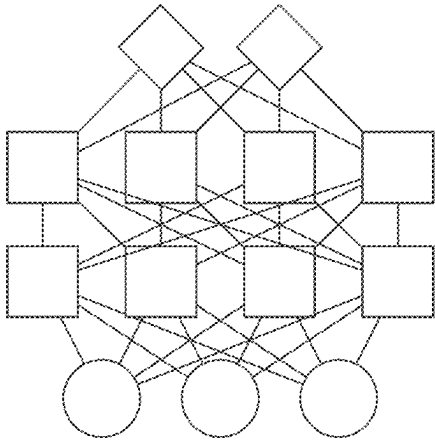


FIG. 8

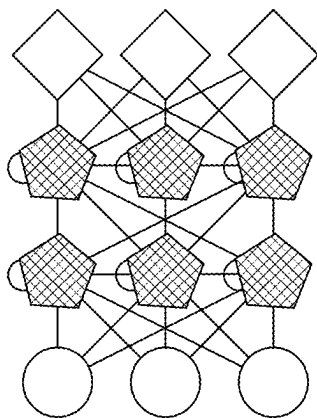


FIG. 9

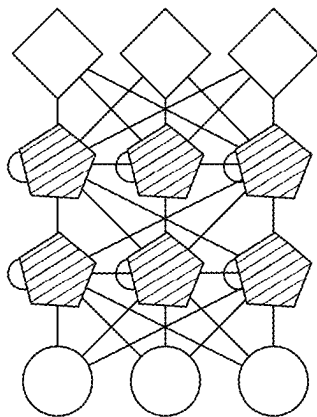


FIG. 10

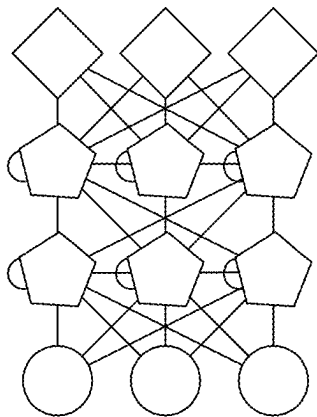


FIG. 11

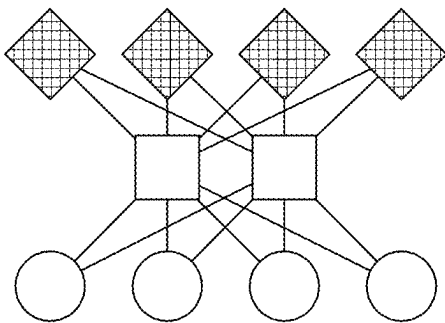


FIG. 12

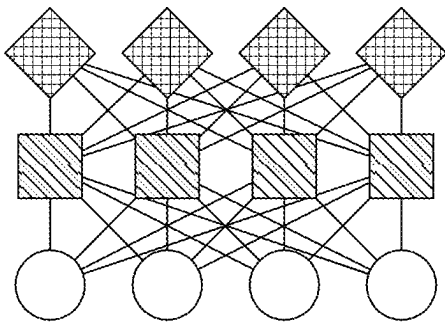


FIG. 13

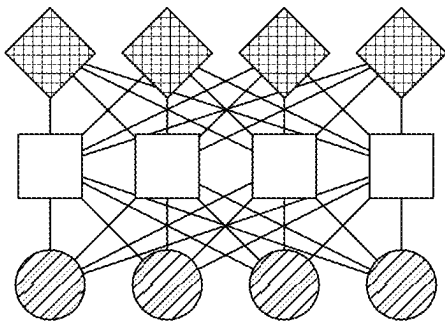


FIG. 14

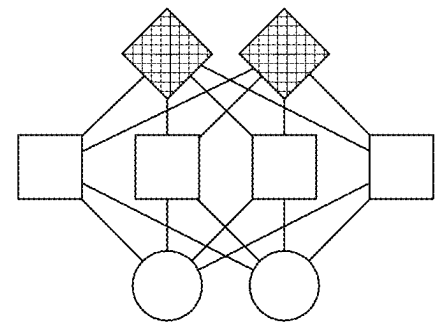


FIG. 15

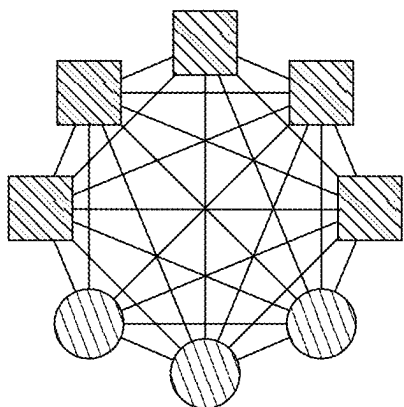


FIG. 18

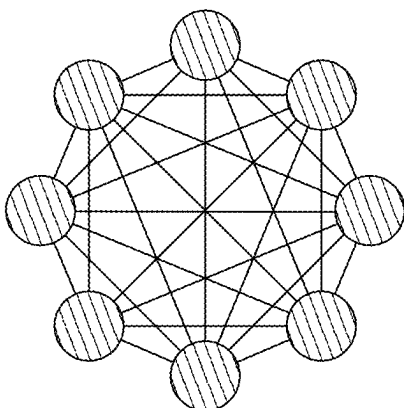


FIG. 17

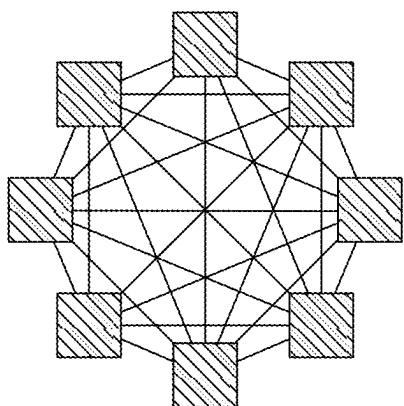


FIG. 16

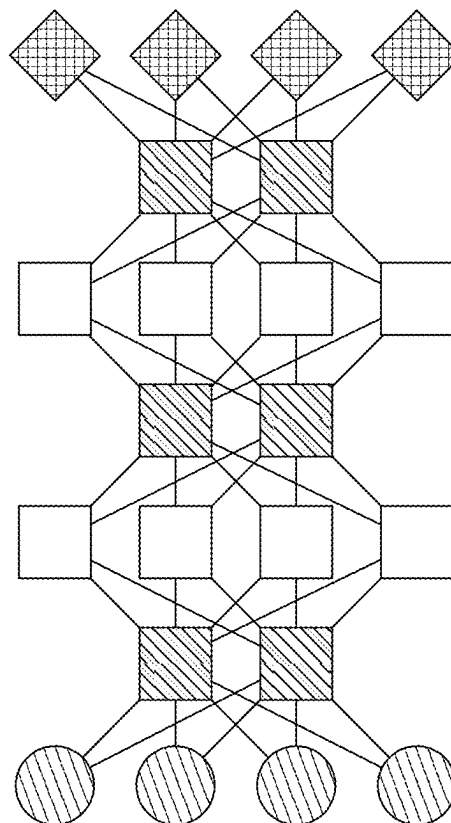


FIG. 20

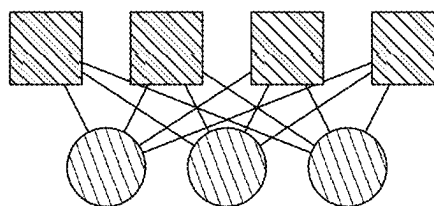


FIG. 19

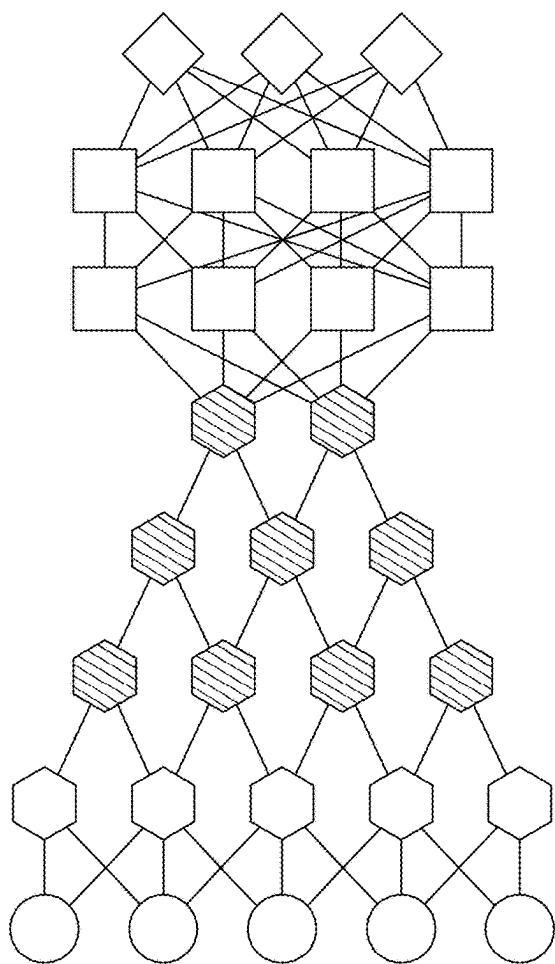


FIG. 21

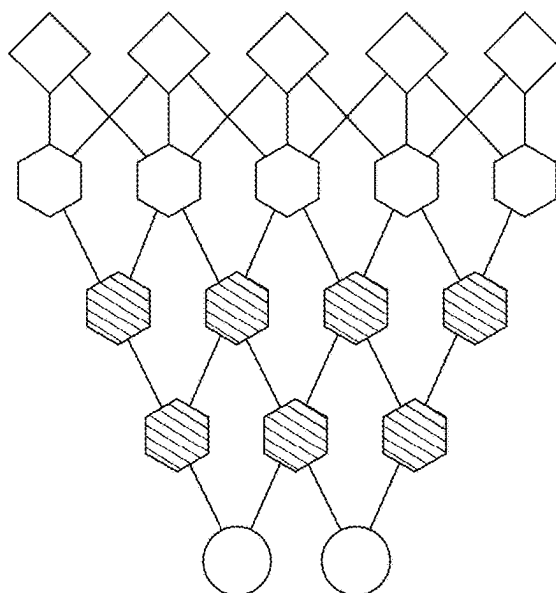


FIG. 22

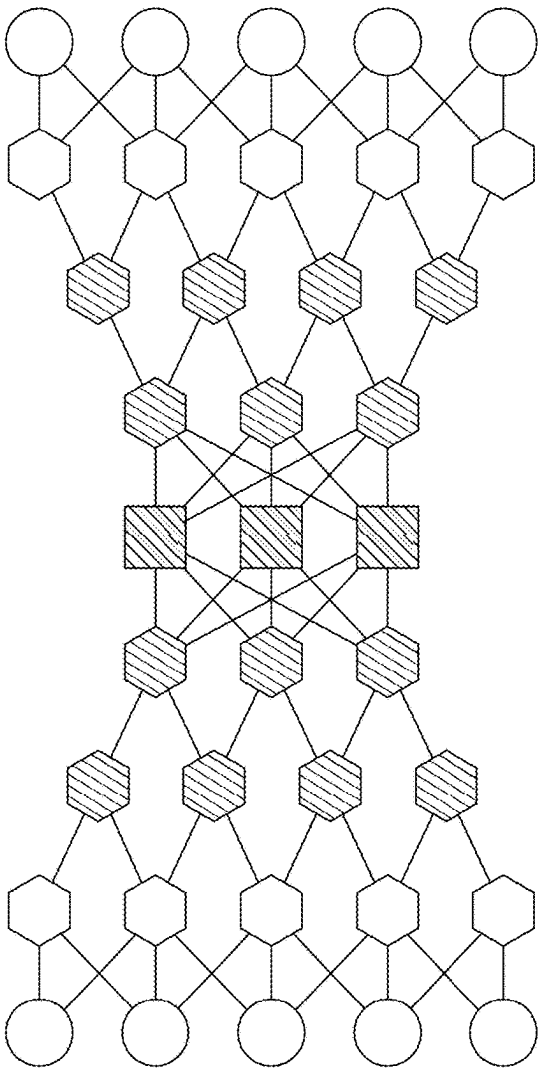


FIG. 23

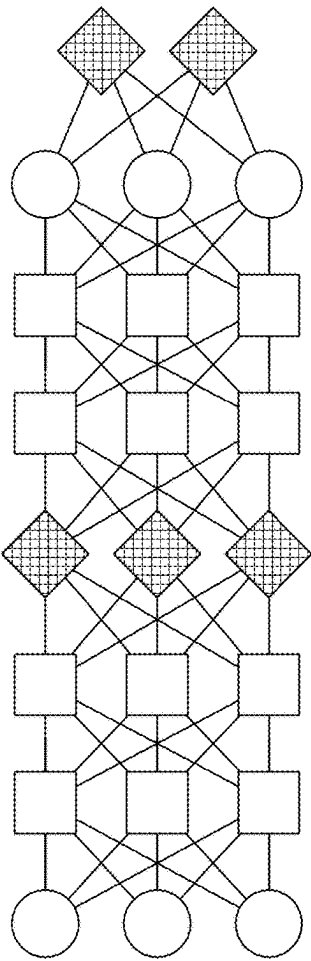


FIG. 24

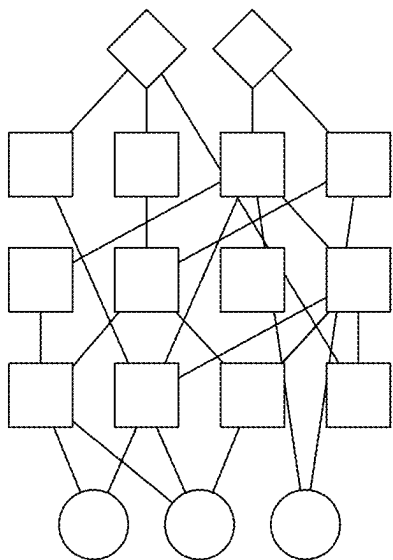


FIG. 26

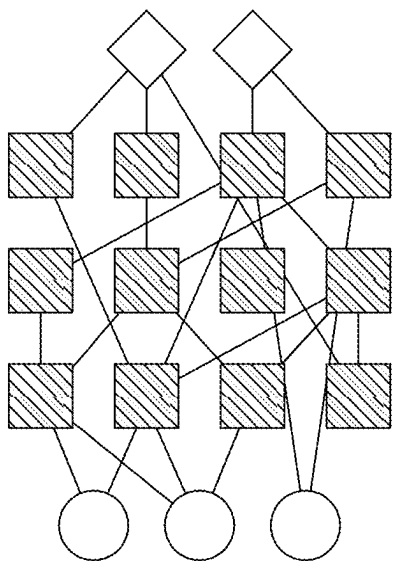


FIG. 25

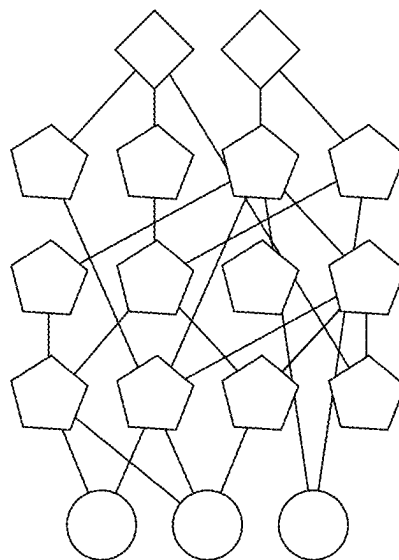


FIG. 27

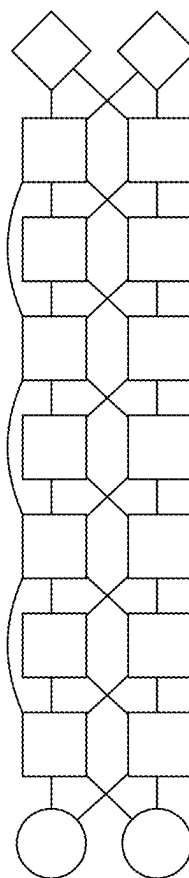


FIG. 28

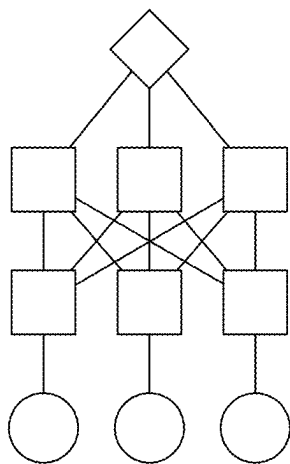


FIG. 30

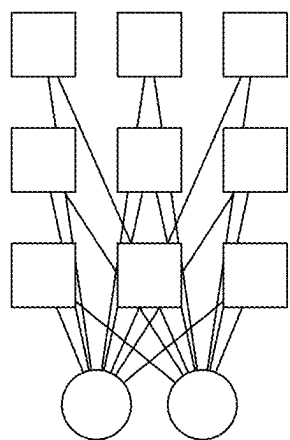


FIG. 29

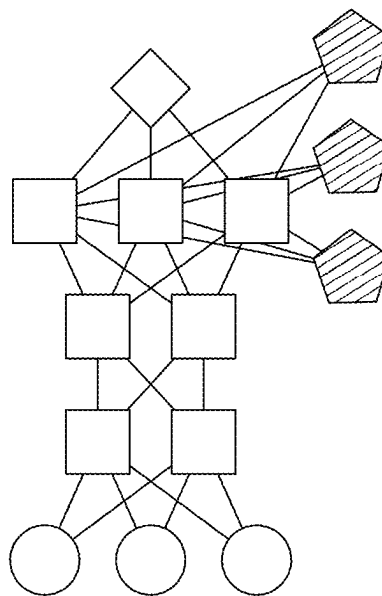


FIG. 31

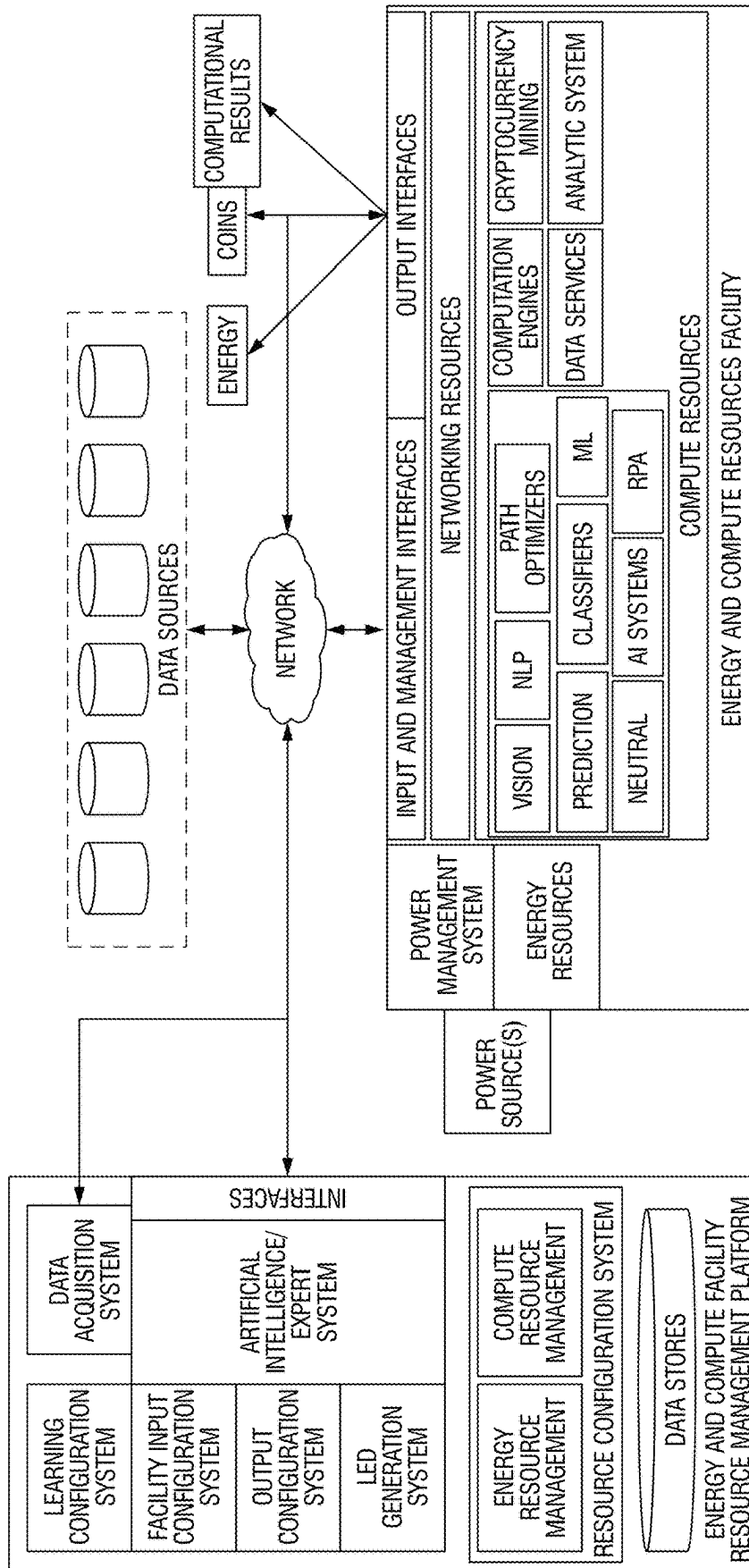


FIG. 32

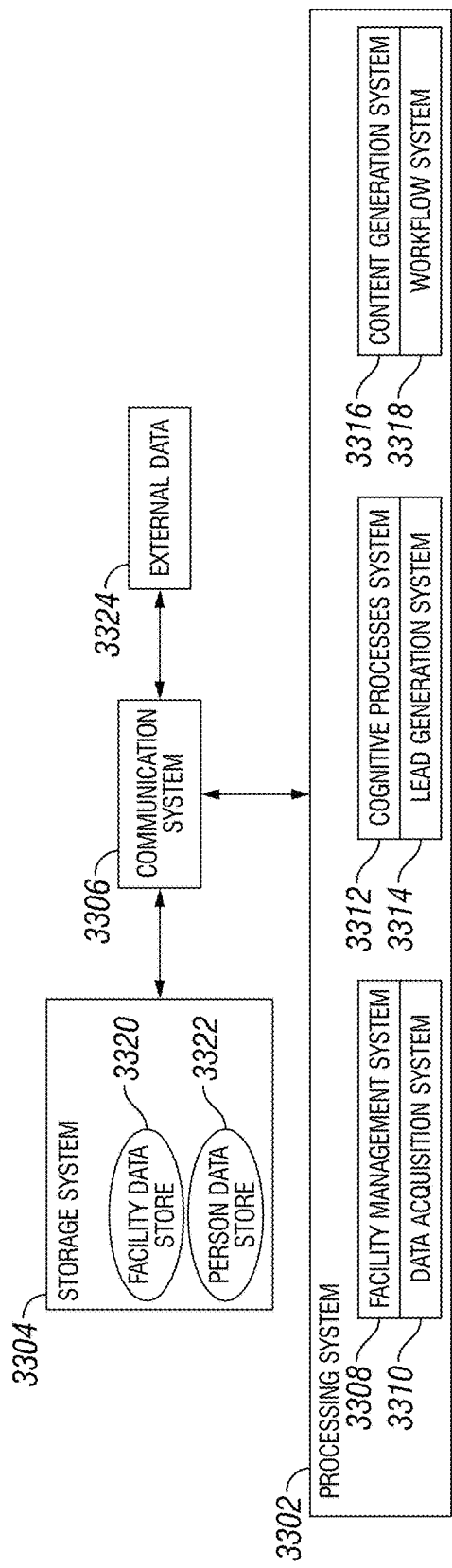


FIG. 33

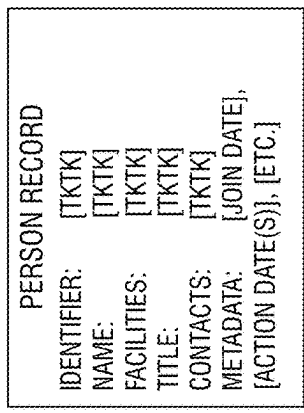


FIG. 35



FIG. 34

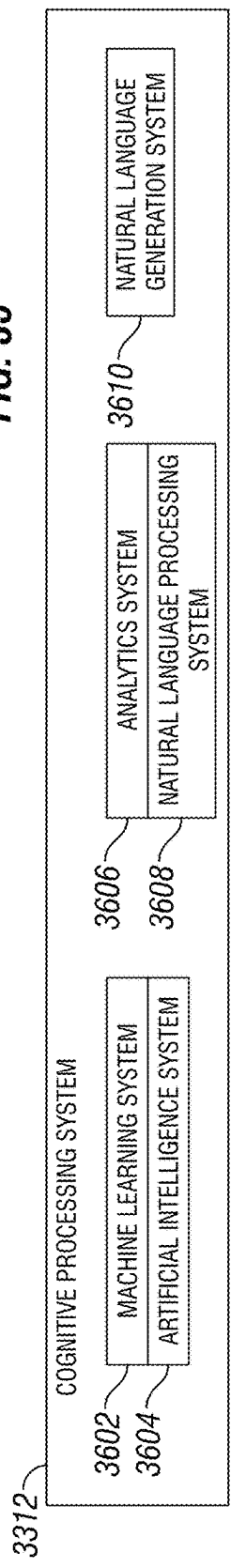
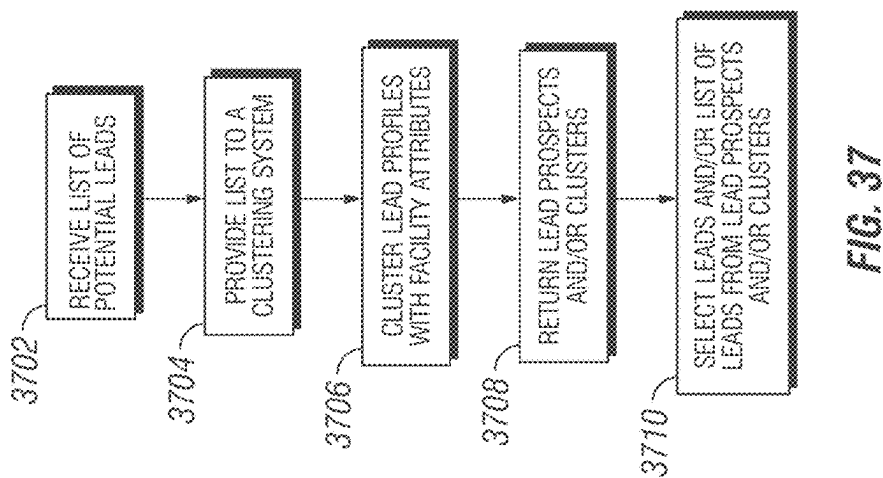
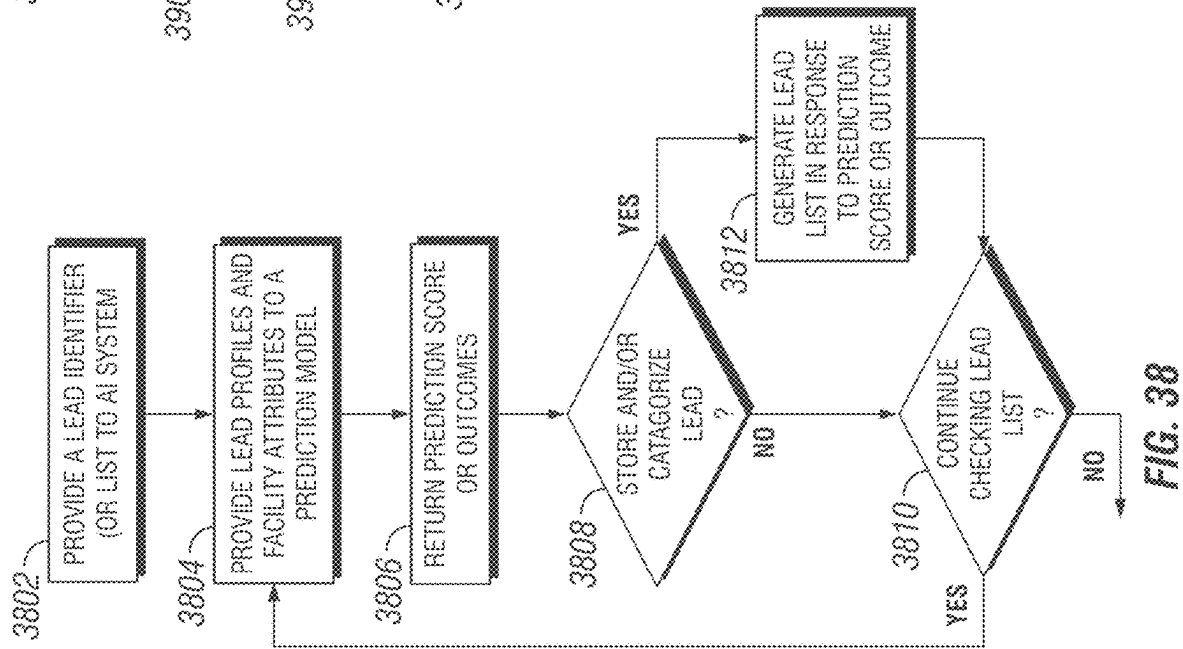
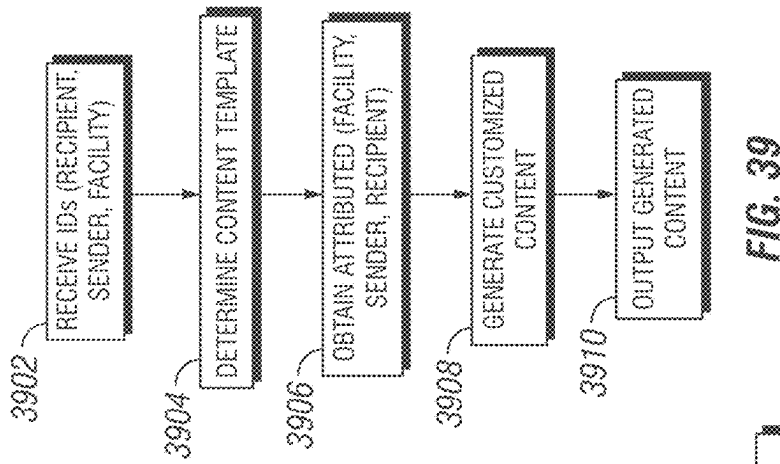


FIG. 36



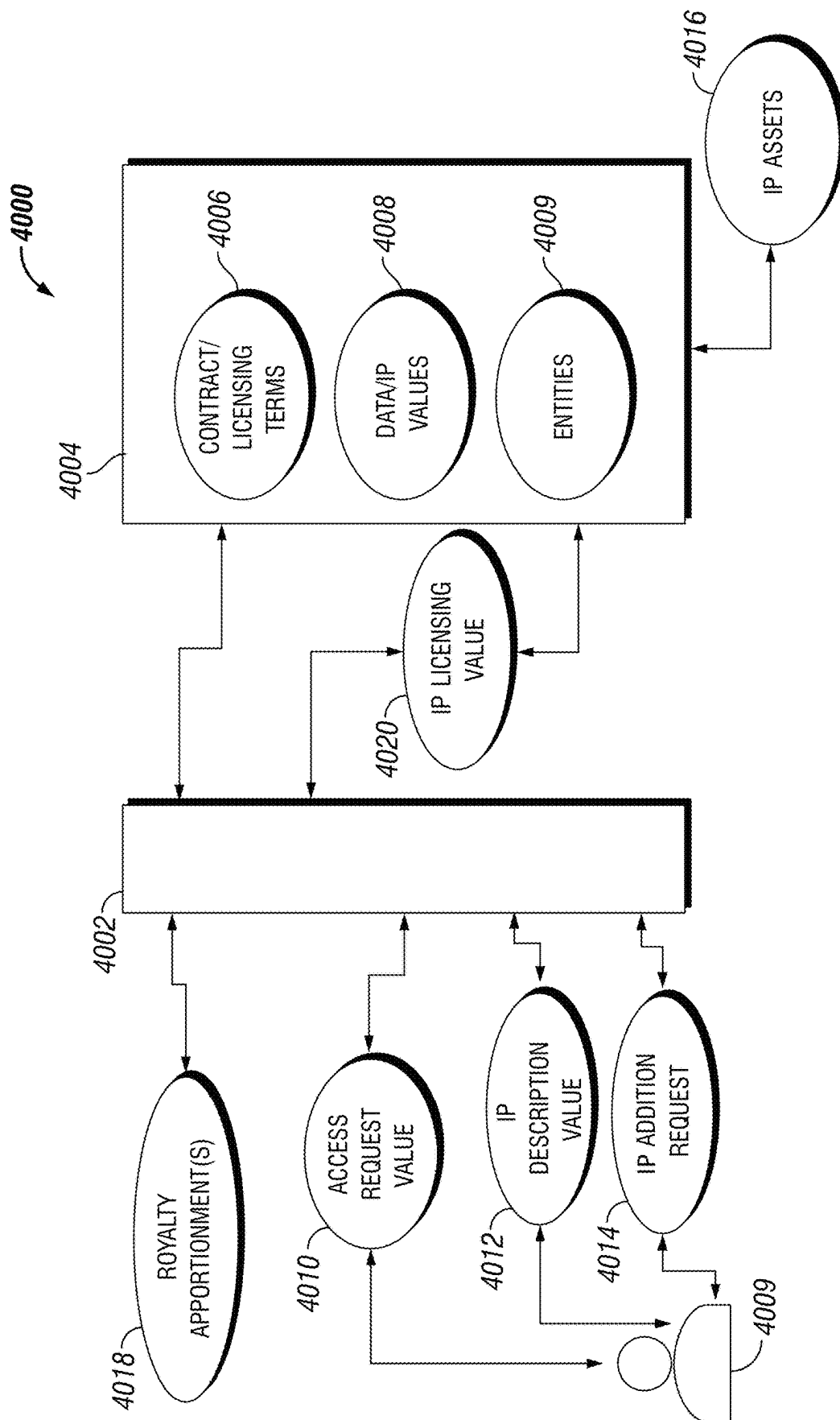


FIG. 40

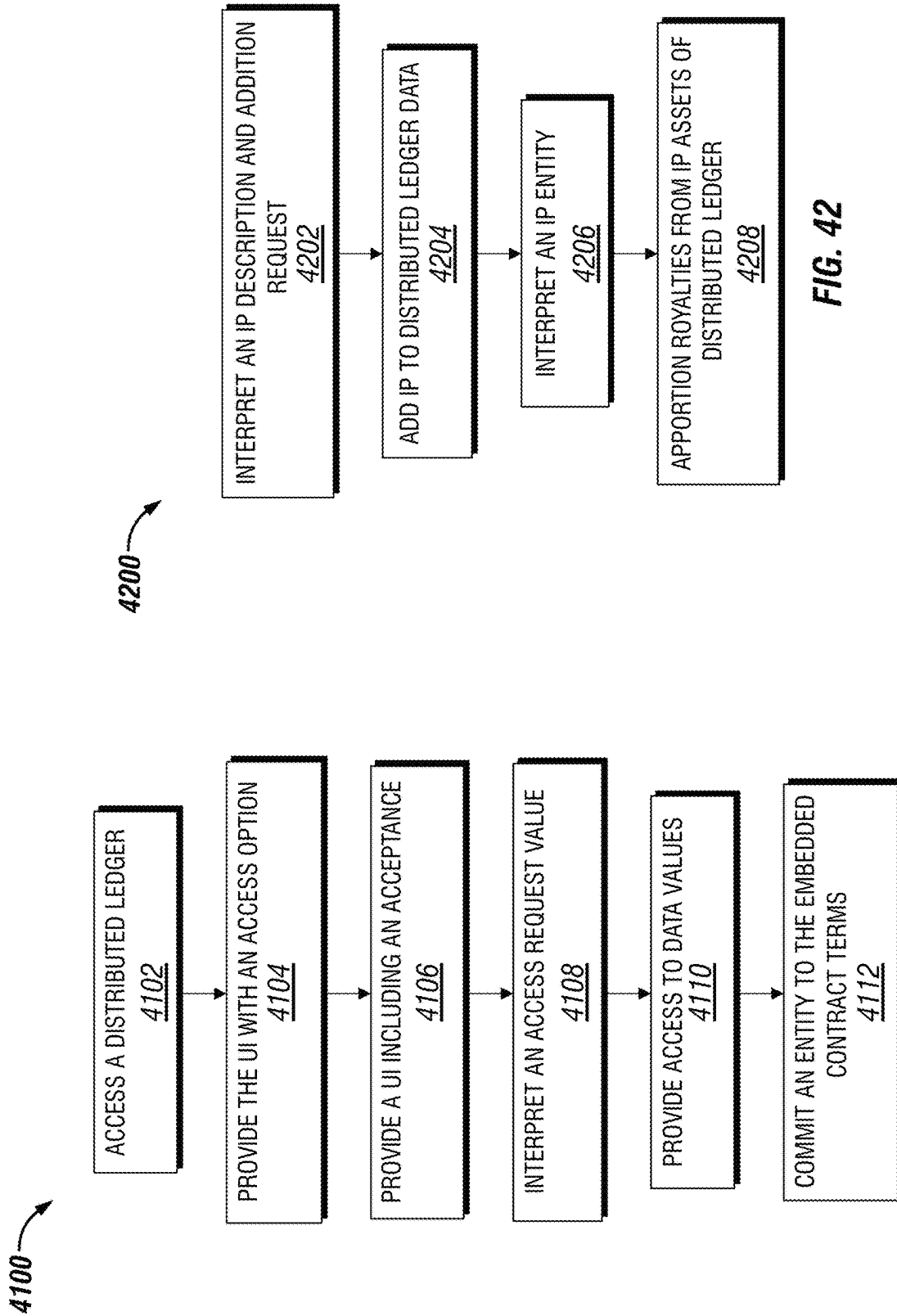


FIG. 41

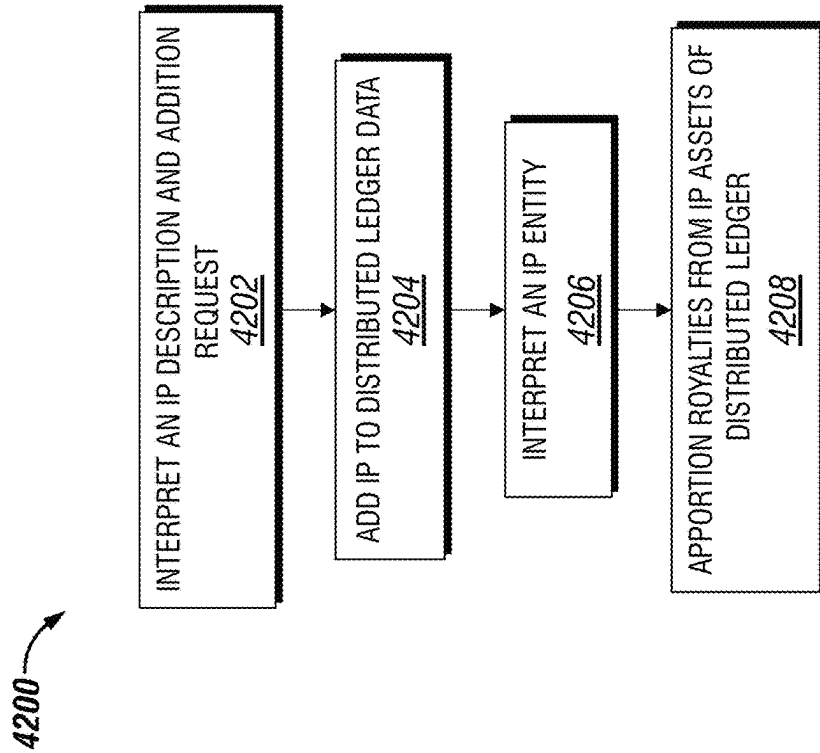


FIG. 42

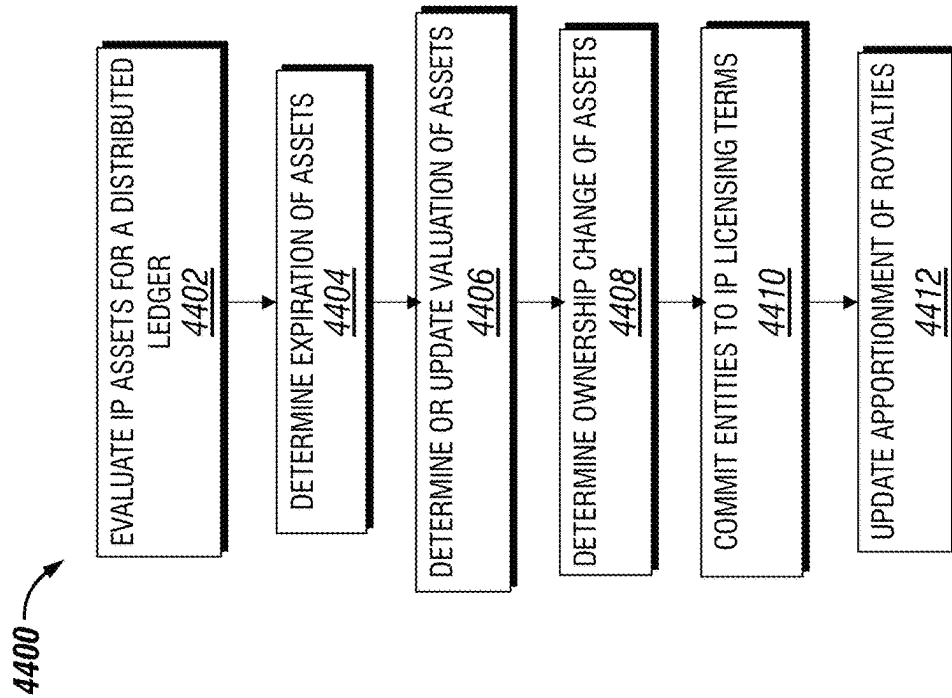


FIG. 44

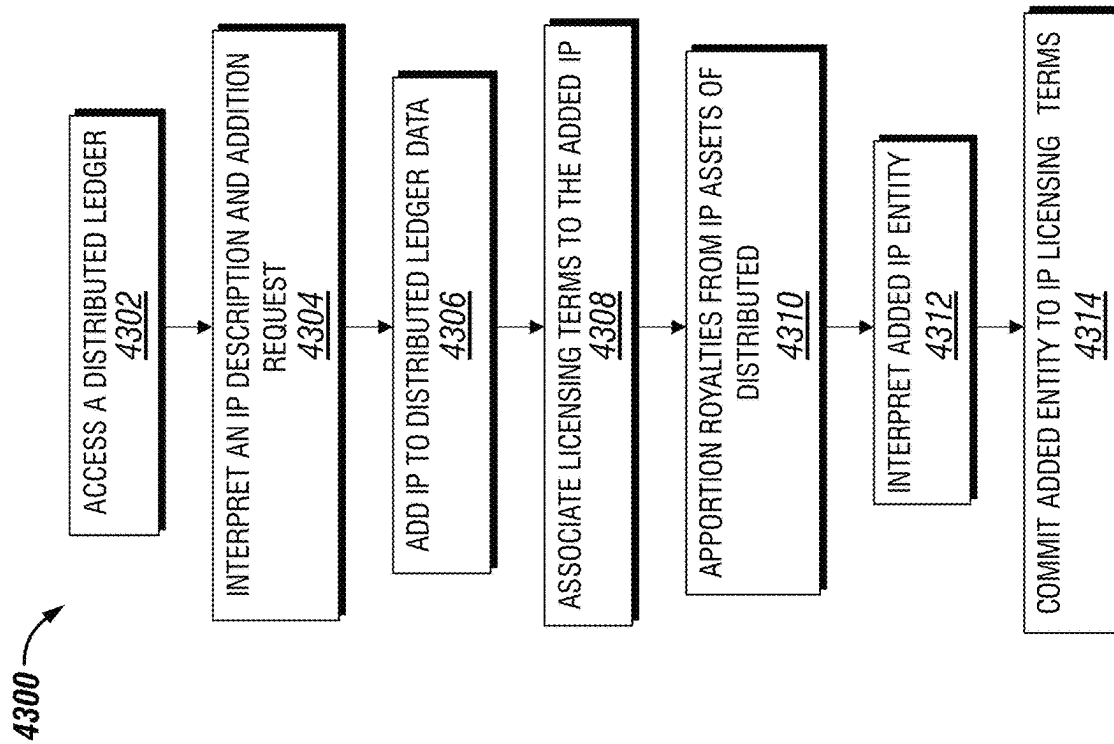


FIG. 43

4500

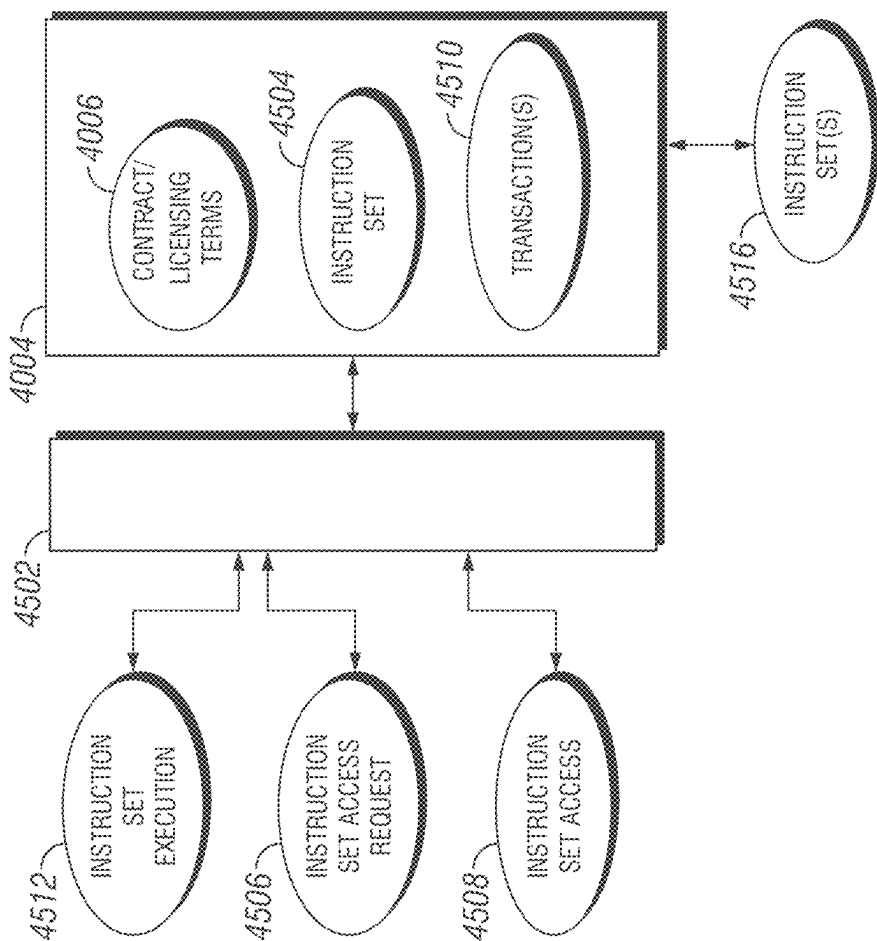


FIG. 45

4600

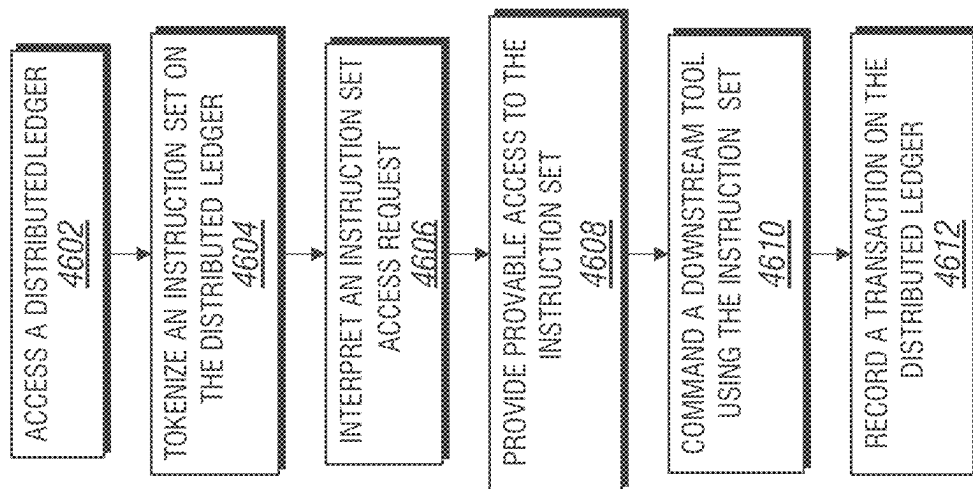


FIG. 46

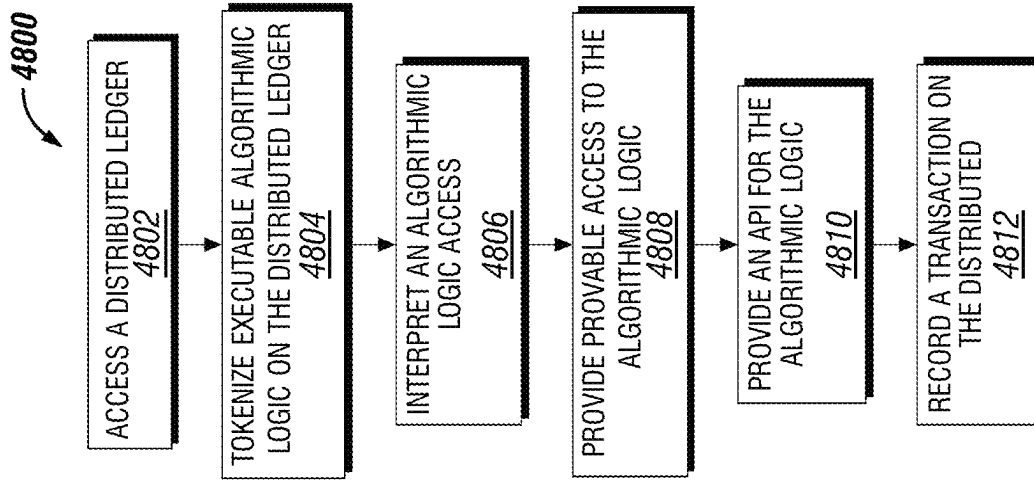


FIG. 48

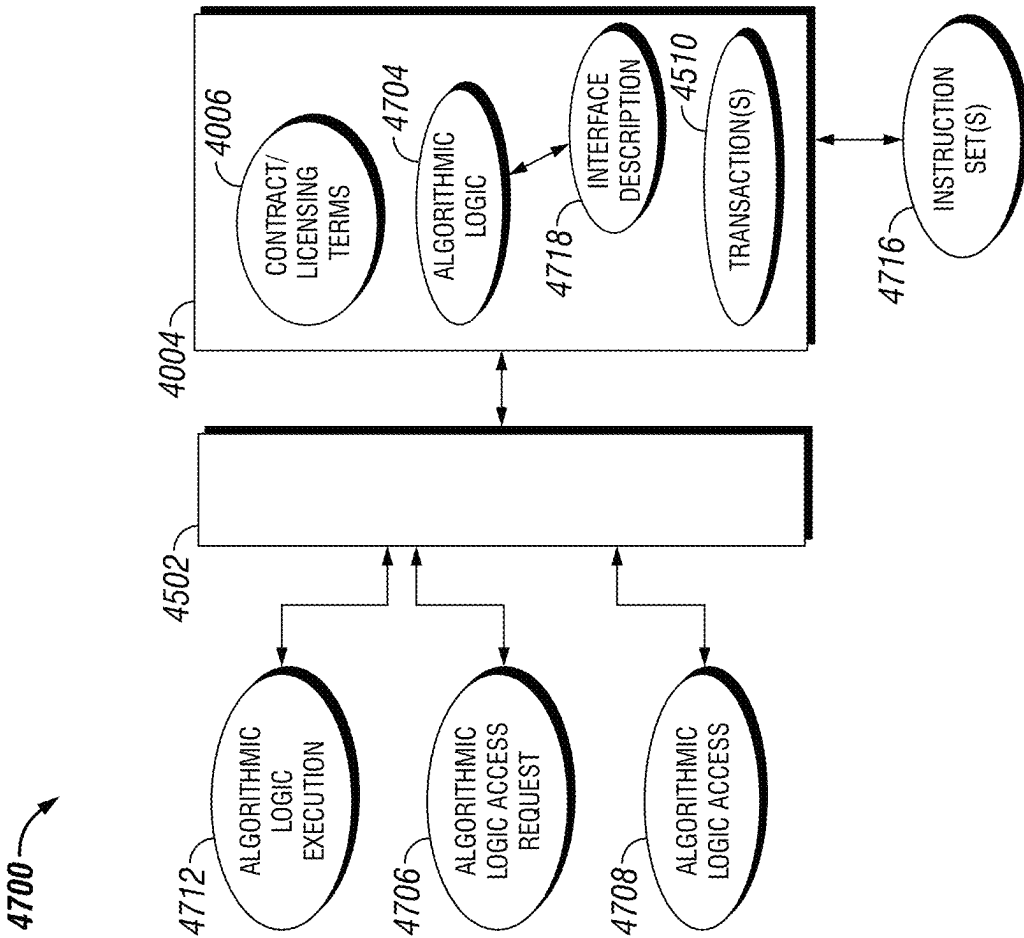


FIG. 47

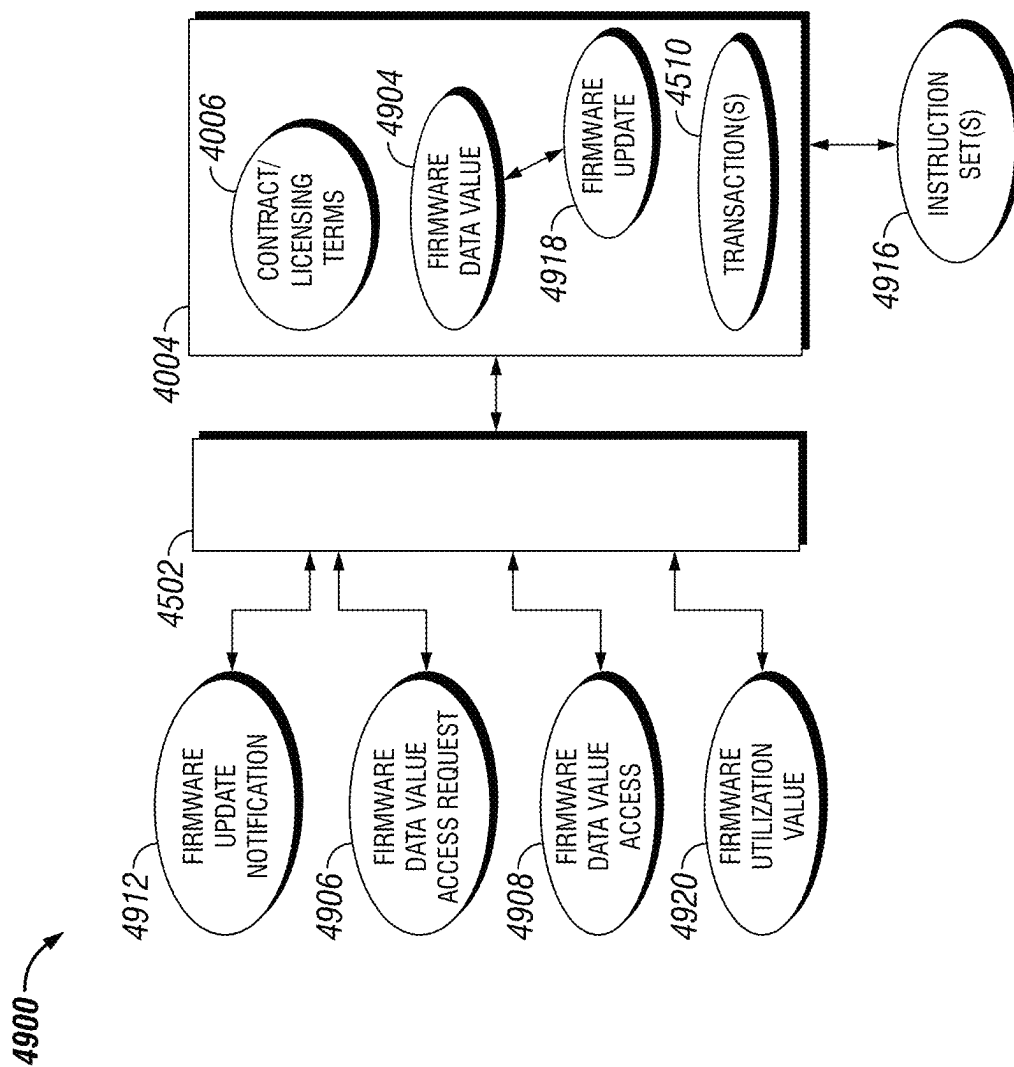


FIG. 49

5000

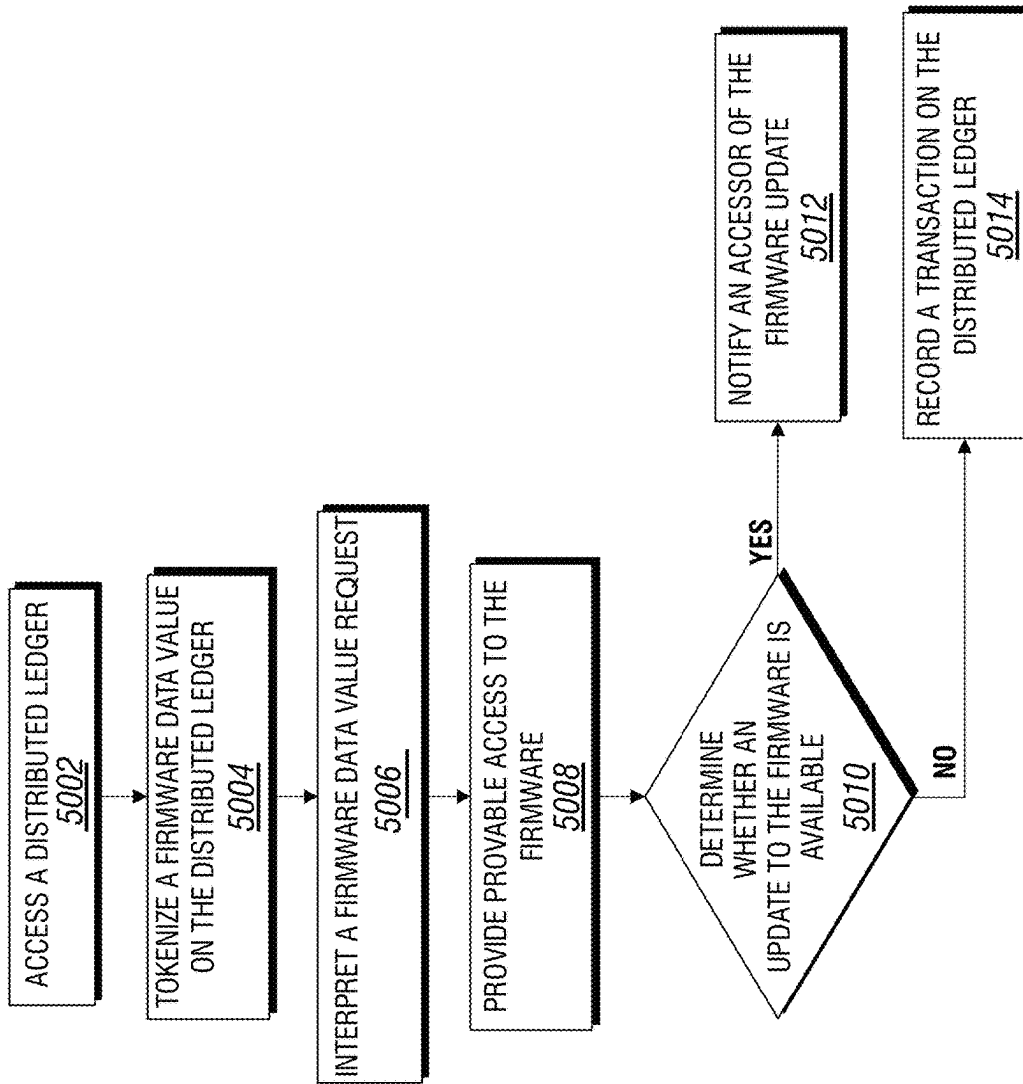


FIG. 50

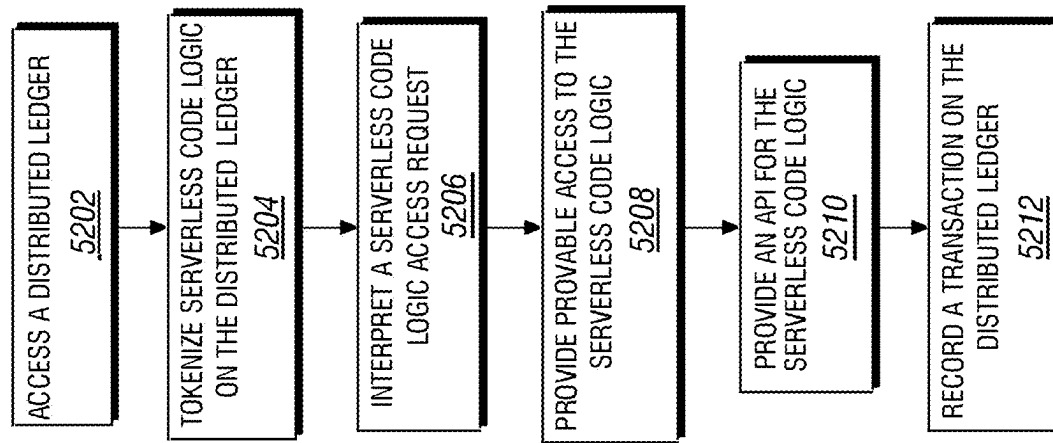


FIG. 52

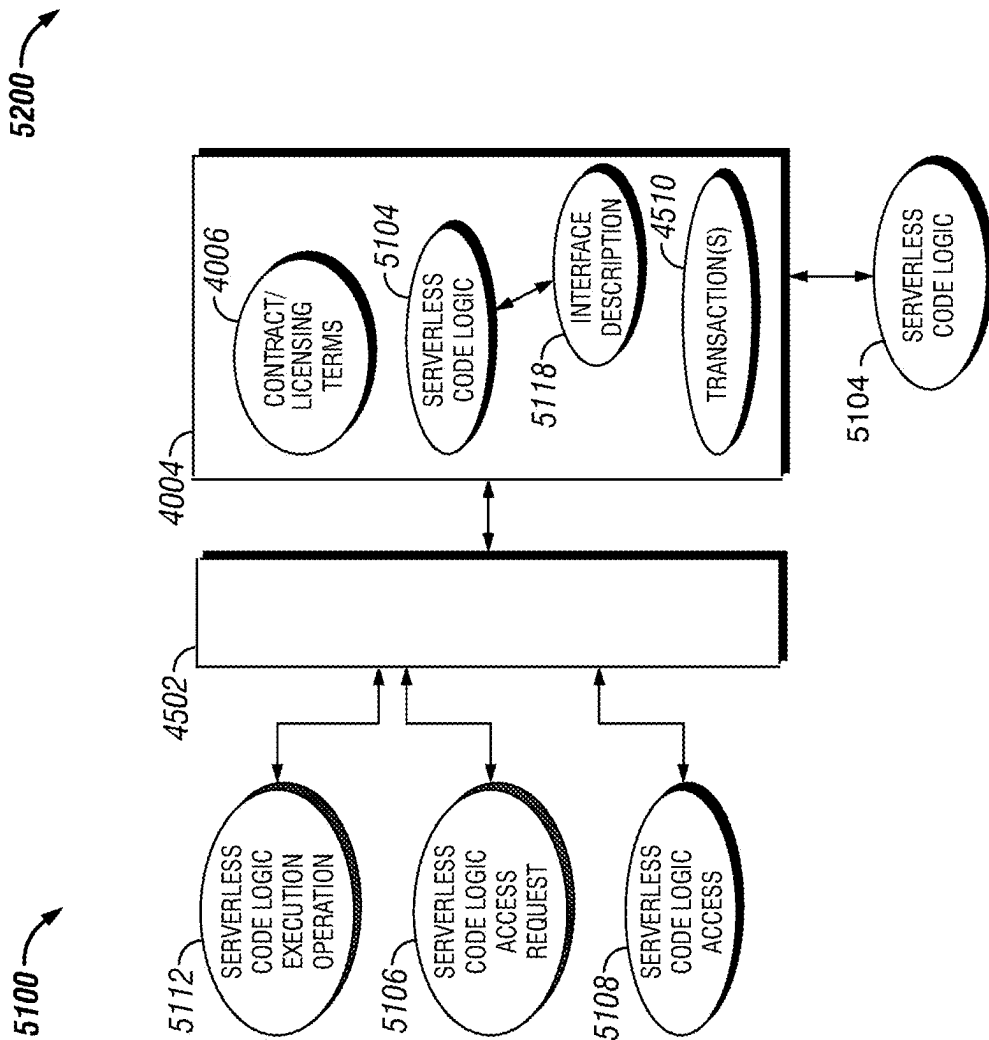


FIG. 51

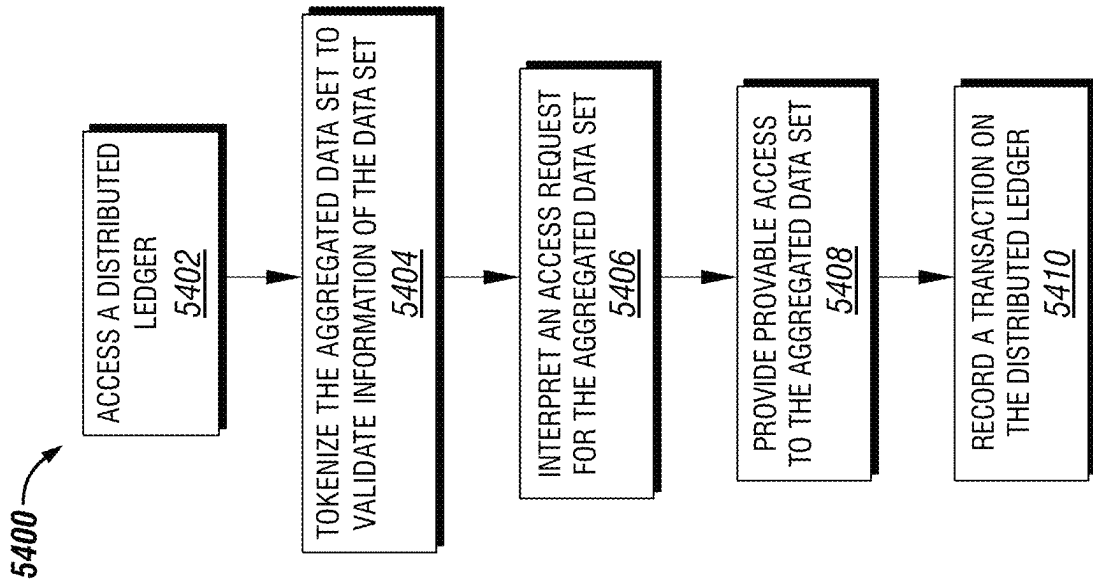


FIG. 54

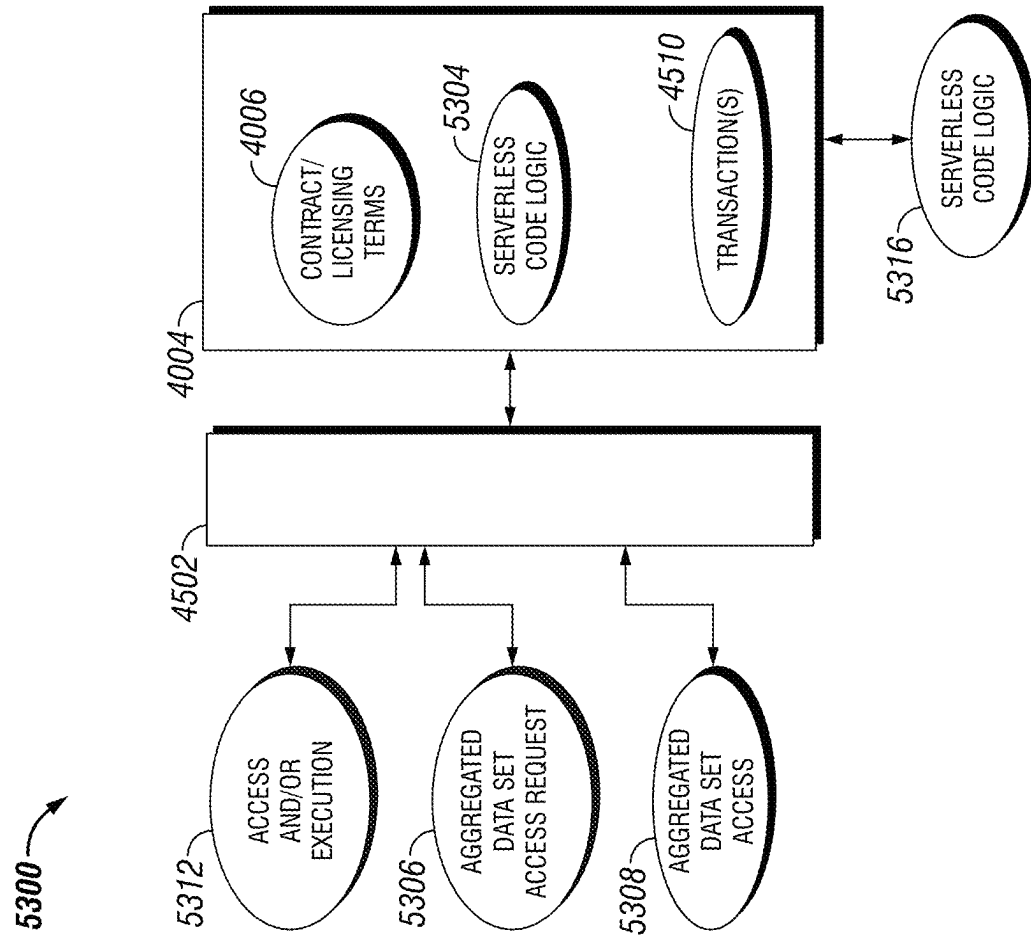


FIG. 53

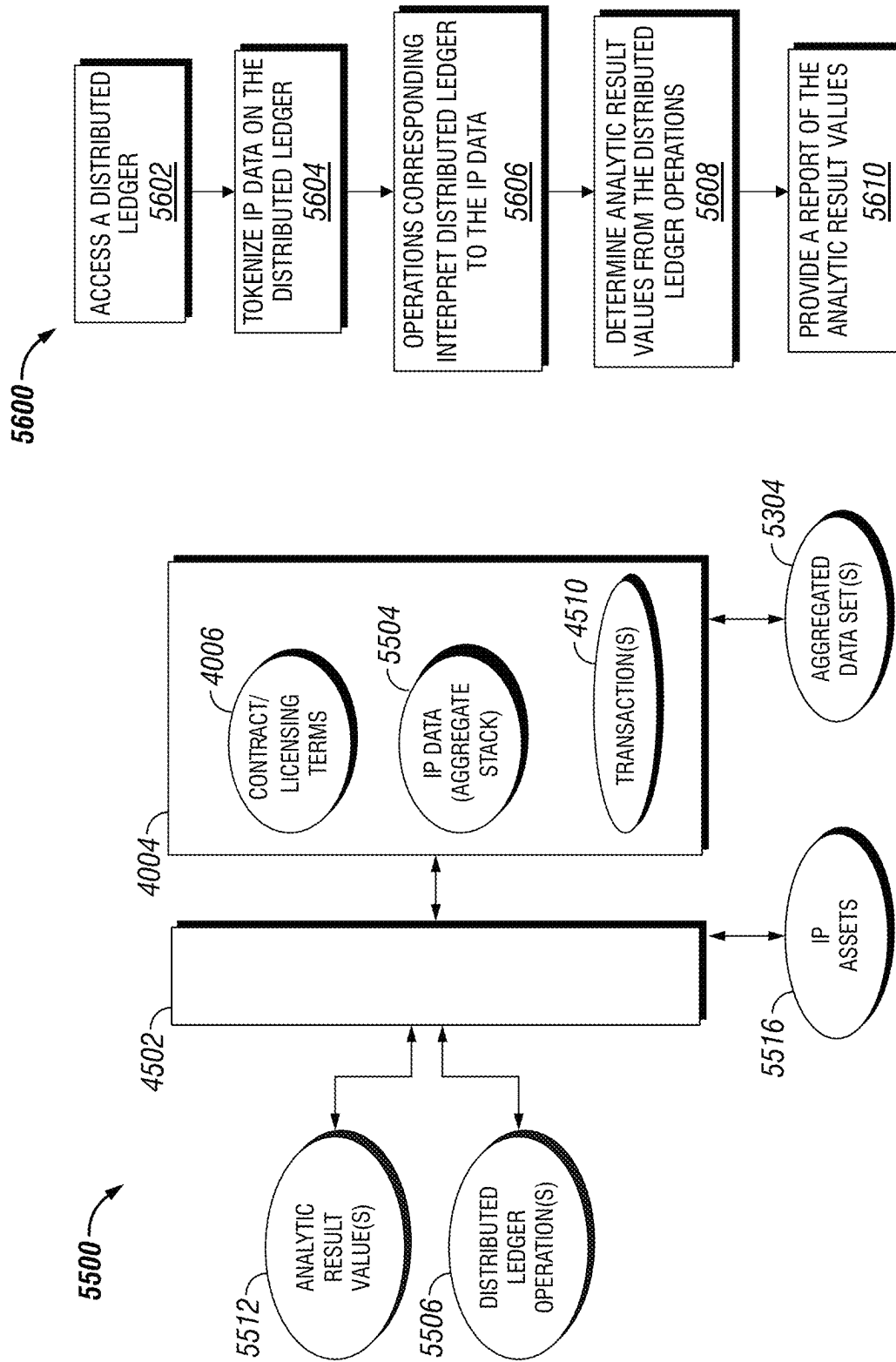


FIG. 56

FIG. 55

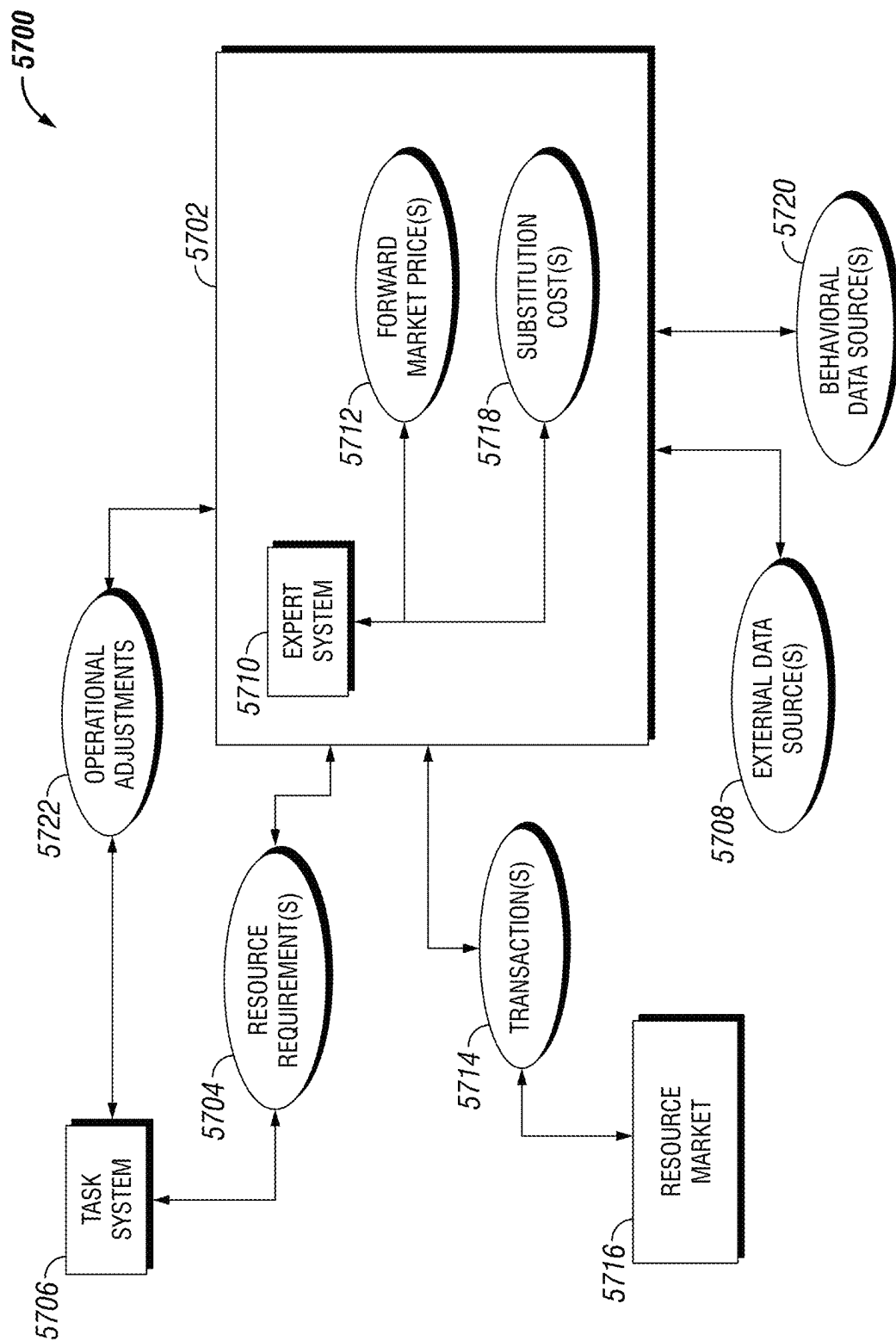
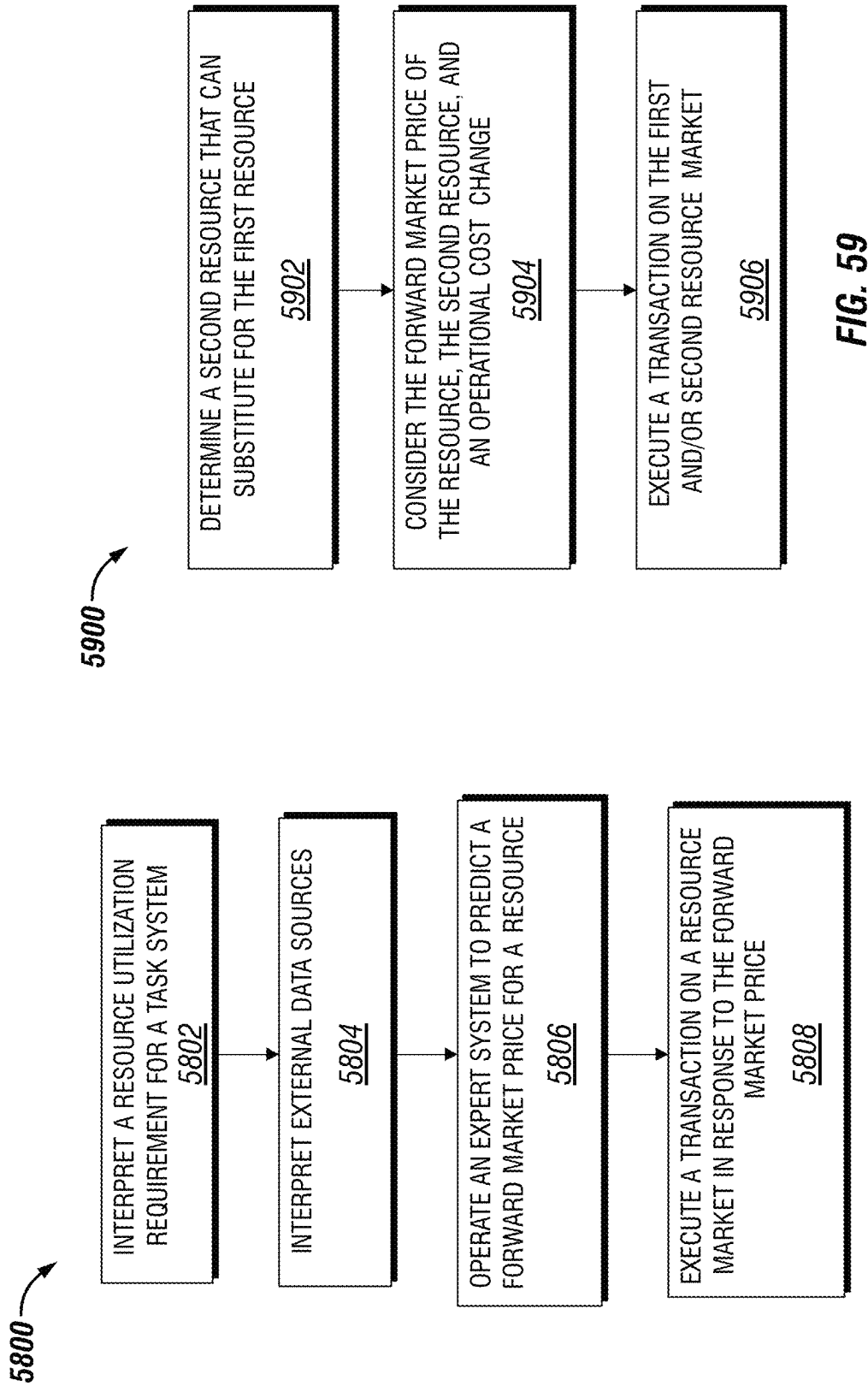


FIG. 57



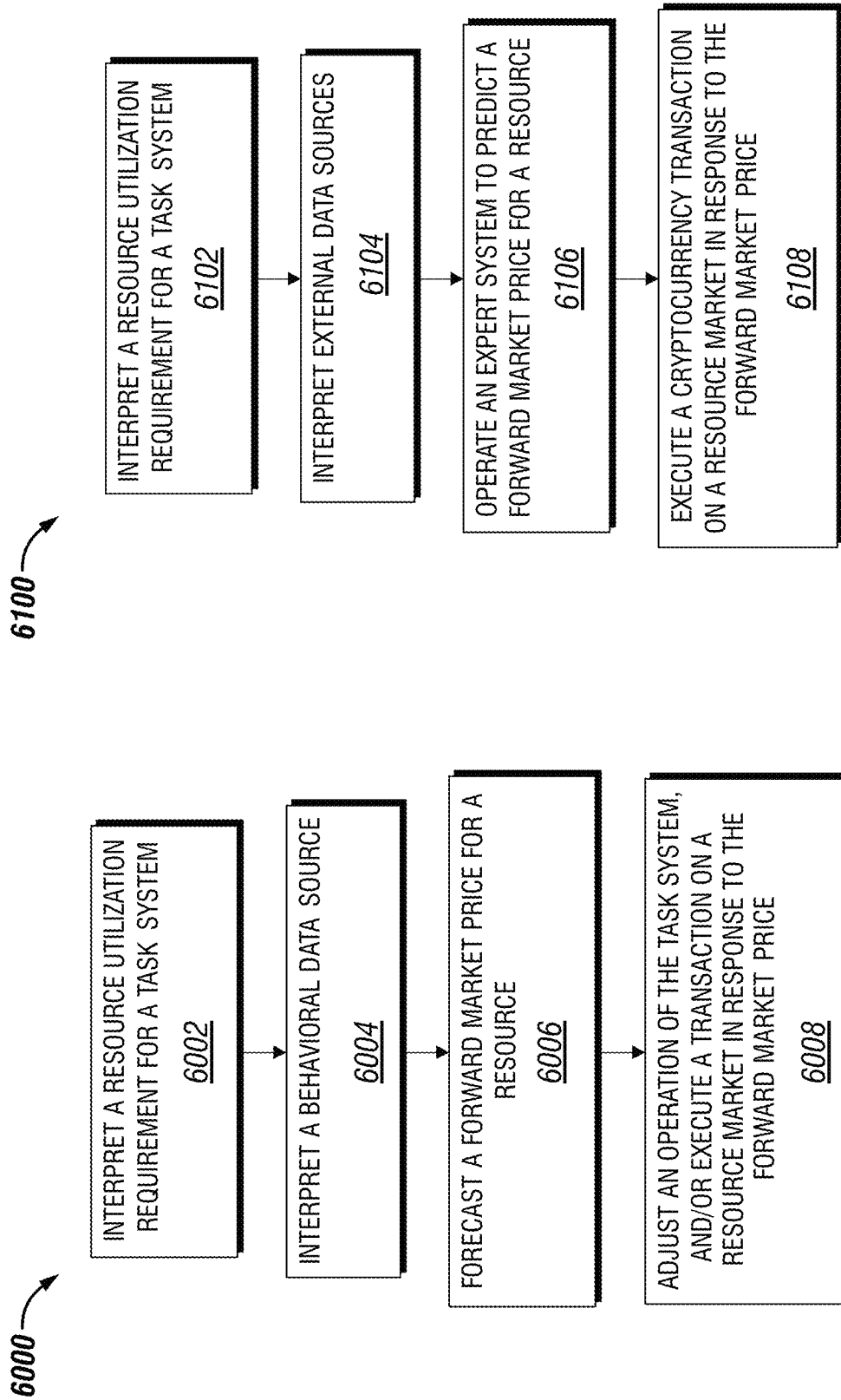


FIG. 60

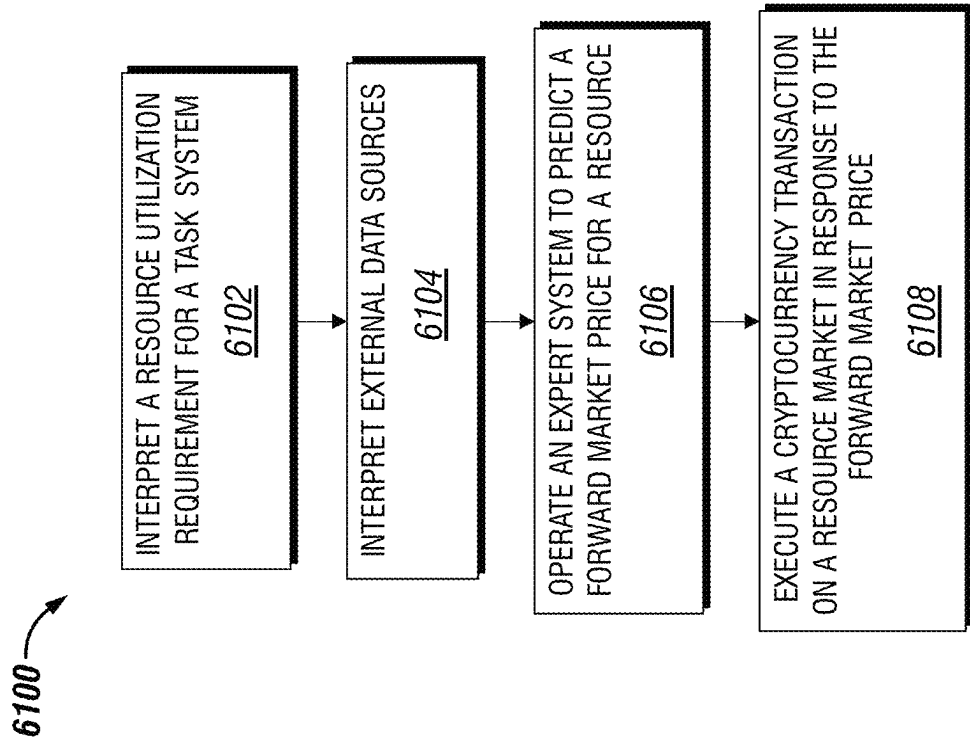


FIG. 61

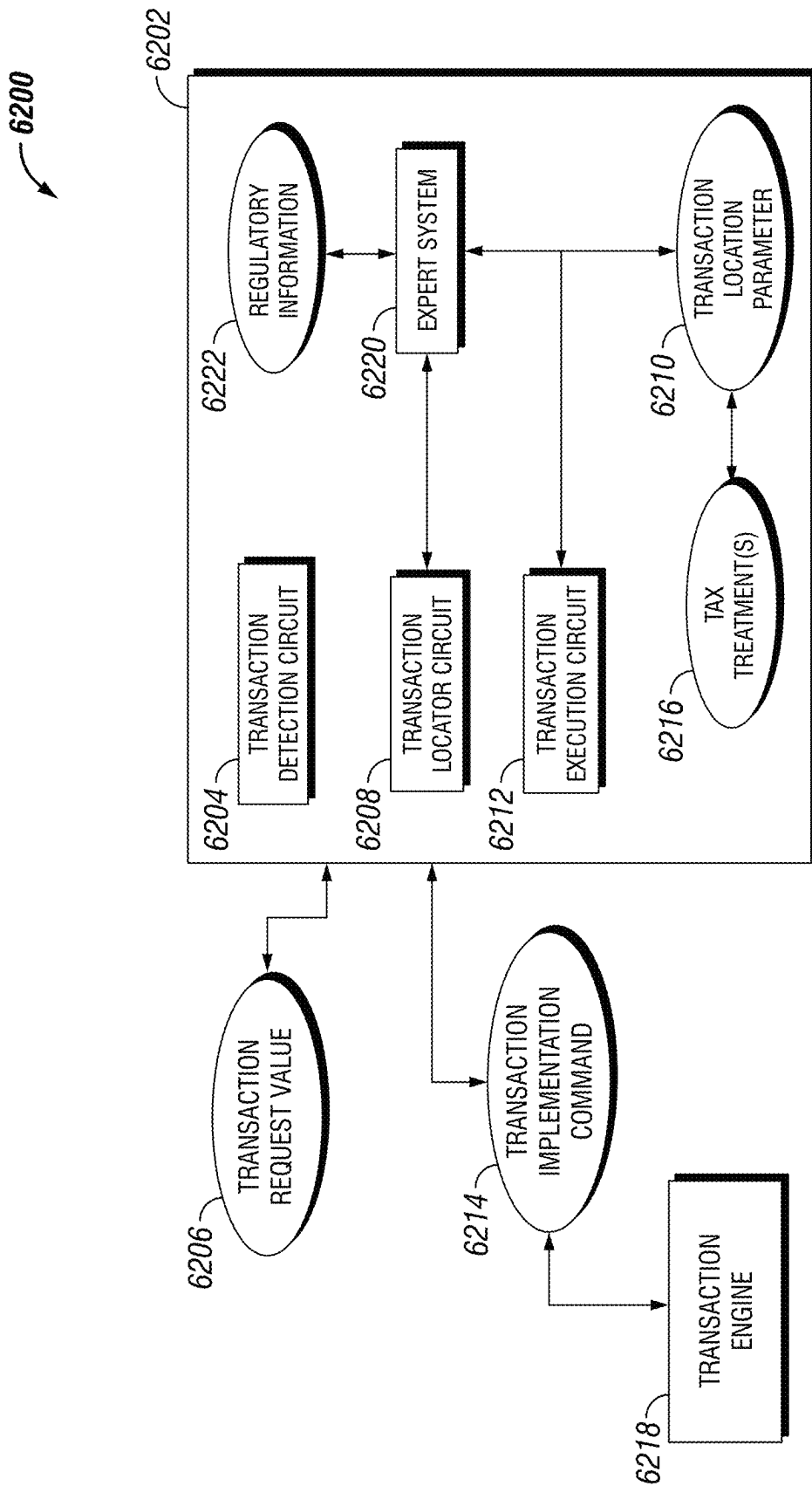
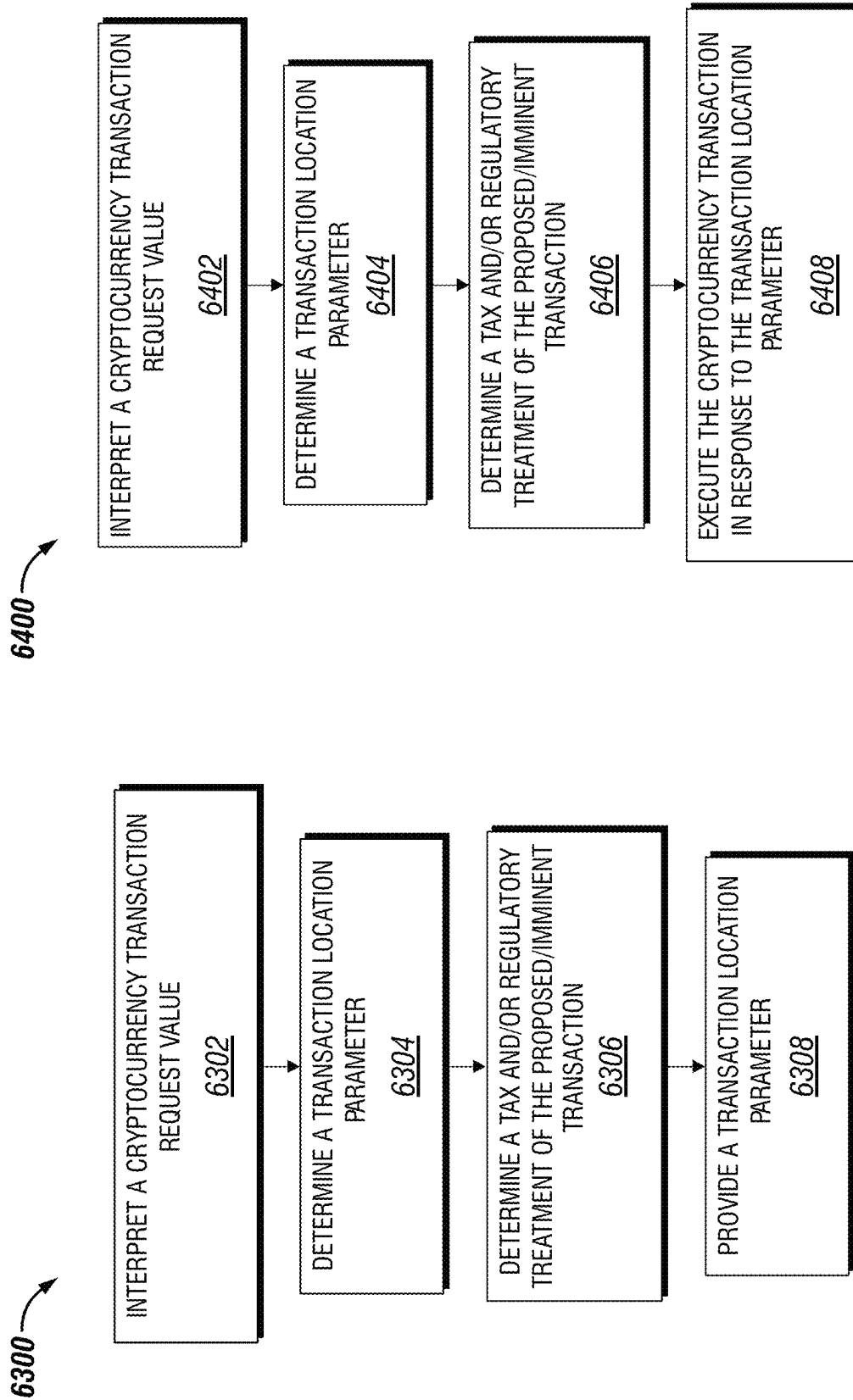


FIG. 62

**FIG. 63****FIG. 64**

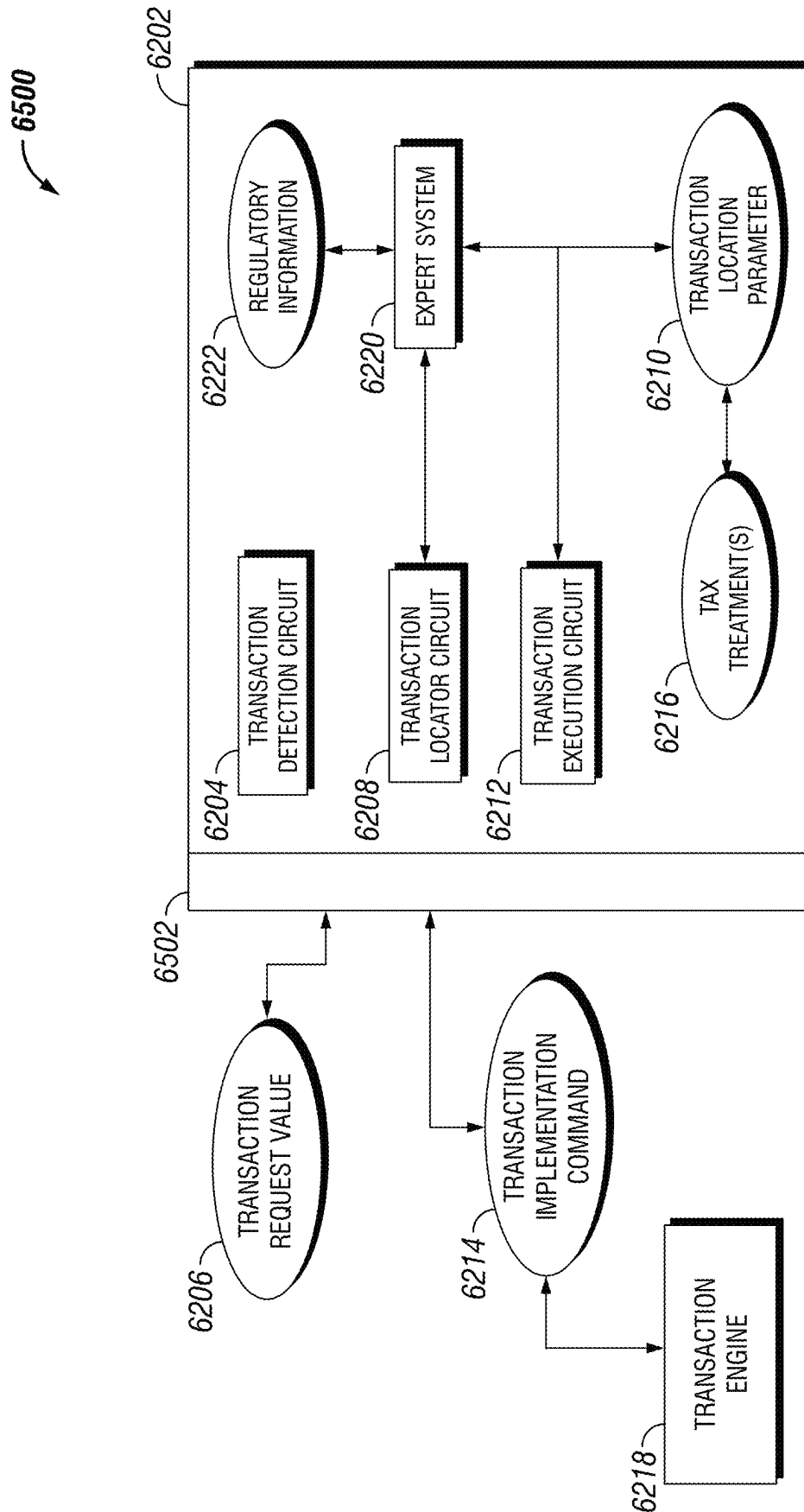
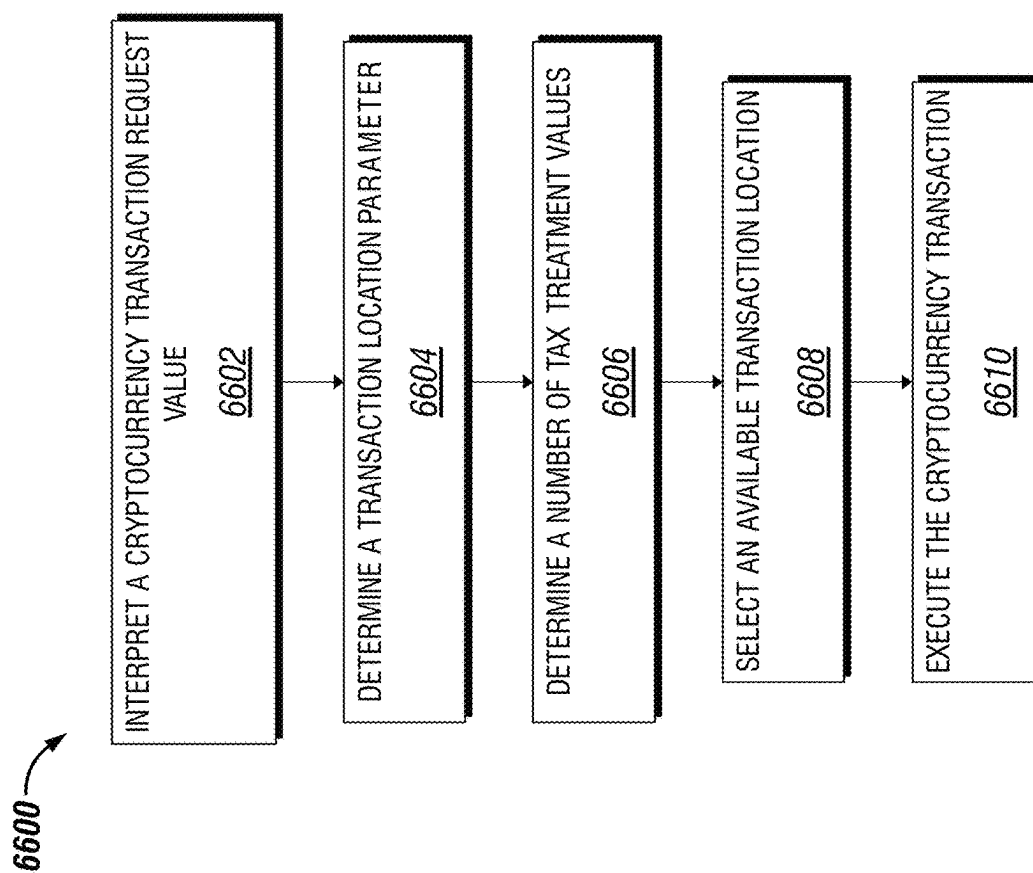


FIG. 65

**FIG. 66**

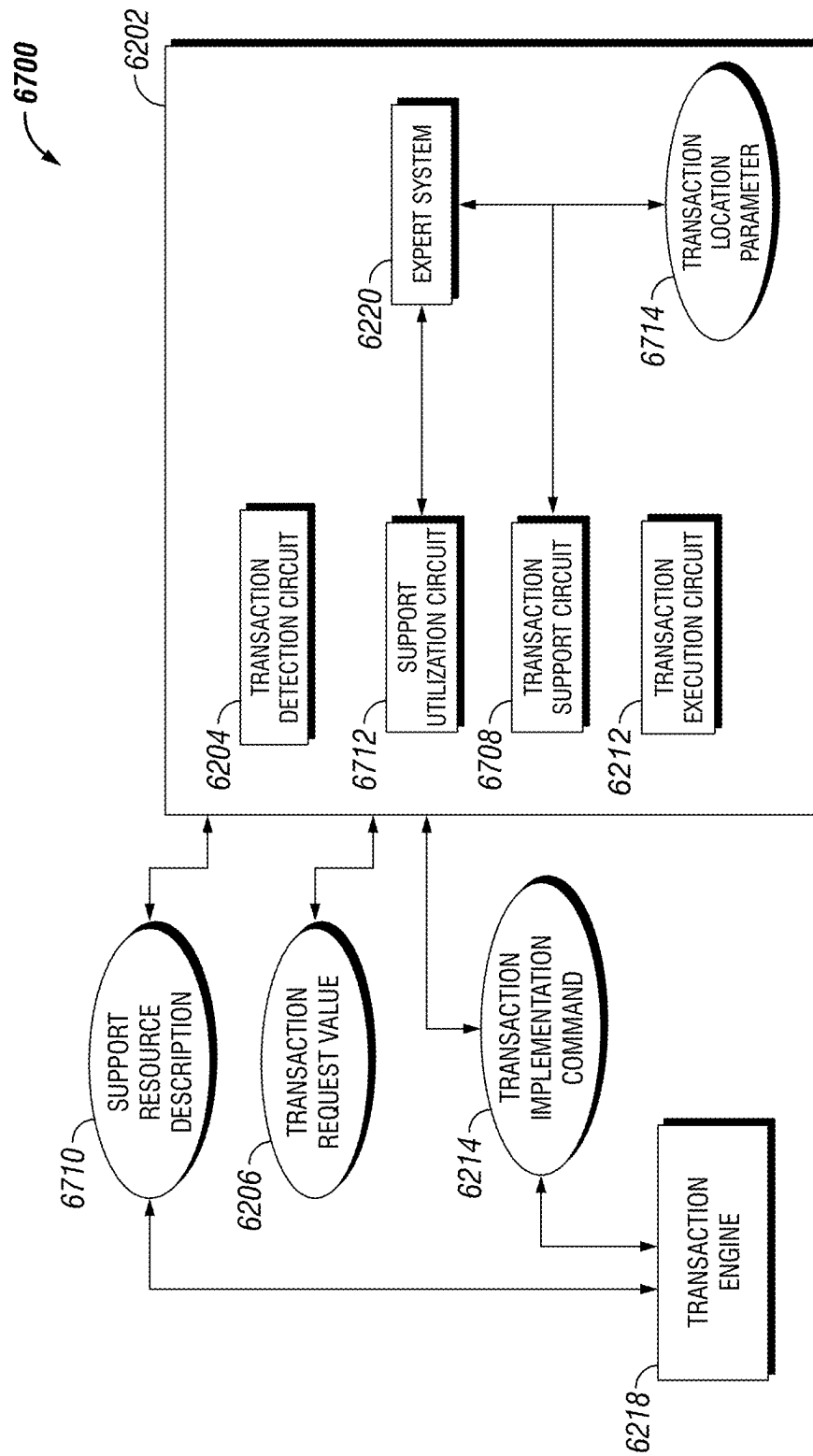


FIG. 67

6800 ↗

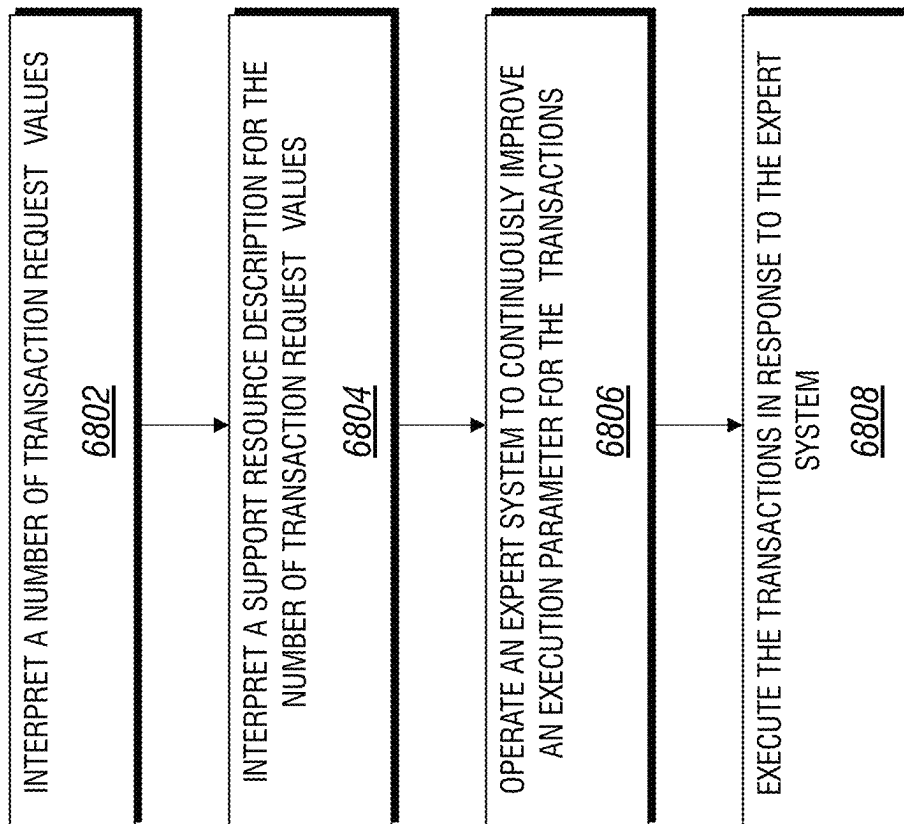


FIG. 68

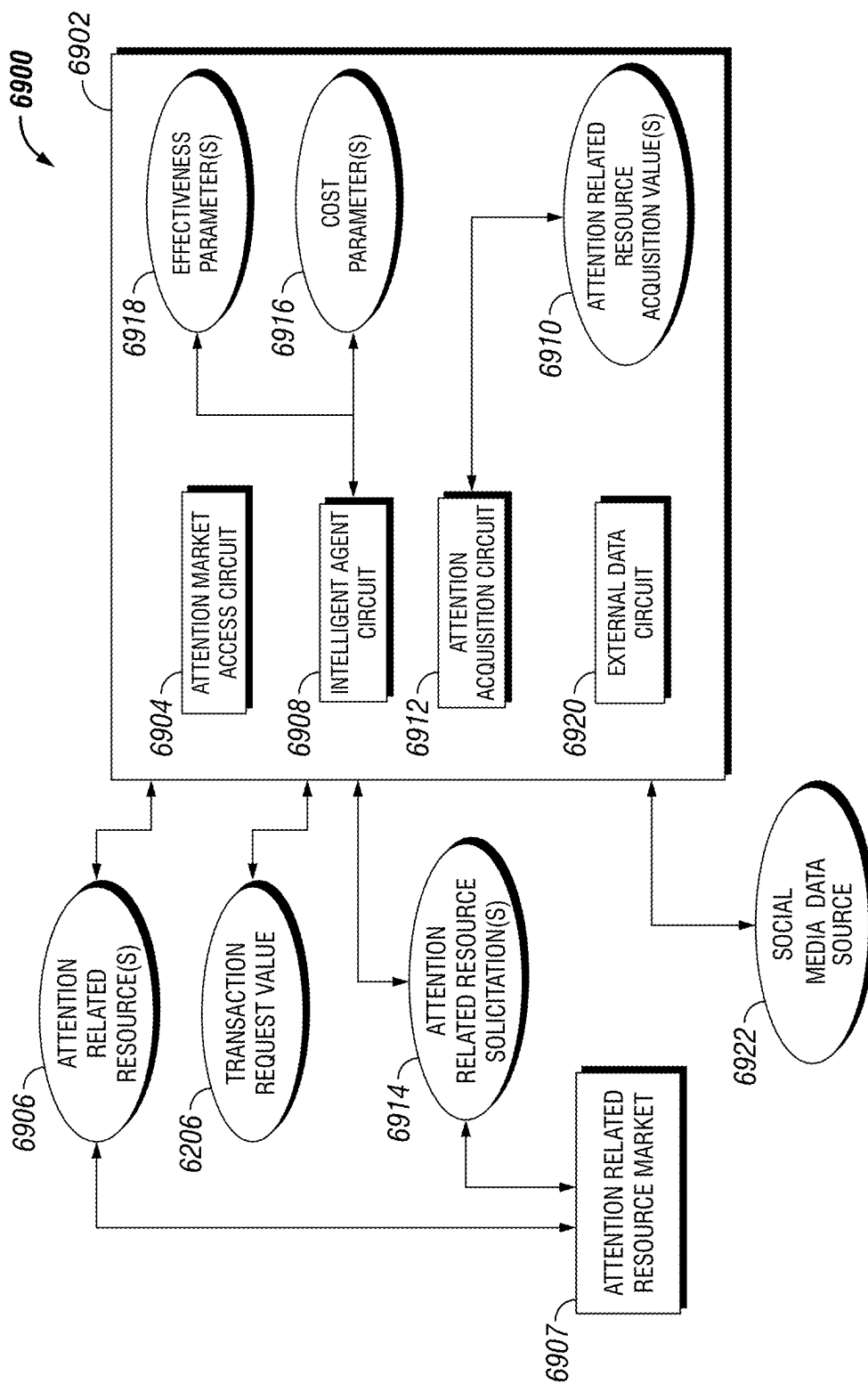


FIG. 69

7000 ↗

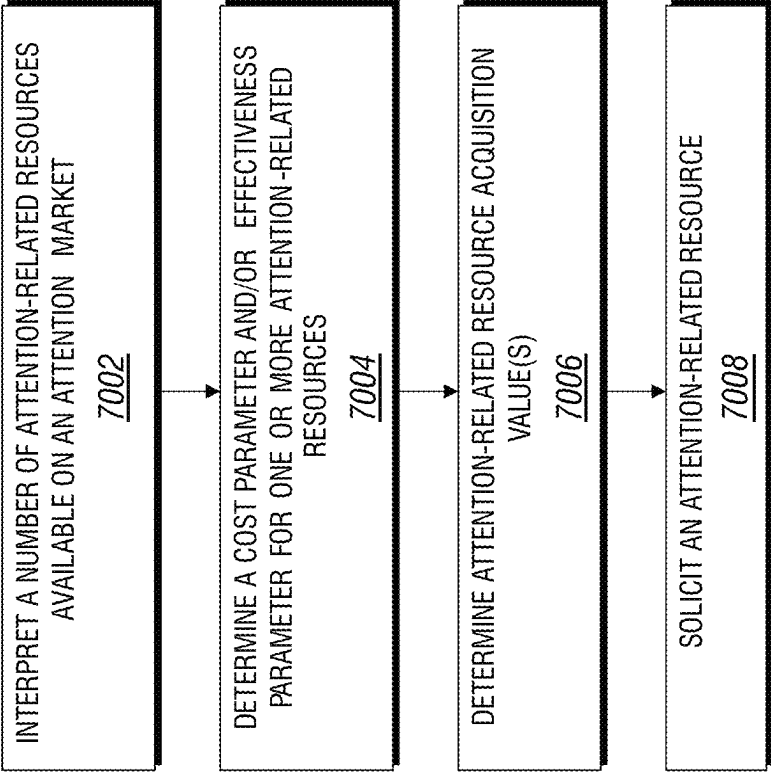


FIG. 70

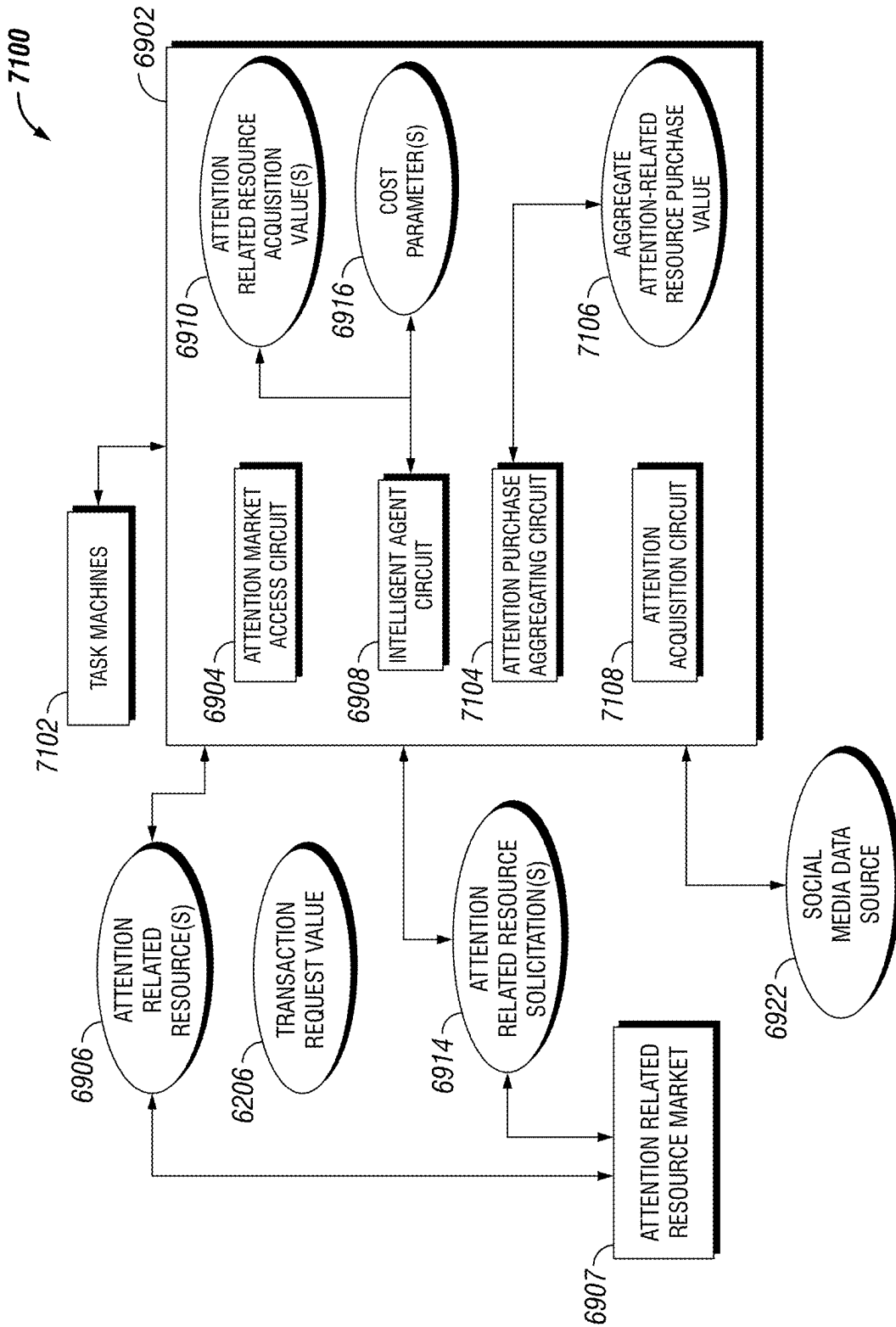


FIG. 71

7200 →

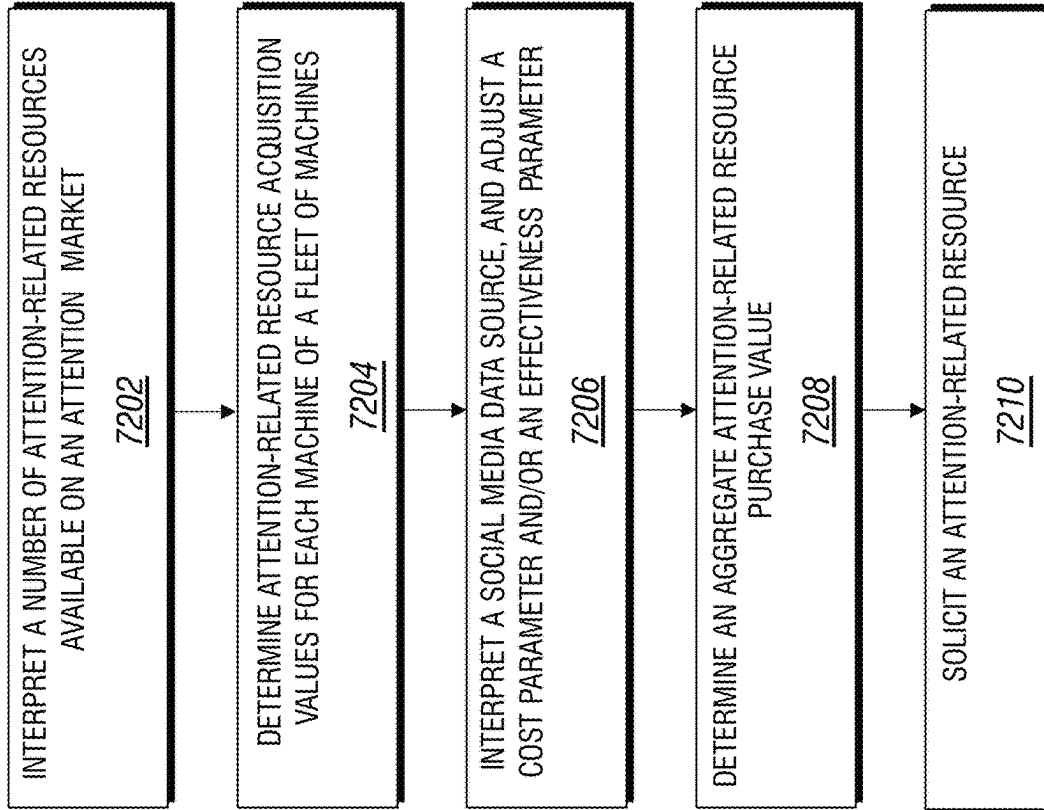


FIG. 72

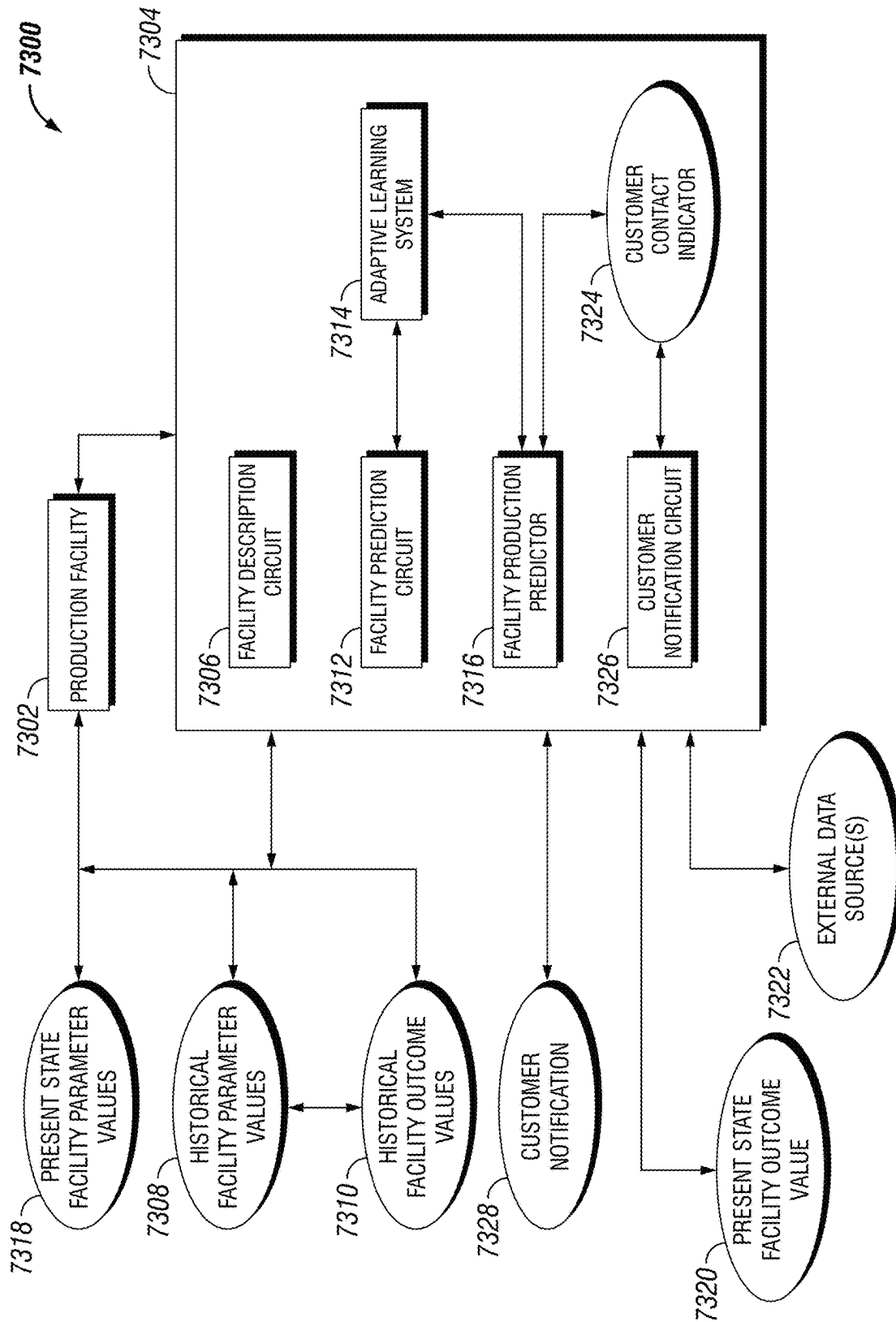


FIG. 73

7400 →

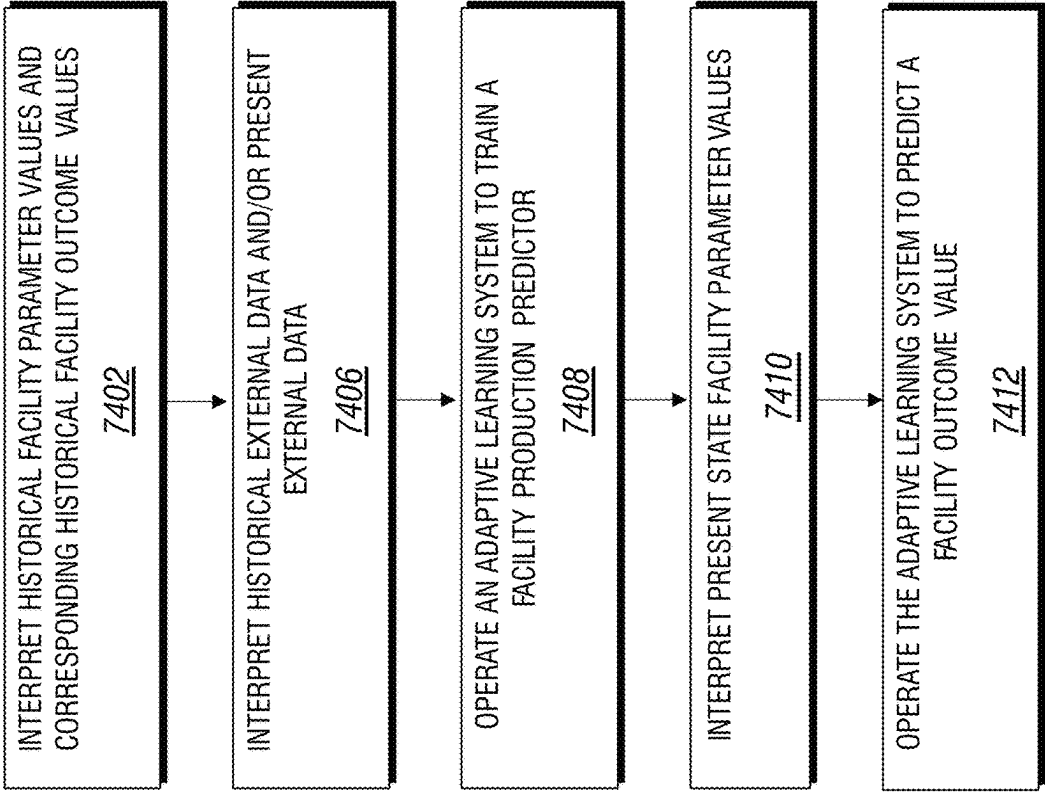


FIG. 74

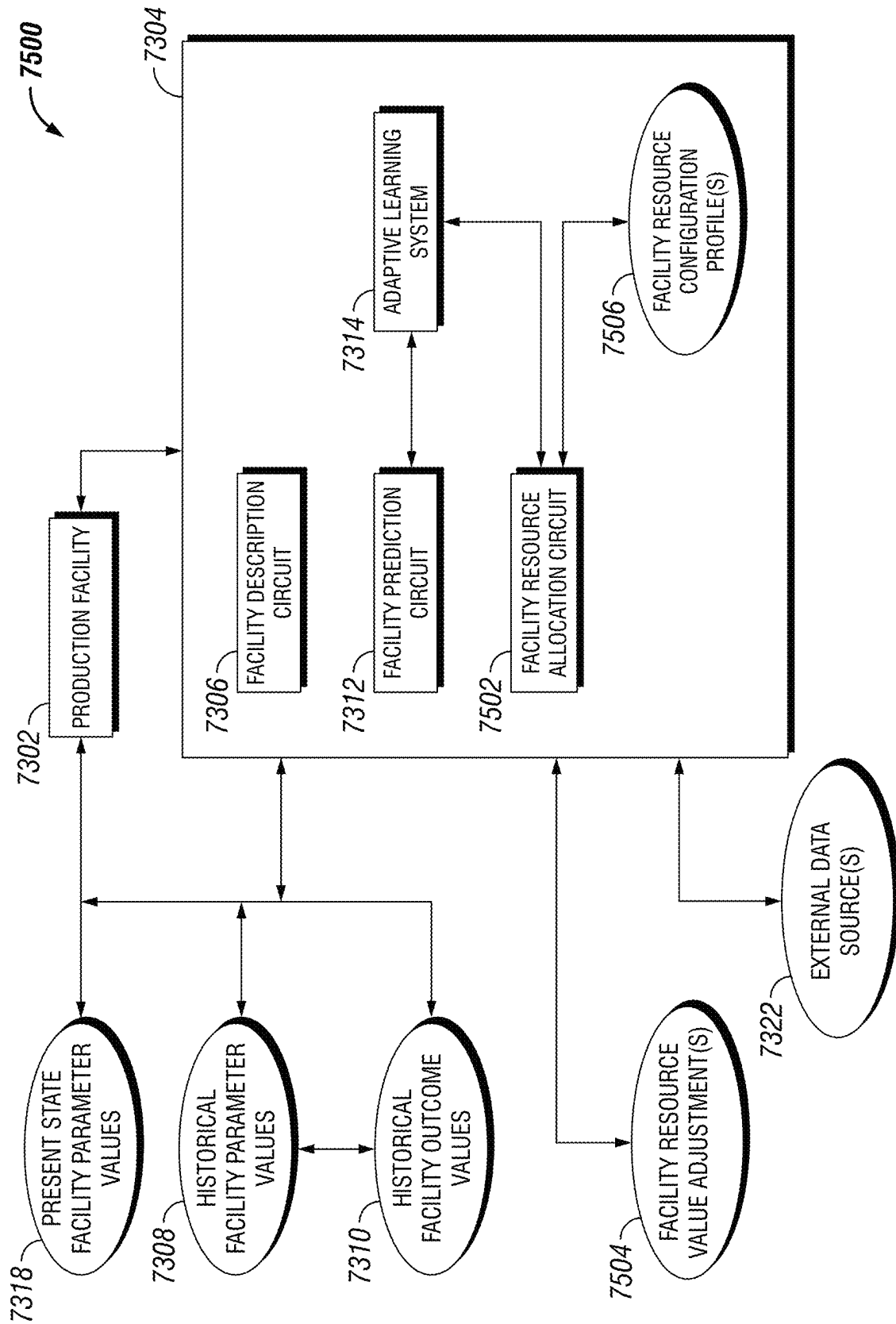


FIG. 75

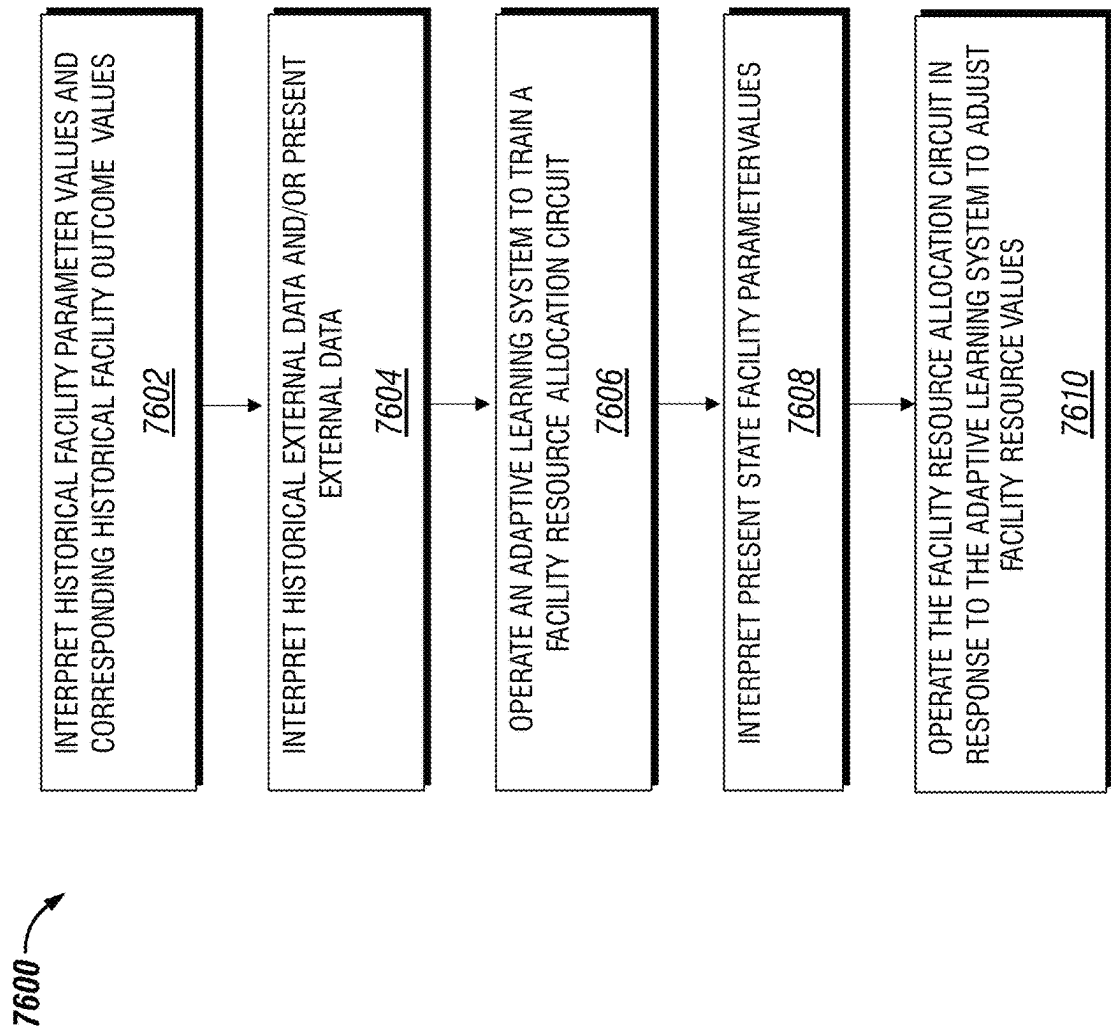
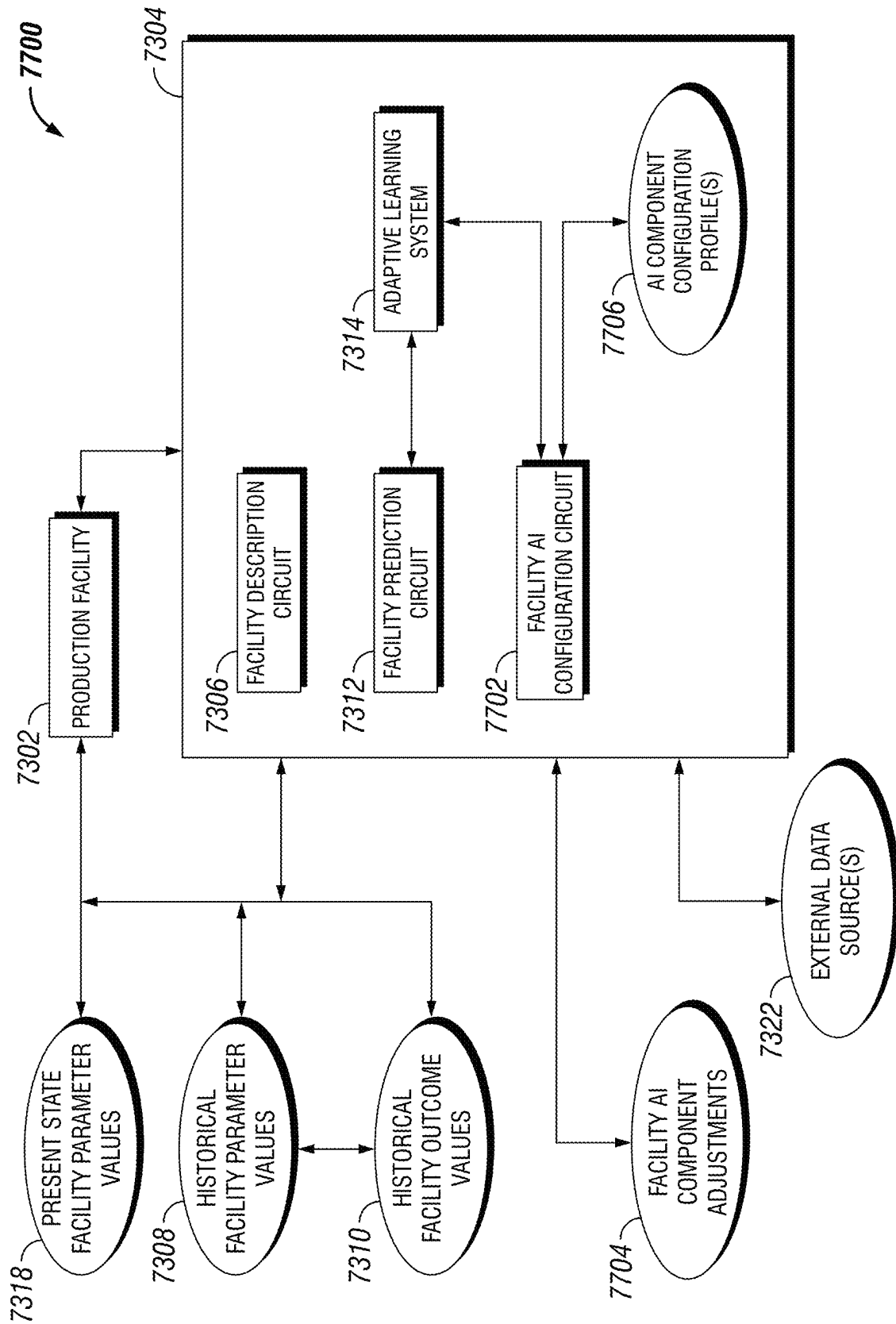


FIG. 76



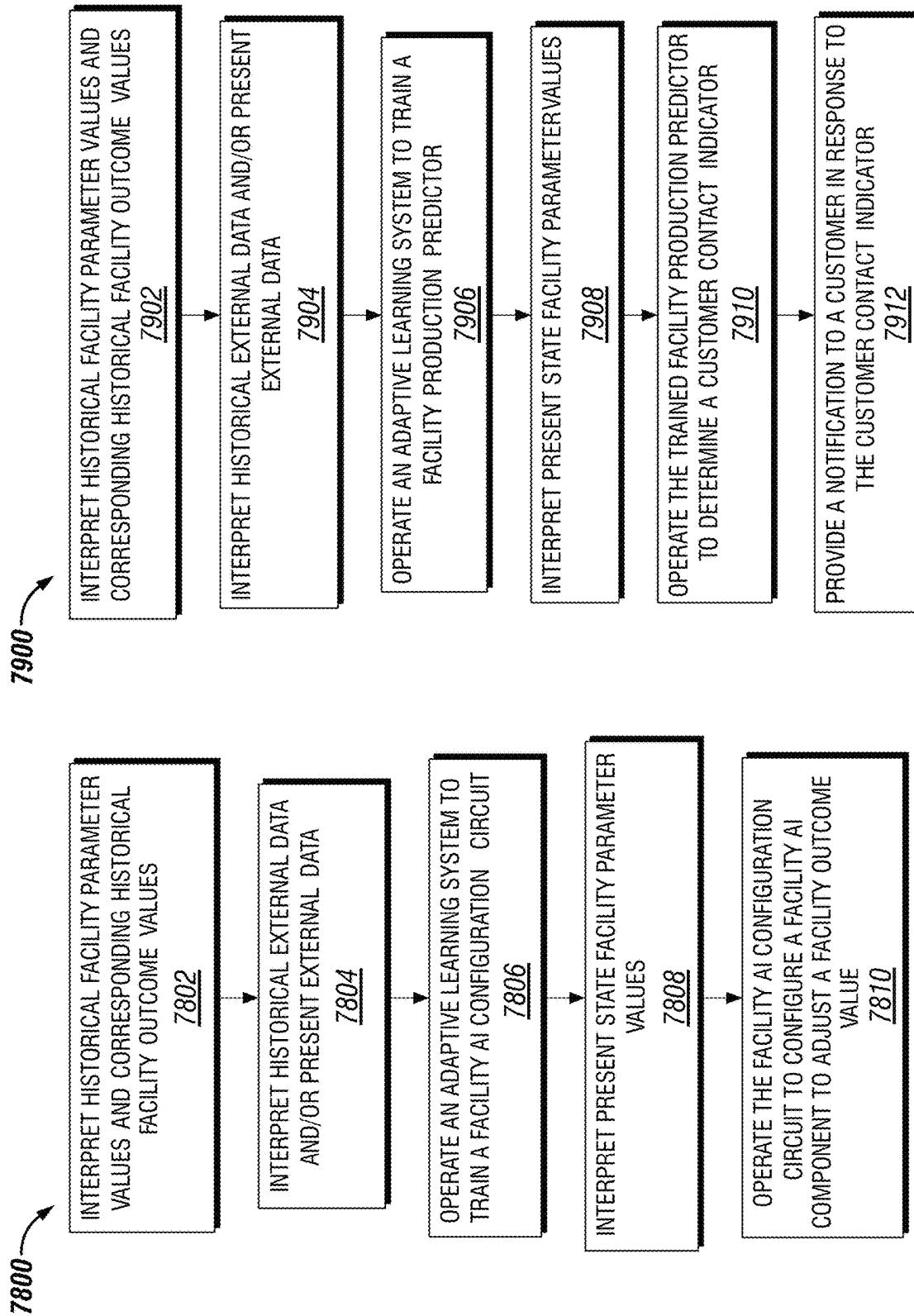


FIG. 78

FIG. 79

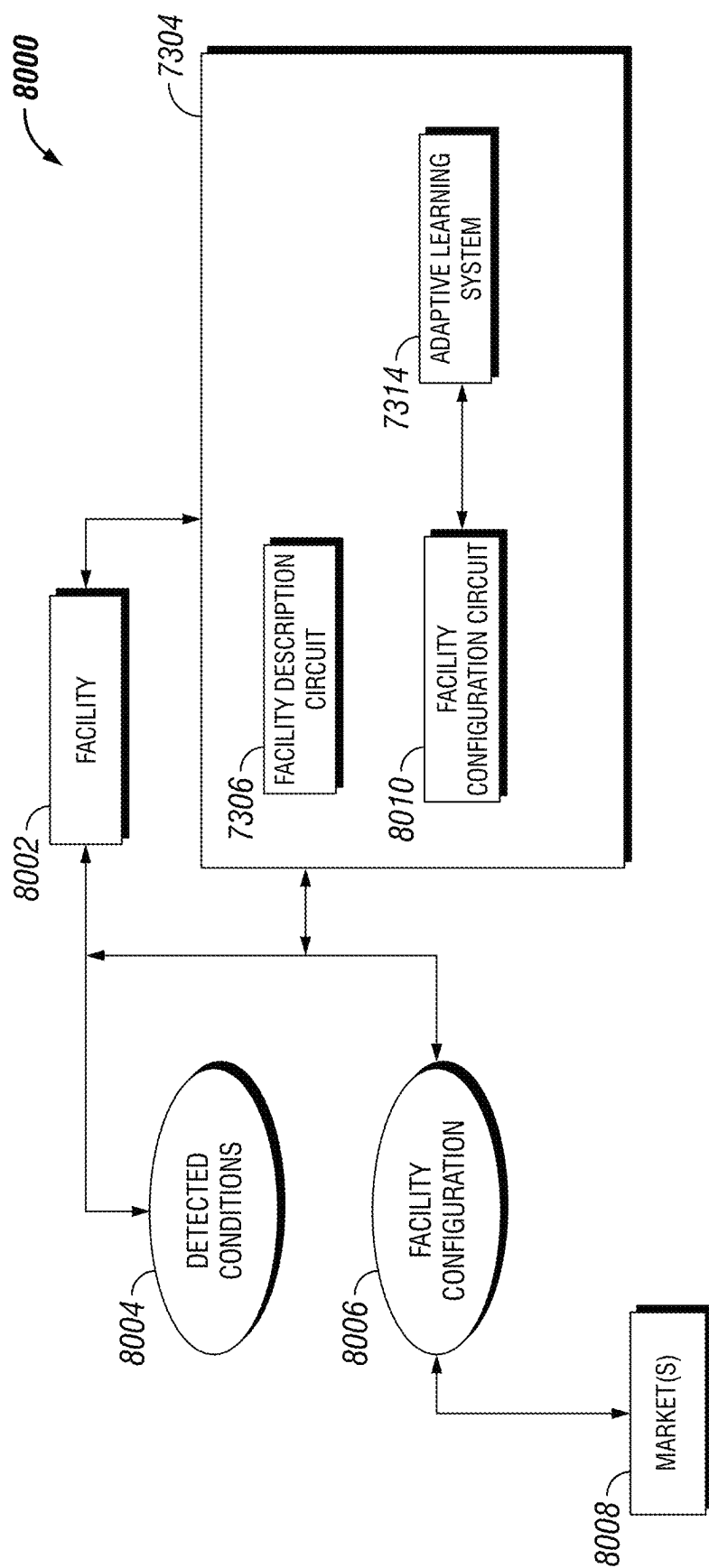


FIG. 80

8100 →

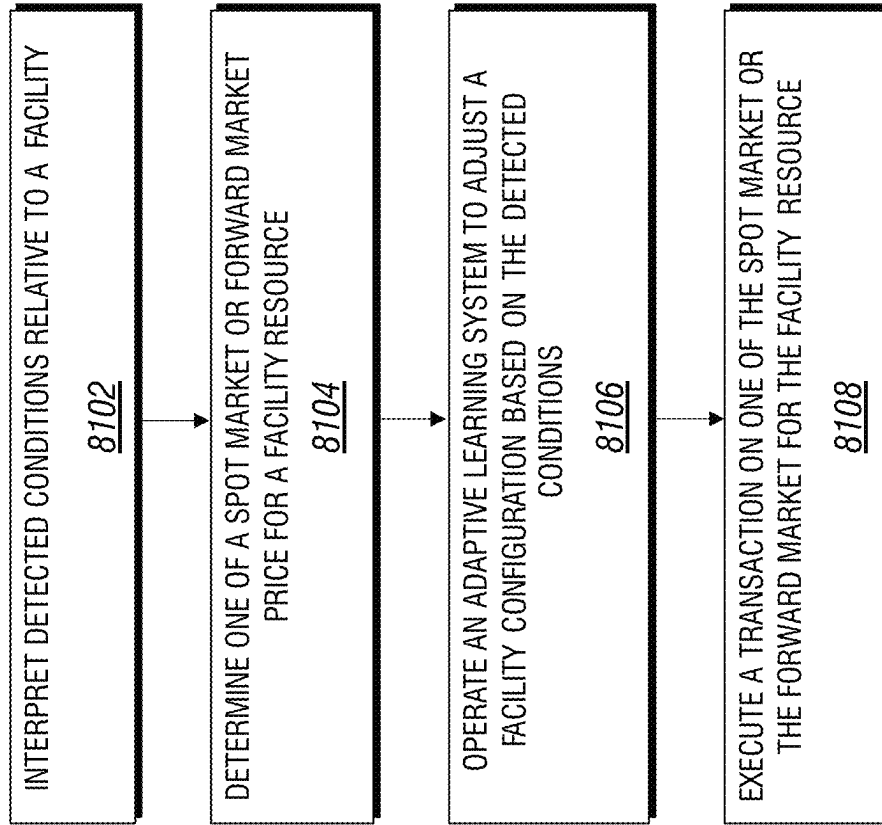


FIG. 81

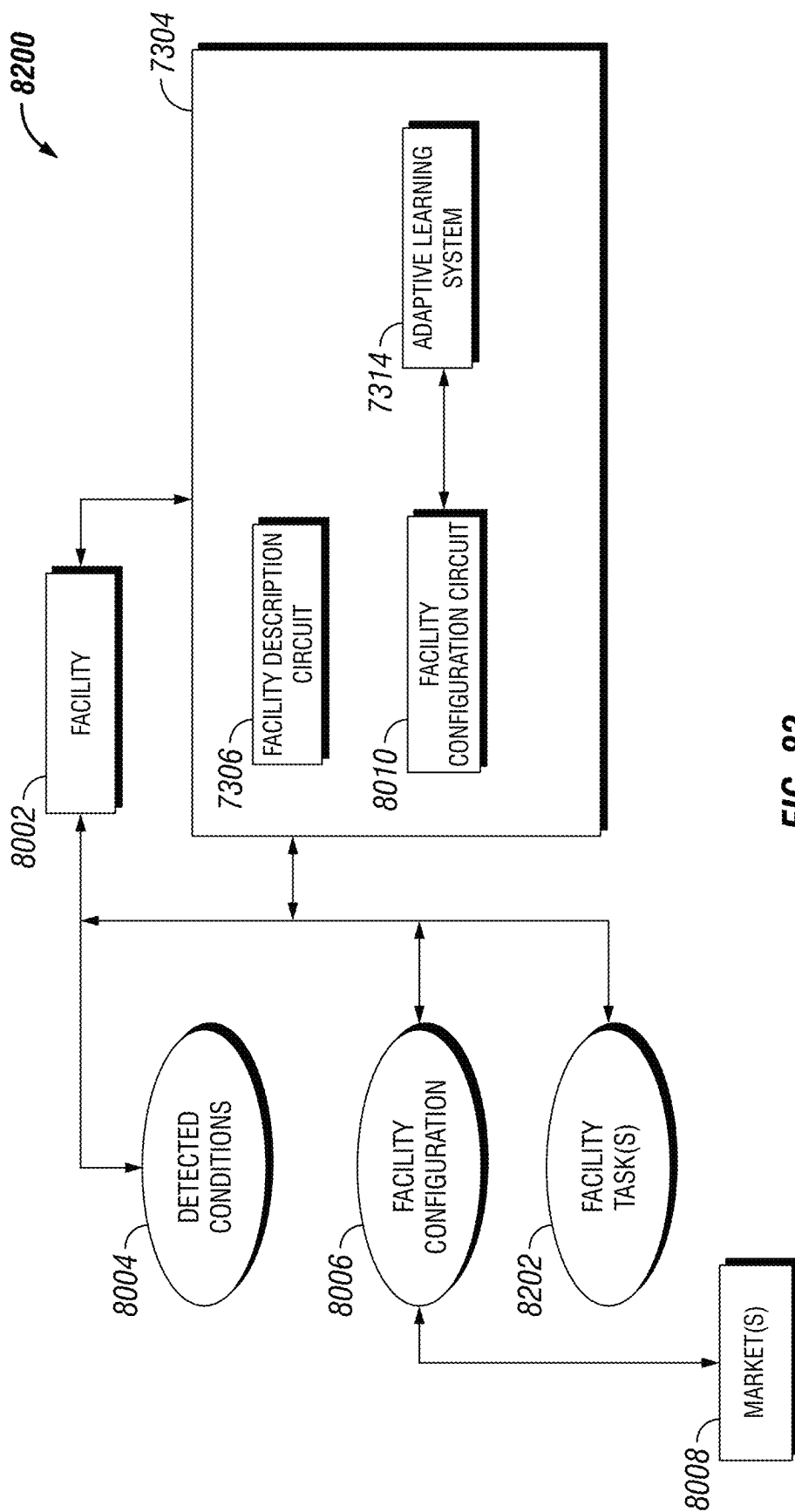


FIG. 82

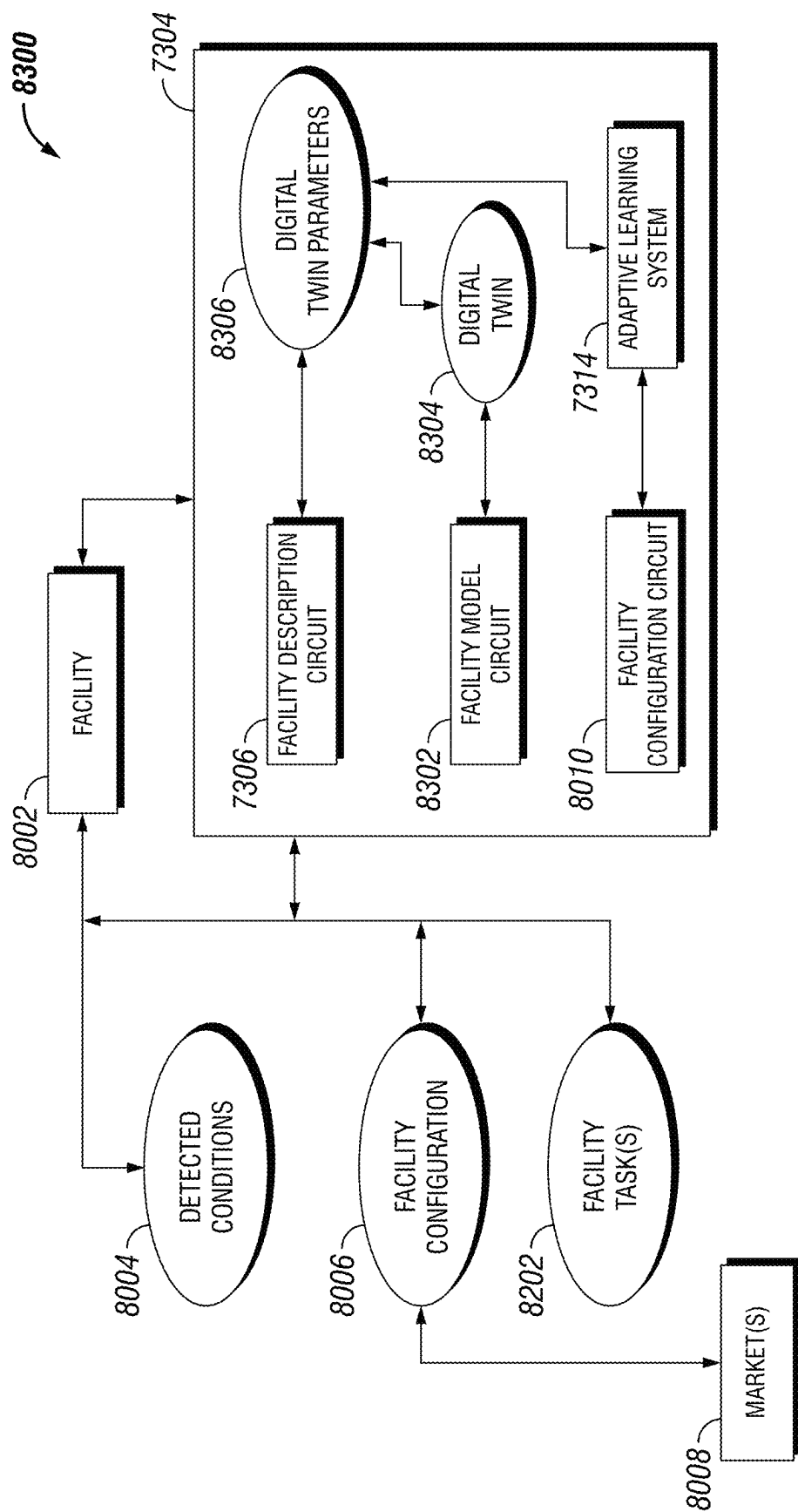


FIG. 83

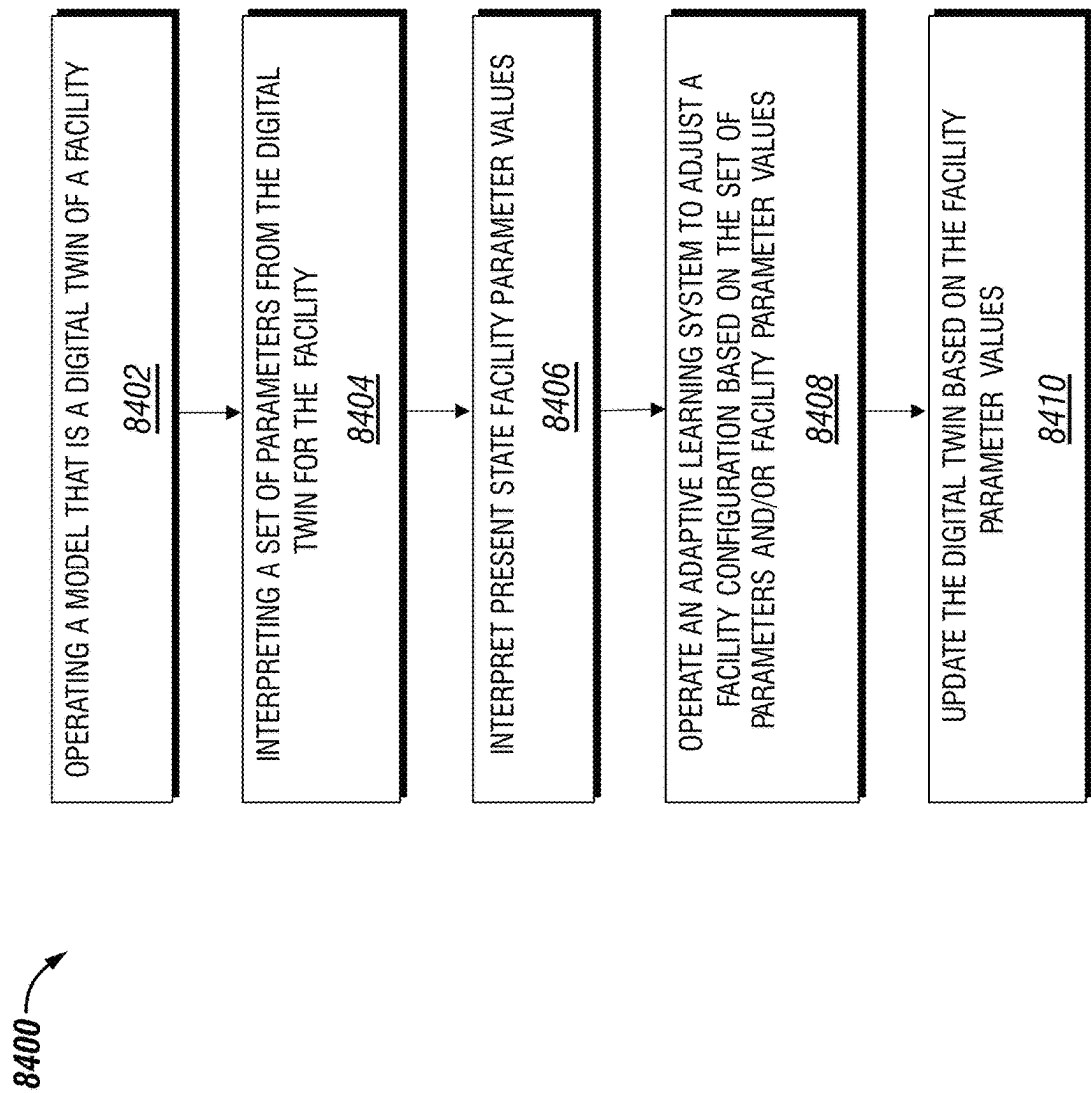


FIG. 84

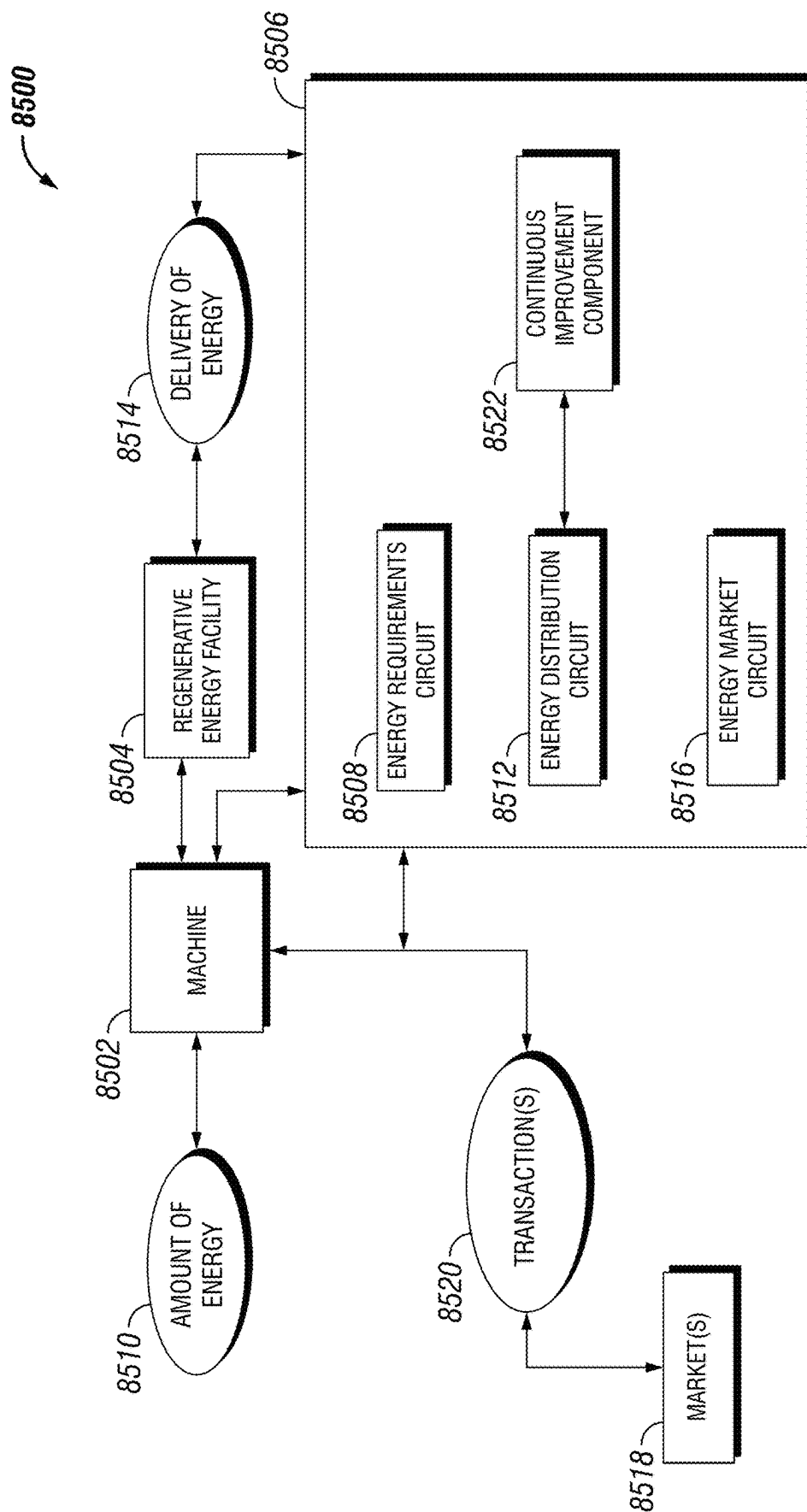


FIG. 85

8600 ↗

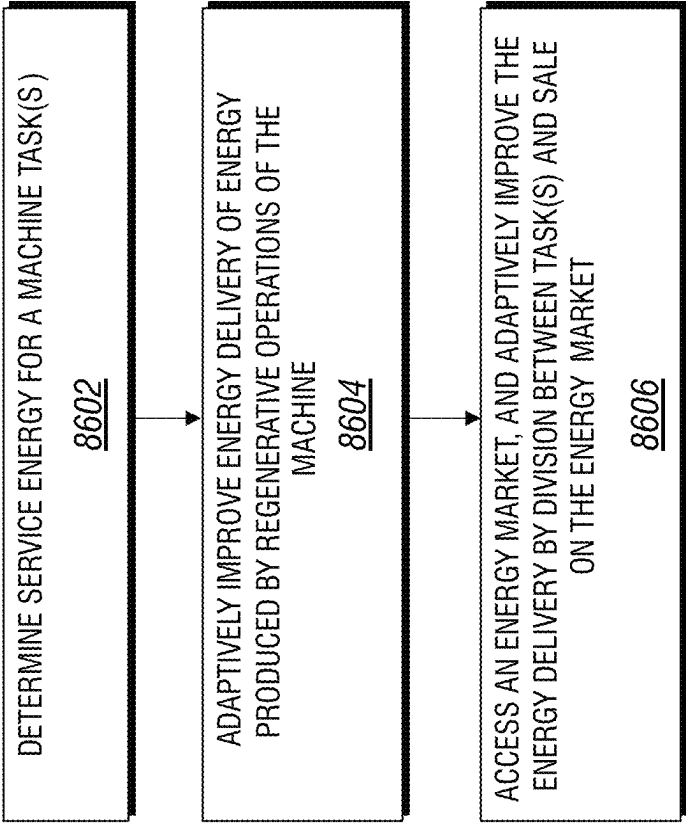


FIG. 86

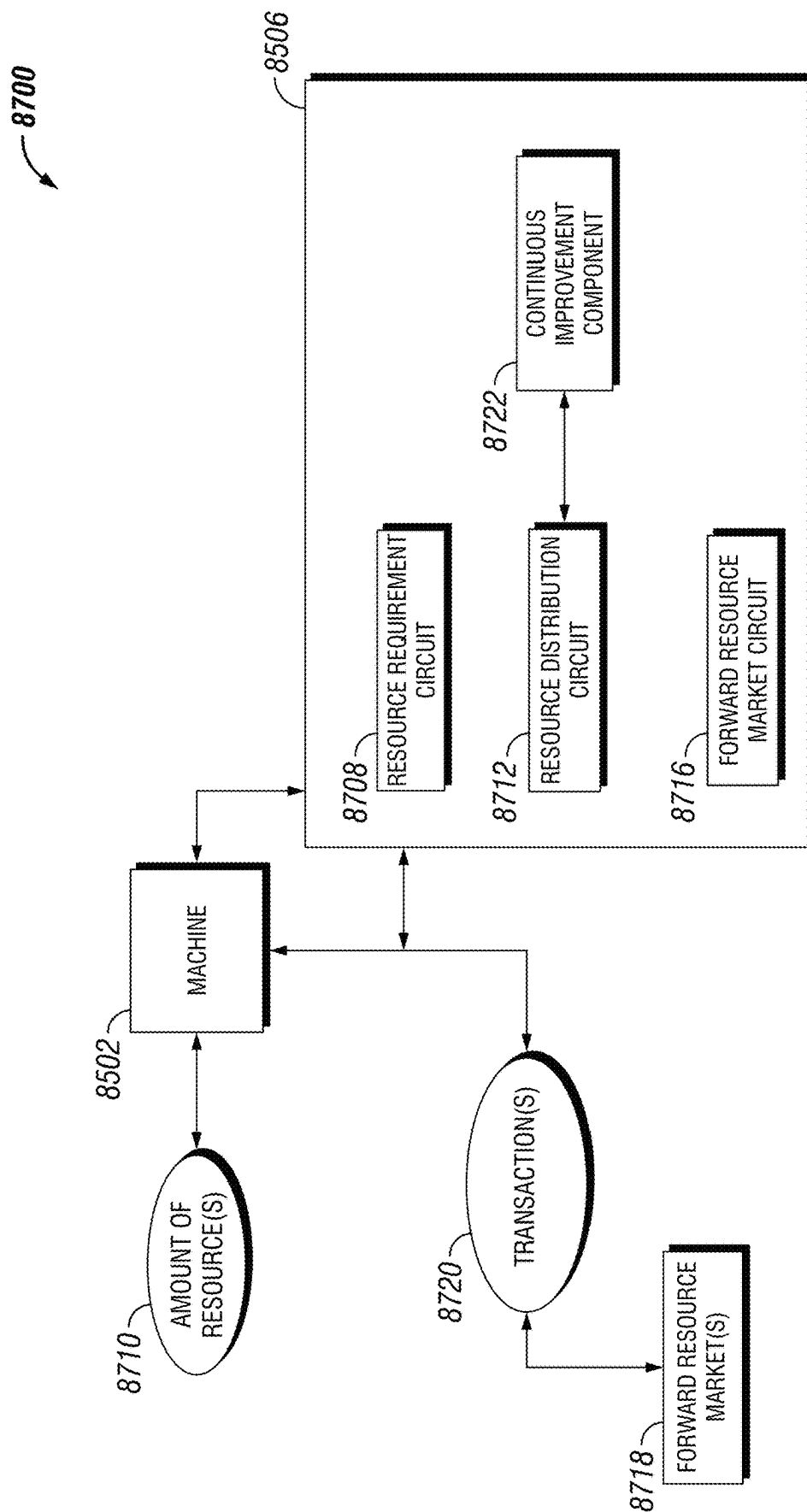


FIG. 87

8800 ↗

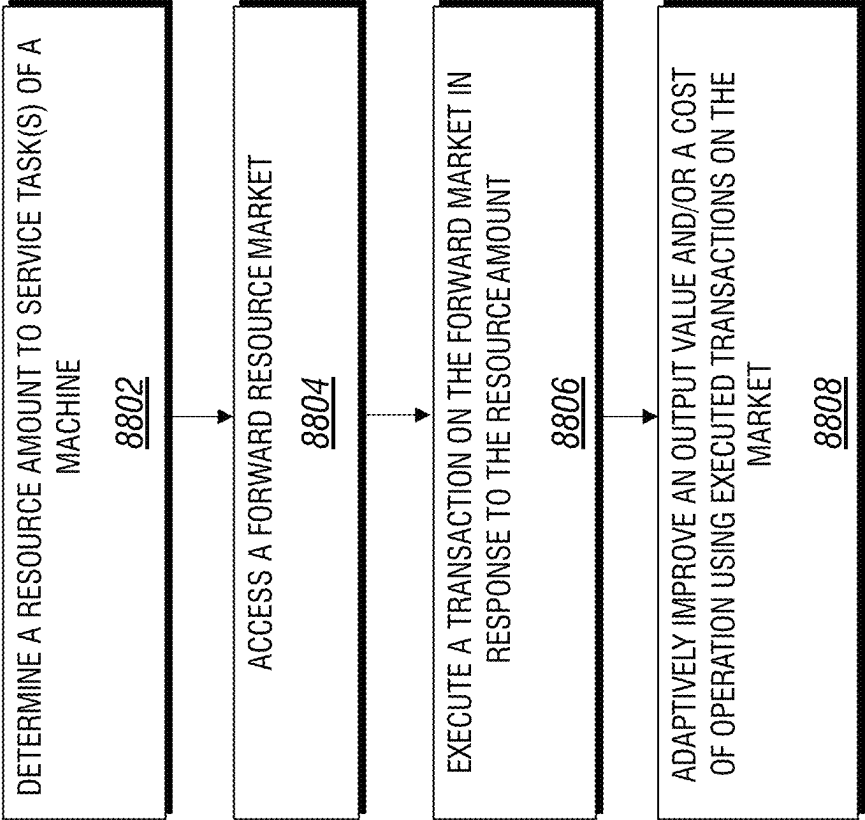


FIG. 88

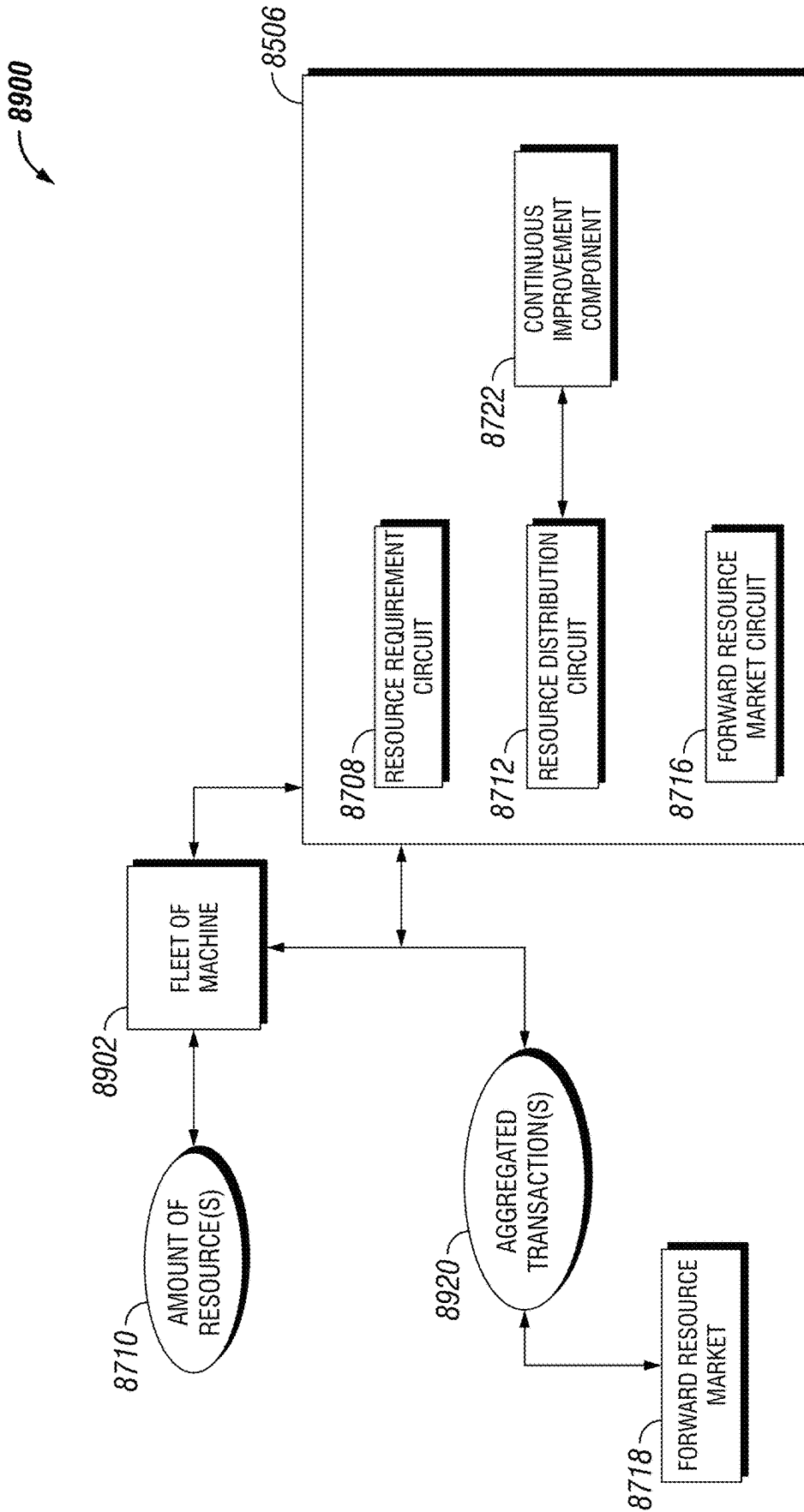


FIG. 89

9000 ↗

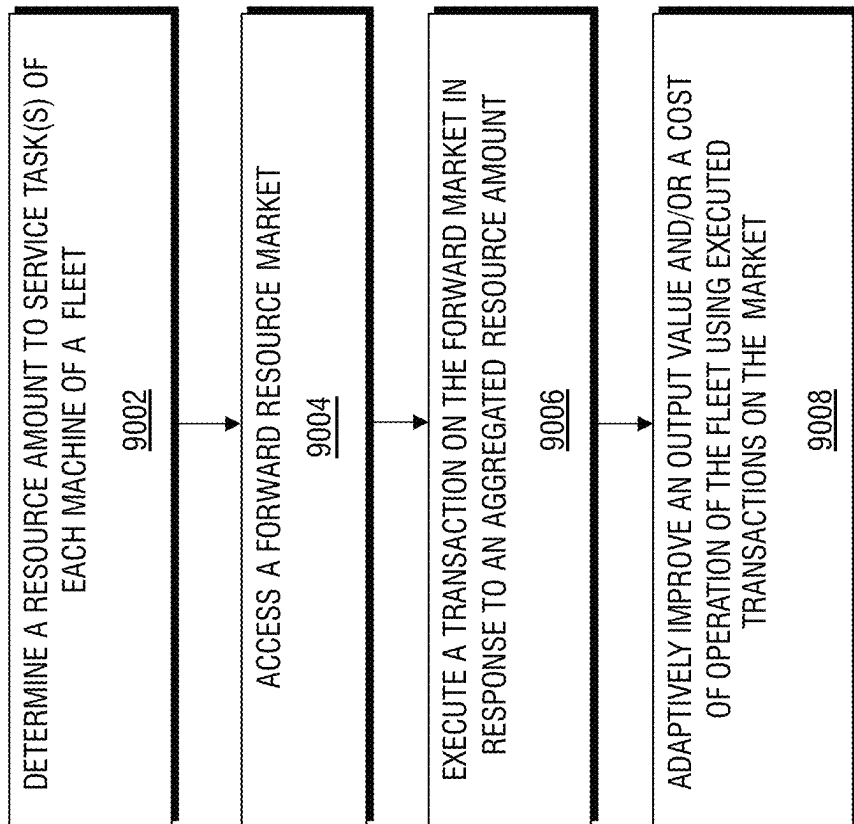


FIG. 90

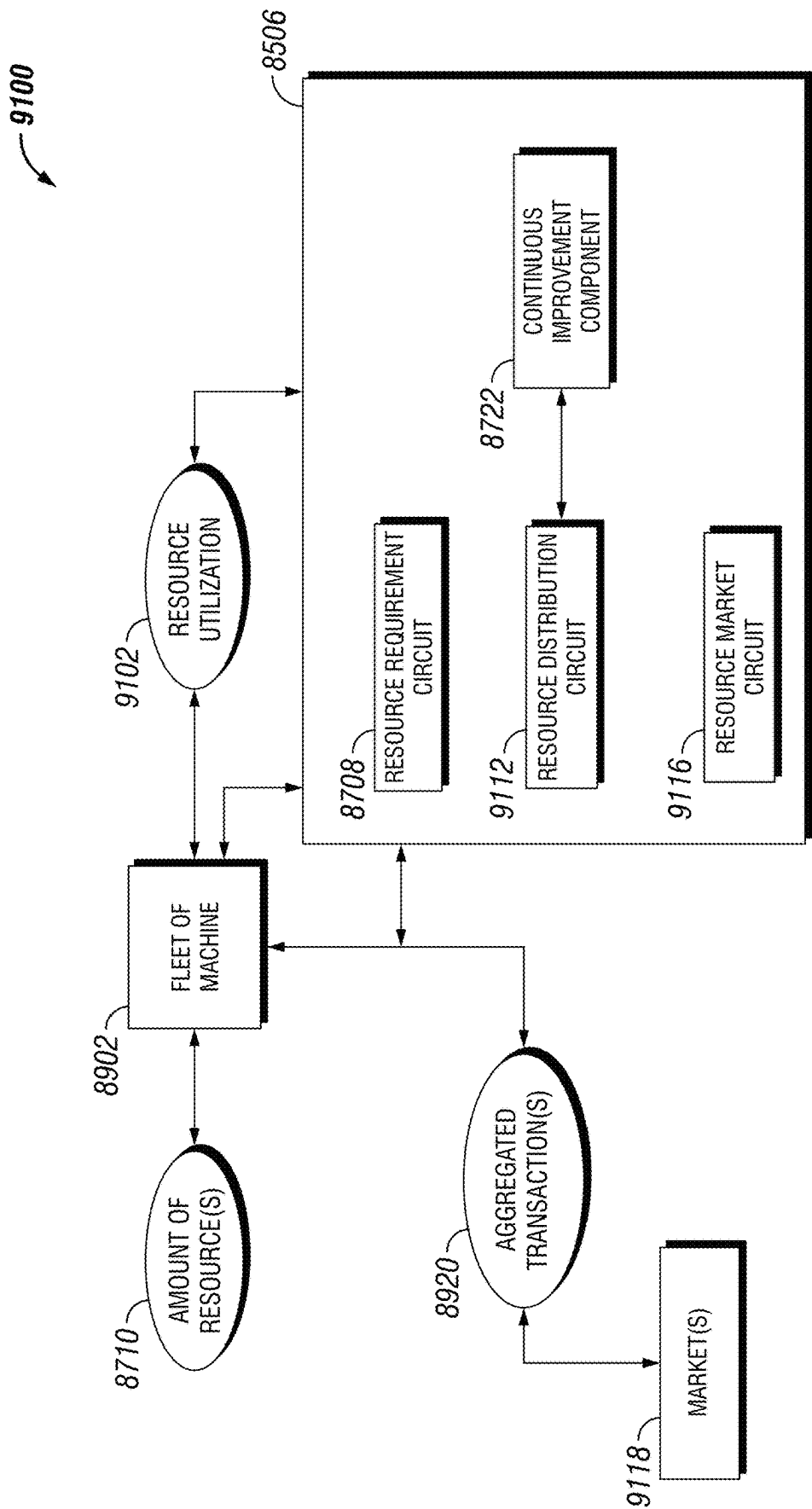


FIG. 91

9200 ↗

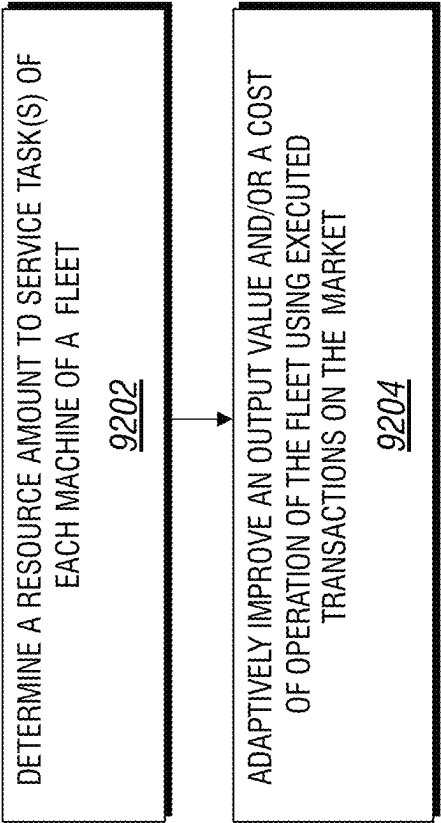


FIG. 92

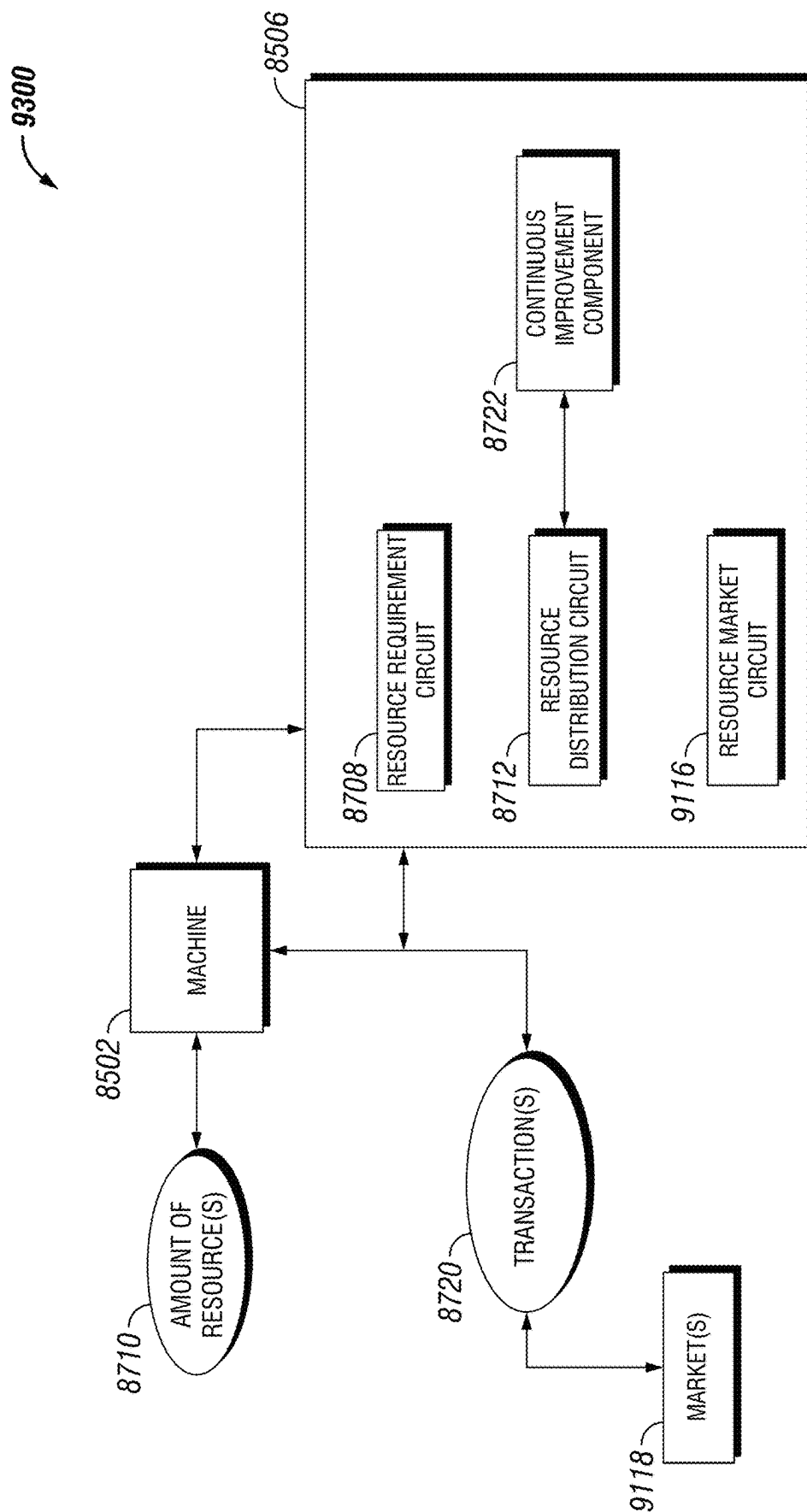


FIG. 93

9400 ↗

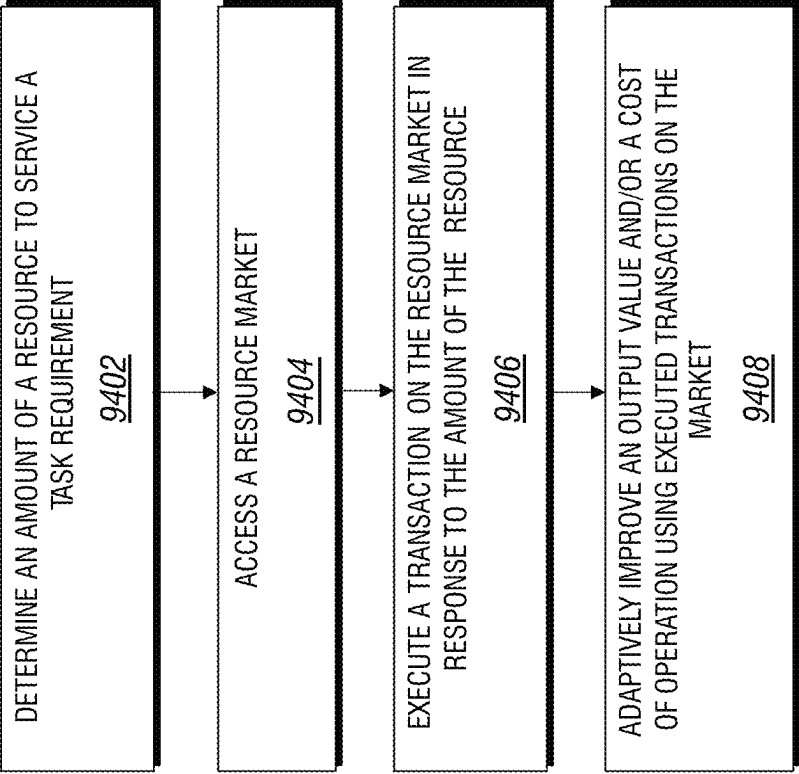


FIG. 94

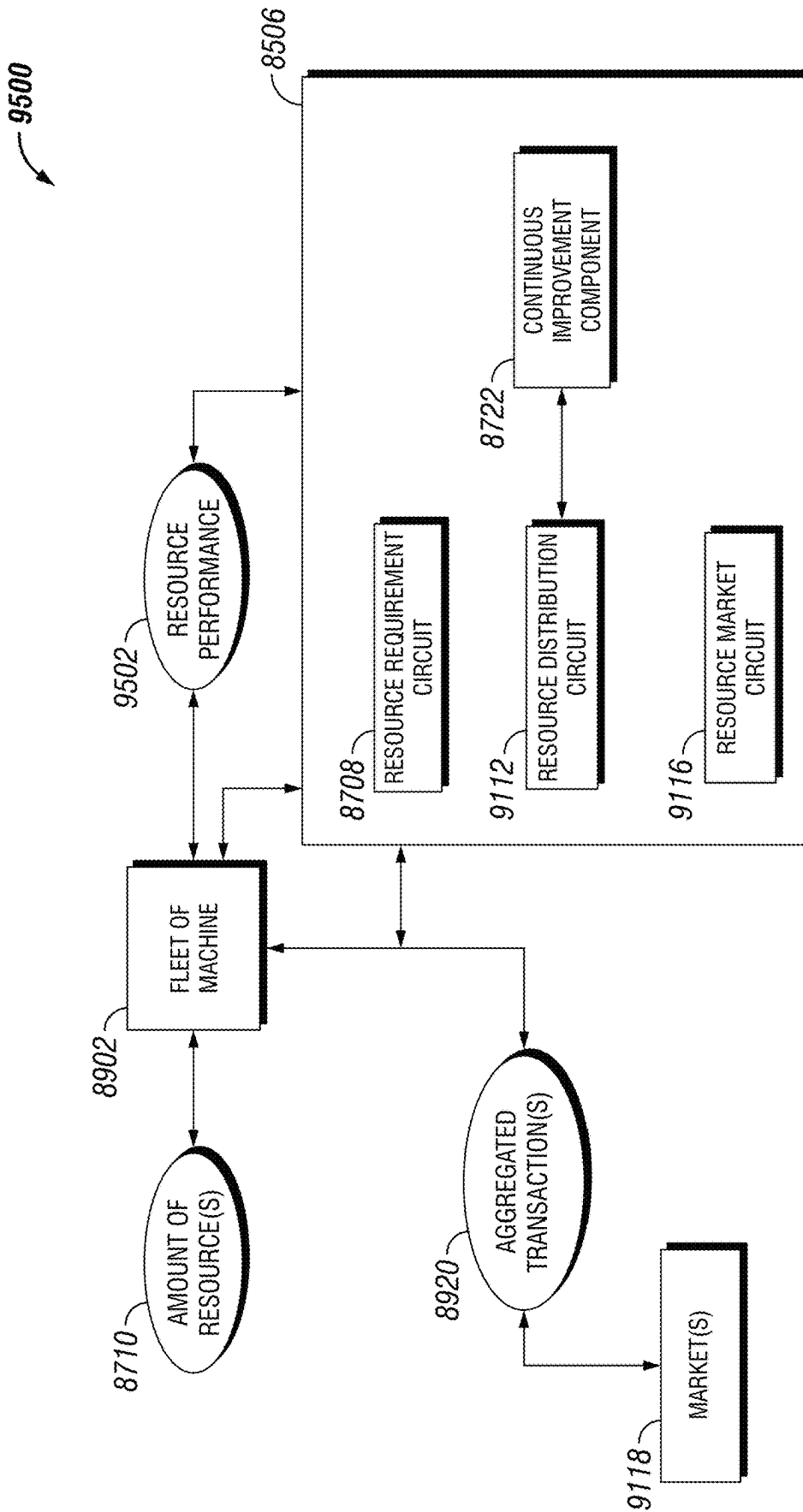
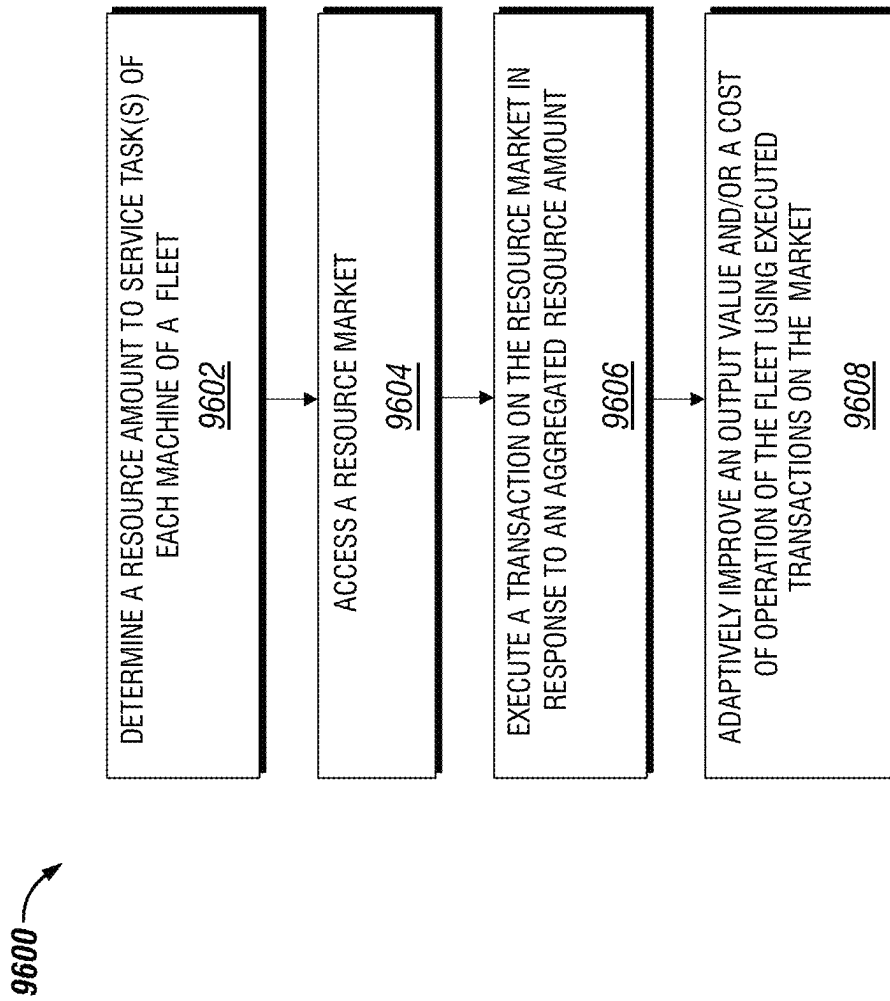


FIG. 95

**FIG. 96**

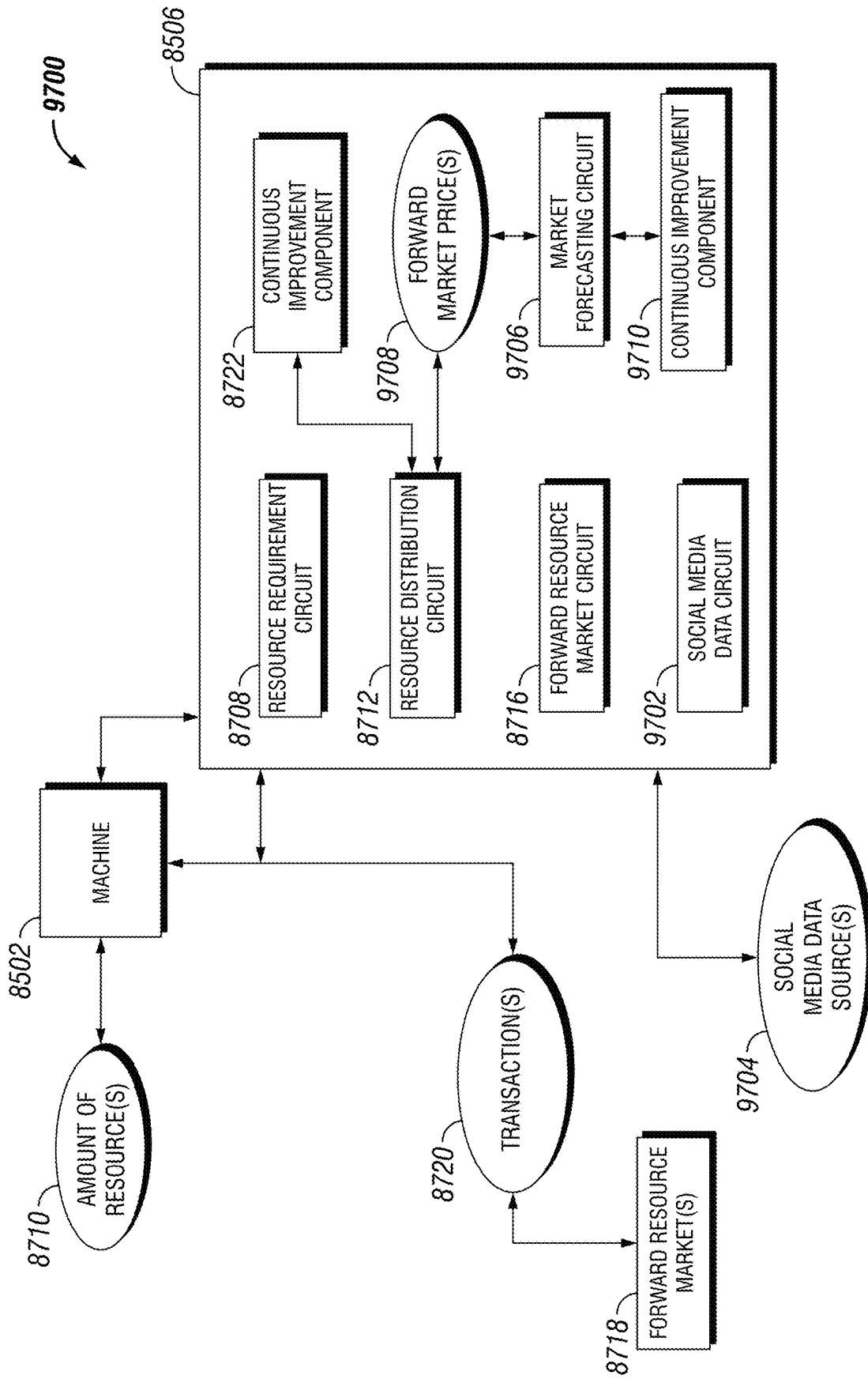


FIG. 97

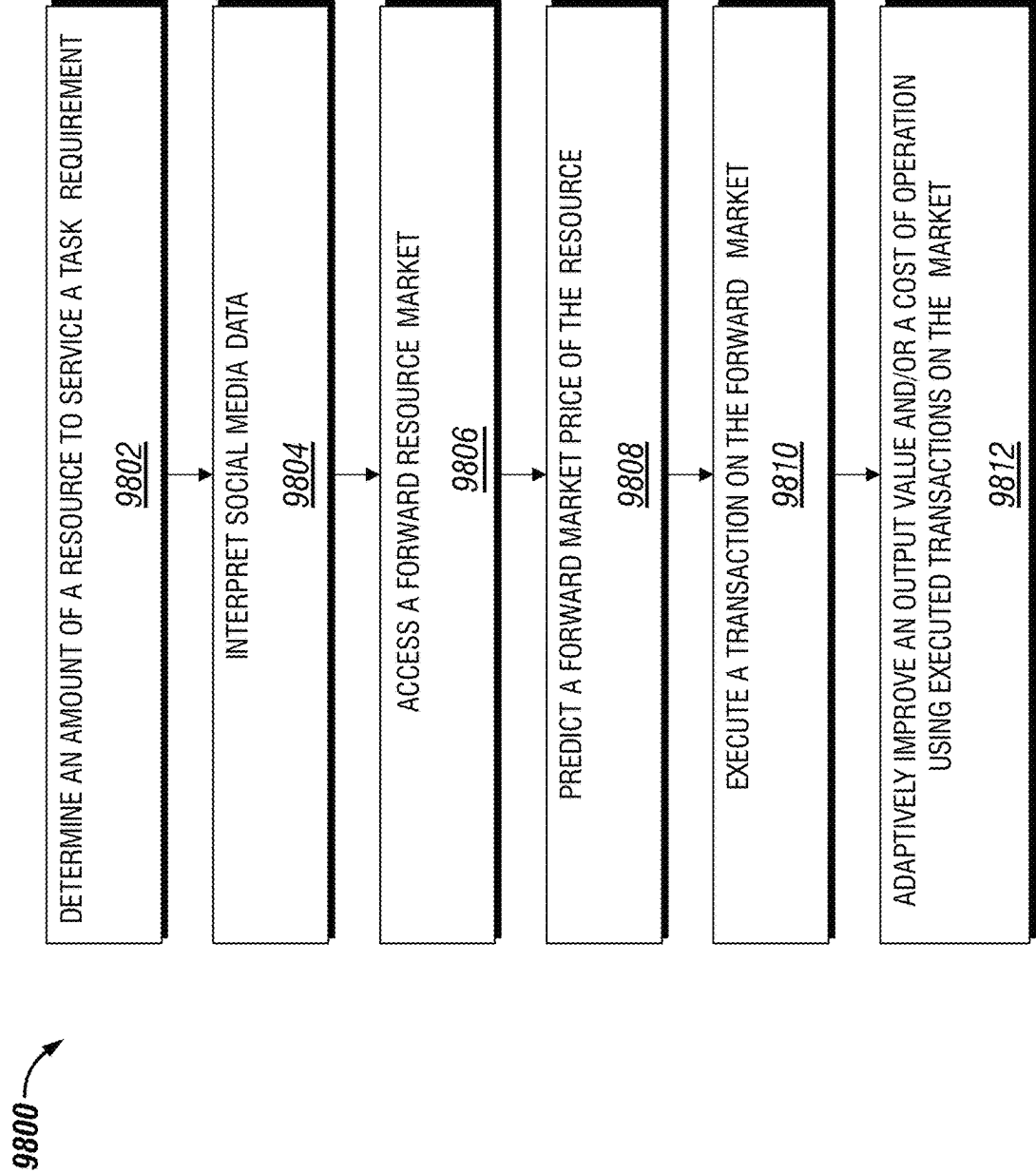


FIG. 98

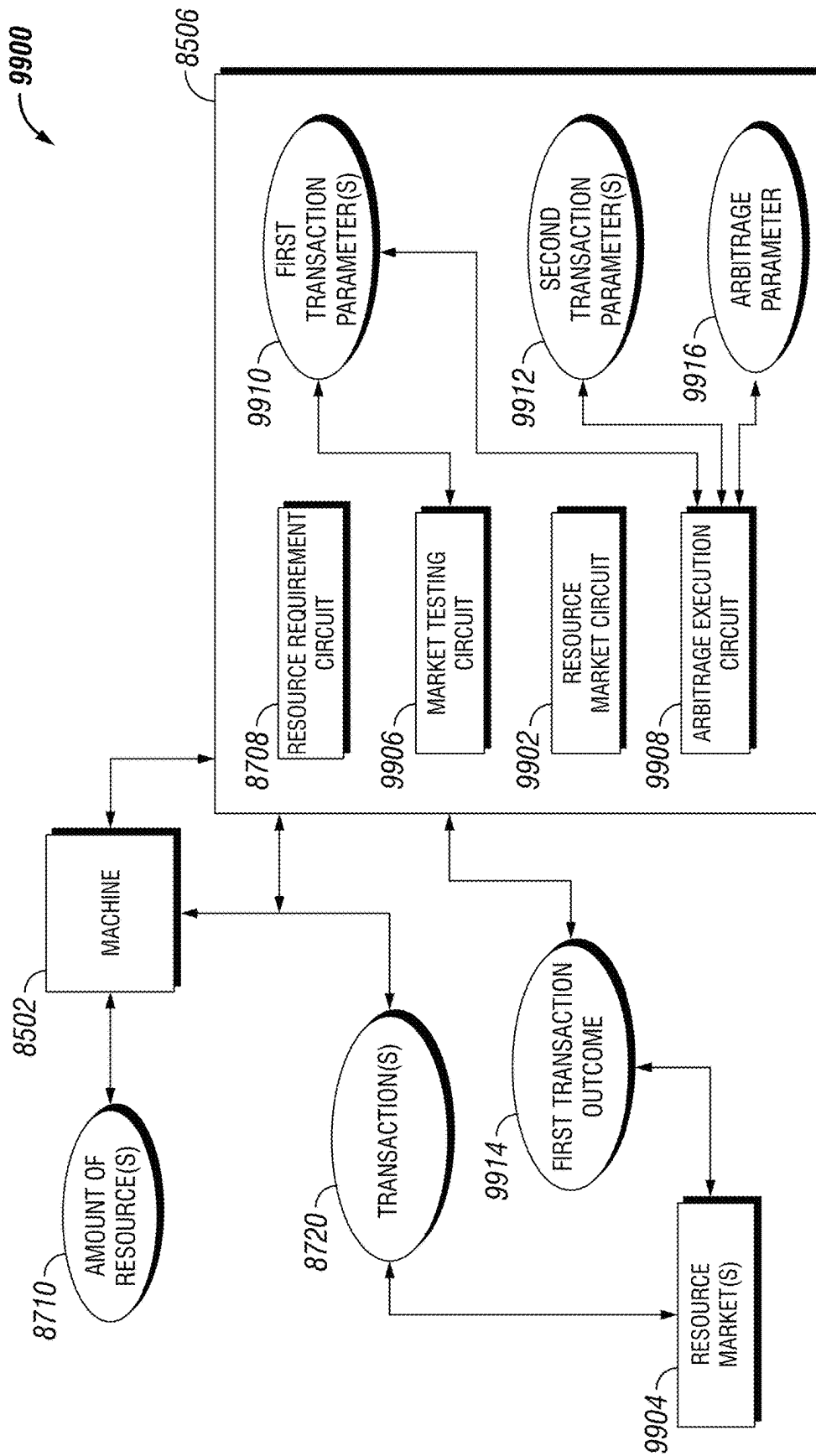


FIG. 99

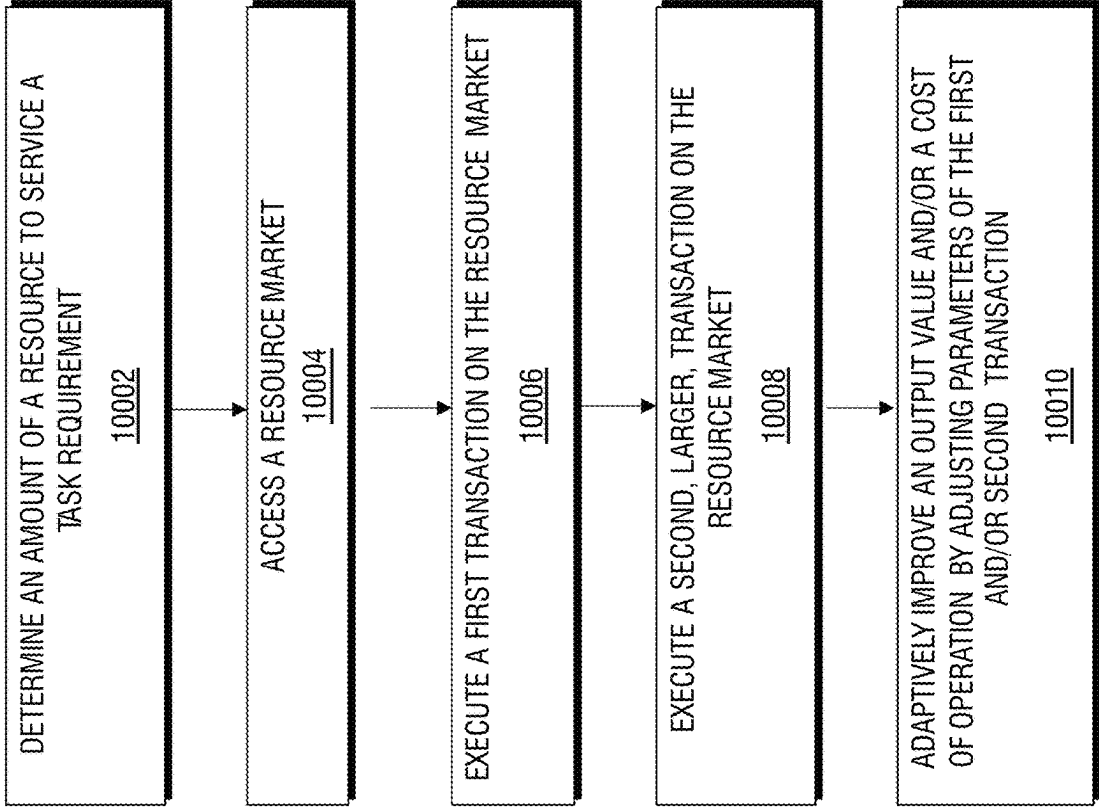


FIG. 100

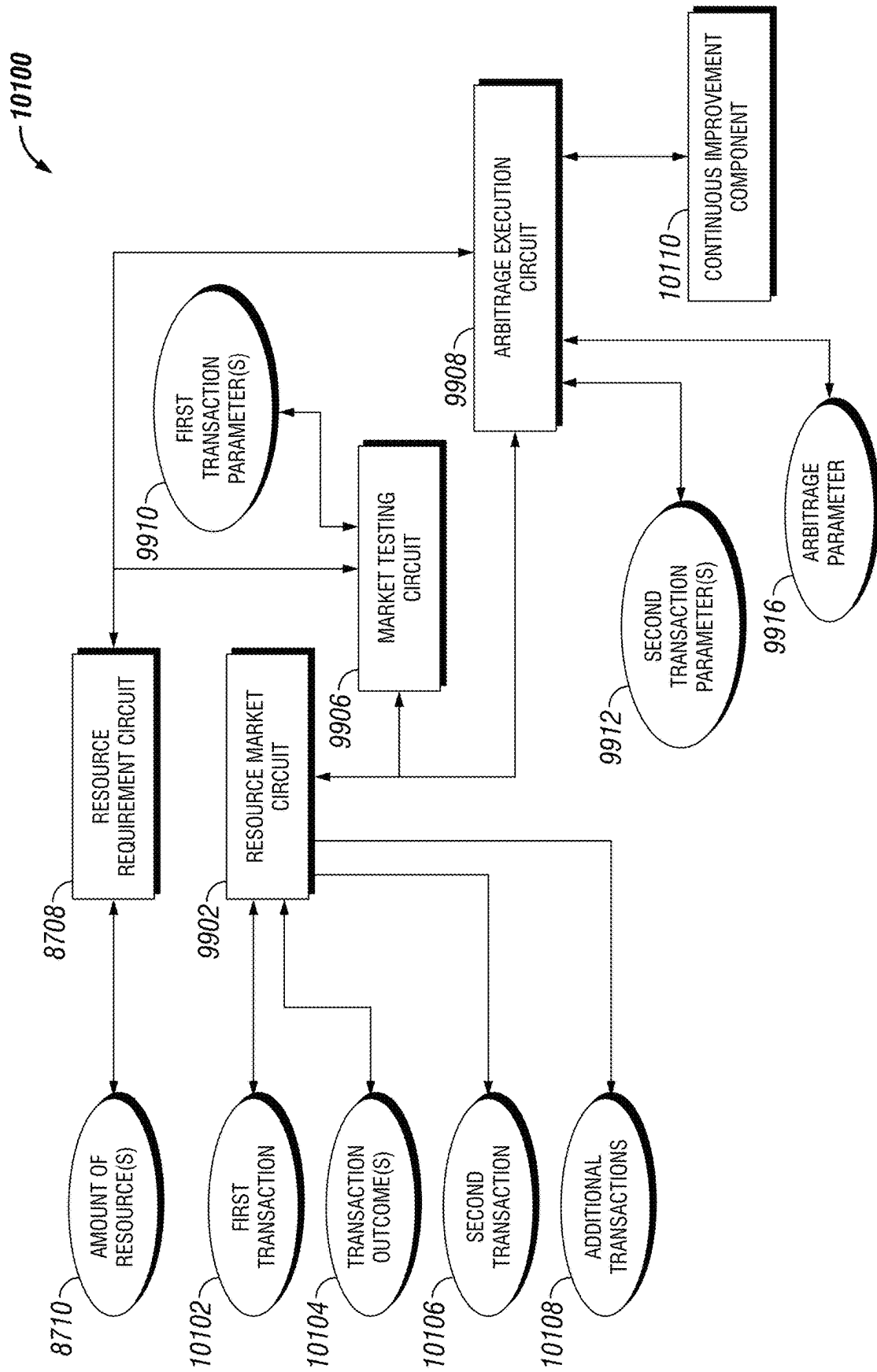


FIG. 101

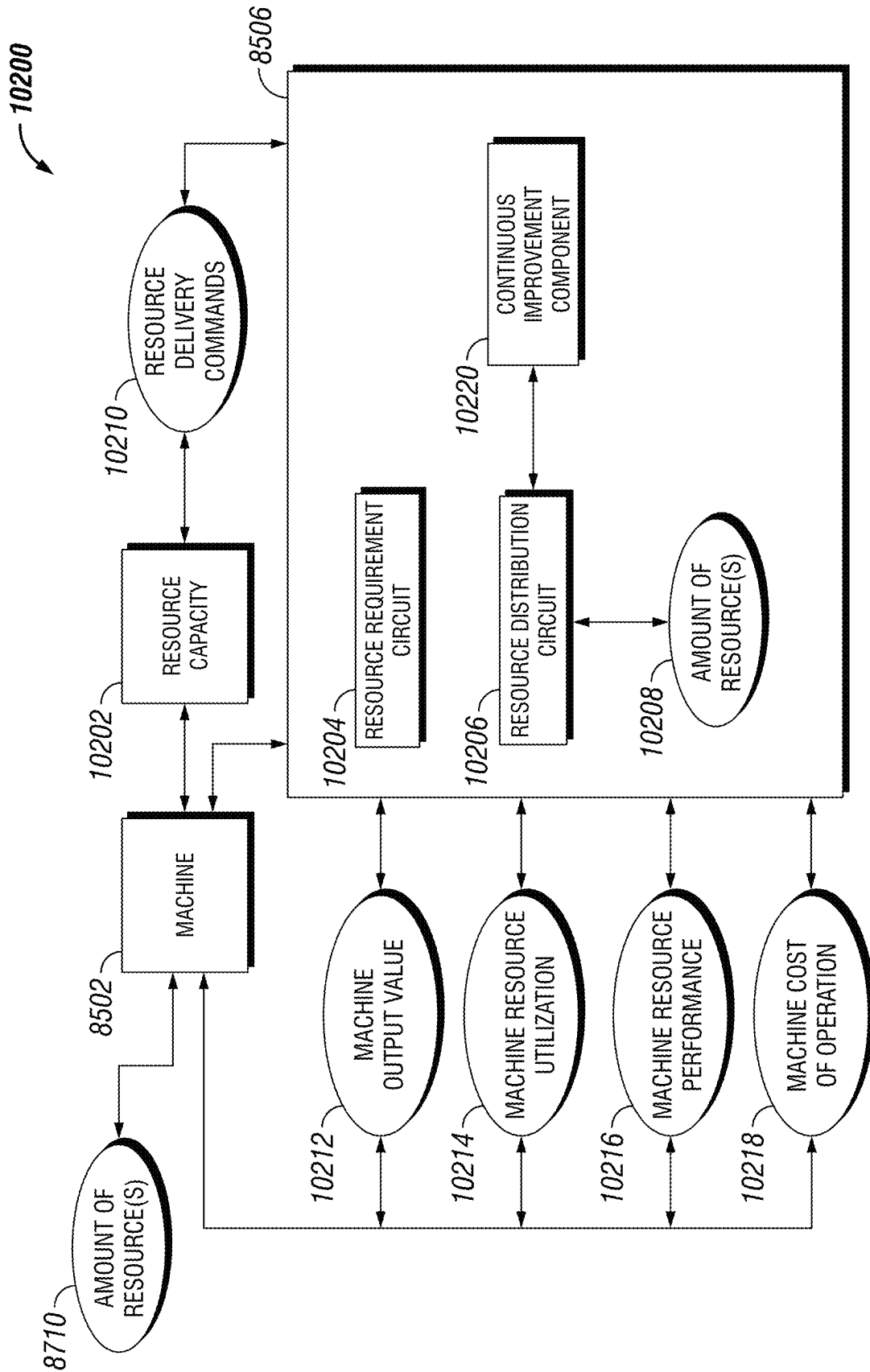
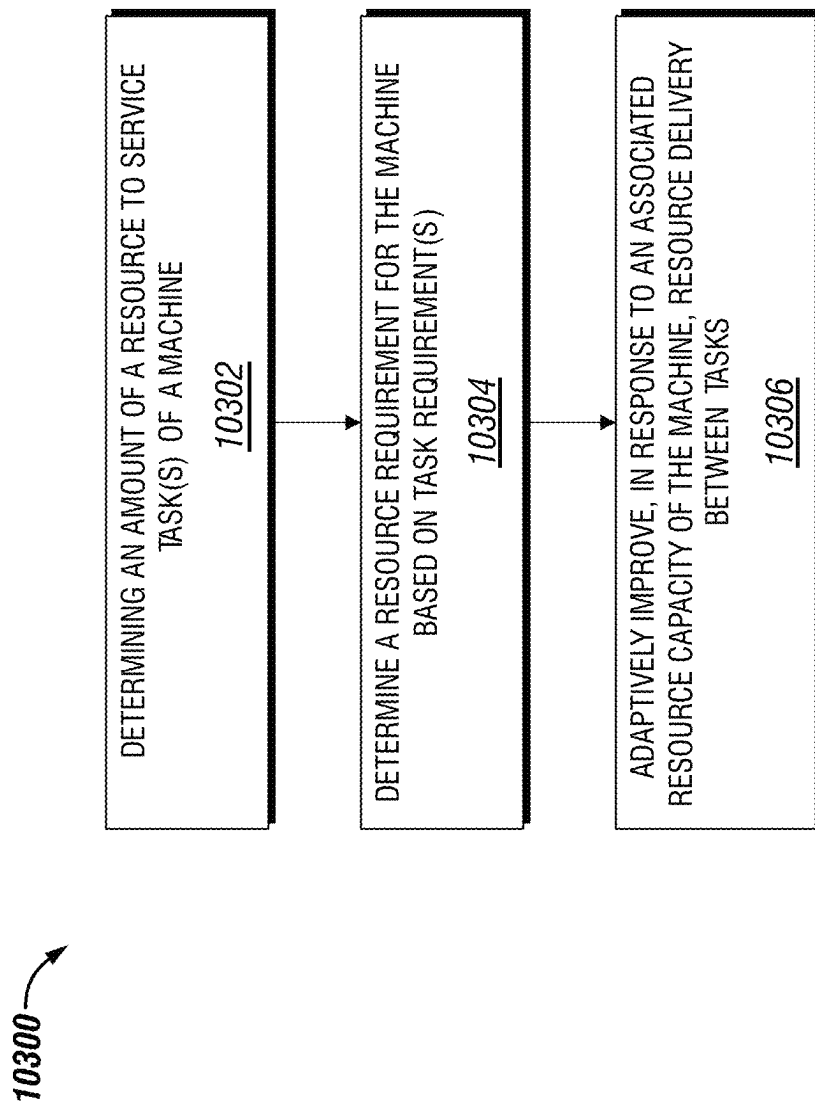


FIG. 102



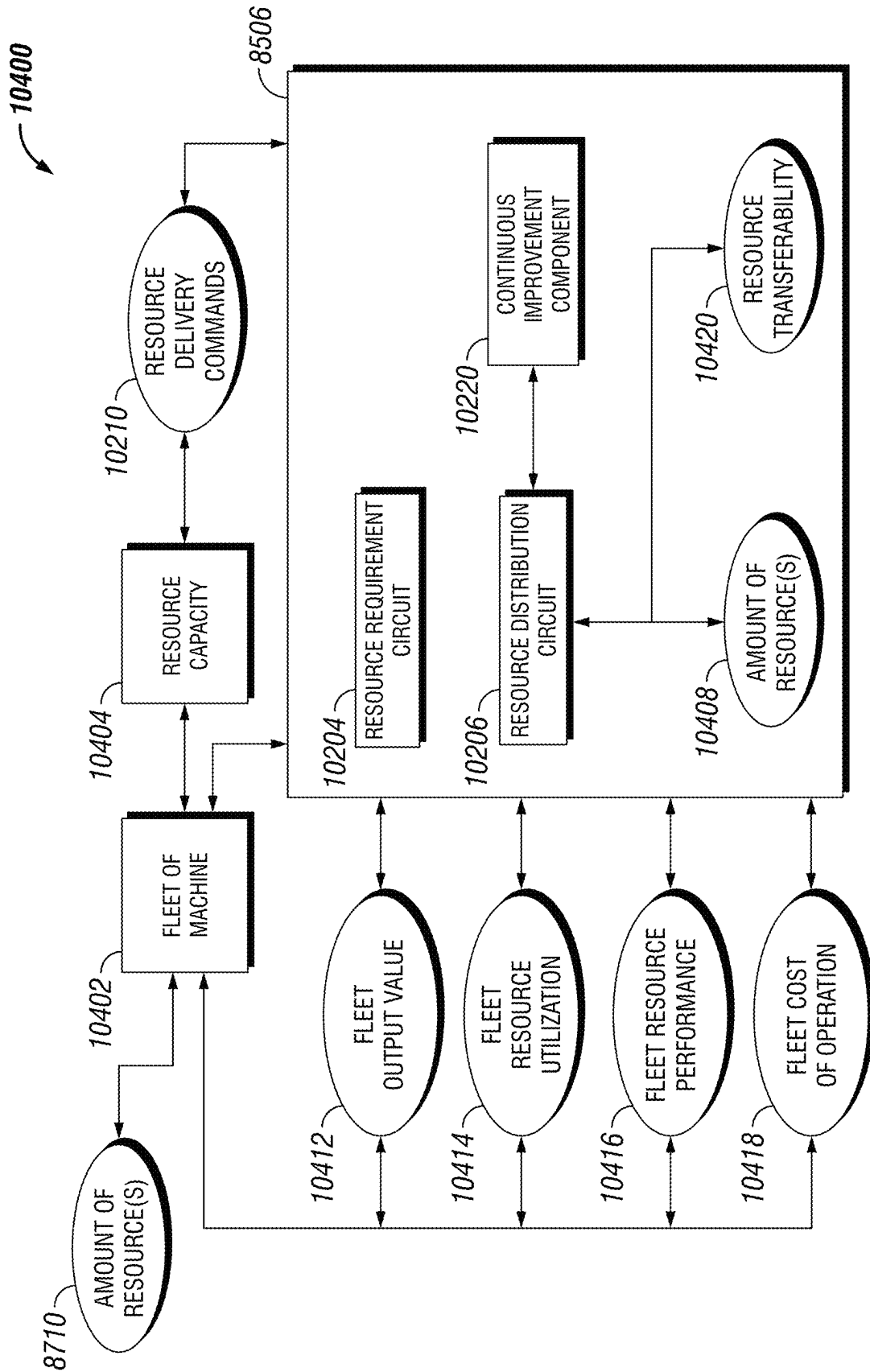
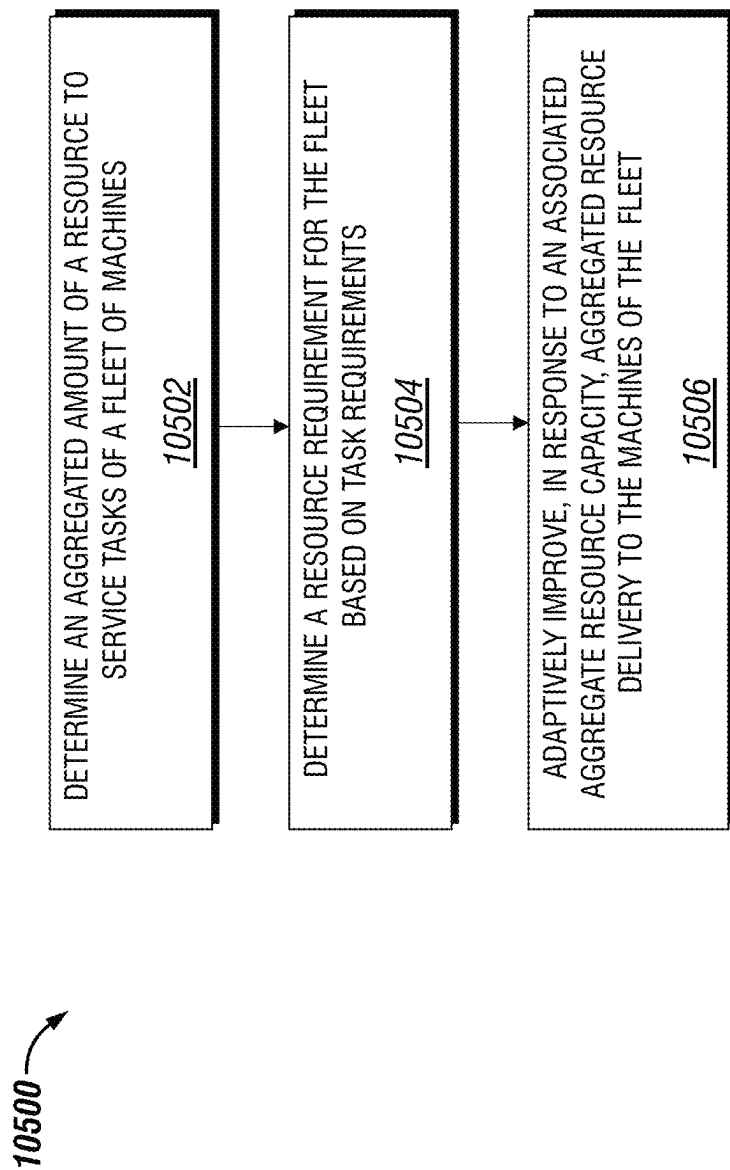


FIG. 104

**FIG. 105**

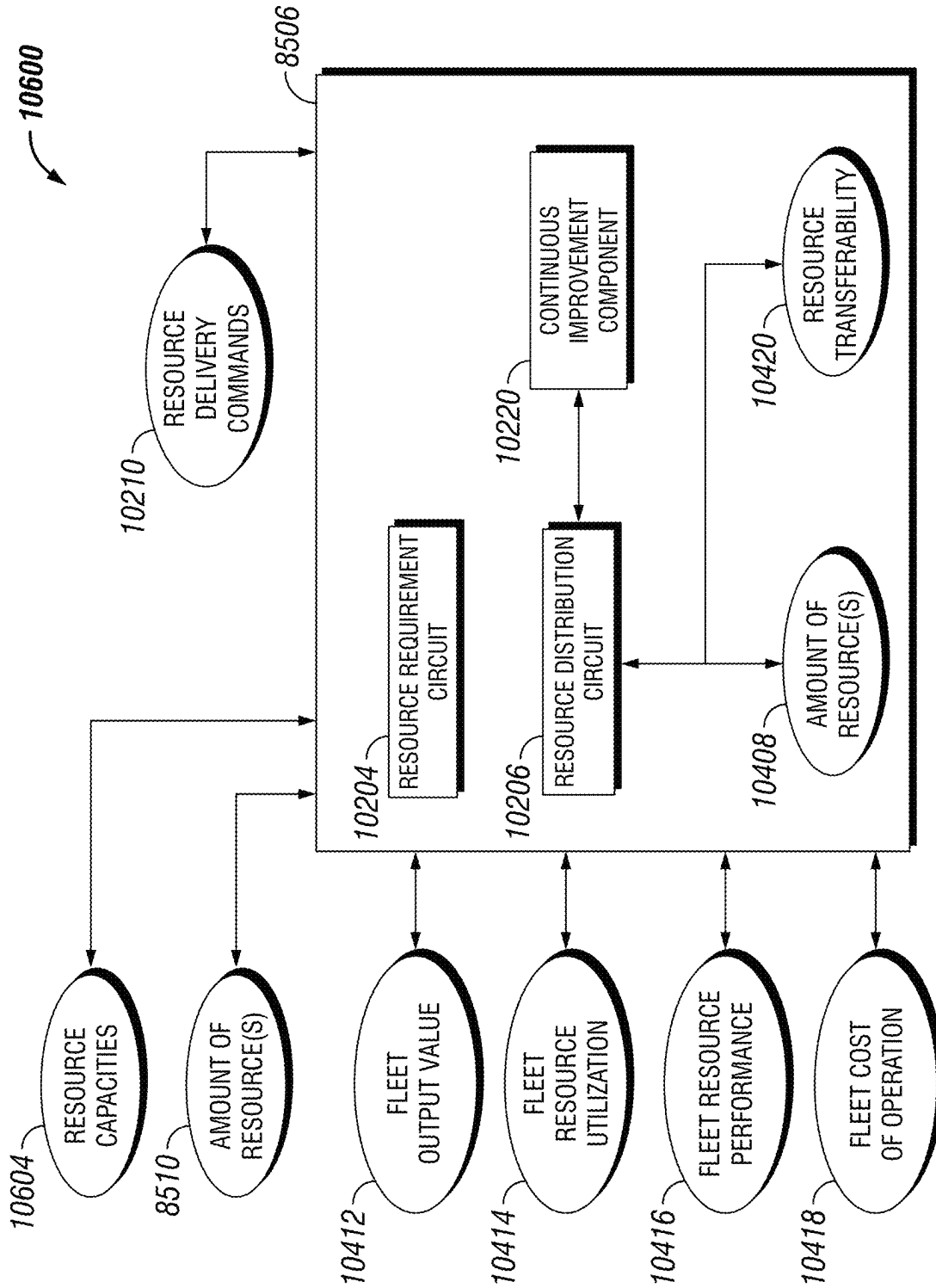


FIG. 106

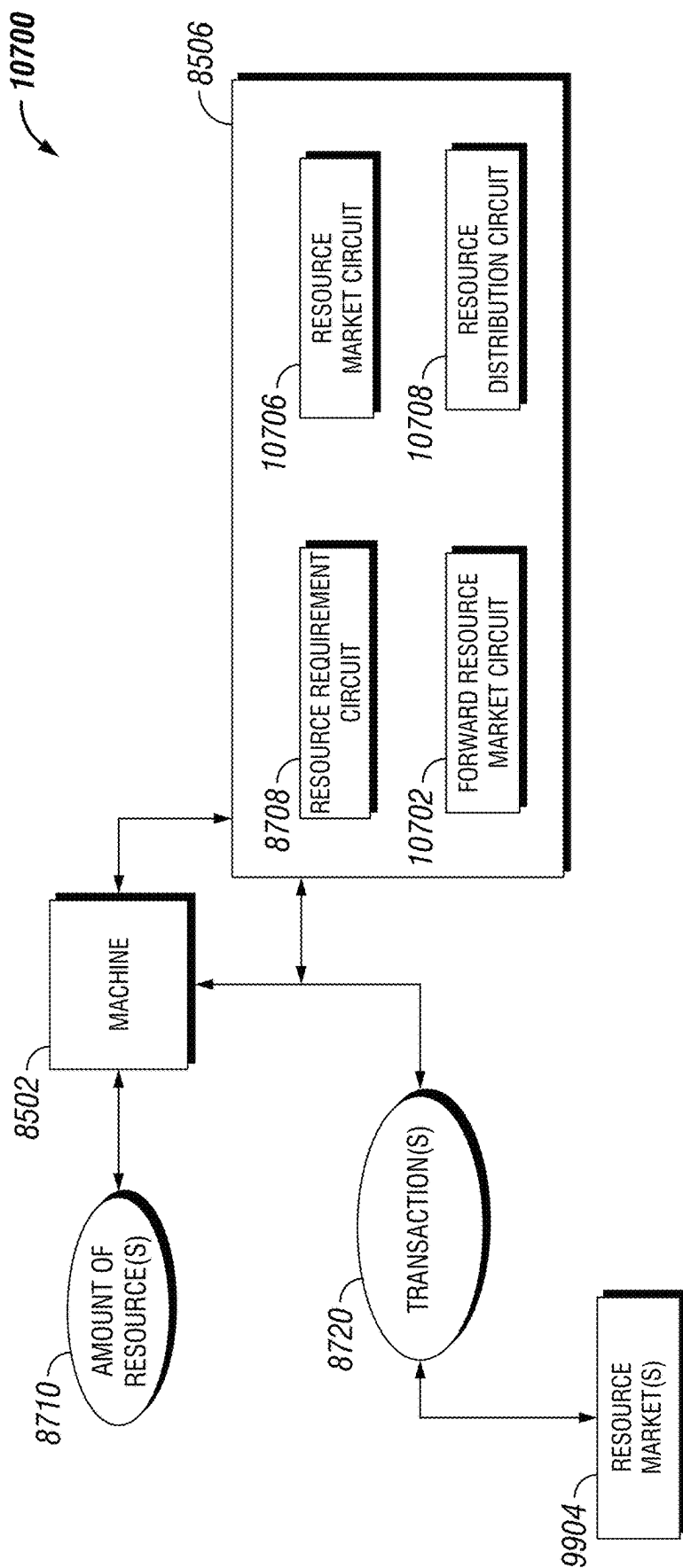


FIG. 107

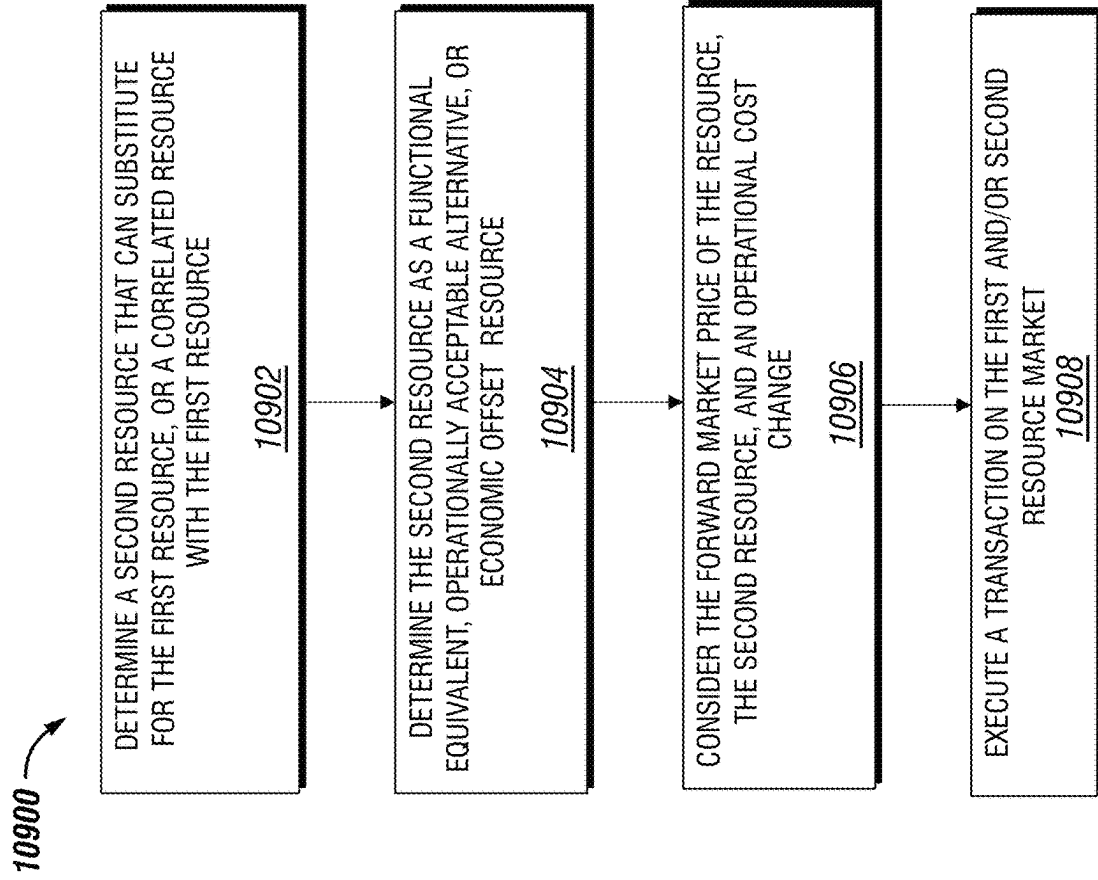


FIG. 109

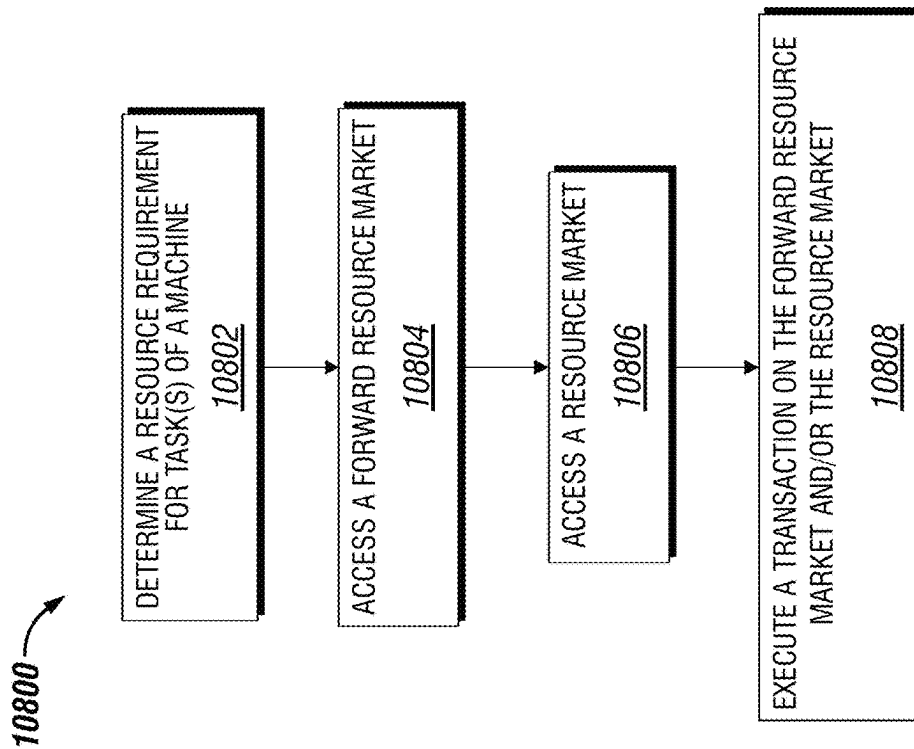


FIG. 108

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SMART CONTRACT MANAGEMENT OF LICENSING AND APPORTIONMENT USING A DISTRIBUTED LEDGER

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/474,767 (SFTX-0004-U11-C03), filed Sep. 14, 2021, and published as U.S. Publication 2022-0004927 A1, entitled “TRANSACTION-ENABLED METHODS FOR PROVIDING PROVABLE ACCESS TO A DISTRIBUTED LEDGER WITH A TOKENIZED INSTRUCTION SET.”

U.S. patent application Ser. No. 17/474,767 (SFTX-0004-U11-C03) is a continuation of U.S. patent application Ser. No. 16/524,620 (SFTX-0004-U11), filed Jul. 29, 2019, entitled “TRANSACTION-ENABLED METHODS FOR PROVIDING PROVABLE ACCESS TO A DISTRIBUTED LEDGER WITH A TOKENIZED INSTRUCTION SET,” granted on Jan. 4, 2022, as U.S. Pat. No. 11,216,750.

U.S. patent application Ser. No. 16/524,620 (SFTX-0004-U11) is a continuation of International Application Serial No. PCT/US2019/030934 (SFTX-0004-WO), filed May 6, 2019, and published as WO 2019/217323, entitled “METHODS AND SYSTEMS FOR IMPROVING MACHINES THAT AUTOMATE EXECUTION OF DISTRIBUTED LEDGER AND OTHER TRANSACTIONS IN SPOT AND FORWARD MARKETS FOR ENERGY, COMPUTE, STORAGE AND OTHER RESOURCES.”

International Application Serial No. PCT/US2019/030934 (SFTX-0004-WO) claims the benefit of priority to the following U.S. Provisional Patent Applications: Ser. No. 62/787,206 (SFTX-0001-P01), filed Dec. 31, 2018, entitled “METHODS AND SYSTEMS FOR IMPROVING MACHINES AND SYSTEMS THAT AUTOMATE EXECUTION OF DISTRIBUTED LEDGER AND OTHER TRANSACTIONS IN SPOT AND FORWARD MARKETS FOR ENERGY, COMPUTE, STORAGE AND OTHER RESOURCES”; Ser. No. 62/667,550 (SFTX-0002-P01), filed May 6, 2018, entitled “METHODS AND SYSTEMS FOR IMPROVING MACHINES AND SYSTEMS THAT AUTOMATE EXECUTION OF DISTRIBUTED LEDGER AND OTHER TRANSACTIONS IN SPOT AND FORWARD MARKETS FOR ENERGY, COMPUTE, STORAGE AND OTHER RESOURCES”; and Ser. No. 62/751,713 (SFTX-0003-P01), filed Oct. 29, 2018, entitled “METHODS AND SYSTEMS FOR IMPROVING MACHINES AND SYSTEMS THAT AUTOMATE EXECUTION OF DISTRIBUTED LEDGER AND OTHER TRANSACTIONS IN SPOT AND FORWARD MARKETS FOR ENERGY, COMPUTE, STORAGE AND OTHER RESOURCES.”

Each of the foregoing applications is incorporated herein by reference in its entirety.

BACKGROUND

Machines and automated agents are increasingly involved in market activities, including for data collection, forecasting, planning, transaction execution, and other activities. This includes increasingly high-performance systems, such as used in high-speed trading. A need exists for methods and systems that improve the machines that enable markets, including for increased efficiency, speed, reliability, and the like for participants in such markets.

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Many markets are increasingly distributed, rather than centralized, with distributed ledgers like Blockchain, peer-to-peer interaction models, and micro-transactions replacing or complementing traditional models that involve centralized authorities or intermediaries. A need exists for improved machines that enable distributed transactions to occur at scale among large numbers of participants, including human participants and automated agents.

Operations on blockchains, such as ones using cryptocurrency, increasingly require energy-intensive computing operations, such as calculating very large hash functions on growing chains of blocks. Systems using proof-of-work, proof-of-stake, and the like have led to “mining” operations by which computer processing power is applied at a large scale in order to perform calculations that support collective trust in transactions that are recorded in blockchains.

Many applications of artificial intelligence also require energy-intensive computing operations, such as where very large neural networks, with very large numbers of interconnections, perform operations on large numbers of inputs to produce one or more outputs, such as a prediction, classification, optimization, control output, or the like.

The growth of the Internet of Things and cloud computing platforms have also led to the proliferation of devices, applications, and connections among them, such that data centers, housing servers and other IT components, consume a significant fraction of the energy consumption of the United States and other developed countries.

As a result of these and other trends, energy consumption has become a major factor in utilization of computing resources, such that energy resources and computing resources (or simply “energy and compute”) have begun to converge from various standpoints, such as requisitioning, purchasing, provisioning, configuration, and management of inputs, activities, outputs and the like. Projects have been undertaken, for example, to place large scale computing resource facilities, such as Bitcoin™ or other cryptocurrency mining operations, in close proximity to large-scale hydropower sources, such as Niagara Falls.

A major challenge for facility owners and operators is the uncertainty involved in optimizing a facility, such as resulting from volatility in the cost and availability of inputs (in particular where less stable renewable resources are involved), variability in the cost and availability of computing and networking resources (such as where network performance varies), and volatility and uncertainty in various end markets to which energy and compute resources can be applied (such as volatility in cryptocurrencies, volatility in energy markets, volatility in pricing in various other markets, and uncertainty in the utility of artificial intelligence in a wide range of applications), among other factors.

A need exists for a flexible, intelligent energy and compute facility that adjust in response to uncertainty and volatility, as well as for an intelligent energy and compute resource management system, such as one that includes capabilities for data collection, storage and processing, automated configuration of inputs, resources and outputs, and learning on a training set of facility outcomes, facility parameters, and data collected from data sources to train an artificial intelligence/machine learning system to optimize various relevant parameters for such a facility.

SUMMARY

Machine learning potentially enables machines that enable or interact with automated markets to develop understanding, such as based on IoT data, social network data, and

other non-traditional data sources, and execute transactions based on predictions, such as by participating in forward markets for energy, compute, advertising and the like. Blockchain and cryptocurrencies may support a variety of automated transactions, and the intersection of blockchain and AI potentially enables radically different transaction infrastructure. As energy is increasingly used for computation, machines that efficiently allocate available energy sources among storage, compute, and base tasks become possible. These and other concepts are addressed by the methods and systems disclosed herein.

The present disclosure describes a method, the method according to one disclosed, non-limiting, embodiment of the present disclosure can include accessing a distributed ledger comprising an instruction set, wherein the instruction set includes an instruction set for a polymer production process, tokenizing the instruction set, interpreting an instruction set access request, in response to the instruction set access request, providing a provable access to the instruction set, providing commands to a production tool of the polymer production process in response to the instruction set access request, and recording a transaction on the distributed ledger in response to the providing commands to the production tool.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the instruction set includes an instruction set for a chemical synthesis subprocess of the polymer production process.

A further embodiment of any of the foregoing embodiments of the present disclosure may further include providing commands to a production tool of the chemical synthesis subprocess of the polymer production process in response to the instruction set access request and recording a transaction on the distributed ledger in response to the providing commands to the production tool of the chemical synthesis subprocess of the polymer production process.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the instruction set includes a field programmable gate array (FPGA) instruction set.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the instruction set further includes an application programming interface (API).

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the distributed ledger includes a smart wrapper for a specific instruction set.

The present disclosure describes a method, the method according to one disclosed, non-limiting, embodiment of the present disclosure can include accessing a distributed ledger including an instruction set, wherein the instruction set includes an instruction set for a chemical synthesis process, tokenizing the instruction set, interpreting an instruction set access request; in response to the instruction set access request, providing a provable access to the instruction set; providing commands to a production tool of the chemical synthesis process in response to the instruction set access request; and recording a transaction on the distributed ledger in response to the providing commands to the production tool.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the instruction set includes a field programmable gate array (FPGA) instruction set.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the instruction set includes an instruction set for a chemical synthesis subprocess for a food preparation process.

A further embodiment of any of the foregoing embodiments of the present disclosure may further include providing commands to a production tool of the chemical synthesis subprocess for the food preparation process in response to the instruction set access request and recording a transaction on the distributed ledger in response to the providing commands to the production tool of the chemical synthesis subprocess for the food preparation process.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the distributed ledger includes instruction ledger operations.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the instruction ledger operations include determining an instruction access probability.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the instruction ledger operations include providing an aggregate view comprising a list of which parties have accessed the instruction set.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the aggregate view further includes how many parties have accessed the instruction set.

The present disclosure describes a method, the method according to one disclosed, non-limiting, embodiment of the present disclosure can include accessing a distributed ledger comprising an instruction set, wherein the instruction set includes an instruction set for a chemical crystal fabrication process, tokenizing the instruction set, interpreting an instruction set access request, in response to the instruction set access request, providing a provable access to the instruction set, providing commands to a production tool of the chemical crystal fabrication process in response to the instruction set access request, and recording a transaction on the distributed ledger in response to the providing commands to the production tool.

A further embodiment of any of the foregoing embodiments of the present disclosure may include situations wherein the instruction set includes a field programmable gate array (FPGA) instruction set.

A further embodiment of any of the foregoing embodiments of the present disclosure may further include enabling a party to submit updates to the instruction set.

A further embodiment of any of the foregoing embodiments of the present disclosure may further include updating the instruction set.

A further embodiment of any of the foregoing embodiments of the present disclosure may further include storing the updated instruction set separately from the distributed ledger.

A further embodiment of any of the foregoing embodiments of the present disclosure may further include providing the updated instruction set to a user.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a schematic diagram of components of a platform for enabling intelligent transactions in accordance with embodiments of the present disclosure.

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FIGS. 2A-2B is a schematic diagram of additional components of a platform for enabling intelligent transactions in accordance with embodiments of the present disclosure.

FIG. 3 is a schematic diagram of additional components of a platform for enabling intelligent transactions in accordance with embodiments of the present disclosure.

FIG. 4 to FIG. 31 are schematic diagrams of embodiments of neural net systems that may connect to, be integrated in, and be accessible by the platform for enabling intelligent transactions including ones involving expert systems, self-organization, machine learning, artificial intelligence and including neural net systems trained for pattern recognition, for classification of one or more parameters, characteristics, or phenomena, for support of autonomous control, and other purposes in accordance with embodiments of the present disclosure.

FIG. 32 is a schematic diagram of components of an environment including an intelligent energy and compute facility, a host intelligent energy and compute facility resource management platform, a set of data sources, a set of expert systems, interfaces to a set of market platforms and external resources, and a set of user or client systems and devices in accordance with embodiments of the present disclosure.

FIG. 33 is a schematic diagram of an energy and computing resource platform in accordance with embodiments of the present disclosure.

FIGS. 34 and 35 are illustrative depictions of data record schema in accordance with embodiments of the present disclosure.

FIG. 36 is a schematic diagram of a cognitive processing system in accordance with embodiments of the present disclosure.

FIG. 37 is a schematic flow diagram of a procedure for selecting leads in accordance with embodiments of the present disclosure.

FIG. 38 is a schematic flow diagram of a procedure for generating a lead list in accordance with embodiments of the present disclosure.

FIG. 39 is a schematic flow diagram of a procedure for generating customized facility attributes in accordance with embodiments of the present disclosure.

FIG. 40 is a schematic diagram of a system including a smart contract wrapper.

FIG. 41 is a schematic flow diagram of a method for executing a smart contract wrapper.

FIG. 42 is a schematic flow diagram of a method for updating an aggregate IP stack.

FIG. 43 is a schematic flow diagram of a method for adding assets and entities.

FIG. 44 is a schematic flow diagram of a method for updating an aggregate IP stack.

FIG. 45 is a schematic diagram of a system for providing verifiable access to an instruction set.

FIG. 46 is a schematic flow diagram of a method for providing verifiable access to an instruction set.

FIG. 47 is a schematic diagram of a system for providing verifiable access to algorithmic logic.

FIG. 48 is a schematic flow diagram of a method for providing verifiable access to executable algorithmic logic.

FIG. 49 is a schematic diagram of a system for providing verifiable access to firmware.

FIG. 50 is a schematic flow diagram of a method for providing verifiable access to firmware.

FIG. 51 is a schematic diagram of a system for providing verifiable access to serverless code logic.

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FIG. 52 is a schematic flow diagram of a method for providing verifiable access to serverless code logic.

FIG. 53 is a schematic diagram of a system for providing verifiable access to an aggregated data set.

FIG. 54 is a schematic flow diagram of a method for providing verifiable access to an aggregated data set.

FIG. 55 is a schematic diagram of a system for analyzing and reporting on an aggregate stack of IP.

FIG. 56 is a schematic flow diagram of a method for analyzing and reporting on an aggregate stack of IP.

FIG. 57 is a schematic diagram of a system for improving resource utilization for a task system.

FIG. 58 is a schematic flow diagram of a method for improving resource utilization for a task system.

FIG. 59 is a schematic flow diagram of a method for improving resource utilization with a substitute resource.

FIG. 60 is a schematic flow diagram of a method for improving resource utilization with behavioral data.

FIG. 61 is a schematic flow diagram of a method for improving resource utilization for a task system.

FIG. 62 is a schematic diagram of a system for improving a cryptocurrency transaction request outcome.

FIG. 63 is a schematic flow diagram of a method for improving a cryptocurrency transaction request outcome.

FIG. 64 is a schematic flow diagram of a method for improving a cryptocurrency transaction request outcome.

FIG. 65 is a schematic diagram of a system for improving a cryptocurrency transaction request outcome.

FIG. 66 is a schematic flow diagram of a method for improving a cryptocurrency transaction request outcome.

FIG. 67 is a schematic diagram of a system for improving execution of cryptocurrency transactions.

FIG. 68 is a schematic flow diagram of a method for improving execution of cryptocurrency transactions.

FIG. 69 is a schematic diagram of a system for improving attention market transaction operations.

FIG. 70 is a schematic flow diagram of a method for improving attention market transaction operations.

FIG. 71 is a schematic diagram of a system for aggregating attention resource acquisition for a fleet.

FIG. 72 is a schematic flow diagram of a method for aggregating attention resource acquisition for a fleet.

FIG. 73 is a schematic diagram of a system to improve production facility outcome predictions.

FIG. 74 is a schematic flow diagram of a method to improve production facility outcome predictions.

FIG. 75 is a schematic diagram of a system to improve a facility resource parameter.

FIG. 76 is a schematic flow diagram of a method to improve a facility resource parameter.

FIG. 77 is a schematic diagram of a system to improve a facility output value.

FIG. 78 is a schematic flow diagram of a method to improve a facility output value.

FIG. 79 is a schematic flow diagram of a method to improve a facility production prediction.

FIG. 80 is a schematic diagram of a system to improve facility resource utilization.

FIG. 81 is a schematic flow diagram of a method to improve facility resource utilization.

FIG. 82 is a schematic diagram of a system to improve facility resource outcomes by adjusting a facility configuration.

FIG. 83 is a schematic diagram of a system for improving facility resource outcomes using a digital twin.

FIG. 84 is a schematic flow diagram of a method for improving facility resource outcomes using a digital twin.

FIG. **85** is a schematic diagram of a system for improving regenerative energy delivery for a facility.

FIG. **86** is a schematic flow diagram of a method for improving regenerative energy delivery for a facility.

FIG. **87** is a schematic diagram of a system for improving resource acquisition for a facility.

FIG. **88** is a schematic flow diagram of a method for improving resource acquisition for a facility.

FIG. **89** is a schematic diagram of a system for improving resource acquisition for a fleet of machines.

FIG. **90** is a schematic flow diagram of a method for improving resource acquisition for a fleet of machines.

FIG. **91** is a schematic diagram of a system for improving resource utilization for a fleet of machines.

FIG. **92** is a schematic flow diagram of a method for improving resource utilization for a fleet of machines.

FIG. **93** is a schematic diagram of a system for improving resource utilization of a machine.

FIG. **94** is a schematic flow diagram of a method for improving resource utilization of a machine.

FIG. **95** is a schematic diagram of a system for improving resource utilization for a fleet of machines.

FIG. **96** is a schematic flow diagram of a method for improving resource utilization for a fleet of machines.

FIG. **97** is a schematic diagram of a system for improving resource utilization for a machine using social media data and a forward resource market.

FIG. **98** is a schematic flow diagram of a method for improving resource utilization for a machine using social media data and a forward resource market.

FIG. **99** is a schematic diagram of a system for improving resource utilization using an arbitrage operation.

FIG. **100** is a schematic flow diagram of a method for improving resource utilization using an arbitrage operation.

FIG. **101** is a schematic diagram of an apparatus for improving resource utilization using an arbitrage operation.

FIG. **102** is a schematic diagram of a system for improving resource distribution for a machine.

FIG. **103** is a schematic flow diagram of a system for improving resource distribution for a machine.

FIG. **104** is a schematic diagram of a system for improving aggregated resource delivery for a fleet of machines.

FIG. **105** is a schematic flow diagram of a method for improving aggregated resource delivery for a fleet of machines.

FIG. **106** is a schematic diagram of an apparatus for improving aggregated resource delivery for a fleet of machines.

FIG. **107** is a schematic diagram of a system for improving resource delivery for a machine using a forward resource market.

FIG. **108** is a schematic flow diagram of a method for improving resource delivery for a machine using a forward resource market.

FIG. **109** is a schematic flow diagram of a method for improving resource delivery with a substitute resource.

DETAILED DESCRIPTION

Referring to FIG. **1**, a set of systems, methods, components, modules, machines, articles, blocks, circuits, services, programs, applications, hardware, software and other elements are provided, collectively referred to herein interchangeably as the system **100** or the platform **100**. The platform **100** enables a wide range of improvements of and for various machines, systems, and other components that enable transactions involving the exchange of value (such as

using currency, cryptocurrency, tokens, rewards or the like, as well as a wide range of in-kind and other resources) in various markets, including current or spot markets **170**, forward markets **130** and the like, for various goods, services, and resources. As used herein, “currency” should be understood to encompass fiat currency issued or regulated by governments, cryptocurrencies, tokens of value, tickets, loyalty points, rewards points, coupons, credits (e.g., regulatory, emissions, and/or industry recognized exchangeable units of credit), abstracted versions of these (e.g., an arbitrary value index between parties understood to be usable in a future transaction or the like), deliverables (e.g., service up-time, contractual delivery of goods or services including time values utilized for averaging or other measures of delivery which may then be exchanged between parties or utilized to offset other exchanged value), and other elements that represent or may be exchanged for value. Resources, such as ones that may be exchanged for value in a marketplace, should be understood to encompass goods, services, natural resources, energy resources, computing resources, energy storage resources, data storage resources, network bandwidth resources, processing resources and the like, including resources for which value is exchanged and resources that enable a transaction to occur (such as necessary computing and processing resources, storage resources, network resources, and energy resources that enable a transaction). Any currency and/or resources may be abstracted to a uniform and/or normalized value scale, and may additionally or alternatively include a time aspect (e.g., calendar date, seasonal correction, and/or appropriate circumstances relevant to the value of the currency and/or resources at the time of a transaction) and/or an exchange rate aspect (e.g., accounting for the value of a currency or resource relative to another similar unit, such as currency exchange rates between countries, discounting of a good or service relative to a most desired or requested configuration of the good or service, etc.).

The platform **100** may include a set of forward purchase and sale machines **110**, each of which may be configured as an expert system or automated intelligent agent for interaction with one or more of the set of spot markets **170** (e.g., reference FIG. **2A**) and forward markets **130**. Enabling the set of forward purchase and sale machines **110** are an intelligent resource purchasing system **164** having a set of intelligent agents for purchasing resources in spot and forward markets; an intelligent resource allocation and coordination engine **168** for the intelligent sale of allocated or coordinated resources, such as compute resources, energy resources, and other resources involved in or enabling a transaction; an intelligent sale engine **172** for intelligent coordination of a sale of allocated resources in spot and futures markets; and an automated spot market testing and arbitrage transaction execution engine **194** for performing spot testing of spot and forward markets, such as with micro-transactions and, where conditions indicate favorable arbitrage conditions, automatically executing transactions in resources that take advantage of the favorable conditions. Each of the engines may use model-based or rule-based expert systems, such as based on rules or heuristics, as well as deep learning systems by which rules or heuristics may be learned over trials involving a large set of inputs. The engines may use any of the expert systems and artificial intelligence capabilities described throughout this disclosure. Interactions within the platform **100**, including of all platform components, and of interactions among them and with various markets, may be tracked and collected, such as by a data aggregation system **144**, such as for aggregating

data on purchases and sales in various marketplaces by the set of machines described herein. Aggregated data may include tracking and outcome data that may be fed to artificial intelligence and machine learning systems, such as to train or supervise the same.

Operations to aggregate information as referenced throughout the present disclosure should be understood broadly. Example operations to aggregate information (e.g., data, purchasing, regulatory information, or any other parameters) include, without limitation: summaries, averages of data values, selected binning of data, derivative information about data (e.g., rates of change, areas under a curve, changes in an indicated state based on the data, exceedance or conformance with a threshold value, etc.), changes in the data (e.g., arrival of new information or a new type of information, information accrued in a defined or selected time period, etc.), and/or categorical descriptions about the data or other information related to the data). It will be understood that the expression of aggregated information can be as desired, including at least as graphical information, a report, stored raw data for utilization in generating displays and/or further use by an artificial intelligence and/or machine learning system, tables, and/or a data stream. In certain embodiments, aggregated data may be utilized by an expert system, an artificial intelligence, and/or a machine learning system to perform various operations described throughout the present disclosure. Additionally or alternatively, expert systems, artificial intelligence, and/or machine learning systems may interact with the aggregated data, including determining which parameters are to be aggregated and/or the aggregation criteria to be utilized. For example, a machine learning system for a system utilizing a forward energy purchasing market may be configured to aggregate purchasing for the system. In the example, the machine learning system may be configured to determine the signal effective parameters to incrementally improve and/or optimize purchasing decisions, and may additionally or alternatively change the aggregation parameters—for example binning criteria for various components of a system (e.g., components that respond in a similar manner from the perspective of energy requirements), determining the time frame of aggregation (e.g., weekly, monthly, seasonal, etc.), and/or changing a type of average, a reference rate for a rate of change of values in the system, or the like. The provided examples are provided for illustration, and are not limiting to any systems or operations described throughout the present disclosure.

The various engines may operate on a range of data sources, including aggregated data from marketplace transactions, tracking data regarding the behavior of each of the engines, and a set of external data sources **182**, which may include social media data sources **180** (such as social networking sites like Facebook™ and Twitter™), Internet of Things (IoT) data sources (including from sensors, cameras, data collectors, appliances, personal devices, and/or instrumented machines and systems), such as IoT sources that provide information about machines and systems that enable transactions and machines and systems that are involved in production and consumption of resources. External data sources **182** may include behavioral data sources, such as automated agent behavioral data sources **188** (such as tracking and reporting on behavior of automated agents that are used for conversation and dialog management, agents used for control functions for machines and systems, agents used for purchasing and sales, agents used for data collection, agents used for advertising, and others), human behavioral data sources (such as data sources tracking online behavior,

mobility behavior, energy consumption behavior, energy production behavior, network utilization behavior, compute and processing behavior, resource consumption behavior, resource production behavior, purchasing behavior, attention behavior, social behavior, and others), and entity behavioral data sources **190** (such as behavior of business organizations and other entities, such as purchasing behavior, consumption behavior, production behavior, market activity, merger and acquisition behavior, transaction behavior, location behavior, and others). The IoT, social and behavioral data from and about sensors, machines, humans, entities, and automated agents may collectively be used to populate expert systems, machine learning systems, and other intelligent systems and engines described throughout this disclosure, such as being provided as inputs to deep learning systems and being provided as feedback or outcomes for purposes of training, supervision, and iterative improvement of systems for prediction, forecasting, classification, automation and control. The data may be organized as a stream of events. The data may be stored in a distributed ledger or other distributed system. The data may be stored in a knowledge graph where nodes represent entities and links represent relationships. The external data sources may be queried via various database query functions. The external data sources **182** may be accessed via APIs, brokers, connectors, protocols like REST and SOAP, and other data ingestion and extraction techniques. Data may be enriched with metadata and may be subject to transformation and loading into suitable forms for consumption by the engines, such as by cleansing, normalization, de-duplication and the like.

The platform **100** may include a set of intelligent forecasting engines **192** for forecasting events, activities, variables, and parameters of spot markets **170**, forward markets **130**, resources that are traded in such markets, resources that enable such markets, behaviors (such as any of those tracked in the external data sources **182**), transactions, and the like. The intelligent forecasting engines **192** may operate on data from the data aggregation system **144** about elements of the platform **100** and on data from the external data sources **182**. The platform may include a set of intelligent transaction engines **136** for automatically executing transactions in spot markets **170** and forward markets **130**. This may include executing intelligent cryptocurrency transactions with an intelligent cryptocurrency execution engine **183** as described in more detail below. The platform **100** may make use of asset of improved distributed ledgers **113** and improved smart contracts **103**, including ones that embed and operate on proprietary information, instruction sets and the like that enable complex transactions to occur among individuals with reduced (or without) reliance on intermediaries. In certain embodiments, the platform **100** may include a distributed processing architecture **146**—for example distributing processing or compute tasks across multiple processing devices, clusters, servers, and/or third-party service devices or cloud devices. These and other components are described in more detail throughout this disclosure. In certain embodiments, one or more aspects of any of the platforms referenced in FIGS. **1** to **3** may be performed by any systems, apparatuses, controllers, or circuits as described throughout the present disclosure. In certain embodiments, one or more aspects of any of the platforms referenced in FIGS. **1** to **3** may include any procedures, methods, or operations described throughout the present disclosure. The example platforms depicted in FIGS. **1** to **3** are illustrative, and any aspects may be omitted or

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altered while still providing one or more benefits as described throughout the present disclosure.

Referring to the block diagrams of FIGS. 2A-2B, further details and additional components of the platform **100** and interactions among them are depicted. The set of forward purchase and sale machines **110** may include a regeneration capacity allocation engine **102** (such as for allocating energy generation or regeneration capacity, such as within a hybrid vehicle or system that includes energy generation or regeneration capacity, a renewable energy system that has energy storage, or other energy storage system, where energy is allocated for one or more of sale on a forward market **130**, sale in a spot market **170**, use in completing a transaction (e.g., mining for cryptocurrency), or other purposes. For example, the regeneration capacity allocation engine **102** may explore available options for use of stored energy, such as sale in current and forward energy markets (e.g., energy forward market **122**, energy market **148**, energy storage forward market **174**, and/or energy storage market **178**) that accept energy from producers, keeping the energy in storage for future use, or using the energy for work (which may include processing work, such as processing activities of the platform like data collection or processing, or processing work for executing transactions, including mining activities for cryptocurrencies). In certain embodiments, the regeneration capacity allocation engine **102** includes a time value of stored energy, for example accounting for energy leakage (e.g., losses over time when stored), future useful work activities that are expected to arise, competing factors that may affect the stored energy (e.g., a release of reservoir water that is expected to occur at a future time for a non-energy purpose), and/or future energy regeneration that can be predicted that may affect the stored energy value proposition (e.g., energy storage will be exceeded if the energy is retained, and/or the value of available useful work activities will change in a relevant time horizon). In certain embodiments, the regeneration capacity allocation engine **102** includes a rate value of stored energy, for example accounting for the incremental cost or benefit of utilizing stored energy rapidly (e.g., a low utilization of energy at the present time is cost effective, but a high utilization of energy at the present time is not cost effective). In certain embodiments, the regeneration capacity allocation engine **102** considers externalities that are outside of the economic considerations of the immediate system. For example, effects on a reservoir or downstream river bed due to energy utilization, grid capacity and/or grid dynamics effects of providing or not providing energy from the energy storage, and/or system effects from providing or not providing energy (e.g., ramping up server utilization with inexpensive energy immediately before a long holiday that may result in overtime pay for service and maintenance personnel), may affect the economic effectiveness of accepting, storing, or utilizing energy). The provided examples are provided for illustration, and are not limiting to any systems or operations described throughout the present disclosure.

The set of forward purchase and sale machines **110** may include an energy purchase and sale machine **104** for purchasing or selling energy, such as in an energy spot market **148** or an energy forward market **122**. The energy purchase and sale machine **104** may use an expert system, neural network or other intelligence to determine timing of purchases, such as based on current and anticipated state information with respect to pricing and availability of energy and based on current and anticipated state information with respect to needs for energy, including needs for energy to perform computing tasks, cryptocurrency mining,

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data collection actions, and other work, such as work done by automated agents and systems and work required for humans or entities based on their behavior. For example, the energy purchase machine may recognize, by machine learning, that a business is likely to require a block of energy in order to perform an increased level of manufacturing based on an increase in orders or market demand and may purchase the energy at a favorable price on a futures market, based on a combination of energy market data and entity behavioral data. Continuing the example, market demand may be understood by machine learning, such as by processing human behavioral data sources **184**, such as social media posts, e-commerce data and the like that indicate increasing demand. The energy purchase and sale machine **104** may sell energy in the energy spot market **148** or the energy forward market **122**. Sale may also be conducted by an expert system operating on the various data sources described herein, including with training on outcomes and human supervision.

The set of forward purchase and sale machines **110** may include a renewable energy credit (REC) purchase and sale machine **108**, which may purchase renewable energy credits, pollution credits, and other environmental or regulatory credits in a spot market **170** or forward market **124** for such credits. Purchasing may be configured and managed by an expert system operating on any of the external data sources **182** or on data aggregated by the set of data aggregation systems **144** for the platform. Renewable energy credits and other credits may be purchased by an automated system using an expert system, including machine learning or other artificial intelligence, such as where credits are purchased with favorable timing based on an understanding of supply and demand that is determined by processing inputs from the data sources. The expert system may be trained on a data set of outcomes from purchases under historical input conditions. The expert system may be trained on a data set of human purchase decisions and/or may be supervised by one or more human operators. The renewable energy credit (REC) purchase and sale machine **108** may also sell renewable energy credits, pollution credits, and other environmental or regulatory credits in a spot market **170** or forward market **124** for such credits. Sale may also be conducted by an expert system operating on the various data sources described herein, including with training on outcomes and human supervision.

The set of forward purchase and sale machines **110** may include an attention purchase and sale machine **112**, which may purchase one or more attention-related resources, such as advertising space, search listing, keyword listing, banner advertisements, participation in a panel or survey activity, participation in a trial or pilot, or the like in a spot market for attention **152** or a forward market for attention **128**. Attention resources may include the attention of automated agents, such as bots, crawlers, dialog managers, and the like that are used for searching, shopping and purchasing. Purchasing of attention resources may be configured and managed by an expert system operating on any of the external data sources **182** or on data aggregated by the set of data aggregation systems **144** for the platform. Attention resources may be purchased by an automated system using an expert system, including machine learning or other artificial intelligence, such as where resources are purchased with favorable timing, such as based on an understanding of supply and demand, that is determined by processing inputs from the various data sources. For example, the attention purchase and sale machine **112** may purchase advertising space in a forward market for advertising based on learning from a wide range of inputs about market conditions,

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behavior data, and data regarding activities of agent and systems within the platform **100**. The expert system may be trained on a data set of outcomes from purchases under historical input conditions. The expert system may be trained on a data set of human purchase decisions and/or may be supervised by one or more human operators. The attention purchase and sale machine **112** may also sell one or more attention-related resources, such as advertising space, search listing, keyword listing, banner advertisements, participation in a panel or survey activity, participation in a trial or pilot, or the like in a spot market for attention **152** or a forward market for attention **128**, which may include offering or selling access to, or attention or, one or more automated agents of the platform **100**. Sale may also be conducted by an expert system operating on the various data sources described herein, including with training on outcomes and human supervision.

The set of forward purchase and sale machines **110** may include a compute purchase and sale machine **114**, which may purchase one or more computation-related resources, such as processing resources, database resources, computation resources, server resources, disk resources, input/output resources, temporary storage resources, memory resources, virtual machine resources, container resources, and others in a spot market for compute **154** or a forward market for compute **132**. Purchasing of compute resources may be configured and managed by an expert system operating on any of the external data sources **182** or on data aggregated by the set of data aggregation systems **144** for the platform. Compute resources may be purchased by an automated system using an expert system, including machine learning or other artificial intelligence, such as where resources are purchased with favorable timing, such as based on an understanding of supply and demand, that is determined by processing inputs from the various data sources. For example, the compute purchase and sale machine **114** may purchase or reserve compute resources on a cloud platform in a forward market for computer resources based on learning from a wide range of inputs about market conditions, behavior data, and data regarding activities of agent and systems within the platform **100**, such as to obtain such resources at favorable prices during surge periods of demand for computing. The expert system may be trained on a data set of outcomes from purchases under historical input conditions. The expert system may be trained on a data set of human purchase decisions and/or may be supervised by one or more human operators. The compute purchase and sale machine **114** may also sell one or more computation-related resources that are connected to, part of, or managed by the platform **100**, such as processing resources, database resources, computation resources, server resources, disk resources, input/output resources, temporary storage resources, memory resources, virtual machine resources, container resources, and others in a spot market for compute **154** or a forward market for compute **132**. Sale may also be conducted by an expert system operating on the various data sources described herein, including with training on outcomes and human supervision.

The set of forward purchase and sale machines **110** may include a data storage purchase and sale machine **118**, which may purchase one or more data-related resources, such as database resources, disk resources, server resources, memory resources, RAM resources, network attached storage resources, storage attached network (SAN) resources, tape resources, time-based data access resources, virtual machine resources, container resources, and others in a spot market for data storage **158** or a forward market for data

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storage **134**. Purchasing of data storage resources may be configured and managed by an expert system operating on any of the external data sources **182** or on data aggregated by the set of data aggregation systems **144** for the platform. Data storage resources may be purchased by an automated system using an expert system, including machine learning or other artificial intelligence, such as where resources are purchased with favorable timing, such as based on an understanding of supply and demand, that is determined by processing inputs from the various data sources. For example, the compute purchase and sale machine **114** may purchase or reserve compute resources on a cloud platform in a forward market for compute resources based on learning from a wide range of inputs about market conditions, behavior data, and data regarding activities of agent and systems within the platform **100**, such as to obtain such resources at favorable prices during surge periods of demand for storage. The expert system may be trained on a data set of outcomes from purchases under historical input conditions. The expert system may be trained on a data set of human purchase decisions and/or may be supervised by one or more human operators. The data storage purchase and sale machine **118** may also sell one or more data storage-related resources that are connected to, part of, or managed by the platform **100** in a spot market for data storage **158** or a forward market for data storage **134**. Sale may also be conducted by an expert system operating on the various data sources described herein, including with training on outcomes and human supervision.

The set of forward purchase and sale machines **110** may include a bandwidth purchase and sale machine **120**, which may purchase one or more bandwidth-related resources, such as cellular bandwidth, Wifi bandwidth, radio bandwidth, access point bandwidth, beacon bandwidth, local area network bandwidth, wide area network bandwidth, enterprise network bandwidth, server bandwidth, storage input/output bandwidth, advertising network bandwidth, market bandwidth, or other bandwidth, in a spot market for bandwidth **160** or a forward market for bandwidth **138**. Purchasing of bandwidth resources may be configured and managed by an expert system operating on any of the external data sources **182** or on data aggregated by the set of data aggregation systems **144** for the platform. Bandwidth resources may be purchased by an automated system using an expert system, including machine learning or other artificial intelligence, such as where resources are purchased with favorable timing, such as based on an understanding of supply and demand, that is determined by processing inputs from the various data sources. For example, the bandwidth purchase and sale machine **120** may purchase or reserve bandwidth on a network resource for a future networking activity managed by the platform based on learning from a wide range of inputs about market conditions, behavior data, and data regarding activities of agent and systems within the platform **100**, such as to obtain such resources at favorable prices during surge periods of demand for bandwidth. The expert system may be trained on a data set of outcomes from purchases under historical input conditions. The expert system may be trained on a data set of human purchase decisions and/or may be supervised by one or more human operators. The bandwidth purchase and sale machine **120** may also sell one or more bandwidth-related resources that are connected to, part of, or managed by the platform **100** in a spot market for bandwidth **160** or a forward market for bandwidth **138**. Sale may also be conducted by an expert

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system operating on the various data sources described herein, including with training on outcomes and human supervision.

The set of forward purchase and sale machines **110** may include a spectrum purchase and sale machine **142**, which may purchase one or more spectrum-related resources, such as cellular spectrum, 3G spectrum, 4G spectrum, LTE spectrum, 5G spectrum, cognitive radio spectrum, peer-to-peer network spectrum, emergency responder spectrum and the like in a spot market for spectrum **162** or a forward market for spectrum **140**. In certain embodiments, a spectrum related resource may relate to a non-wireless communication protocol, such as frequency stacking on a hard line (e.g., a copper wire or optical fiber). Purchasing of spectrum resources may be configured and managed by an expert system operating on any of the external data sources **182** or on data aggregated by the set of data aggregation systems **144** for the platform. Spectrum resources may be purchased by an automated system using an expert system, including machine learning or other artificial intelligence, such as where resources are purchased with favorable timing, such as based on an understanding of supply and demand, that is determined by processing inputs from the various data sources. For example, the spectrum purchase and sale machine **142** may purchase or reserve spectrum on a network resource for a future networking activity managed by the platform based on learning from a wide range of inputs about market conditions, behavior data, and data regarding activities of agent and systems within the platform **100**, such as to obtain such resources at favorable prices during surge periods of demand for spectrum. The expert system may be trained on a data set of outcomes from purchases under historical input conditions. The expert system may be trained on a data set of human purchase decisions and/or may be supervised by one or more human operators. The spectrum purchase and sale machine **142** may also sell one or more spectrum-related resources that are connected to, part of, or managed by the platform **100** in a spot market for spectrum **162** or a forward market for spectrum **140**. Sale may also be conducted by an expert system operating on the various data sources described herein, including with training on outcomes and human supervision.

In embodiments, the intelligent resource allocation and coordination engine **168**, including the intelligent resource purchasing system **164**, the intelligent sale engine **172** and the automated spot market testing and arbitrage transaction execution engine **194**, may provide coordinated and automated allocation of resources and coordinated execution of transactions across the various forward markets **130** and spot markets **170** by coordinating the various purchase and sale machines, such as by an expert system, such as a machine learning system (which may model-based or a deep learning system, and which may be trained on outcomes and/or supervised by humans). For example, the allocation and coordination engine **168** may coordinate purchasing of resources for a set of assets and coordinated sale of resources available from a set of assets, such as a fleet of vehicles, a data center of processing and data storage resources, an information technology network (on premises, cloud, or hybrids), a fleet of energy production systems (renewable or non-renewable), a smart home or building (including appliances, machines, infrastructure components and systems, and the like thereof that consume or produce resources), and the like.

The platform **100** may incrementally improve or optimize allocation of resource purchasing, sale and utilization based on data aggregated in the platform, such as by tracking

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activities of various engines and agents, as well as by taking inputs from external data sources **182**. In embodiments, outcomes may be provided as feedback for training the intelligent resource allocation and coordination engine **168**, such as outcomes based on yield, profitability, optimization of resources, optimization of business objectives, satisfaction of goals, satisfaction of users or operators, or the like. For example, as the energy for computational tasks becomes a significant fraction of an enterprise's energy usage, the platform **100** may learn to optimize how a set of machines that have energy storage capacity allocate that capacity among computing tasks (such as for cryptocurrency mining, application of neural networks, computation on data and the like), other useful tasks (that may yield profits or other benefits), storage for future use, or sale to the provider of an energy grid. The platform **100** may be used by fleet operators, enterprises, governments, municipalities, military units, first responder units, manufacturers, energy producers, cloud platform providers, and other enterprises and operators that own or operate resources that consume or provide energy, computation, data storage, bandwidth, or spectrum. The platform **100** may also be used in connection with markets for attention, such as to use available capacity of resources to support attention-based exchanges of value, such as in advertising markets, micro-transaction markets, and others.

Operations to optimize, as used throughout the present disclosure, should be understood broadly. In certain embodiments, operations to optimize include operations to improve outcomes, including incremental and/or iterative improvements. In certain embodiments, optimization can include operations to improve outcomes until a threshold improvement level is reached (e.g., a success criterion is met, further improvements are below a threshold level of improvement, a particular outcome is improved by a threshold amount, etc.). In certain embodiments, optimization may be performed utilizing a cost and benefit analysis, where cost is in actual currency, a normalized cost value, a cost index configured to describe the resources, time, and/or lost opportunity of a particular action, or any other cost description. In certain embodiments, benefits may be in actual currency, a normalized benefit value, a benefit index, or any other measure or description of the benefit of a particular action. In certain embodiments, other parameters such as the time value and/or time trajectory of costs or benefits may be included in the optimization—for example as a limiting value (e.g., optimization is the best value after 5 minutes of computations) and/or as a factor (e.g., a growing cost or shrinking benefit is applied as optimization analyses progress) in the optimization process. Any operations utilizing artificial intelligence, expert systems, machine learning, and/or any other systems or operations described throughout the present disclosure that incrementally improve, iteratively improve, and/or formally optimize parameters are understood as examples of optimization and/or improvement herein. One of skill in the art, having the benefit of the present disclosure and information ordinarily available when contemplating a particular system, can readily determine parameters and criteria for optimization for a particular system. Certain considerations that may be relevant to a particular system include, without limitation: the cost of resource utilization including time values and/or time trajectories of those costs; the benefits of the action goals (e.g., selling resources, completing calculations, providing bandwidth, etc.) including time values and/or time trajectories of those benefits; seasonal, periodic, and/or episodic effects on the availability of resources and/or the demand for

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resources; costs of capitalization for a system and/or for a servicing system (e.g., costs to add computing resources, and/or costs for a service provider to add computing resources); operating costs for utilization of resources, including time values, time trajectories, and externalities such as personnel, maintenance, and incremental utilization of service life for resources; capacity of resource providers and/or cost curves for resource utilization; diminishing return curves and/or other external effects for benefit provisions (e.g., the 100th unit of computation may pay less than the 50th unit of computation for a particular system, and/or the ability to provide 100 units of computation may open other markets and/or allow for servicing of a different customer base than the ability to provide only 50 units of computation); and/or risk factors related to resource utilization (e.g., increasing data storage at a single location may increase risk over distributed data; increasing throughput of a system may change the risks, such as increased traffic, higher operating points for systems, increased risk of regulatory violations, or the like).

Referring still to FIGS. 2A-2B, the platform 100 may include a set of intelligent forecasting engines 192 that forecast one or more attributes, parameters, variables, or other factors, such as for use as inputs by the set of forward purchase and sale machines, the intelligent transaction engines 136 (such as for intelligent cryptocurrency execution) or for other purposes. Each of the set of intelligent forecasting engines 192 may use data that is tracked, aggregated, processed, or handled within the platform 100, such as by the data aggregation system 144, as well as input data from external data sources 182, such as social media data sources 180, automated agent behavioral data sources 188, human behavioral data sources 184, entity behavioral data sources 190 and IoT data sources 198. These collective inputs may be used to forecast attributes, such as using a model (e.g., Bayesian, regression, or other statistical model), a rule, or an expert system, such as a machine learning system that has one or more classifiers, pattern recognizers, and predictors, such as any of the expert systems described throughout this disclosure. In embodiments, the set of intelligent forecasting engines 192 may include one or more specialized engines that forecast market attributes, such as capacity, demand, supply, and prices, using particular data sources for particular markets. These may include an energy price forecasting engine 215 that bases its forecast on behavior of an automated agent, a network spectrum price forecasting engine 217 that bases its forecast on behavior of an automated agent, a REC price forecasting engine 219 that bases its forecast on behavior of an automated agent, a compute price forecasting engine 221 that bases its forecast on behavior of an automated agent, a network spectrum price forecasting engine 223 that bases its forecast on behavior of an automated agent. In each case, observations regarding the behavior of automated agents, such as ones used for conversation, for dialog management, for managing electronic commerce, for managing advertising and others may be provided as inputs for forecasting to the engines. The intelligent forecasting engines 192 may also include a range of engines that provide forecasts at least in part based on entity behavior, such as behavior of business and other organizations, such as marketing behavior, sales behavior, product offering behavior, advertising behavior, purchasing behavior, transactional behavior, merger and acquisition behavior, and other entity behavior. These may include an energy price forecasting engine 225 using entity behavior, a network spectrum price forecasting engine 227 using entity behavior, a REC price forecasting engine 229 using entity

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behavior, a compute price forecasting engine 231 using entity behavior, and a network spectrum price forecasting engine 233 using entity behavior.

The intelligent forecasting engines 192 may also include a range of engines that provide forecasts at least in part based on human behavior, such as behavior of consumers and users, such as purchasing behavior, shopping behavior, sales behavior, product interaction behavior, energy utilization behavior, mobility behavior, activity level behavior, activity type behavior, transactional behavior, and other human behavior. These may include an energy price forecasting engine 235 using human behavior, a network spectrum price forecasting engine 237 using human behavior, a REC price forecasting engine 239 using human behavior, a compute price forecasting engine 241 using human behavior, and a network spectrum price forecasting engine 243 using human behavior.

Referring still to FIGS. 2A-2B, the platform 100 may include a set of intelligent transaction engines 136 that automate execution of transactions in forward markets 130 and/or spot markets 170 based on determination that favorable conditions exist, such as by the intelligent resource allocation and coordination engine 168 and/or with use of forecasts from the intelligent forecasting engines 192. The intelligent transaction engines 136 may be configured to automatically execute transactions, using available market interfaces, such as APIs, connectors, ports, network interfaces, and the like, in each of the markets noted above. In embodiments, the intelligent transaction engines may execute transactions based on event streams that come from external data sources 182, such as IoT data sources 198 and social media data sources 180. The engines may include, for example, an IoT forward energy transaction engine 195 and/or an IoT compute market transaction engine 106, either or both of which may use data 185 from the Internet of Things (IoT) to determine timing and other attributes for market transaction in a market for one or more of the resources described herein, such as an energy market transaction, a compute resource transaction or other resource transaction. IoT data 185 may include instrumentation and controls data for one or more machines (optionally coordinated as a fleet) that use or produce energy or that use or have compute resources, weather data that influences energy prices or consumption (such as wind data influencing production of wind energy), sensor data from energy production environments, sensor data from points of use for energy or compute resources (such as vehicle traffic data, network traffic data, IT network utilization data, Internet utilization and traffic data, camera data from work sites, smart building data, smart home data, and the like), and other data collected by or transferred within the Internet of Things, including data stored in IoT platforms and of cloud services providers like Amazon, IBM, and others. The intelligent transaction engines 136 may include engines that use social data to determine timing of other attributes for a market transaction in one or more of the resources described herein, such as a social data forward energy transaction engine 199 and/or a social data for compute market transaction engine 116. Social data may include data from social networking sites (e.g., Facebook™, YouTube™, Twitter™, Snapchat™, Instagram™, and others), data from websites, data from e-commerce sites, and data from other sites that contain information that may be relevant to determining or forecasting behavior of users or entities, such as data indicating interest or attention to particular topics, goods or services, data indicating activity types and levels, such as may be observed by machine processing of image data showing individuals

engaged in activities, including travel, work activities, leisure activities, and the like. Social data may be supplied to machine learning, such as for learning user behavior or entity behavior, and/or as an input to an expert system, a model, or the like, such as one for determining, based on the social data, the parameters for a transaction. For example, an event or set of events in a social data stream may indicate the likelihood of a surge of interest in an online resource, a product, or a service, and compute resources, bandwidth, storage, or like may be purchased in advance (avoiding surge pricing) to accommodate the increased interest reflected by the social data market predictor **186**.

Referring to FIG. 3, the platform **100** may include capabilities for transaction execution that involve one or more distributed ledgers **113** and one or more smart contracts **103**, where the distributed ledgers **113** and smart contracts **103** are configured to enable specialized transaction features for specific transaction domains. One such domain is intellectual property, which transactions are highly complex, involving licensing terms and conditions that are somewhat difficult to manage, as compared to more straightforward sales of goods or services. In embodiments, a smart contract wrapper, such as wrapper aggregating intellectual property **105**, is provided, using a distributed ledger, wherein the smart contract embeds IP licensing terms for intellectual property that is embedded in the distributed ledger and wherein executing an operation on the distributed ledger provides access to the intellectual property and commits the executing party to the IP licensing terms. Licensing terms for a wide range of goods and services, including digital goods like video, audio, video game, video game element, music, electronic book and other digital goods may be managed by tracking transactions involving them on a distributed ledger, whereby publishers may verify a chain of licensing and sublicensing. The distributed ledger may be configured to add each licensee to the ledger, and the ledger may be retrieved at the point of use of a digital item, such as in a streaming platform, to validate that licensing has occurred.

In embodiments, an improved distributed ledger is provided with the smart contract wrapper, such as an IP wrapper **105**, container, smart contract or similar mechanism for aggregating intellectual property licensing terms, wherein a smart contract wrapper on the distributed ledger allows an operation on the ledger to add intellectual property to an aggregate stack of intellectual property. In many cases, intellectual property builds on other intellectual property, such as where software code is derived from other code, where trade secrets or know-how **109** for elements of a process are combined to enable a larger process, where patents covering sub-components of a system or steps in a process are pooled, where elements of a video game include sub-component assets from different creators, where a book contains contributions from multiple authors, and the like. In embodiments, a smart IP wrapper aggregates licensing terms for different intellectual property items (including digital goods, including ones embodying different types of intellectual property rights, and transaction data involving the item), as well as optionally one or more portions of the item corresponding to the transaction data, are stored in a distributed ledger that is configured to enable validation of agreement to the licensing terms (such as at a point of use) and/or access control to the item. In certain embodiments, a smart IP wrapper may include sub-licenses, dependent licenses, verification of ownership and chain of title, and/or any other features that ensure that a license is valid and is able to be used. In embodiments, a royalty apportionment wrapper **115** may be provided in a system having a distrib-

uted ledger for aggregating intellectual property licensing terms, wherein a smart contract wrapper on the distributed ledger allows an operation on the ledger to add intellectual property and to agree to an apportionment of royalties among the parties in the ledger. Thus, a ledger may accumulate contributions to the ledger along with evidence of agreement to the apportionment of any royalties among the contributors of the IP that is embedded in and/or controlled by the ledger. The ledger may record licensing terms and automatically vary them as new contributions are made, such as by one or more rules. For example, contributors may be given a share of a royalty stack according to a rule, such as based on a fractional contribution, such as based on lines of code contributed, a number and/or value of effective operations contributed from a set of operations performed by one or more computer programs, a valuation contribution from a particular IP element into a larger good or service provided under the license or license group, lines of authorship, contribution to components of a system, and the like. In embodiments, a distributed ledger may be forked into versions that represent varying combinations of sub-components of IP, such as to allow users to select combinations that are of most use, thereby allowing contributors who have contributed the most value to be rewarded. Variation and outcome tracking may be iteratively improved, such as by machine learning. In certain embodiments, operations on a distributed ledger may include updating the licensing terms, valuations, and/or royalty shares according to external data, such as litigation and/or administrative decisions (e.g., from a patent or trademark office) that may affect intellectual property assets (e.g., increasing a validity estimate, determining an asset is invalid or unenforceable, and/or creating a determined valuation for the asset), changes of ownership, expiration and/or aging of assets, and/or changing of asset status (e.g., a patent application issuing as a patent).

In embodiments, a distributed ledger is provided for aggregating intellectual property licensing terms, wherein a smart contract wrapper on the distributed ledger allows an operation on the ledger to add intellectual property to an aggregate stack of intellectual property and/or to determine the relationship of the contributed intellectual property to the aggregate stack and to royalty generating elements related to the aggregate stack such as goods or services sold using the licensing terms. In certain embodiments, operations on the ledger update the relationships of various elements of intellectual property in the aggregate stack in response to additions to the stack—for example where a newly contributed element of intellectual property replaces an older one for certain goods or services, and/or changes the value proposition for intellectual property elements already in the aggregate stack.

In embodiments, the platform **100** may have an improved distributed ledger for aggregating intellectual property licensing terms, wherein a smart contract wrapper on the distributed ledger allows an operation on the ledger to commit a party to a contract term via an IP transaction wrapper **119** of the ledger. This may include operations involving cryptocurrencies, tokens, or other operations, as well as conventional payments and in-kind transfers, such as of various resources described herein. The ledger may accumulate evidence of commitments to IP transactions by parties, such as entering into royalty terms, revenue sharing terms, IP ownership terms, warranty and liability terms, license permissions and restrictions, field of use terms, and many others. In certain embodiments, the ledger may accumulate transactional data between parties which may include costs and/or payments in any form, including abstract or

indexed valuations that may be converted to currency and/or traded goods or services at a later time. In certain embodiments, the ledger may additionally or alternatively include geographic information (e.g., where a transaction occurred, where contractual acceptance is deemed to have occurred, where goods or services were performed/delivered, and/or where related data is stored), entity information (e.g., which entities, sub-entities, and/or affiliates are involved in licenses and transactions), and/or time information (e.g., when acceptance occurs, when licensing and updates occur, when goods and services are ordered, when contractual terms or payments are committed, and/or when contractual terms or payments are delivered). It can be seen that the use of improved distributed ledgers throughout the disclosure supports numerous improvements over previously known systems, including at least improved management of licensing agreements, tax management, contract management, data security, regulatory compliance, confidence that the agreed terms are correct on the merits, and confidence that the agreed terms are implemented properly.

In embodiments, improved distributed ledgers may include ones having a tokenized instruction set, such that operation on the distributed ledger provides provable access to the instruction set. A party wishing to share permission to know how, a trade secret or other valuable instructions may thus share the instruction set via a distributed ledger that captures and stores evidence of an action on the ledger by a third party, thereby evidencing access and agreement to terms and conditions of access. In embodiments, the platform **100** may have a distributed ledger that tokenizes executable algorithmic logic **121**, such that operation on the distributed ledger provides provable access to the executable algorithmic logic. A variety of instruction sets may be stored by a distributed ledger, such as to verify access and verify agreement to terms (such as smart contract terms). In embodiments, instruction sets that embody trade secrets may be separated into sub-components, so that operations must occur on multiple ledgers to get (provable) access to a trade secret. This may permit parties wishing to share secrets, such as with multiple sub-contractors or vendors, to maintain provable access control, while separating components among different vendors to avoid sharing an entire set with a single party. Various kinds of executable instruction sets may be stored on specialized distributed ledgers that may include smart wrappers for specific types of instruction sets, such that provable access control, validation of terms, and tracking of utilization may be performed by operations on the distributed ledger (which may include triggering access controls within a content management system or other systems upon validation of actions taken in a smart contract on the ledger. In embodiments, the platform **100** may have a distributed ledger that tokenizes a 3D printer instruction set **123**, such that operation on the distributed ledger provides provable access to the instruction set.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set for a coating process **125**, such that operation on the distributed ledger provides provable access to the instruction set.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set for a semiconductor fabrication process **129**, such that operation on the distributed ledger provides provable access to the fabrication process.

In embodiments, the platform **100** may have a distributed ledger that tokenizes a firmware program **131**, such that operation on the distributed ledger provides provable access to the firmware program.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set for an FPGA **133**, such that operation on the distributed ledger provides provable access to the FPGA.

In embodiments, the platform **100** may have a distributed ledger that tokenizes serverless code logic **135**, such that operation on the distributed ledger provides provable access to the serverless code logic.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set for a crystal fabrication system **139**, such that operation on the distributed ledger provides provable access to the instruction set.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set for a food preparation process **141**, such that operation on the distributed ledger provides provable access to the instruction set.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set for a polymer production process **143**, such that operation on the distributed ledger provides provable access to the instruction set.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set for chemical synthesis process **145**, such that operation on the distributed ledger provides provable access to the instruction set.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set for a biological production process **149**, such that operation on the distributed ledger provides provable access to the instruction set.

In embodiments, the platform **100** may have a distributed ledger that tokenizes a trade secret with an expert wrapper **151**, such that operation on the distributed ledger provides provable access to the trade secret and the wrapper provides validation of the trade secret by the expert. An interface may be provided by which an expert accesses the trade secret on the ledger and verifies that the information is accurate and sufficient to allow a third party to use the secret.

In embodiments, the platform **100** may have a distributed ledger that includes instruction ledger operation analytics **159**, for example providing aggregate views **155** of a trade secret into a chain that proves which and how many parties have viewed the trade secret. Views may be used to allocate value to creators of the trade secret, to operators of the platform **100**, or the like. In embodiments, the platform **100** may have a distributed ledger that determines an instruction access probability **157**, such as a chance that an instruction set or other IP element has been accessed, will be accessed, and/or will be accessed in a given time frame (e.g., the next day, next week, next month, etc.).

In embodiments, the platform **100** may have a distributed ledger that tokenizes an instruction set **111**, such that operation on the distributed ledger provides provable access (e.g., presented as views **155**) to the instruction set **111** and execution of the instruction set **161** on a system results in recording a transaction in the distributed ledger.

In embodiments, the platform **100** may have a distributed ledger that tokenizes an item of intellectual property and a reporting system that reports an analytic result based on the operations performed on the distributed ledger or the intellectual property, for example using the instruction ledger operations analytics. In certain embodiments, analytics may additionally or alternatively be provided for any distributed ledger and data stored thereon, such as IP, algorithmic logic, or any other distributed ledger operations described throughout the present disclosure.

In embodiments, the platform **100** may have a distributed ledger that aggregates a set of instructions, where an opera-

tion on the distributed ledger adds at least one instruction to a pre-existing set of instructions **161** to provide a modified set of instructions.

Referring still to FIG. 3, an intelligent cryptocurrency execution engine **183** may provide intelligence for the timing, location and other attributes of a cryptocurrency transaction, such as a mining transaction, an exchange transaction, a storage transaction, a retrieval transaction, or the like. Cryptocurrencies like Bitcoin™ are increasingly widespread, with specialized coins having emerged for a wide variety of purposes, such as exchanging value in various specialized market domains. Initial offerings of such coins, or ICOs, are increasingly subject to regulations, such as securities regulations, and in some cases to taxation. Thus, while cryptocurrency transactions typically occur within computer networks, jurisdictional factors may be important in determining where, when and how to execute a transaction, store a cryptocurrency, exchange it for value. In embodiments, intelligent cryptocurrency execution engine **183** may use features embedded in or wrapped around the digital object representing a coin, such as features that cause the execution of transactions in the coin to be undertaken with awareness of various conditions, including geographic conditions, regulatory conditions, tax conditions, market conditions, IoT data for cryptotransaction **295** and social data for cryptotransaction **193**, and the like.

In embodiments, the platform **100** may include a tax aware coin **165** or smart wrapper for a cryptocurrency coin that directs execution of a transaction involving the coin to a geographic location based on tax treatment of at least one of the coin and the transaction in the geographic location.

In embodiments, the platform **100** may include a location-aware coin **169** or smart wrapper that enables a self-executing cryptocurrency coin that commits a transaction upon recognizing a location-based parameter that provides favorable tax treatment.

In embodiments, the platform **100** may include an expert system or AI agent for tax-aware coin usage **171** that uses machine learning to optimize the execution of cryptocurrency transactions based on tax status. Machine learning may use one or more models or heuristics, such as populated with relevant jurisdictional tax data, may be trained on a training set of human trading operations, may be supervised by human supervisors, and/or may use a deep learning technique based on outcomes over time, such as when operating on a wide range of internal system data and external data sources **182** as described throughout this disclosure.

In embodiments, the platform **100** may include regulation aware coin **173** having a coin, a smart wrapper, and/or an expert system that aggregates regulatory information covering cryptocurrency transactions and automatically selects a jurisdiction for an operation based on the regulatory information. Machine learning may use one or more models or heuristics, such as populated with relevant jurisdictional regulatory data, may be trained on a training set of human trading operations, may be supervised by human supervisors, and/or may use a deep learning technique based on outcomes over time, such as when operating on a wide range of internal system data and external data sources **182** as described throughout this disclosure.

In embodiments, the platform **100** may include an energy price-aware coin **175**, wrapper, or expert system that uses machine learning to optimize the execution of a cryptocurrency transaction based on real time energy price information for an available energy source. Cryptocurrency transactions, such as coin mining and blockchain operations, may

be highly energy intensive. An energy price-aware coin may be configured to time such operations based on energy price forecasts, such as with one or more of the intelligent forecasting engines **192** described throughout this disclosure.

In embodiments, the platform **100** may include an energy source aware coin **179**, wrapper, or expert system that uses machine learning to optimize the execution of a cryptocurrency transaction based on an understanding of available energy sources to power computing resources to execute the transaction. For example, coin mining may be performed only when renewable energy sources are available. Machine learning for optimization of a transaction may use one or more models or heuristics, such as populated with relevant energy source data (such as may be captured in a knowledge graph, which may contain energy source information by type, location and operating parameters), may be trained on a training set of input-output data for human-initiated transactions, may be supervised by human supervisors, and/or may use a deep learning technique based on outcomes over time, such as when operating on a wide range of internal system data and external data sources **182** as described throughout this disclosure.

In embodiments, the platform **100** may include a charging cycle aware coin **181**, wrapper, or an expert system that uses machine learning to optimize charging and recharging cycle of a rechargeable battery system to provide energy for execution of a cryptocurrency transaction. For example, a battery may be discharged for a cryptocurrency transaction only if a minimum threshold of battery charge is maintained for other operational use, if re-charging resources are known to be readily available, or the like. Machine learning for optimization of charging and recharging may use one or more models or heuristics, such as populated with relevant battery data (such as may be captured in a knowledge graph, which may contain energy source information by type, location and operating parameters), may be trained on a training set of human operations, may be supervised by human supervisors, and/or may use a deep learning technique based on outcomes over time, such as when operating on a wide range of internal system data and external data sources **182** as described throughout this disclosure.

Optimization of various intelligent coin operations may occur with machine learning that is trained on outcomes, such as financial profitability. Any of the machine learning systems described throughout this disclosure may be used for optimization of intelligent cryptocurrency transaction management.

In embodiments, compute resources, such as those mentioned throughout this disclosure, may be allocated to perform a range of computing tasks, both for operations that occur within the platform **100**, ones that are managed by the platform, and ones that involve the activities, workflows and processes of various assets that may be owned, operated or managed in conjunction with the platform, such as sets or fleets of assets that have or use computing resources. Examples of compute tasks include, without limitation, cryptocurrency mining, distributed ledger calculations and storage, forecasting tasks, transaction execution tasks, spot market testing tasks, internal data collection tasks, external data collection, machine learning tasks, and others. As noted above, energy, compute resources, bandwidth, spectrum, and other resources may be coordinated, such as by machine learning, for these tasks. Outcome and feedback information may be provided for the machine learning, such as outcomes

for any of the individual tasks and overall outcomes, such as yield and profitability for business or other operations involving the tasks.

In embodiments, networking resources, such as those mentioned throughout this disclosure, may be allocated to perform a range of networking tasks, both for operations that occur within the platform 100, ones that are managed by the platform, and ones that involve the activities, workflows and processes of various assets that may be owned, operated or managed in conjunction with the platform, such as sets or fleets of assets that have or use networking resources. Examples of networking tasks include cognitive network coordination, network coding, peer bandwidth sharing (including, for example cost-based routing, value-based routing, outcome-based routing and the like), distributed transaction execution, spot market testing, randomization (e.g., using genetic programming with outcome feedback to vary network configurations and transmission paths), internal data collection and external data collection. As noted above, energy, compute resources, bandwidth, spectrum, and other resources may be coordinated, such as by machine learning, for these networking tasks. Outcome and feedback information may be provided for the machine learning, such as outcomes for any of the individual tasks and overall outcomes, such as yield and profitability for business or other operations involving the tasks.

In embodiments, data storage resources, such as those mentioned throughout this disclosure, may be allocated to perform a range of data storage tasks, both for operations that occur within the platform 100, ones that are managed by the platform, and ones that involve the activities, workflows and processes of various assets that may be owned, operated or managed in conjunction with the platform, such as sets or fleets of assets that have or use networking resources. Examples of data storage tasks include distributed ledger storage, storage of internal data (such as operational data with the platform), cryptocurrency storage, smart wrapper storage, storage of external data, storage of feedback and outcome data, and others. As noted above, data storage, energy, compute resources, bandwidth, spectrum, and other resources may be coordinated, such as by machine learning, for these data storage tasks. Outcome and feedback information may be provided for the machine learning, such as outcomes for any of the individual tasks and overall outcomes, such as yield and profitability for business or other operations involving the tasks.

In embodiments, smart contracts, such as ones embodying terms relating to intellectual property, trade secrets, know how, instruction sets, algorithmic logic, and the like may embody or include contract terms, which may include terms and conditions for options, royalty stacking terms, field exclusivity, partial exclusivity, pooling of intellectual property, standards terms (such as relating to essential and non-essential patent usage), technology transfer terms, consulting service terms, update terms, support terms, maintenance terms, derivative works terms, copying terms, and performance-related rights or metrics, among many others.

In embodiments where an instruction set is embodied in digital form, such as contained in or managed by a distributed ledger transactions system, various systems may be configured with interfaces that allow them to access and use the instruction sets. In embodiments, such systems may include access control features that validate proper licensing by inspection of a distributed ledger, a key, a token, or the like that indicates the presence of access rights to an instruction set. Such systems that execute distributed instruction sets may include systems for 3D printing, crystal

fabrication, semiconductor fabrication, coating items, producing polymers, chemical synthesis and biological production, among others.

Networking capabilities and network resources should be understood to include a wide range of networking systems, components and capabilities, including infrastructure elements for 3G, 4G, LTE, 5G and other cellular network types, access points, routers, and other WiFi elements, cognitive networking systems and components, mobile networking systems and components, physical layer, MAC layer and application layer systems and components, cognitive networking components and capabilities, peer-to-peer networking components and capabilities, optical networking components and capabilities, and others.

Referring to FIG. 4 through FIG. 31, embodiments of the present disclosure, including ones involving expert systems, self-organization, machine learning, artificial intelligence, and the like, may benefit from the use of a neural net, such as a neural net trained for pattern recognition, for classification of one or more parameters, characteristics, or phenomena, for support of autonomous control, and other purposes. References to a neural net throughout this disclosure should be understood to encompass a wide range of different types of neural networks, machine learning systems, artificial intelligence systems, and the like, such as feed forward neural networks, radial basis function neural networks, self-organizing neural networks (e.g., Kohonen self-organizing neural networks), recurrent neural networks, modular neural networks, artificial neural networks, physical neural networks, multi-layered neural networks, convolutional neural networks, hybrids of neural networks with other expert systems (e.g., hybrid fuzzy logic—neural network systems), Autoencoder neural networks, probabilistic neural networks, time delay neural networks, convolutional neural networks, regulatory feedback neural networks, radial basis function neural networks, recurrent neural networks, Hopfield neural networks, Boltzmann machine neural networks, self-organizing map (SOM) neural networks, learning vector quantization (LVQ) neural networks, fully recurrent neural networks, simple recurrent neural networks, echo state neural networks, long short-term memory neural networks, bi-directional neural networks, hierarchical neural networks, stochastic neural networks, genetic scale RNN neural networks, committee of machines neural networks, associative neural networks, physical neural networks, instantaneously trained neural networks, spiking neural networks, neocognitron neural networks, dynamic neural networks, cascading neural networks, neuro-fuzzy neural networks, compositional pattern-producing neural networks, memory neural networks, hierarchical temporal memory neural networks, deep feed forward neural networks, gated recurrent unit (GRU) neural networks, auto encoder neural networks, variational auto encoder neural networks, denoising auto encoder neural networks, sparse auto-encoder neural networks, Markov chain neural networks, restricted Boltzmann machine neural networks, deep belief neural networks, deep convolutional neural networks, deconvolutional neural networks, deep convolutional inverse graphics neural networks, generative adversarial neural networks, liquid state machine neural networks, extreme learning machine neural networks, echo state neural networks, deep residual neural networks, support vector machine neural networks, neural Turing machine neural networks, and/or holographic associative memory neural networks, or hybrids or combinations of the foregoing, or combinations with other expert systems, such as rule-based systems, model-based systems (including ones based on physical models,

statistical models, flow-based models, biological models, biomimetic models, and the like).

In embodiments, FIGS. 5 through 31 depict exemplary neural networks and FIG. 4 depicts a legend showing the various components of the neural networks depicted throughout FIGS. 5 to 31. FIG. 4 depicts various neural net components depicted in cells that are assigned functions and requirements. In embodiments, the various neural net examples may include (from top to bottom in the example of FIG. 4): back fed data/sensor input cells, data/sensor input cells, noisy input cells, and hidden cells. The neural net components also include probabilistic hidden cells, spiking hidden cells, output cells, match input/output cells, recurrent cells, memory cells, different memory cells, kernels, and convolution or pool cells.

In embodiments, FIG. 5 depicts an exemplary perceptron neural network that may connect to, integrate with, or interface with the platform 100. The platform may also be associated with further neural net systems such as a feed forward neural network (FIG. 6), a radial basis neural network (FIG. 7), a deep feed forward neural network (FIG. 8), a recurrent neural network (FIG. 9), a long/short term neural network (FIG. 10), and a gated recurrent neural network (FIG. 11). The platform may also be associated with further neural net systems such as an auto encoder neural network (FIG. 12), a variational neural network (FIG. 13), a denoising neural network (FIG. 14), a sparse neural network (FIG. 15), a Markov chain neural network (FIG. 16), and a Hopfield network neural network (FIG. 17). The platform may further be associated with additional neural net systems such as a Boltzmann machine neural network (FIG. 18), a restricted BM neural network (FIG. 19), a deep belief neural network (FIG. 20), a deep convolutional neural network (FIG. 21), a deconvolutional neural network (FIG. 22), and a deep convolutional inverse graphics neural network (FIG. 23). The platform may also be associated with further neural net systems such as a generative adversarial neural network (FIG. 24), a liquid state machine neural network (FIG. 25), an extreme learning machine neural network (FIG. 26), an echo state neural network (FIG. 27), a deep residual neural network (FIG. 28), a Kohonen neural network (FIG. 29), a support vector machine neural network (FIG. 30), and a neural Turing machine neural network (FIG. 31).

The foregoing neural networks may have a variety of nodes or neurons, which may perform a variety of functions on inputs, such as inputs received from sensors or other data sources, including other nodes. Functions may involve weights, features, feature vectors, and the like. Neurons may include perceptrons, neurons that mimic biological functions (such as of the human senses of touch, vision, taste, hearing, and smell), and the like. Continuous neurons, such as with sigmoidal activation, may be used in the context of various forms of neural net, such as where back propagation is involved.

In many embodiments, an expert system or neural network may be trained, such as by a human operator or supervisor, or based on a data set, model, or the like. Training may include presenting the neural network with one or more training data sets that represent values, such as sensor data, event data, parameter data, and other types of data (including the many types described throughout this disclosure), as well as one or more indicators of an outcome, such as an outcome of a process, an outcome of a calculation, an outcome of an event, an outcome of an activity, or the like. Training may include training in optimization, such as training a neural network to optimize one or more systems based on one or more optimization approaches, such as

Bayesian approaches, parametric Bayes classifier approaches, k-nearest-neighbor classifier approaches, iterative approaches, interpolation approaches, Pareto optimization approaches, algorithmic approaches, and the like. Feedback may be provided in a process of variation and selection, such as with a genetic algorithm that evolves one or more solutions based on feedback through a series of rounds.

In embodiments, a plurality of neural networks may be deployed in a cloud platform that receives data streams and other inputs collected (such as by mobile data collectors) in one or more transactional environments and transmitted to the cloud platform over one or more networks, including using network coding to provide efficient transmission. In the cloud platform, optionally using massively parallel computational capability, a plurality of different neural networks of various types (including modular forms, structure-adaptive forms, hybrids, and the like) may be used to undertake prediction, classification, control functions, and provide other outputs as described in connection with expert systems disclosed throughout this disclosure. The different neural networks may be structured to compete with each other (optionally including use evolutionary algorithms, genetic algorithms, or the like), such that an appropriate type of neural network, with appropriate input sets, weights, node types and functions, and the like, may be selected, such as by an expert system, for a specific task involved in a given context, workflow, environment process, system, or the like.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a feed forward neural network, which moves information in one direction, such as from a data input, like a data source related to at least one resource or parameter related to a transactional environment, such as any of the data sources mentioned throughout this disclosure, through a series of neurons or nodes, to an output. Data may move from the input nodes to the output nodes, optionally passing through one or more hidden nodes, without loops. In embodiments, feed forward neural networks may be constructed with various types of units, such as binary McCulloch-Pitts neurons, the simplest of which is a perceptron.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a capsule neural network, such as for prediction, classification, or control functions with respect to a transactional environment, such as relating to one or more of the machines and automated systems described throughout this disclosure.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a radial basis function (RBF) neural network, which may be preferred in some situations involving interpolation in a multi-dimensional space (such as where interpolation is helpful in optimizing a multi-dimensional function, such as for optimizing a data marketplace as described here, optimizing the efficiency or output of a power generation system, a factory system, or the like, or other situation involving multiple dimensions. In embodiments, each neuron in the RBF neural network stores an example from a training set as a "prototype." Linearity involved in the functioning of this neural network offers RBF the advantage of not typically suffering from problems with local minima or maxima.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a radial basis function (RBF) neural network, such as one that employs a distance criterion with respect to a center (e.g., a Gaussian function). A radial basis function

may be applied as a replacement for a hidden layer, such as a sigmoidal hidden layer transfer, in a multi-layer perceptron. An RBF network may have two layers, such as where an input is mapped onto each RBF in a hidden layer. In embodiments, an output layer may comprise a linear combination of hidden layer values representing, for example, a mean predicted output. The output layer value may provide an output that is the same as or similar to that of a regression model in statistics. In classification problems, the output layer may be a sigmoid function of a linear combination of hidden layer values, representing a posterior probability. Performance in both cases is often improved by shrinkage techniques, such as ridge regression in classical statistics. This corresponds to a prior belief in small parameter values (and therefore smooth output functions) in a Bayesian framework. RBF networks may avoid local minima, because the only parameters that are adjusted in the learning process are the linear mapping from hidden layer to output layer. Linearity ensures that the error surface is quadratic and therefore has a single minimum. In regression problems, this may be found in one matrix operation. In classification problems, the fixed non-linearity introduced by the sigmoid output function may be handled using an iteratively re-weighted least squares function or the like. RBF networks may use kernel methods such as support vector machines (SVM) and Gaussian processes (where the RBF is the kernel function). A non-linear kernel function may be used to project the input data into a space where the learning problem may be solved using a linear model.

In embodiments, an RBF neural network may include an input layer, a hidden layer, and a summation layer. In the input layer, one neuron appears in the input layer for each predictor variable. In the case of categorical variables, $N-1$ neurons are used, where N is the number of categories. The input neurons may, in embodiments, standardize the value ranges by subtracting the median and dividing by the interquartile range. The input neurons may then feed the values to each of the neurons in the hidden layer. In the hidden layer, a variable number of neurons may be used (determined by the training process). Each neuron may consist of a radial basis function that is centered on a point with as many dimensions as a number of predictor variables. The spread (e.g., radius) of the RBF function may be different for each dimension. The centers and spreads may be determined by training. When presented with the vector of input values from the input layer, a hidden neuron may compute a Euclidean distance of the test case from the neuron's center point and then apply the RBF kernel function to this distance, such as using the spread values. The resulting value may then be passed to the summation layer. In the summation layer, the value coming out of a neuron in the hidden layer may be multiplied by a weight associated with the neuron and may add to the weighted values of other neurons. This sum becomes the output. For classification problems, one output is produced (with a separate set of weights and summation units) for each target category. The value output for a category is the probability that the case being evaluated has that category. In training of an RBF, various parameters may be determined, such as the number of neurons in a hidden layer, the coordinates of the center of each hidden-layer function, the spread of each function in each dimension, and the weights applied to outputs as they pass to the summation layer. Training may be used by clustering algorithms (such as k-means clustering), by evolutionary approaches, and the like.

In embodiments, a recurrent neural network may have a time-varying, real-valued (more than just zero or one) acti-

vation (output). Each connection may have a modifiable real-valued weight. Some of the nodes are called labeled nodes, some output nodes, and others hidden nodes. For supervised learning in discrete time settings, training sequences of real-valued input vectors may become sequences of activations of the input nodes, one input vector at a time. At each time step, each non-input unit may compute its current activation as a nonlinear function of the weighted sum of the activations of all units from which it receives connections. The system may explicitly activate (independent of incoming signals) some output units at certain time steps.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a self-organizing neural network, such as a Kohonen self-organizing neural network, such as for visualization of views of data, such as low-dimensional views of high-dimensional data. The self-organizing neural network may apply competitive learning to a set of input data, such as from one or more sensors or other data inputs from or associated with a transactional environment, including any machine or component that relates to the transactional environment. In embodiments, the self-organizing neural network may be used to identify structures in data, such as unlabeled data, such as in data sensed from a range of data sources about or sensors in or about in a transactional environment, where sources of the data are unknown (such as where events may be coming from any of a range of unknown sources). The self-organizing neural network may organize structures or patterns in the data, such that they may be recognized, analyzed, and labeled, such as identifying market behavior structures as corresponding to other events and signals.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a recurrent neural network, which may allow for a bi-directional flow of data, such as where connected units (e.g., neurons or nodes) form a directed cycle. Such a network may be used to model or exhibit dynamic temporal behavior, such as involved in dynamic systems, such as a wide variety of the automation systems, machines and devices described throughout this disclosure, such as an automated agent interacting with a marketplace for purposes of collecting data, testing spot market transactions, execution transactions, and the like, where dynamic system behavior involves complex interactions that a user may desire to understand, predict, control and/or optimize. For example, the recurrent neural network may be used to anticipate the state of a market, such as one involving a dynamic process or action, such as a change in state of a resource that is traded in or that enables a marketplace of transactional environment. In embodiments, the recurrent neural network may use internal memory to process a sequence of inputs, such as from other nodes and/or from sensors and other data inputs from or about the transactional environment, of the various types described herein. In embodiments, the recurrent neural network may also be used for pattern recognition, such as for recognizing a machine, component, agent, or other item based on a behavioral signature, a profile, a set of feature vectors (such as in an audio file or image), or the like. In a non-limiting example, a recurrent neural network may recognize a shift in an operational mode of a marketplace or machine by learning to classify the shift from a training data set consisting of a stream of data from one or more data sources of sensors applied to or about one or more resources.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability

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may use a modular neural network, which may comprise a series of independent neural networks (such as ones of various types described herein) that are moderated by an intermediary. Each of the independent neural networks in the modular neural network may work with separate inputs, accomplishing subtasks that make up the task the modular network as whole is intended to perform. For example, a modular neural network may comprise a recurrent neural network for pattern recognition, such as to recognize what type of machine or system is being sensed by one or more sensors that are provided as input channels to the modular network and an RBF neural network for optimizing the behavior of the machine or system once understood. The intermediary may accept inputs of each of the individual neural networks, process them, and create output for the modular neural network, such as an appropriate control parameter, a prediction of state, or the like.

Combinations among any of the pairs, triplets, or larger combinations, of the various neural network types described herein, are encompassed by the present disclosure. This may include combinations where an expert system uses one neural network for recognizing a pattern (e.g., a pattern indicating a problem or fault condition) and a different neural network for self-organizing an activity or work flow based on the recognized pattern (such as providing an output governing autonomous control of a system in response to the recognized condition or pattern). This may also include combinations where an expert system uses one neural network for classifying an item (e.g., identifying a machine, a component, or an operational mode) and a different neural network for predicting a state of the item (e.g., a fault state, an operational state, an anticipated state, a maintenance state, or the like). Modular neural networks may also include situations where an expert system uses one neural network for determining a state or context (such as a state of a machine, a process, a work flow, a marketplace, a storage system, a network, a data collector, or the like) and a different neural network for self-organizing a process involving the state or context (e.g., a data storage process, a network coding process, a network selection process, a data marketplace process, a power generation process, a manufacturing process, a refining process, a digging process, a boring process, or other process described herein).

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a physical neural network where one or more hardware elements is used to perform or simulate neural behavior. In embodiments, one or more hardware neurons may be configured to stream voltage values, current values, or the like that represent sensor data, such as to calculate information from analog sensor inputs representing energy consumption, energy production, or the like, such as by one or more machines providing energy or consuming energy for one or more transactions. One or more hardware nodes may be configured to stream output data resulting from the activity of the neural net. Hardware nodes, which may comprise one or more chips, microprocessors, integrated circuits, programmable logic controllers, application-specific integrated circuits, field-programmable gate arrays, or the like, may be provided to optimize the machine that is producing or consuming energy, or to optimize another parameter of some part of a neural net of any of the types described herein. Hardware nodes may include hardware for acceleration of calculations (such as dedicated processors for performing basic or more sophisticated calculations on input data to provide outputs, dedicated processors for filtering or compressing data, dedicated processors for de-

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compressing data, dedicated processors for compression of specific file or data types (e.g., for handling image data, video streams, acoustic signals, thermal images, heat maps, or the like), and the like. A physical neural network may be embodied in a data collector, including one that may be reconfigured by switching or routing inputs in varying configurations, such as to provide different neural net configurations within the data collector for handling different types of inputs (with the switching and configuration optionally under control of an expert system, which may include a software-based neural net located on the data collector or remotely). A physical, or at least partially physical, neural network may include physical hardware nodes located in a storage system, such as for storing data within a machine, a data storage system, a distributed ledger, a mobile device, a server, a cloud resource, or in a transactional environment, such as for accelerating input/output functions to one or more storage elements that supply data to or take data from the neural net. A physical, or at least partially physical, neural network may include physical hardware nodes located in a network, such as for transmitting data within, to or from an industrial environment, such as for accelerating input/output functions to one or more network nodes in the net, accelerating relay functions, or the like. In embodiments of a physical neural network, an electrically adjustable resistance material may be used for emulating the function of a neural synapse. In embodiments, the physical hardware emulates the neurons, and software emulates the neural network between the neurons. In embodiments, neural networks complement conventional algorithmic computers. They are versatile and may be trained to perform appropriate functions without the need for any instructions, such as classification functions, optimization functions, pattern recognition functions, control functions, selection functions, evolution functions, and others.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a multilayered feed forward neural network, such as for complex pattern classification of one or more items, phenomena, modes, states, or the like. In embodiments, a multilayered feed forward neural network may be trained by an optimization technique, such as a genetic algorithm, such as to explore a large and complex space of options to find an optimum, or near-optimum, global solution. For example, one or more genetic algorithms may be used to train a multilayered feed forward neural network to classify complex phenomena, such as to recognize complex operational modes of machines, such as modes involving complex interactions among machines (including interference effects, resonance effects, and the like), modes involving non-linear phenomena, modes involving critical faults, such as where multiple, simultaneous faults occur, making root cause analysis difficult, and others. In embodiments, a multilayered feed forward neural network may be used to classify results from monitoring of a marketplace, such as monitoring systems, such as automated agents, that operate within the marketplace, as well as monitoring resources that enable the marketplace, such as computing, networking, energy, data storage, energy storage, and other resources.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a feed-forward, back-propagation multi-layer perceptron (MLP) neural network, such as for handling one or more remote sensing applications, such as for taking inputs from sensors distributed throughout various transactional environments. In embodiments, the MLP neural network may be used for classification of transactional environments

and resource environments, such as spot markets, forward markets, energy markets, renewable energy credit (REC) markets, networking markets, advertising markets, spectrum markets, ticketing markets, rewards markets, compute markets, and others mentioned throughout this disclosure, as well as physical resources and environments that produce them, such as energy resources (including renewable energy environments, mining environments, exploration environments, drilling environments, and the like, including classification of geological structures (including underground features and above ground features), classification of materials (including fluids, minerals, metals, and the like), and other problems. This may include fuzzy classification.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a structure-adaptive neural network, where the structure of a neural network is adapted, such as based on a rule, a sensed condition, a contextual parameter, or the like. For example, if a neural network does not converge on a solution, such as classifying an item or arriving at a prediction, when acting on a set of inputs after some amount of training, the neural network may be modified, such as from a feed forward neural network to a recurrent neural network, such as by switching data paths between some subset of nodes from unidirectional to bi-directional data paths. The structure adaptation may occur under control of an expert system, such as to trigger adaptation upon occurrence of a trigger, rule or event, such as recognizing occurrence of a threshold (such as an absence of a convergence to a solution within a given amount of time) or recognizing a phenomenon as requiring different or additional structure (such as recognizing that a system is varying dynamically or in a non-linear fashion). In one non-limiting example, an expert system may switch from a simple neural network structure like a feed forward neural network to a more complex neural network structure like a recurrent neural network, a convolutional neural network, or the like upon receiving an indication that a continuously variable transmission is being used to drive a generator, turbine, or the like in a system being analyzed.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use an autoencoder, autoassociator or Diabolo neural network, which may be similar to a multilayer perceptron (MLP) neural network, such as where there may be an input layer, an output layer and one or more hidden layers connecting them. However, the output layer in the auto-encoder may have the same number of units as the input layer, where the purpose of the MLP neural network is to reconstruct its own inputs (rather than just emitting a target value). Therefore, the auto encoders may operate as an unsupervised learning model. An auto encoder may be used, for example, for unsupervised learning of efficient codings, such as for dimensionality reduction, for learning generative models of data, and the like. In embodiments, an auto-encoding neural network may be used to self-learn an efficient network coding for transmission of analog sensor data from a machine over one or more networks or of digital data from one or more data sources. In embodiments, an auto-encoding neural network may be used to self-learn an efficient storage approach for storage of streams of data.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a probabilistic neural network (PNN), which, in embodiments, may comprise a multi-layer (e.g., four-layer) feed forward neural network, where layers may include input layers, hidden layers, pattern/summation layers and an

output layer. In an embodiment of a PNN algorithm, a parent probability distribution function (PDF) of each class may be approximated, such as by a Parzen window and/or a non-parametric function. Then, using the PDF of each class, the class probability of a new input is estimated, and Bayes' rule may be employed, such as to allocate it to the class with the highest posterior probability. A PNN may embody a Bayesian network and may use a statistical algorithm or analytic technique, such as Kernel Fisher discriminant analysis technique. The PNN may be used for classification and pattern recognition in any of a wide range of embodiments disclosed herein. In one non-limiting example, a probabilistic neural network may be used to predict a fault condition of an engine based on collection of data inputs from sensors and instruments for the engine.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a time delay neural network (TDNN), which may comprise a feed forward architecture for sequential data that recognizes features independent of sequence position. In embodiments, to account for time shifts in data, delays are added to one or more inputs, or between one or more nodes, so that multiple data points (from distinct points in time) are analyzed together. A time delay neural network may form part of a larger pattern recognition system, such as using a perceptron network. In embodiments, a TDNN may be trained with supervised learning, such as where connection weights are trained with back propagation or under feedback. In embodiments, a TDNN may be used to process sensor data from distinct streams, such as a stream of velocity data, a stream of acceleration data, a stream of temperature data, a stream of pressure data, and the like, where time delays are used to align the data streams in time, such as to help understand patterns that involve understanding of the various streams (e.g., changes in price patterns in spot or forward markets).

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a convolutional neural network (referred to in some cases as a CNN, a ConvNet, a shift invariant neural network, or a space invariant neural network), wherein the units are connected in a pattern similar to the visual cortex of the human brain. Neurons may respond to stimuli in a restricted region of space, referred to as a receptive field. Receptive fields may partially overlap, such that they collectively cover the entire (e.g., visual) field. Node responses may be calculated mathematically, such as by a convolution operation, such as using multilayer perceptrons that use minimal preprocessing. A convolutional neural network may be used for recognition within images and video streams, such as for recognizing a type of machine in a large environment using a camera system disposed on a mobile data collector, such as on a drone or mobile robot. In embodiments, a convolutional neural network may be used to provide a recommendation based on data inputs, including sensor inputs and other contextual information, such as recommending a route for a mobile data collector. In embodiments, a convolutional neural network may be used for processing inputs, such as for natural language processing of instructions provided by one or more parties involved in a workflow in an environment. In embodiments, a convolutional neural network may be deployed with a large number of neurons (e.g., 100,000, 500,000 or more), with multiple (e.g., 4, 5, 6 or more) layers, and with many (e.g., millions) of parameters. A convolutional neural net may use one or more convolutional nets.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability

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may use a regulatory feedback network, such as for recognizing emergent phenomena (such as new types of behavior not previously understood in a transactional environment).

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a self-organizing map (SOM), involving unsupervised learning. A set of neurons may learn to map points in an input space to coordinates in an output space. The input space may have different dimensions and topology from the output space, and the SOM may preserve these while mapping phenomena into groups.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a learning vector quantization neural net (LVQ). Prototypical representatives of the classes may parameterize, together with an appropriate distance measure, in a distance-based classification scheme.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use an echo state network (ESN), which may comprise a recurrent neural network with a sparsely connected, random hidden layer. The weights of output neurons may be changed (e.g., the weights may be trained based on feedback). In embodiments, an ESN may be used to handle time series patterns, such as, in an example, recognizing a pattern of events associated with a market, such as the pattern of price changes in response to stimuli.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a Bi-directional, recurrent neural network (BRNN), such as using a finite sequence of values (e.g., voltage values from a sensor) to predict or label each element of the sequence based on both the past and the future context of the element. This may be done by adding the outputs of two RNNs, such as one processing the sequence from left to right, the other one from right to left. The combined outputs are the predictions of target signals, such as ones provided by a teacher or supervisor. A bi-directional RNN may be combined with a long short-term memory RNN.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a hierarchical RNN that connects elements in various ways to decompose hierarchical behavior, such as into useful subprograms. In embodiments, a hierarchical RNN may be used to manage one or more hierarchical templates for data collection in a transactional environment.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a stochastic neural network, which may introduce random variations into the network. Such random variations may be viewed as a form of statistical sampling, such as Monte Carlo sampling.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a genetic scale recurrent neural network. In such embodiments, an RNN (often an LSTM) is used where a series is decomposed into a number of scales where every scale informs the primary length between two consecutive points. A first order scale consists of a normal RNN, a second order consists of all points separated by two indices and so on. The Nth order RNN connects the first and last node. The outputs from all the various scales may be treated as a committee of members, and the associated scores may be used genetically for the next iteration.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a committee of machines (CoM), comprising a

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collection of different neural networks that together “vote” on a given example. Because neural networks may suffer from local minima, starting with the same architecture and training, but using randomly different initial weights often gives different results. A CoM tends to stabilize the result.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use an associative neural network (ASNN), such as involving an extension of a committee of machines that combines multiple feed forward neural networks and a k-nearest neighbor technique. It may use the correlation between ensemble responses as a measure of distance amid the analyzed cases for the kNN. This corrects the bias of the neural network ensemble. An associative neural network may have a memory that may coincide with a training set. If new data become available, the network instantly improves its predictive ability and provides data approximation (self-learns) without retraining. Another important feature of ASNN is the possibility to interpret neural network results by analysis of correlations between data cases in the space of models.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use an instantaneously trained neural network (ITNN), where the weights of the hidden and the output layers are mapped directly from training vector data.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a spiking neural network, which may explicitly consider the timing of inputs. The network input and output may be represented as a series of spikes (such as a delta function or more complex shapes). SNNs may process information in the time domain (e.g., signals that vary over time, such as signals involving dynamic behavior of markets or transactional environments). They are often implemented as recurrent networks.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a dynamic neural network that addresses nonlinear multivariate behavior and includes learning of time-dependent behavior, such as transient phenomena and delay effects. Transients may include behavior of shifting market variables, such as prices, available quantities, available counterparties, and the like.

In embodiments, cascade correlation may be used as an architecture and supervised learning algorithm, supplementing adjustment of the weights in a network of fixed topology. Cascade-correlation may begin with a minimal network, then automatically trains and add new hidden units one by one, creating a multi-layer structure. Once a new hidden unit has been added to the network, its input-side weights may be frozen. This unit then becomes a permanent feature-detector in the network, available for producing outputs or for creating other, more complex feature detectors. The cascade-correlation architecture may learn quickly, determine its own size and topology, and retain the structures it has built even if the training set changes and requires no back-propagation.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a neuro-fuzzy network, such as involving a fuzzy inference system in the body of an artificial neural network. Depending on the type, several layers may simulate the processes involved in a fuzzy inference, such as fuzzification, inference, aggregation and defuzzification. Embedding a fuzzy system in a general structure of a neural net as the

benefit of using available training methods to find the parameters of a fuzzy system.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a compositional pattern-producing network (CPPN), such as a variation of an associative neural network (ANN) that differs the set of activation functions and how they are applied. While typical ANNs often contain only sigmoid functions (and sometimes Gaussian functions), CPPNs may include both types of functions and many others. Furthermore, CPPNs may be applied across the entire space of possible inputs, so that they may represent a complete image. Since they are compositions of functions, CPPNs in effect encode images at infinite resolution and may be sampled for a particular display at whatever resolution is optimal.

This type of network may add new patterns without re-training. In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a one-shot associative memory network, such as by creating a specific memory structure, which assigns each new pattern to an orthogonal plane using adjacently connected hierarchical arrays.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a hierarchical temporal memory (HTM) neural network, such as involving the structural and algorithmic properties of the neocortex. HTM may use a biomimetic model based on memory-prediction theory. HTM may be used to discover and infer the high-level causes of observed input patterns and sequences.

In embodiments, methods and systems described herein that involve an expert system or self-organization capability may use a holographic associative memory (HAM) neural network, which may comprise an analog, correlation-based, associative, stimulus-response system. Information may be mapped onto the phase orientation of complex numbers. The memory is effective for associative memory tasks, generalization and pattern recognition with changeable attention.

In embodiments, various embodiments involving network coding may be used to code transmission data among network nodes in a neural net, such as where nodes are located in one or more data collectors or machines in a transactional environment.

In embodiments, one or more of the controllers, circuits, systems, data collectors, storage systems, network elements, or the like as described throughout this disclosure may be embodied in or on an integrated circuit, such as an analog, digital, or mixed signal circuit, such as a microprocessor, a programmable logic controller, an application-specific integrated circuit, a field programmable gate array, or other circuit, such as embodied on one or more chips disposed on one or more circuit boards, such as to provide in hardware (with potentially accelerated speed, energy performance, input-output performance, or the like) one or more of the functions described herein. This may include setting up circuits with up to billions of logic gates, flip-flops, multiplexers, and other circuits in a small space, facilitating high-speed processing, low power dissipation, and reduced manufacturing cost compared with board-level integration. In embodiments, a digital IC, typically a microprocessor, digital signal processor, microcontroller, or the like may use Boolean algebra to process digital signals to embody complex logic, such as involved in the circuits, controllers, and other systems described herein. In embodiments, a data collector, an expert system, a storage system, or the like may be embodied as a digital integrated circuit, such as a logic

IC, memory chip, interface IC (e.g., a level shifter, a serializer, a deserializer, and the like), a power management IC and/or a programmable device; an analog integrated circuit, such as a linear IC, RF IC, or the like, or a mixed signal IC, such as a data acquisition IC (including A/D converters, D/A converter, digital potentiometers) and/or a clock/timing IC.

With reference to FIG. 32, the environment includes an intelligent energy and compute facility (such as a large scale facility hosting many compute resources and having access to a large energy source, such as a hydropower source), as well as a host intelligent energy and compute facility resource management platform, referred to in some cases for convenience as the energy and information technology platform (with networking, data storage, data processing and other resources as described herein), a set of data sources, a set of expert systems, interfaces to a set of market platforms and external resources, and a set of user (or client) systems and devices.

A facility may be configured to access an inexpensive (at least during some time periods) power source (such as a hydropower dam, a wind farm, a solar array, a nuclear power plant, or a grid), to contain a large set of networked information technology resources, including processing units, servers, and the like that are capable of flexible utilization (such as by switching inputs, switching configurations, switching programming and the like), and to provide a range of outputs that can also be flexibly configured (such as passing through power to a smart grid, providing computational results (such as for cryptocurrency mining, artificial intelligence, or analytics). A facility may include a power storage system, such as for large scale storage of available power.

Example features and operations of an intelligent energy and compute facility resource management platform are described herein. In operation, a user can access the energy and information technology platform to initiate and manage a set of activities that involve optimizing energy and computing resources among a diverse set of available tasks. Energy resources may include hydropower, nuclear power, wind power, solar power, grid power and the like, as well as energy storage resources, such as batteries, gravity power, kinetic energy storage, pressurized fluids, and storage using chemical and/or thermal techniques, such as energy storage in molten salts. Computing resources may include GPUs, FPGAs, servers, chips, ASICs, processors, data storage media, networking resources, and many others. Available tasks may include cryptocurrency hash processing, expert system processing, computer vision processing, NLP, path optimization, applications of models such as for analytics, etc.

In embodiments, the platform may include various subsystems that may be implemented as micro services, such that other subsystems of the system access the functionality of a subsystem providing a micro service via application programming interface API. In some embodiments, the various services that are provided by the subsystems may be deployed in bundles that are integrated, such as by a set of APIs. Examples of the subsystems are described in greater detail with respect to FIG. 33.

The External Data Sources can include any system or device that can provide data to the platform. Examples of external data sources can include market data sources (e.g., for financial markets, commercial markets (including e-commerce), advertising markets, energy markets, telecommunication markets, and many others), government or regulatory data sources, industry specific data sources, subscrip-

tion based services accessing proprietary or public information, and/or news data sources. The energy and computing resource platform accesses external data sources via a network (e.g., the Internet) in any suitable manner (e.g., crawlers, extract-transform-load (ETL) systems, gateways, brokers, application programming interfaces (APIs), spiders, distributed database queries, and the like).

A facility, in the present example, is a facility that has an energy resource (e.g., a hydro power resource) and a set of compute resources (e.g., a set of flexible computing resources that can be provisioned and managed to perform computing tasks, such as GPUs, FPGAs and many others, a set of flexible networking resources that can similarly be provisioned and managed, such as by adjusting network coding protocols and parameters), and the like.

User and client systems and devices can include any system or device that may consume one or more computing or energy resource made available by the energy and computing resource platform. Examples include cryptocurrency systems (e.g., for Bitcoin and other cryptocurrency mining operations), expert and artificial intelligence systems (such as neural networks and other systems, such as for computer vision, natural language processing, path determination and optimization, pattern recognition, deep learning, supervised learning, decision support, and many others), energy management systems (such as smart grid systems), and many others. User and client systems may include user devices, such as smartphones, tablet computer devices, laptop computing devices, personal computing devices, smart televisions, gaming consoles, and the like.

FIG. 33 illustrates an example energy and computing resource platform according to some embodiments of the present disclosure. In embodiments, the energy and computing resource platform may include a processing system 3302, a storage system 3304, and a communication system 3306.

The processing system 3302 may include one or more processors and memory. The processors may operate in an individual or distributed manner. The processors may be in the same physical device or in separate devices, which may or may not be located in the same facility. The memory may store computer-executable instructions that are executed by the one or more processors. In embodiments, the processing system 3302 may execute the facility management system 3308, the data acquisition system 3310, the cognitive processing system 3312, the lead generation system 3314, the content generation system 3316, and the workflow system 3318.

The storage system 3304 may include one or more computer-readable storage mediums. The computer-readable storage mediums may be located in the same physical device or in separate devices, which may or may not be located in the same facility, which may or may not be located in the same facility. The computer-readable storage mediums may include flash devices, solid-state memory devices, hard disk drives, and the like. In embodiments, the storage system 3304 stores one or more of a facility data store 3320, a person data store 3322, and/or include data stores for any other type of data. The data stores are depicted separately for clarity of the description, but may be stored in the same or a distinct physical location or device, and/or a given data store may be distributed across physical locations or devices.

The communication system 3306 may include one or more transceivers and/or network devices that are configured to effectuate wireless or wired communication with one or more external devices, including user devices and/or

servers, via a network (e.g., the Internet and/or a cellular network). In certain embodiments, the communication system 3306 provides access to external data 3324, the internet, web-based resources, a LAN, a WAN, and/or other systems or devices. The communication system 3306 may implement any suitable communication protocol. For example, the communication system 3306 may implement an IEEE 801.11 wireless communication protocol and/or any suitable cellular communication protocol to effectuate wireless communication with external devices via a wireless network.

An example energy and computing resource management platform discovers, provisions, manages and optimizes energy and compute resources using artificial intelligence and expert systems with sensitivity to market and other conditions by learning on a set of outcomes. An example energy and computer resource management platform discovers and facilitates cataloging of resources, optionally by user entry and/or automated detection (including peer detection). In certain embodiments, an energy and computing resource management platform implements a graphical user interface to receive relevant information regarding the energy and compute resources that are available. For example, a “digital twin” may be created of an energy and compute facility that allows modeling, prediction and the like. In certain embodiments, an energy and computing resource management platform generates a set of data records that define the facility or a set of facilities under common ownership or operation by a host. The data records may have any suitable schema. In some embodiments (e.g., FIG. 34), the facility data records may include a facility identifier (e.g., a unique identifier that corresponds to the facility), a facility type (e.g., energy system and capabilities, compute systems and capabilities, networking systems and capabilities), facility attributes (e.g., name of the facility, name of the facility initiator, description of the facility, keywords of the facility, goals of the facility, timing elements, schedules, and the like), participants/potential participants in the facility (e.g., identifiers of owners, operators, hosts, service providers, consumers, clients, users, workers, and others), and any suitable metadata (e.g., creation date, launch date, scheduled requirements and the like). An example energy and computer resource management platform generates content, such as a document, message, alert, report, webpage and/or application page based on the contents of the data record. For example, an example energy and computer resource management platform obtains the data record of the facility and populates a webpage template with the data (or selected portions of the data) contained therein. In addition, an energy and computer resource management platform can implement management of existing facilities, update the data record of a facility, determine, predict, and/or estimate outcomes (e.g., energy produced, compute tasks completed, processing outcomes achieved, financial outcomes achieved, service levels met and many others), and send of information (e.g., updates, alerts, requests, instructions, and the like) to individuals and systems.

Data Acquisition Systems can acquire various types of data from different data sources and organize that data into one or more data structures. In embodiments, the data acquisition system receives data from users via a user interface (e.g., user types in profile information). In embodiments, the data acquisition system can retrieve data from passive electronic sources and/or external data. In embodiments, the data acquisition system can implement crawlers to crawl different websites or applications. In embodiments, the data acquisition system can implement an API to retrieve data from external data sources or user devices (e.g., various

contact lists from user's phone or email account). In embodiments, the data acquisition system can structure the obtained data into appropriate data structures. In embodiments, the data acquisition system generates and maintains person records based on data collected regarding individuals. In embodiments, a person datastore stores person records. In some of these embodiments, the person datastore may include one or more databases, indexes, tables, and the like. Each person record may correspond to a respective individual and may be organized according to any suitable schema.

FIG. 35 illustrates an example schema of a person record. In the example, each person record may include a unique person identifier (e.g., username or value), and may define all data relating to a person, including a person's name, facilities they are a part of or associated with (e.g., a list of facility identifiers), attributes of the person (age, location, job, company, role, skills, competencies, capabilities, education history, job history, and the like), a list of contacts or relationships of the person (e.g., in a role hierarchy or graph), and any suitable metadata (e.g., date joined, dates actions were taken, dates input was received, and the like).

In embodiments, the data acquisition system generates and maintains one or more graphs based on the retrieved data. In some embodiments, a graph datastore may store the one or more graphs. The graph may be specific to a facility or may be a global graph. The graph may be used in many different applications (e.g., identifying a set of roles, such as for authentication, for approvals, and the like for persons, or identifying system configurations, capabilities, or the like, such as hierarchies of energy producing, computing, networking, or other systems, subsystems and/or resources).

In embodiments, a graph may be stored in a graph database, where data is stored in a collection of nodes and edges. In some embodiments, a graph has nodes representing entities and edges representing relationships, each node may have a node type (also referred to as an entity type) and an entity value, each edge may have a relationship type and may define a relationship between two entities. For example, a person node may include a person ID that identifies the individual represented by the node and a company node may include a company identifier that identifies a company. A "works for" edge that is directed from a person node to a company node may denote that the person represented by the edge node works for the company represented by the company node. In another example, a person node may include a person ID that identifies the individual represented by the node and a facility node may include a facility identifier that identifies a facility. A "manages" edge that is directed from a person node to a facility node may denote that the person represented by the person node is a manager of the facility represented by the facility node. Furthermore in embodiments, an edge or node may contain or reference additional data. For example, a "manages" edge may include a function that indicates a specific function within a facility that is managed by a person. The graph(s) can be used in a number of different applications, which are discussed with respect to the cognitive processing system.

In embodiments, validated Identity information may be imported from one or more identity information providers, as well as data from LinkedIn™ and other social network sources regarding data acquisition and structuring data. In embodiments, the data acquisition system may include an identity management system (not shown in Figs) of the platform may manage identity stitching, identity resolution, identity normalization, and the like, such as determining where an individual represented across different social net-

working sites and email contacts is in fact the same person. In embodiments, the data acquisition system may include a profile aggregation system (not shown in Figs) that finds and aggregates disparate pieces of information to generate a comprehensive profile for a person. The profile aggregation system may also deduplicate individuals.

The cognitive processing system may implement one or more of machine learning processes, artificial intelligence processes, analytics processes, natural language processing processes, and natural language generation processes. FIG. 36 illustrates an example cognitive processing system according to some embodiments of the present disclosure. In this example, the cognitive processing system may include a machine learning system 3602, an artificial intelligence (AI) system 3604, an analytics system 3606, a natural language processing system 3608, and a natural language generation system 3610.

In embodiments, the machine learning system 3602 may train models, such as predictive models (e.g., various types of neural networks, regression based models, and other machine-learned models). In embodiments, training can be supervised, semi-supervised, or unsupervised. In embodiments, training can be done using training data, which may be collected or generated for training purposes.

An example machine learning system 3602 trains a facility output model. A facility output model (or prediction model) may be a model that receive facility attributes and outputs one or more predictions regarding the production or other output of a facility. Examples of predictions may be the amount of energy a facility will produce, the amount of processing the facility will undertake, the amount of data a network will be able to transfer, the amount of data that can be stored, the price of a component, service or the like (such as supplied to or provided by a facility), a profit generated by accomplishing a given tasks, the cost entailed in performing an action, and the like. In each case, the machine learning system optionally trains a model based on training data. In embodiments, the machine learning system may receive vectors containing facility attributes (e.g., facility type, facility capability, objectives sought, constraints or rules that apply to utilization of resources or the facility, or the like), person attributes (e.g., role, components managed, and the like), and outcomes (e.g., energy produced, computing tasks completed, and financial results, among many others). Each vector corresponds to a respective outcome and the attributes of the respective facility and respective actions that led to the outcome. The machine learning system takes in the vectors and generates predictive model based thereon. In embodiments, the machine learning system may store the predictive models in the model datastore.

In embodiments, training can also be done based on feedback received by the system, which is also referred to as "reinforcement learning." In embodiments, the machine learning system may receive a set of circumstances that led to a prediction (e.g., attributes of facility, attributes of a model, and the like) and an outcome related to the facility and may update the model according to the feedback.

In embodiments, training may be provided from a training data set that is created by observing actions of a set of humans, such as facility managers managing facilities that have various capabilities and that are involved in various contexts and situations. This may include use of robotic process automation to learn on a training data set of interactions of humans with interfaces, such as graphical user interfaces, of one or more computer programs, such as dashboards, control systems, and other systems that are used to manage an energy and compute management facility.

In embodiments, an artificial intelligence (AI) system leverages predictive models to make predictions regarding facilities. Examples of predictions include ones related to inputs to a facility (e.g., available energy, cost of energy, cost of compute resources, networking capacity and the like, as well as various market information, such as pricing information for end use markets), ones related to components or systems of a facility (including performance predictions, maintenance predictions, uptime/downtime predictions, capacity predictions and the like), ones related to functions or workflows of the facility (such as ones that involved conditions or states that may result in following one or more distinct possible paths within a workflow, a process, or the like), ones related to outputs of the facility, and others. In embodiments, the AI system receives a facility identifier. In response to the facility identifier, the AI system may retrieve attributes corresponding to the facility. In some embodiments, the AI system may obtain the facility attributes from a graph. Additionally or alternatively, the AI system may obtain the facility attributes from a facility record corresponding to the facility identifier, and the person attributes from a person record corresponding to the person identifier.

Examples of additional attributes that can be used to make predictions about a facility or a related process of system include: related facility information; owner goals (including financial goals); client goals; and many more additional or alternative attributes. In embodiments, the AI system may output scores for each possible prediction, where each prediction corresponds to a possible outcome. For example, in using a prediction model used to determine a likelihood that a hydroelectric source for a facility will produce 5 MW of power, the prediction model can output a score for a “will produce” outcome and a score for a “will not produce” outcome. The AI system may then select the outcome with the highest score as the prediction. Alternatively, the AI system may output the respective scores to a requesting system.

In embodiments, a clustering system clusters records or entities based on attributes contained herein. For example, similar facilities, resources, people, clients, or the like may be clustered. The clustering system may implement any suitable clustering algorithm. For example, when clustering people records to identify a list of customer leads corresponding to resources that can be sold by a facility, the clustering system may implement k-nearest neighbors clustering, whereby the clustering system identifies k people records that most closely relate to the attributes defined for the facility. In another example, the clustering system may implement k-means clustering, such that the clustering system identifies k different clusters of people records, whereby the clustering system or another system selects items from the cluster.

In embodiments, an analytics system may perform analytics relating to various aspects of the energy and computing resource platform. The analytics system may analyze certain communications to determine which configurations of a facility produce the greatest yield, what conditions tend to indicate potential faults or problems, and the like.

FIG. 37 depicts example operations of a lead generation system 3314 to generate a lead list. Lead generation system 3314 receives 3702 a list of potential leads (such as for consumers of available products or resources). The lead generation system 3314 may provide 3704 the list of leads to the clustering system. The clustering system clusters 3706 the profile of the lead with the clusters of facility attributes to identify one or more clusters. In embodiments, the clustering system returns 3708 a list of lead prospects. In

other embodiments, the clustering system returns 3708 the clusters, and the lead generation system selects 3710 the list of leads from the cluster to which a prospect belongs.

FIG. 38 depicts example operations of a lead generation system 3314 to determine facility outputs for leads identified in the list of leads. In embodiments, the lead generation system 3314 provides 3802 a lead identifier of a respective lead to the AI system 3604. The AI system may then obtain the lead attributes of the lead and facility attributes of the facility and may feed 3804 the respective attributes into a prediction model. The prediction model returns 3806 a prediction, which may be scores associated with each possible outcome, or a single predicted outcome that was selected based on its respective score (e.g., the outcome having the highest score). Based on the prediction score or outcome, the lead may be stored and/or categorized (e.g. operation 3808). The lead generation system may iterate (e.g., operation 3810) in this manner for each lead in the lead list. For example, the lead generation system may generate leads (e.g., operation 3812) that are consumers of compute capabilities, energy capabilities, predictions and forecasts, optimization results, and others.

In embodiments the lead generation system provides the facility operator or host of the systems with an indicator of the reason why a lead may be willing to engage the facility, such as, for example, that the lead is an intensive user of computing resources, such as to forecast behavior of a complex, multi-variable market, or to mine for cryptocurrency.

In embodiments, a content generation system of the platform generates content for a contact event, such as an email, text message, or a post to a network, or a machine-to-machine message, such as communicating via an API or a peer-to-peer system. In embodiments, the content is customized using artificial intelligence based on the attributes of the facility, attributes of a recipient (e.g., based on the profile of a person, the role of a person, or the like), and/or relating to the project or activity to which the facility relates. The content generation system may be seeded with a set of templates, which may be customized, such as by training the content generation system on a training set of data created by human writers, and which may be further trained by feedback based on outcomes tracked by the platform, such as outcomes indicating success of particular forms of communication in generating donations to a facility, as well as other indicators as noted throughout this disclosure. The content generation system may customize content based on attributes of the facility, a project, and/or one or more people, and the like. For example, a facility manager may receive short messages about events related to facility operations, including codes, acronyms and jargon, while an outside consumer of outputs from the facility may receive a more formal report relating to the same event.

FIG. 39 depicts example operations of a content generation system 3316 to generate personalized content. The content generation system receives 3902 a recipient id, a sender id (which may be a person or a system, among others), and a facility id. The content generation system may determine 3904 the appropriate content template to use based on the relationships among the recipient, sender and facility and/or based on other considerations (e.g., a recipient who is a busy manager is more likely to respond to less formal messages or more formal messages). The content generation system may provide the template (or an identifier thereof) to the natural language generation system, along with the recipient id, the sender id, and the facility id. The natural language generation system may obtain 3906 facility

attributes based on the facility id, and person attributes corresponding to the recipient or sender based on their identities. The natural language generation system may then generate **3908** the personalized or customized content based on the selected template, the facility parameters, and/or other attributes of the various types described herein. The natural language generation system may output **3910** the generated content to the content generation system.

In embodiments, a person, such as a facility manager, may approve the generated content provided by the content generation system and/or make edits to the generated content, then send the content, such as via email and/or other channels. In embodiments, the platform tracks the contact event.

In embodiments, the workflow management system may support various workflows associated with a facility, such as including interfaces of the platform by which a facility manager may review various analytic results, status information, and the like. In embodiments, the workflow management system tracks the operation of a post-action follow-up module to ensure that the correct follow-up messages are automatically, or under control of a facility agent using the platform, sent to appropriate individuals, systems and/or services.

In the various embodiments, various elements are included for a workflow for each of an energy project, a compute project (e.g., cryptocurrency and/or AI) and hybrids.

Transactions, as described herein, may include financial transactions using various forms of currency, including fiat currencies supported by governments, cryptocurrencies, tokens or points (such as loyalty points and rewards points offered by airlines, hospitality providers, and many other businesses), and the like. Transactions may also be understood to encompass a wide range of other transactions involving exchanges of value, including in-kind transactions that involve the exchange of resources. Transactions may include exchanges of currencies of various types, including exchanges between currencies and in-kind resources. Resources exchanged may include goods, services, compute resources, energy resources, network bandwidth resources, natural resources, and the like. Transactions may also include ones involving attention resources, such as by prospective counterparties in transactions, such as consumers of goods, services and other, who may be humans or, in some situations, may be other consumers, such as intelligent (e.g., AI-based agents).

Certain features of the present disclosure are referenced as a compute task herein. The term compute task should be understood broadly. In certain embodiments, and without limitation to any other aspect of the present disclosure, a compute task includes any one or more of: execution of one or more computer readable instructions by a processor; intermediate storage of commands for execution (e.g., in a cache or buffer); operations to store, communicate, or perform calculations on data; and/or processing of data for error checking, formatting, compression, decompression, configuring packets, or the like. In certain embodiments, and without limitation to any other aspect of the present disclosure, a compute task includes any one or more of: cryptocurrency mining operations, distributed ledger calculations, transaction execution operations, internal/external data collection operations, and/or digital transformation of data elements, models, or the like. In certain embodiments, compute resources include any device configured to support compute tasks at least during certain operating conditions of a system, including, without limitation: processor(s), co-

processor(s), memory caches, random access memory (RAM), buses. In certain embodiments, compute resources may be provided by a single device and/or distributed across multiple devices. In certain embodiments, compute resources for a system may be an aggregate of devices, potentially distributed and in communication within a single hardware system, through a network (e.g., a LAN, WAN, Wi-Fi, or other communicative coupling system), through an intranet, and/or through the internet.

Certain features of the present disclosure are referenced as a network task herein. The term network task should be understood broadly. In certain embodiments, and without limitation to any other aspect of the present disclosure, a network task includes any one or more of: communicating an element of data to a network device (e.g., a packet, data for a packet, and/or metadata or other information about the data); configuring data for a network communication (e.g., compiling into one or more packets; structuring, dividing, compressing, or combining the data for network communication); caching, buffering, or otherwise storing data related to network operations; transmitting data from one device to another device (e.g., using a wired or wireless transmitting/receiving device and a communication protocol); and/or performing operations to register or unregister a device from a group of devices (e.g., in a mesh network, peer-to-peer network, or other network configuration). In certain embodiments, and without limitation to any other aspect of the present disclosure, a network task includes any one or more of: cognitive coordination of network assets; peer bandwidth sharing; transaction execution; spot market testing; internal/external data collection; advanced analytics (e.g., of data access, stored data, user or accessor interactions, etc.); smart contract operations; connected insurance operations; and/or distributed ledger storage. In certain embodiments, any operations performed by a network device, and/or performed to support network communications by a network device, are contemplated as network tasks herein. In certain embodiments, network resources include any device configured to support network tasks at least during certain operating conditions of a system, including, without limitation: networking adapters; networking processors or sub-processors; memory caches or buffers; communication links (e.g., ports, connectors, wires, transmitters, and/or receivers); network infrastructure such as routers, repeaters, hardware comprising a LAN, WAN, intranet, and/or internet; and/or aggregated or abstracted aspects of a network such as bandwidth or availability of any communication system or communication channel.

It can be seen that, in certain embodiments, a task may be considered a compute task for one system or purpose, and/or a network task for another system or purpose. Further, a given device may be considered a compute resource for one system or purpose, and/or a network resource for another system or purpose. In certain embodiments, a given device on a system may be considered a compute resource under certain operating conditions and/or for certain considerations, and the given device on the system may be considered a compute resource under other operating conditions and/or for certain other considerations. For example, a given processor may be configured to perform operations to execute computer readable instructions, and therefore be available as a computing resource for determinations made by platform **100** in a first context, and the same processor may be configured to support network communications (e.g., packaging data, performing network coding, or other network support operations), and therefore also be available as a network resource for the platform **100** in a second

context. In another example, a platform 100 may be performing operations to improve and/or optimize computing and/or network resource consumption for a system having multiple processors in communication over a network. In the example, the platform 100 may consider the various processors separately from the network resources—for example distributing the computing tasks across processors, and calculating the incurred network resource consumption separately. Additionally or alternatively, in the example, the platform 100 may abstract the network resource consumption associated with distributing computing tasks across processors as processor resource consumption, thereby assigning the associated networking resources that support distribution of processing as processing resources.

One of skill in the art, having the benefit of the present disclosure and information ordinarily available when contemplating a particular system, can readily determine which tasks are compute tasks, which tasks are network tasks, which resources are compute resources, and which resources are network resources for the particular system and at which operating conditions of the system. In certain embodiments, for example where improvement or optimization operations are considering both compute and network resource optimizations, a particular system may allow the operations of the platform 100 to determine, or to adjust, which tasks are compute and network tasks, and/or which resources are compute and network resources. Certain considerations for determining which tasks/resources are compute/network tasks/resources include, without limitation: the limiting aspects of the particular system, including the limiting aspect of the system with time and/or operating condition; the system parameters to be improved or optimized; the desired authority to be given to AI, machine learning, expert system, or other adaptive devices in the system; the cost drivers in the system for various devices or features (e.g., infrastructure; support; additional communication paths; upgrades to operating systems, protocols, or firmware, etc.); the priorities for system improvement between capital investment, operating costs, energy consumption, etc.; and/or the capacity limitations in the system, including present and future capacity, and/or capacities based on time and/or operating condition.

Certain features of the present disclosure are referenced as a data storage task herein. The term data storage task should be understood broadly. In certain embodiments, without limitation to any other aspect of the present disclosure, a data storage task is a task associated with the storage of data for access at a later time, and/or to support the ability to access the data at a later time. Data storage tasks can include, without limitation: operations to communicate data to a storage device; operations to retrieve stored data from a storage device; operations to store data on the storage device; operations to config. the data for storage or retrieval (e.g., setting or verifying authorizations, performing compression or decompression, formatting the data, and/or summarizing or simplifying the data); operations to move data from one storage to another (e.g., moving data between short-term, intermediate term, and long-term storage; and/or transferring data from one data storage location to another to support improvements or optimizations, such as moving less accessed data to a lower cost storage location, etc.); and/or operations to delete stored data. Example and non-limiting data storage resources include: data storage devices of any type and storage medium; and/or communication devices and/or processing to support data storage devices. It can be seen that, in certain embodiments, a task may be considered a compute task for one system or purpose, a network task for

another system or purpose, and/or a data storage task for another system or purpose. Further, a given device may be considered a compute resource for one system or purpose, a network resource for another system or purpose, and/or a data storage resource for another system or purpose.

Certain features of the present disclosure are referenced as a core task herein. The term core task should be understood broadly. In certain embodiments, without limitation to any other aspect of the present disclosure, a core task is a task associated with a system or facility that relates to the function or purpose of that system or facility. A few examples include, without limitation: a core task for a manufacturing facility relates to the manufacturing operations of the facility; a core task for a chemical production plant relates to the chemical production operations of the facility; a core task for an autonomous vehicle relates to the operations of the vehicle; and/or a core task for an insurance provider relates to the provision and service of insurance products of the provider. In certain embodiments, a core task includes any related tasks for the facility, which may or may not be critical or primary tasks for the facility. For example, a manufacturing facility may operate a system to track recycling operations, manage parking, and/or tracking the schedules for an intra-company softball league for the manufacturing facility. In certain embodiments, a core task is any task performed for the merits of the underlying facility, where some increment of data associated to the task is available, or becomes available, to a platform 100 for consideration in supporting one or more aspects of the task. In certain embodiments, a task may be a core task for certain systems and/or operating conditions, and another type of task (e.g., a compute task, a network task, and/or a data storage task) for other systems and/or other operating conditions. For example, communication of employee e-mails may be a core task for supporting a manufacturing facility, and may additionally or alternatively be a network task, compute task, and/or data storage task. In a further example, communication of employee e-mails may be a core task during certain operating periods (e.g., during working hours, for each employee during that employee's shift period, etc.), and may be a network task, compute task, and/or data storage task during other operating periods (e.g., during off-hours archiving periods).

Certain features of the present disclosure are referenced as forward markets herein. The term forward market should be understood broadly, and includes any market that provides for trading of any type of resource scheduled for future delivery of the resource. A forward market contemplates formal markets, such as energy trading, commodity trading, compute resource trading, data storage trading, network bandwidth trading, and/or spectrum trading markets whereby parties can access the markets and purchase or sell resources (e.g., in a set quantity for a set delivery time). Additionally or alternatively, a forward market contemplates an informal market, where parties set forth a mechanism to trade or commit resources that are to be delivered at a later time. Trading may be performed in any currency, or based on in-kind contributions, and a forward market may be a mechanism for actual delivery of resources as scheduled, or a mechanism for trading on the future value of resources without actual delivery being contemplated (e.g., with some other mechanism that tends to bring the future price in to the spot price as the time for each forward looking period approaches). In certain embodiments, a forward market may be privately operated, and/or operated as a service where a platform 100 sets up the market, or communicates with the market. In certain embodiments, as described throughout the

present disclosure, transactions on the forward market may be captured in a distributed ledger.

Certain features of the present disclosure are referenced as spot markets herein. The term spot market should be understood broadly, and includes any market that provides for trading of any type of resource at a price based on the current trading price of the resource for immediate delivery. A spot market contemplates formal markets and/or informal markets. Trading on a spot market may be performed in any currency, or based on in-kind contributions. In certain embodiments, a spot market may be privately operated, and/or operated as a service where a platform 100 sets up the market, or communicates with the market. In certain embodiments, as described throughout the present disclosure, transactions on the spot market may be captured in a distributed ledger.

Certain features of the present disclosure are referenced as purchasing or sale of one or more resources, including at least: energy, energy credits, network bandwidth (e.g., communication capacity), spectrum and/or spectrum allocation (e.g., certain frequency bandwidths, including potentially transmission rates, transmission power, and/or geographical limitations); compute resources (or capacity); network resources (or capacity); data storage resources (or capacity); and/or energy storage resources (or capacity). A purchase or sale, as utilized herein, includes any transaction wherein an amount of a resource or other commitment (e.g., an element of intellectual property (IP), an IP license, a service, etc.) is traded for a unit of currency of any type and/or an amount of another resource or commitment. In certain embodiments, a purchase or sale may be of the same type of resource or commitment, for example where energy for one time period (e.g., immediate delivery, or a first future time period) is traded for energy at another time period (e.g., a second future time period, which is distinct from the immediate delivery or the first future time period). In certain embodiments, one side of the purchase or sale includes a currency of any type, including at least a sovereign currency, a cryptocurrency, and/or an arbitrary agreed upon currency (e.g., specific to a private market or the like).

Certain features of the present disclosure are referenced as a machine herein. The term machine, as utilized herein, should be understood broadly. In certain embodiments, a machine includes any component related to a facility having at least one associated task, which may be a core task, a compute task, a network task, a data storage task, and/or an energy storage task. In certain embodiments, a machine includes any component related to a facility that utilizes at least one resource, which may be an energy resource, a compute resource, a network resource, and/or a data storage resource. In certain embodiments, a machine includes any one or more aspects of any controller, AI implementing device, machine learning implementing device, deep learning implementing device, neural network implementing device, distributed ledger implementing or accessing device, intelligent agent, a circuit configured to perform any operations described throughout the present disclosure, and/or a market (forward and/or spot) implementing or accessing device as described throughout the present disclosure. In certain embodiments, a machine is operatively and/or communicatively coupled to one or more facility components, market(s), distributed ledger(s), external data, internal data, resources (of any type), and/or one or more other machines within a system. In certain embodiments, two or more machines be provided with at least one aspect of cooperation between the machines, forming a fleet of machines. In certain embodiments, two machines may cooperate for cer-

tain aspects of a system or in certain operating conditions of the system, and thereby form a fleet of machines for those aspects or operating conditions, but may be separate individually operating machines for other aspects or operating conditions. In certain embodiments, machines forming a part of a fleet of machines may be associated with (e.g., positioned at, communicatively coupled to, and/or operatively coupled to) the same facility, or distinct facilities. In certain embodiments, a machine may be associated with more than one facility, and/or associated with different facilities at different times or operating conditions.

Certain aspects of the present disclosure are referenced as energy credits herein. The term energy credits, as utilized herein, should be understood broadly. In certain embodiments, an energy credit is a regulatory, industry agreed, or other indicia of energy utilization that is tracked for a particular purpose, such as CO₂ emissions, greenhouse gas emissions, and/or any other emissions measure. In certain embodiments, an energy credit may be “negative” (e.g., relating to increased emissions) or “positive” (e.g., relating to reduced emissions). In certain embodiments, energy credits may relate to particular components (e.g., automobiles of a certain power rating or application, computing related energy utilization, etc.) and/or generic energy utilization (e.g., without regard to the specific application utilizing the energy). In certain embodiments, energy credits may relate to taxation schemes, emissions control schemes, industry agreement schemes, and/or certification schemes (e.g., voluntary, involuntary, standards-related, or the like). In certain embodiments, an energy credit includes any indicia of energy utilization where verified tracking (e.g., for reporting purposes) of that indicia can be utilized to increment or decrement value for a facility, facility owner, or facility operator. Non-limiting examples include: an entity subject to a regulatory requirement for reporting emissions; and/or an entity reporting emissions performance in a public format (e.g., an annual report).

Certain aspects of the present disclosure are referenced as collective optimization. Collective optimization, as used herein, includes the improvement and/or optimization of multiple aspects of a system (e.g., multiple machines, multiple components of a facility, multiple facilities, etc.) together as an optimized or improved system. It will be understood that collective optimization may occur within more than one dimension—for example a collectively optimized or improved system may have a higher overall energy consumption than before operations to collectively optimize or improve, but have improvement in some other aspect (e.g., utilization of energy credits, lower cost of operation, superior product or outcome delivery, lower network utilization, lower compute resource usage, lower data storage usage, etc.).

Certain aspects of the present disclosure are referenced as social media data sources. Social media data sources include, without limitation: information publicly available on any social media site or other mass media platform (e.g., from comments sections of news articles; review sections of an online retailer; publicly available aspects of profiles, comments, and/or reactions of entities on social media sites; etc.); proprietary information properly obtained from any social media site or other mass media platform (e.g., purchased information, information available through an accepted terms of use, etc.); and the like. In certain embodiments, social media data sources include cross-referenced and/or otherwise aligned data from multiple sources—for example where a comment from one site is matched with a profile from another site, data is matched with a member list

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from a professional group membership, data is matched from a company website, etc. In certain embodiments, social media data sources include cross-referenced and/or otherwise aligned data from other data sources, such as IoT data sources, automated agent behavioral data sources, business entity data sources, human behavioral data sources, and/or any other data source accessible to a machine, platform 100, or other device described throughout the present disclosure.

Certain aspects of the present disclosure reference determining (and/or optimizing, improving, reducing, etc.) the utilization or consumption of energy or resources. Determining the utilization or consumption of energy or resources should be understood broadly, and may include consideration of a time value of the consumption, and/or an event-related value of the consumption (e.g., calendar events such as holidays or weekends, and/or specific real-time events such as news related events, industry related events, events related to specific geographical areas, and the like). In certain embodiments, the utilization or consumption of energy or resources may include consideration of the type of energy or resource (e.g., coal-generated electricity versus wind-generated electricity), the source of the energy or resource (e.g., the geographical origin of the energy available, the entity providing a compute resource, etc.), the total capacity of the energy or resource (e.g., within a facility or group of facilities, from a third-party, etc.), and/or non-linear considerations of the cost of the energy or resource (e.g., exceeding certain thresholds, the likely cost behavior in a market responsive to a purchase event, etc.).

Certain aspects of the present disclosure reference performing operations to implement an arbitrage strategy. An arbitrage strategy as utilized herein should be understood broadly. An arbitrage strategy includes any strategy structured to favorably utilize a differential between a present value of a resource and a predicted future value of the resource. In certain embodiments, implementing an arbitrage strategy includes a determination that a given value (either a present value on a spot market, or a future value for at least one time frame) of the resource is abnormally low or high relative to an expected or anticipated value, and to execute operations to either purchase or sell the resource and benefit from the abnormal value. In certain embodiments, an arbitrage strategy is implemented as a portion of an overall optimization or improvement operation. For example, in certain embodiments implementing the arbitrage strategy may push the overall system away from the otherwise optimum value (e.g., buying or selling more of a resource than the improved or optimized system would otherwise perform), and the benefits of the implementation of the arbitrage strategy are considered within the value of the entire system. In certain embodiments, an arbitrage strategy is implemented as a standalone transaction (e.g., for a system that is not presently operating any core tasks, and/or a system where implementing an arbitrage strategy is the core task), and the arbitrage strategy is implemented as the primary, or the only, system level improvement or optimization.

Certain aspects of the present disclosure are referenced as a small transaction, and/or a rapidly executed transaction. A small transaction as utilized herein references a transaction that is small enough to limit the risk of the transaction to a threshold level, where the threshold level is either a level that is below an accepted cost of the transaction, or below a disturbance level (e.g., a financial disturbance, an operational disturbance, etc.) for the system. For example, wherein an implementation of an arbitrage strategy includes a small transaction for an energy resource, the small trans-

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action may be selected to be small enough such that the amount of energy bought or sold does not change the basic operational equilibrium of the system under current operating conditions, and/or such that the amount of potential loss from the transaction is below a threshold value (e.g., an arbitrage fund, an operating cash amount, or the like). In certain embodiments, the small transaction is selected to be large enough to test the arbitrage opportunity determination—for example a large enough transaction that the execution of the transaction will occur in a similar manner (e.g., not likely to be absorbed by a broker, having an expected similarity in execution speed, and/or having an expected similarity in successful execution likelihood) to a planned larger trade to be performed. It will be understood that more than one small transaction, potentially of increasing size, may be performed before a larger transaction is performed, and/or that a larger transaction may be divided into one or more portions. A rapidly executed transaction includes any transaction that is expected to have a rapid time constant with regard to the expected time frame of the arbitrage opportunity. For example, where a price anomaly is expected to persist for one hour, a rapidly executed transaction may be a transaction expected to clear in much less than one hour (e.g., less than half of the hour, to provide time to execute the larger transaction). In another example, where a price anomaly is expected to persist for 10 minutes, a rapidly executed transaction may be a transaction expected to clear in much less than 10 minutes. It will be understood that any machine, AI component, machine learning component, deep learning component, expert system, controller, and/or any other adaptive component described throughout the present disclosure may adaptively improve the size, timing, and/or number of small transactions and large transactions as a part of improving or optimizing an implementation of an arbitrage strategy. Additionally or alternatively, any parameters of the arbitrage determination, such as the expected value of the arbitrage opportunity and/or the expected persistence time of the arbitrage opportunity, may be adaptively improved.

Certain aspects of the present disclosure are referenced as a token, and/or certain operations of the present disclosure are referenced as tokenizing one or more aspects of data or other parameters. Tokens, and/or operations to tokenize, should be understood broadly, and include operations and/or data utilized to abstract underlying data and/or to provide confirmable or provable access to the underlying data. Without limitation to any other aspect of the present disclosure, tokens include wrapper data that corresponds to underlying data values, hashes or hashing operations, surrogate values, and/or compartmentalized data. Tokenization operations may include hashing, wrapping, or other data separation and/or compartmentalization operations, and may further include authorization operations such as verification of a user or other interacting party, including verification checks based on IP addresses, login interfaces, and/or verifications based on characteristics of the user or other interacting party that are accessible to the tokenizing system. In certain embodiments, a token may include certain aspects of the underlying or tokenized data (e.g., headers, titles, publicly available information, and/or metadata), and/or a token may be entirely abstracted from the underlying or tokenized data. In certain embodiments, tokens may be utilized to provide access to encrypted or isolated data, and/or to confirm that access to the encrypted or isolated data has been provided, or that the data has been accessed.

Certain aspects of the present disclosure reference provable access (e.g., to data, instruction sets, and/or IP assets).

Provable access, as utilized herein, should be understood broadly. In certain embodiments, provable access includes a recordation of actual access to the data, for example recording a data value demonstrating the data was accessed, and may further include user or accessor information such as usernames, e-mail addresses, IP addresses, geographic locations, time stamps, and/or which portions of the data were accessed. In certain embodiments, provable access includes a recordation of the availability of the data to a user or potential accessor, and may further include user or accessor information such as usernames, e-mail addresses, IP addresses, geographic locations, time frames or stamps, and/or which portions of the data were available for access. In certain embodiments, provable access includes storing the recordation on a system (e.g., on a distributed ledger, and/or in a memory location available to any controller, machine, or other intelligent operating entity as described throughout the present disclosure). In certain embodiments, provable access includes providing the user or accessor of the data with a data value such that the user or access or is able to demonstrate the access or access availability. In certain embodiments, a data value and/or distributed ledger entry forming a portion of the provable access may be encrypted, tokenized, or otherwise stored in a manner whereby the provable access can be verified, but may require an encryption key, login information, or other operation to determine the access or access availability from the data value or distributed ledger entry.

Certain aspects of the present disclosure are referenced as an instruction set and/or as executable algorithmic logic. An instruction set and/or executable algorithmic logic as referenced herein should be understood broadly. In certain embodiments, an instruction set or executable algorithmic logic includes descriptions of certain operations (e.g., flow charts, recipes, pseudo-code, and/or formulas) to perform the underlying operations—for example an instruction set for a process may include a description of the process that may be performed to implement the process. In certain embodiments, an instruction set or executable algorithmic logic includes portions of certain operations, for example proprietary, trade secret, calibration values, and/or critical aspects of a process, where the remainder of the process may be generally known, publicly available, or provided separately from the portions of the process provided as an instruction set or executable algorithmic logic. In certain embodiments, an instruction set or executable algorithmic logic may be provided as a black box, whereby the user or accessor of the instruction set or executable algorithmic logic may not have access to the actual steps or descriptions, but may otherwise have enough information to implement the instruction set or executable algorithmic logic. For example, and without limitation, a black box instruction set or executable algorithmic logic may have a description of the inputs and outputs of the process, enabling the user or accessor to include the instruction set or executable algorithmic logic into a process (e.g., as a module of executable instructions stored in a computer readable medium, and/or as an input to a machine responsive to the black box operations) without having access to the actual operations performed in the instruction set or the executable algorithmic logic.

Certain aspects of the present disclosure are referenced as a distributed ledger. A distributed ledger, as referenced herein, should be understood broadly. Without limiting any other aspect of the present disclosure, a distributed ledger includes any data values that are provided in a manner to be stored in distributed locations (e.g., stored in multiple

memory locations across a number of systems or devices), such that individual members of the distributed system can add data values to the set of data values, and where the distributed system is configured to verify that the added data values are consistent with the entire set of data values, and then to update the entire set of data values thereby updating the distributed ledger. A block chain is an example implementation of a distributed ledger, where a critical mass (e.g., more than half) of the distributed memory locations create agreement on the data values in the distributed ledger, thereby creating an updated version of the distributed ledger. In certain embodiments, a distributed ledger may include a recordation of transactions, stored data, stored licensing terms, stored contract terms and/or obligations, stored access rights and/or access events to data on the ledger, and/or stored instruction sets, access rights, and/or access events to the instruction sets. In certain embodiments, aspects of the data on a distributed ledger may be stored in a separate location, for example with the distributed ledger including a pointer or other identifying location to the underlying data (e.g., an instruction set stored separately from the distributed ledger). In certain embodiments, an update to the separately stored data on the distributed ledger may include an update to the separately stored data, and an update to the pointer or other identifying location on the distributed ledger, thereby updating the separately stored data as referenced by the distributed ledger. In certain embodiments, a wrapper or other interface object with the distributed ledger may facilitate updates to data in the distributed ledger or referenced by the distributed ledger, for example where a party submits an updated instruction set, and where the wrapper stores the updated instruction set separately from the distributed ledger, and updates the pointer or identifying location on the distributed ledger to access the updated instruction set, thereby creating a modified instruction set (or other data).

Certain aspects of the present disclosure are referenced as a wrapper, expert wrapper, a smart wrapper, and/or a smart contract wrapper. A wrapper, as referenced herein, should be understood broadly. Without limitation to any other aspect of the present disclosure, a wrapper references any interfacing system, circuit, and/or computer executable instructions providing an interface between the wrapped object (e.g., data values and/or a distributed ledger) and any system, circuit, machine, user, and/or accessor of the wrapped object. A wrapper, in certain embodiments, provides additional functionality for the wrapped object, user interfaces, API, and/or any other capabilities to implement operations described herein. In certain embodiments, a wrapper can provide for access authorization, access confirmation, data formatting, execution of agreement terms, updating of agreement terms, data storage, data updating, creation and/or control of metadata, and/or any other operations as described throughout the present disclosure. In certain embodiments, parameters of the wrapper (e.g., authorized users, data in a stack of data, creation of new data stacks, adjustments to contract terms, policies, limitations to total numbers of users or data values, etc.) may be configurable by a super user, an authorized user, an owner, and/or an administrator of the wrapper, and/or parameters of the wrapper may be accessible within the wrapped object (e.g., as data values stored in a distributed ledger which, when updated, change certain parameters of the wrapper). An expert wrapper or a smart wrapper includes, without limitation, any wrapper that includes or interacts with an expert system, an AI system, a ML system, and/or an adaptive system.

Certain aspects of the present disclosure are referenced as IP licensing terms and/or contract terms herein. IP licensing terms, as used herein, should be understood broadly. Without limitation to any other aspect of the present disclosure, IP licensing terms include permissions to access and/or utilize any element of IP. For example, and IP licensing term may include an identification of the IP element (e.g., a trade secret; a patent and/or claims of a patent; an image, media element, written description, or other copyrighted data element; and/or proprietary information), a description of the access or usage terms (e.g., how the IP element may be utilized by the accessor), a description of the scope of the utilization (e.g., time frames, fields of use, volume or other quantitative limits, etc.), a description of rights relating to the IP element (e.g., derivative works, improvements, etc.), and/or a description of sub-licensing rights (e.g., provisions for suppliers, customers, affiliates, or other third parties that may interact with the user or accessor in a manner related to the IP element). In certain embodiments, IP licensing terms may include a description of the exclusivity or non-exclusivity provided in relation to an IP element. In certain embodiments, IP licensing terms may relate to open source terms, educational use, non-commercial use, or other IP utilization terms that may or may not relate to a commercial transaction or an exchange of monetary/currency value, but may nevertheless provide for limitations to the use of the IP element for the user or accessor. Without limitation to any other aspect of the present disclosure, contract terms include any one or more of: options (e.g., short and/or put options relating to any transaction, security, or other tradeable assets); field exclusivity; royalty stacking (e.g., distribution of royalties between a group of owners and/or beneficiaries); partial exclusivity (e.g., by fields-of-use, geographic regions, and/or transaction types); pools (e.g., shared or aggregated IP stacks, data pools, and/or resource pools); standard terms; technology transfer terms; performance-related rights or metrics; updates to any of the foregoing; and/or user selections (e.g., which may include further obligations to the user and/or costs to the user) of any of the foregoing.

Certain aspects herein are described as a task system. A task system includes any component, device, and/or group of components or devices that performs a task (or a portion of a task) of any kind described throughout the present disclosure, including without limitation any type of task described for a machine. The task may have one or more associated resource requirements, such as energy consumption, energy storage, data storage, compute requirements, networking requirements, and/or consumption of associated credits or currency (e.g., energy credits, emissions credits, etc.). In certain embodiments, the resource utilization of the task may be negative (e.g., consumption of the resource) or positive (e.g., regeneration of energy, deletion of data, etc.), and may further include intermediate values or time trajectories of the resource utilization (e.g., data storage requirements that vary over an operating period for the task, energy storage requirements that may fill or deplete over an operating period, etc.). The determination of any resource requirement for a task herein should be understood broadly, and may be determined according to published information from the task system (e.g., according to a current load, energy consumption, etc.), determined according to a scheduled or defined value (e.g., entered by an operator, administrator, and/or provided as a communication by a controller associated with the task system), and/or may be determined over time, such as by observing operating histories of the task system. In certain embodiments, expert systems and/or

machine learning components may be applied to determine resource requirements for a task system—for example determining relationships between any available data inputs and the resource requirements for upcoming tasks, which allows for continuous improvement of resource requirement determinations, and further allows for training of the system to determine which data sources are likely to be predictive of resource requirements (e.g., calendar date, periodic cycles, customer orders or other business indicators, related industry indicators, social media events and/or other current events that tend to drive resource requirements for a particular task, etc.).

Referencing FIG. 40, an example system 4000 is a transaction-enabling system including a smart contract wrapper 4002. The contract wrapper 4002 may be configured to access a distributed ledger 4004 including a number of embedded contract and/or IP licensing terms 4006 and a number of data values 4008, and to interpret an access request value 4010 for the data values 4008. In response to the access request value 4010, the contract wrapper provides access to at least a portion of the data values 4008, and commits an entity 4009 (e.g., a subscribing user, a customer, and/or a prospective customer for the data) providing the access request value 4010 to at least one of the embedded contract and/or IP licensing terms 4006. The contract wrapper 4002 may be embodied as computer readable instructions that provide a user interface between a user and the distributed ledger, and may further be implemented as a web interface, an API, a computer program product, or the like. The contract wrapper 4002 is operationally positioned between one or more users 4009 and the distributed ledger 4004 and data values 4008, but the contract wrapper 4002 may be stored and function in a separate physical location from the distributed ledger 4004, data values 4008, and/or the users 4009. Additionally or alternatively, the contract wrapper 4002 may be distributed across multiple devices and/or locations.

An example embodiment includes the data values 4008 including intellectual property (IP) data corresponding to a plurality of IP assets 4016, such as a listing, description, and/or summary information for patents, trade secrets, or other proprietary information, and the embedded contract and/or IP licensing terms 4006 include a number of intellectual property (IP) licensing terms (e.g., usage rights, fields of use, limitations, time frames, royalty rates, and the like) for the corresponding IP assets. In certain embodiments, the data values 4008 may include the IP assets 4016 (e.g., proprietary information, recipes, instructions, or the like), and/or the data values 4008 may correlate to IP assets 4016 stored elsewhere, and may further include sufficient information for a user to understand what is represented in the IP assets 4016. The example contract wrapper 4002 may further commit the entity 4009 providing the access request value 4010 to corresponding contract and/or IP licensing terms 4006 for accessed ones of the IP assets 4016—for example only committing the user to terms for assets 4016 that are agreed upon, accessed, and/or utilized (e.g., committed contract terms).

An example contract wrapper 4002 is further configured to interpret an IP description value 4012 and an IP addition request 4014, and to add additional IP data to the data values 4008 in response to the IP description value 4012 and the IP addition request 4014, where the additional IP data includes IP data corresponding to an additional IP asset. For example, the contract wrapper 4002 may accept an IP description value 4012 from a user (e.g., a document, reference number, or the like), and respond to the IP addition request 4014 to

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add information to the data values **4008** consistent with the IP description value **4012**, thereby adding one or more IP assets **4016** to the data values **4008**. In certain embodiments, the contract wrapper **4002** may further provide a user interface to interact with the user **4009** or other entity adding the IP asset, which may include determining permissions to add an asset, and/or consent or approval from the user or other parties. In certain embodiments, consent or approval may be performed through rules, an intelligent system, or the like, for example to ensure that IP assets being added are of a selected type, quality, valuation, or other have other selected characteristics.

An example contract wrapper **4002** accesses a number of owning entities corresponding to the IP assets of the data values **4008**. The example contract wrapper **4002** apportions royalties **4018** generated from the IP assets corresponding to the data values **4008** in response to the corresponding IP license terms **4006**, such as apportionment based on asset valuations, asset counts, and/or any agreed upon apportionment parameters. In certain embodiments, the contract wrapper **4002** adds an IP asset to an aggregate stack of IP assets based on an IP addition request **4014**, and updates the apportionment of royalties **4018** based upon the owning entities **4009** and IP assets **4016** for the aggregate stack after the addition of the IP asset. In certain embodiments, the contract wrapper **4002** is configured to commit the entity adding the IP asset to the IP licensing terms **4006**, and/or the IP licensing terms **4006** as amended with the addition of the new IP assets.

An example transaction-enabling system **4000** including a smart contract wrapper **4002**, the smart contract wrapper according to one disclosed non-limiting embodiment of the present disclosure may be configured to access a distributed ledger includes a plurality of intellectual property (IP) licensing terms **4006** corresponding to a plurality of IP assets **4016**, wherein the plurality of IP assets **4016** include an aggregate stack of IP, and to interpret an IP description value **4012** and an IP addition request **4014**, and, in response to the IP addition request **4014** and the IP description value **4012**, to add an IP asset to the aggregate stack of IP. An example smart contract wrapper **4002** interprets an IP licensing value **4020** corresponding to the IP description value **4012**, and to add the IP licensing value **4020** to the plurality of IP licensing terms **4006** in response to the IP description value and the IP addition request. The IP licensing value **4020** may be determined from input by the user **4009**, automated or machine learning improved operations performed on indicia of the added IP asset (e.g., according to valuation algorithms such as markets affected by the IP asset, value contributions of the IP asset, participation of the IP asset into industry standard systems or operations, references to the IP asset by other IP assets, and the like), and/or may further depend upon the role or permissions of the user **4009**. In certain embodiments, a first user **4009** adds the IP asset to the IP assets **4016**, and a second user **4009** provides additional data utilized to determine the IP licensing value **4020**. An example smart contract wrapper **4002** further associates one or more contract and/or IP licensing terms **4006** to the added IP asset. In certain embodiments, one or more IP assets are stored within the data values **4008**, and/or are referenced to a separate data store having the IP assets **4016**. An example aggregate stack of IP further includes a reference to the data store for one or more IP assets **4016**.

Referencing FIG. **41**, an example procedure **4100** to execute a smart contract wrapper is depicted. The procedure **4100** includes an operation **4102** to access a distributed ledger including a number of embedded contract terms and

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a number of data values, an operation **4104** to provide a user interface (UI) including an access option, an operation **4106** to provide a user interface including acceptance input portion, and an operation **4108** to interpret an access request value for the data values. The example procedure **4100** further includes an operation **4110** to provide access to at least a portion of the plurality of data values, and an operation **4112** to commit an entity providing the access request value to at least one of the plurality of embedded contract terms.

An example procedure may include providing the entity providing the access request value with a user interface including a contract acceptance input, and where the providing access and committing the entity is in response to a user input on the user interface. An example procedure may include the data values including IP data (e.g., IP elements, or information corresponding to IP elements), where the embedded contract terms include IP licensing terms for the corresponding IP assets. An example procedure further includes the operation **4112** to commit an entity providing the access request value to corresponding IP licensing terms for accessed IP assets.

Referencing FIG. **42**, an example procedure **4200** includes an operation **4202** to interpret an IP description value and an IP addition request, and an operation **4204** to add additional IP data to a plurality of data values on a distributed ledger in response to the IP description value and the IP addition request, wherein the additional IP data includes IP data corresponding to an additional IP asset. An example procedure **4200** further includes an operation **4206** to further interpret an IP addition entity (e.g., the entity adding the IP data, and/or an entity designated as an owner of the additional IP asset by the entity adding the IP data), and an operation **4208** to apportion royalties from the plurality of IP assets to the plurality of owning entities in response to the corresponding IP licensing terms, and/or further in response to the additional data and the IP addition entity.

Referencing FIG. **43**, an example procedure **4300** includes an operation **4302** to access a distributed ledger, an operation **4304** to interpret an IP description and an IP addition request, an operation **4306** to add IP asset(s) to the distributed ledger in response to the IP description and the IP addition request, an operation **4308** to associate one or more licensing terms to the added IP, and an operation **4310** to apportion royalties from IP assets of the distributed ledger to owning entities of the IP assets. An example procedure **4300** further includes an operation **4312** to interpret an added IP entity (e.g., as a change to an existing IP asset, and/or an entity owning an added IP asset), and an operation **4314** to commit the added IP entity to one or more of the IP licensing terms.

Referencing FIG. **44**, an example procedure **4400** includes an operation **4402** to evaluate IP assets for a distributed ledger, an operation **4404** to determine that one or more assets have expired (e.g., according to the terms of the licensing agreements, according to a date defined with the IP asset, and/or according to external data such as a database updated in response to a court ruling or other decision about the asset). The example procedure **4400** further includes an operation **4406** to determine or update a valuation of the IP assets, an operation **4408** to determine whether an ownership change has occurred for one or more of the IP assets, an operation **4410** to commit updated entities to the IP licensing terms, and/or an operation **4412** to update apportionment of royalties to owning entities in response to an asset expiration, asset valuation change,

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and/or a change in an owning entity for one or more IP assets. In certain embodiments, an example procedure **4400** includes an operation to provide a user interface to a new owning entity of at least one of the IP assets, and an operation to commit a new owning entity to one or more of the IP licensing terms in response to a user input on the user interface.

In certain embodiments, IP assets described herein include a listing of IP assets, an aggregated stack of IP assets, and/or any other organization of IP assets. In certain embodiments, IP assets may be grouped, sub-grouped, clustered, or organized in any other manner, and licensing terms may be associated, in whole or part, with the groups, sub-groups, and/or clusters of IP assets. In certain embodiments, a number of IP assets may be within a first aggregate stack for a first purpose (e.g., a particular field of use, type of accessing entity, etc.), but within separate aggregated stacks for a second purpose (e.g., a different field of use, type of accessing entity, etc.).

Referencing FIG. **45**, an example transaction-enabling system **4500** includes a controller **4502**, the controller **4502** according to one disclosed non-limiting embodiment of the present disclosure may be configured to: access a distributed ledger **4004** including an instruction set **4504**, tokenize the instruction set, interpret an instruction set access request **4506**, and, in response to the instruction set access request **4506**, provide a provable access **4508** to the instruction set. In certain embodiments, the controller **4502** provides provable access **4508** by recording a value on the distributed ledger **4004**, providing a user **4009** with a token demonstrating access, and/or records a transaction **4510** on the distributed ledger **4004** indicating the access. In certain embodiments, the instruction set **4504** may include an instruction set for any one or more of the following: a coating process, a 3D printer (e.g., a build file, diagram, and/or operational instructions), a semiconductor fabrication process, a field programmable gate array (FPGA) instruction set, a food preparation instruction set (e.g., an industrial process, a formula, a recipe, etc.), a polymer production instruction set, a chemical synthesis instruction set, a biological production process instruction set, and/or a crystal fabrication system.

An example controller **4502** further interprets an execution operation **4512** of the instruction set, and records a transaction **4510** on the distributed ledger **4004** in response to the execution operation **4512**. In certain embodiments, interpreting an execution operation **4512** includes determining that a user has accessed the instruction set **4504** sufficiently to determine a process described in the instruction set **4504**, determining that a user, the controller, or another aspect of the system **4500** has provided instructions to a device responsive to the instruction set **4504**, and/or receiving a confirmation, data value, or other communication indicating that the instruction set **4504** has been executed. In certain embodiments, one or more instruction sets **4504** stored on the distributed ledger **4004** may be at least partially stored in a separate data store of instructions **4516**, where the distributed ledger **4004** may store references, partial instructions, summaries, or the like, and access the separate data store of instructions **4516** as needed. In certain embodiments, one or more instruction sets **4504** may be stored on the distributed ledger **4004**. In certain embodiments, access to the instruction set(s) **4504** may be provided in accordance with one or more contract terms **4006**, and/or may be provided in response to committing a user or accessing entity to the one or more contract terms **4006**.

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Referencing FIG. **46**, an example procedure **4600** includes an operation **4602** to access a distributed ledger including an instruction set, an operation **4604** to tokenize the instruction set, and an operation **4606** to interpret an instruction set access request. The example procedure **4600** includes, in response to the instruction set access request, an operation **4608** to provide provable access to the instruction set. In certain embodiments, the instruction set may include an instruction set for any one or more of the following: a coating process, a 3D printer (e.g., a build file, diagram, and/or operational instructions), a semiconductor fabrication process, a field programmable gate array (FPGA) instruction set, a food preparation instruction set (e.g., an industrial process, a formula, a recipe, etc.), a polymer production instruction set, a chemical synthesis instruction set, a biological production process instruction set, and/or a crystal fabrication system. An example procedure **4600** further includes an operation **4610** to command a downstream tool (e.g., a production tool for a process, an industrial machine, a controller for an industrial process, a 3D printing machine, etc.) using the instruction set. In certain embodiments, the instruction set includes an FPGA instruction set. In certain embodiments, the procedure **4600** includes an operation to determine that an execution operation related to the instruction set has occurred, and/or an operation **4612** to record a transaction on the distributed ledger in response to an access of the instruction set, operation **4610** to command a downstream tool using the instruction set, and/or an execution operation related to the instruction set.

Referencing FIG. **47**, an example transaction-enabling system **4700** includes a controller **4502**, the controller **4502** according to one disclosed non-limiting embodiment of the present disclosure may be configured to: access a distributed ledger **4004** including executable algorithmic logic **4704**, tokenize the executable algorithmic logic, interpret an executable algorithmic logic access request **4706**, and, in response to the executable algorithmic logic access request **4706**, provide a provable access **4708** to the executable algorithmic logic. In certain embodiments, the controller **4502** provides provable access **4708** by recording a value on the distributed ledger **4004**, providing a user **4009** with a token demonstrating access, and/or records a transaction **4510** on the distributed ledger **4004** indicating the access. In certain embodiments, the executable algorithmic logic **4704** may include logical descriptions, computer readable and executable code in any form including source code and/or assembly language, a black box executable code or function (e.g., having an API, embedded code, and/or a code block for a specified program such as Matlab™, Simulink™, LabView™, Java™, or the like). In certain embodiments, the algorithmic logic **4704** further includes, and/or is communicated with, an interface description **4718** (e.g., providing input and output values, ranges, and/or formats; time values such as sampling rates, time constants, or the like; and/or parameter descriptions including required values, optional values, flags or enable/disable settings for features, and/or may further include or be communicated with documentation, instructions, or the like for the algorithmic logic **4704**. In certain embodiments, the interface description **4718** may be provided as an API to the algorithmic logic **4704**, and/or may be provided to support an API implementation to access the algorithmic logic.

An example controller **4502** further interprets an execution operation **4712** of the algorithmic logic, and records a transaction **4510** on the distributed ledger **4004** in response to the execution operation **4712**. In certain embodiments, interpreting an execution operation **4712** includes determin-

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ing that a user has accessed the algorithmic logic **4704** sufficiently to determine a process described in the algorithmic logic **4704**, determining that a user, the controller, or another aspect of the system **4700** has provided execution instructions to a device responsive to the algorithmic logic **4704**, and/or receiving a confirmation, data value, or other communication indicating that the algorithmic logic **4704** has been executed, downloaded, and/or copied. In certain embodiments, one or more algorithmic logic elements stored on the distributed ledger **4004** may be at least partially stored in a separate data store **4716**, where the distributed ledger **4004** may store references, partial instructions, documentation, interface descriptions **4718**, summaries, or the like, and access the separate data store **4716** of algorithmic logic elements as needed. In certain embodiments, one or more algorithmic logic **4704** elements may be stored on the distributed ledger **4004**. In certain embodiments, access to the algorithmic logic **4704** elements may be provided in accordance with one or more contract terms **4006**, and/or may be provided in response to committing a user or accessing entity to the one or more contract terms **4006**.

Referencing FIG. **48**, an example procedure **4800** includes an operation **4802** to access a distributed ledger including executable algorithmic logic, an operation **4804** to tokenize the executable algorithmic logic, and an operation **4806** to interpret an access request for the executable algorithmic logic. An example procedure **4800** further includes, in response to the access request, an operation **4808** to provide provable access to the executable algorithmic logic. In certain embodiments, the procedure **4800** includes an operation **4810** to provide an interface for the algorithmic logic (e.g., as an API, and/or any other interface description provided throughout the present disclosure). An example procedure **4800** includes an operation to interpret an execution operation of the executable algorithmic logic, and/or an operation **4812** to record a transaction on the distributed ledger in response to the execution operation.

Referencing FIG. **49**, an example transaction-enabling system **4900** includes a controller **4502**, where the controller may be configured to access a distributed ledger **4004** including a firmware data value **4904**, to tokenize the firmware data value, and to interpret an access request **4906** for the firmware data value. The example controller **4502** may be further configured to, in response to the access request **4906**, provide a provable access **4908** to a firmware corresponding to the firmware data value **4904**. In certain embodiments, the controller **4502** provides provable access **4908** by recording a value on the distributed ledger **4004**, providing a user **4009** with a token demonstrating access, and/or records a transaction **4510** on the distributed ledger **4004** indicating the access. In certain embodiments, the firmware data value **4904** may include the corresponding firmware (e.g., as stored code, an installation package, etc.), and/or may include a reference to the firmware and/or related metadata (e.g., dates, versions, size of the firmware, applicable components, etc.). In certain embodiments, one or more aspects of the firmware or the firmware data value(s) **4904** may be stored in a separate data store **4916** accessible to the controller **4502** and/or the distributed ledger **4004**.

An example controller **4502** is further configured to determine that a firmware update **4918** has occurred for a firmware data value **4904**, and to provide an update notification **4912** to an accessor of the firmware data value **4904** in response to the firmware update **4918**—for example to ensure that a current user or accessor receives (or chooses not to receive) the updated firmware data value **4904**, and/or to notify a previous user or accessor that an update of the

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firmware data value **4904** has occurred. An example controller **4502** is further configured to interpret a firmware utilization value **4920** (e.g., a download operation, installation operation, and/or execution operation of the firmware data value **4904**), and/or may further record a transaction **4510** on the distributed ledger **4004** in response to the firmware utilization value **4920**. In certain embodiments, the firmware data value **4904** may include firmware for a component of a production process, and/or firmware for a production tool. Example and non-limiting production tools include tools for a process such as: a coating process, a 3D printing process, a semiconductor fabrication process, a food preparation process, a polymer production process, a chemical synthesis process, a biological production process, and/or a crystal fabrication process. In certain embodiments, the firmware data value **4904** may include firmware for a compute resource and/or firmware for a networking resource.

Referencing FIG. **50**, an example procedure **5000** includes an operation **5002** to access a distributed ledger including a firmware data value, an operation **5004** to tokenize the firmware data value on the distributed ledger, an operation **5006** to interpret an access request for the firmware data value, and, in response to the access request, an operation **5008** to provide a provable access to the firmware corresponding to the firmware data value. The example procedure **5000** further includes an operation **5010** to determine whether an update to the firmware is available. In response to the operation **5010** determining “YES”, the procedure **5000** includes an operation **5012** to notify an accessor of the firmware update. In response to the operation **5010** determining “NO”, and/or following the notification operation **5012**, the example procedure **5000** includes an operation **5014** to record a transaction on the distributed ledger. The operation **5014** may be responsive to access, downloading, executing, updating, and/or installing of the firmware and/or a firmware asset.

Referencing FIG. **51**, an example transaction-enabling system **5100** includes a controller **4502**, which may be configured to access a distributed ledger **4004** including serverless code logic **5104**, to tokenize the serverless code logic, and to interpret an access request **5106** for the serverless code logic. The example controller **4502** is further configured to, in response to the access request **5106**, provide a provable access **5108** to the serverless code logic. In certain embodiments, the controller **4502** provides provable access **5108** by recording a value on the distributed ledger **4004**, providing a user **4009** with a token demonstrating access, and/or records a transaction **4510** on the distributed ledger **4004** indicating the access. In certain embodiments, the serverless code logic **5104** may include logical descriptions, computer readable and executable code in any form including source code and/or assembly language, and/or a black box executable code embodying the serverless code logic. In certain embodiments, the serverless code logic **5104** further includes, and/or is communicated with, an interface description **5118** (e.g., providing interface parameters; formatting; and/or parameter descriptions including required values, optional values, flags or enable/disable settings for features, and/or may further include or be communicated with documentation, instructions, or the like for the serverless code logic **5104**). In certain embodiments, the interface description **5118** may be provided as an API to the serverless code logic **5104**, and/or may be provided to support an API implementation to access the serverless code logic **5104**.

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An example controller **4502** further interprets an execution operation **5112** of the serverless code logic **5104**, and records a transaction **4510** on the distributed ledger **4004** in response to the execution operation **5112**. In certain embodiments, interpreting an execution operation **5112** of the serverless code logic includes determining that a user has accessed the serverless code logic **5104**, determining that a user, the controller, or another aspect of the transaction-enabling system **5100** has provided execution instructions to a device responsive to the serverless code logic **5104**, and/or receiving a confirmation, data value, or other communication indicating that the serverless code logic **5104** has been executed, downloaded, and/or copied. In certain embodiments, one or more serverless code logic elements stored on the distributed ledger **4004** may be at least partially stored in a separate data store **5116**, where the distributed ledger **4004** may store references, partial instructions, documentation, interface descriptions **5118**, summaries, or the like, and access the separate data store **5116** of serverless code logic elements as needed. In certain embodiments, one or more serverless code logic **5104** elements may be stored on the distributed ledger **4004**. In certain embodiments, access to the serverless code logic **5104** elements may be provided in accordance with one or more contract terms **4006**, and/or may be provided in response to committing a user or accessing entity to the one or more contract terms **4006**.

Referencing FIG. **52**, an example procedure **5200** includes an operation **5202** to access a distributed ledger, an operation **5204** to tokenize serverless code logic on the distributed ledger, and an operation **5206** to interpret an access request for the serverless code logic. The example procedure **5200** includes an operation **5208**, in response to the access request, to provide a provable access to the serverless code logic. An example procedure **5200** further includes an operation **5210** to provide an API for the serverless code logic, and/or an operation **5212** to record a transaction on the distributed ledger in response to an access operation and/or an execution operation of the serverless code logic.

Referencing FIG. **53**, an example transaction-enabling system **5300** includes a controller **4502**, where the controller **4502** is configured to access a distributed ledger **4004** including an aggregated data set **5304**, to interpret an access request **5306** for the aggregated data set, and, in response to the access request **5306**, to provide a provable access **5308** to the aggregated data set **5304**. In certain embodiments, the provable access **5308** includes which parties have accessed the aggregated data set, how many parties have accessed the aggregated data set **5304**, how many times each party has accessed the aggregated data set **5304**, and/or which portions of the aggregated data set **5304** have been accessed. The aggregated data set **5304** may be stored fully or partially on the distributed ledger **4004**, and/or may reference all or a portion of the aggregated data set **5304** on a separate data store **5316**.

In certain embodiments, the distributed ledger **4004** includes a block chain, and in certain further embodiments the aggregated data set **5304** includes a trade secret and/or proprietary information. In certain embodiments, the system **5300** includes an expert wrapper (e.g., operated by controller **4502**) for the distributed ledger **4004**, where the expert wrapper tokenizes the aggregated data set **5304** and/or validates a trade secret and/or proprietary information of the aggregated data set **5304**. In certain embodiments, the distributed ledger **4004** includes a set of instructions (e.g., as part of the aggregated data set **5304**, and/or as a separate data store on or in communication with the distributed ledger

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4004), and the controller **4502** is further configured to interpret an instruction update value, and to update the set of instructions in response to the access request **5306** and/or the instruction update value. In certain embodiments, the updated set of instructions are updated on the distributed ledger **4004**, and/or further updated by pushing the updated instruction set to a user or previous accessor of the instruction set(s). In certain embodiments, the system **5300** further includes a smart wrapper for the distributed ledger (e.g., operated by the controller **4502**), where the smart wrapper is configured to allocate a number of sub-sets of instructions to the distributed ledger **4004** as the aggregated data set **5304**, and to manage access to the number of sub-sets of instructions in response to the access request **5306**. In certain further embodiments, the controller **4502** is further configured to interpret an access **5312** (e.g., by receiving and/or responding to the access request **5306**), and to record a transaction **4510** on the distributed ledger **4004** in response to the access. In certain embodiments, the controller **4502** is configured to interpret an execution **5312** of one of the number of sub-sets of instructions (e.g., by determining executable instructions have been accessed, by providing a command to a production tool or industrial component in response to the access request **5306**, and/or via any other execution determinations described throughout the present disclosure). In certain further embodiments, the controller **4502** is further configured to record a transaction **4510** on the distributed ledger **4004** in response to the execution **5312** of the one of the number of sub-sets of instructions.

Referencing FIG. **54**, an example procedure **5400** includes an operation **5402** to access a distributed ledger including an aggregated data set, an operation **5404** to tokenize the aggregated data set to validate information (e.g., trade secret or proprietary information) of the aggregated data set, and an operation **5406** to interpret an access request for the aggregated data set. The example procedure **5400** further includes an operation **5408** to provide provable access to the aggregated data set. In certain embodiments, operation **5408** to provide provable access to the aggregated data set includes determining which parties have accessed the aggregated data set, how many parties have accessed the aggregated data set, how many times specific parties have accessed the aggregated data set, and/or combinations of these with portions of the aggregated data set. In certain embodiments, the aggregated data set may include a number of sub-sets of instructions, and the procedure **5400** may further include interpreting an update to one of the sub-sets of instructions, updating the aggregated data set in response to the update(s), and/or pushing an update to a user and/or a previous accessor of the instructions. In certain embodiments, the procedure **5400** includes operating an expert wrapper to tokenize the aggregated data set and to validate the data of the aggregated data set. In certain embodiments, the procedure **5400** includes operating a smart wrapper to manage access to the aggregated data set. In certain embodiments, the procedure **5400** includes an operation **5410** to record a transaction on the distributed ledger in response to an access of the aggregated data set, an execution operation of at least a portion of the aggregated data set, and/or an update operation related to the aggregated data set.

Referencing FIG. **55**, an example transaction-enabling system **5500** includes a controller **4502**, where the controller **4502** is configured to access a distributed ledger **4004** including intellectual property (IP) data **5504** corresponding to a number of IP assets **5516**, where the number of IP assets include an aggregate stack of IP data **5504**. The controller **4502** is further configured to tokenize the IP data, to interpret

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a distributed ledger operation **5506** corresponding to at least one of the number of IP assets **5516**, and to determine an analytic result value **5512** in response to the distributed ledger operation **5506** and the tokenized IP data, and provide a report of the analytic result value **5512**. In certain embodiments, the IP assets **5516** are the aggregate stack of IP, and in certain embodiments, the IP data **5504** include a list defining the aggregate stack of IP. In certain embodiments, one or more IP assets **5516** may be stored within the IP data **5504**, and/or may be referenced within the IP data **5504** and stored separately (within the distributed ledger **4004** or on an IP asset **5516** data store in communication with the distributed ledger **4004** and/or a wrapper for the distributed ledger **4004**).

Example and non-limiting distributed ledger operations **5506** include operations such as: accessing IP data **5504**; executing a process utilizing IP data **5504**; adding IP data **5504** corresponding to an additional IP asset to the aggregate stack of IP; and/or removing IP data **5504** corresponding from the aggregate stack of IP. In certain further embodiments, distributed ledger operations **5506** include one or more operations such as: changing an owner of an IP asset **5516**; installing or executing firmware or executable logic corresponding to IP data **5504**; and/or discontinuing access to the IP data.

Example and non-limiting analytic result values **5512** include result values such as: a number of access events corresponding to at least one of the plurality of IP assets **5516**; statistical information corresponding to access events for one or more IP assets **5516**; a distribution, frequency, or other description of access events for the IP assets **5516**; a distribution, frequency, or other description of installation or execution events for the IP assets; access times and/or processing times corresponding to one or more of the IP assets; and/or unique entity access, execution, or installation events for one or more of the IP assets. In certain embodiments, analytic result values **5512** include summaries, statistical analyses (e.g., averages, groupings, determination of outliers, etc.), ordering (e.g., high-to-low volume access rates, revenue values, etc.), timing (e.g., time since a most recent access, installation, execution, or update), bucketing descriptions (e.g., monthly, weekly, by volume or revenue category, etc.) of any of the foregoing, and/or trending descriptions of any of the foregoing.

Referencing FIG. **56**, an example procedure **5600** includes an operation **5602** to access a distributed ledger including a number of IP data corresponding to a number of IP assets, wherein the number of IP assets include an aggregate stack of IP, an operation **5604** to tokenize the IP data, and an operation **5606** to interpret distributed ledger operation(s) corresponding to at least one of the plurality of IP assets. The example procedure **5600** further includes an operation **5608** to determine an analytic result value in response to the distributed ledger operation(s) and the tokenized IP data, and an operation **5610** to provide a report of the analytic result value. In certain embodiments, the operation **5610** includes providing the report to a user, an entity owning at least one of the IP assets, an entity considering a purchase of access to at least one of the IP assets, an operator or administrator for a controller **4502** (and/or a smart wrapper for the distributed ledger) and/or the distributed ledger, and/or a support professional related to the controller **4502**, the distributed ledger, or the IP assets (e.g., an accounting professional, a tax professional, an IT support professional, a server administrator, an IP portfolio manager, a regulatory officer, etc.). The information within the report, the type of report available, and/or the underlying set of

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distributed ledger operations and/or related IP assets utilized to inform the report may be restricted and/or may default to certain parameters depending upon the target entity for the report and/or the entity requesting the report. In certain embodiments, the procedure **5600** may further include recording a transaction on the distributed ledger in response to the operation **5610** to provide the report—for example to record the instance of the report occurring and/or the related parameters for the report, to provide a provable record that the report was provided, and/or to provide for a transaction and/or payment for the report (e.g., providing additional information to a customer, potential customer, and/or asset owner).

Referencing FIG. **57**, an example transaction-enabling system **5700** includes a controller **5702** configured to interpret a resource utilization requirement **5704** for a task system **5706** having at least one task, such as a compute task, a network task, or a core task. The example controller **5702** is further configured to interpret a number of external data sources **5708**, where the external data sources **5708** include at least one data source outside of the task system **5706**. The example controller **5702** operates an expert system **5710** that predicts a forward market price **5712** in response to the resource utilization requirement **5704** and the external data sources **5708**, and that executes a transaction **5714** on a resource market **5716** in response to the predicted forward market price **5712**. Example and non-limiting external data sources **5708** include one or more sources such as IoT data sources and social media data sources. Example and non-limiting operations to predict the forward market price include determining indicators on social media that relate to forward market prices (e.g., conversation types, keyword occurrences, sharing topic indicators, and/or event indicators such as cultural, sporting, or internet events), determining external data indicators that relate to forward market prices (e.g., data from industrial systems that participate in a market that may affect resource prices, news events that may affect resource prices, published or proprietary data such as orders, economic indicators, demographic indicators, or the like that may affect resource prices). All examples for external data described herein are non-limiting examples to illustrate certain principles of the present disclosure.

In certain embodiments, the resource requirement(s) **5704** relate to a compute resource, a network bandwidth resource, a spectrum resource, a data storage resource, an energy resource, and/or an energy credit resource. In certain embodiments, the resource market **5716** may be a forward market and/or a spot market for a resource. In certain embodiments, the transactions **5714** may be on a forward market and/or a spot market. In certain embodiments, the transactions **5714** may include a purchase or a sale of the resource, and may further include combinations (e.g., a purchase on a spot market and a sale on a forward market).

An example system **5700** includes the resource utilization requirement **5704** including a requirement for a first resource, and where the forward market price **5712** is a forward price prediction for the first resource, for a second resource, and/or for both resources. For example, the second resource may be a resource that can be substituted for the first resource. The substitution may be direct, such as where multiple types of fuel available to power a system, or where multiple task delivery components are available that consume distinct types of resources. Additionally or alternatively, the substitution may be indirect, such as where operational changes can trade out one type of resource utilization for another—e.g., trading compute resources for

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network resources (e.g., outsourcing compute tasks, effectively making them network tasks for the task system), changing a process to reduce a type of resource (e.g., operating a less efficient algorithm that results in increased data storage but reduced computing; operating a process at a lower rate or temperature, reducing energy usage for the direct operation, but increasing energy usage due to the increased time of operating a facility performing the process, and further where the energy usage for the direct operation and for operating the facility may be distinct types of energy). The examples of the first resource and second resource are non-limiting and provided for illustration. An example system 5700 further includes the controller 5702 configured to operate the expert system 5710 to determine a substitution cost 5718 of the second resource, and to execute the transaction 5714 in response to the substitution cost 5718—which may include purchasing or selling the first resource and/or the second resource, or both, and/or varying the transactions 5714 over time. In certain embodiments, the substitution cost 5718 is determined from the forward market price(s) 5712 of the first and second resource, the spot market price(s) of the first and second resource, and/or from other system costs or effects that may result from utilization changes between the first and second resource, such as operational change costs to the task system 5706 (e.g., time to complete tasks; facility changes such as personnel and/or equipment changes due to operating with either the first or second resources; and/or secondary effects such as consumption of energy credits, exceedance of a capacity of a component of the task system, and/or changes to a quality of products or services provided by the task system 5706).

Referencing FIG. 58, an example procedure 5800 includes an operation 5802 to interpret a resource utilization requirement for a task system having at least one of a compute task, a network task, or a core task; an operation 5804 to interpret a number of external data sources, where the number of external data sources include at least one data source outside of the task system. The example procedure 5800 further includes an operation 5806 to operate an expert system (and/or AI or machine learning system) to predict a forward market price for a resource in response to the resource utilization requirement and the number of external data sources; and an operation 5808 to execute a transaction on a resource market in response to the predicted forward market price.

Referencing FIG. 59, an example procedure 5900 includes an operation 5902 to determine a second resource that can substitute for the first resource, an operation 5904 to consider the forward market price of one or more resources, and/or to further consider the operational cost change between the first and second resources, and an operation 5906 to execute a transaction on a resource market of the first resource or the second resource in response to the substitution cost and/or operational cost change between the first and second resources.

With further reference to FIG. 57, the example system 5700 includes the controller 5702 configured to interpret a resource utilization requirement 5704 for a task system 5706, and to interpret a behavioral data source 5720; to operate a machine (e.g., an expert system 5710, and/or an AI or machine learning component) to forecast a forward market price 5712 for a resource in response to the resource utilization requirement 5704 and the behavioral data source 5720, and to perform one of adjusting an operation of the task system (e.g., providing operational adjustments 5722) or executing a transaction 5714 in response to the forecast of the forward market price 5712 for the resource. Example and

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non-limiting behavioral data sources 5720 include data sources such as: an automated agent behavioral data source, a human behavioral data source, and/or a business entity behavioral data source. In certain embodiments, the controller 5702 is further configured to operation the machine to provide operational adjustments 5722 such as: adjusting operations of the task system 5706 to increase or reduce the resource utilization requirement 5704; adjusting operations of the task system 5706 to time shift at least a portion of the resource utilization requirement 5704; adjusting operations of the task system 5706 to substitute utilization of a first resource for utilization of a second resource; and/or accessing an external provider to provide at least a portion of at least one of the compute task, the network task, or the core task.

Referencing FIG. 60, an example procedure 6000 includes an operation 6002 to interpret a resource utilization requirement for a task system, an operation 6004 to interpret a behavioral data source, an operation 6006 to operate a machine to forecast a forward market value for a resource in response to the resource utilization requirement and the behavioral data source, and an operation 6008 to adjust an operation of the task system and/or to execute a transaction in response to the forecast of the forward market value for the resource. In certain embodiments, procedure 6000 further includes operations such as those described in reference to FIG. 59 to further consider substitution costs for a first and second resource, and to perform operations 6008 further in response to the substitution costs for the first and second resource and adjust an operation of the task system and/or execute a transaction in response to the forward market price.

With further reference to FIG. 57, an example system 5700 includes the controller 5702 configured to interpret a resource utilization requirement 5704 for a task system 5706, to interpret a number of external data sources 5708, to operate an expert system 5710 to predict a forward market price 5712 for a resource in response to the resource utilization requirement 5704 and the number of external data sources, and to execute a cryptocurrency transaction (e.g., transaction 5714) on a resource market 5716 in response to the predicted forward market price.

With reference to FIG. 61, an example procedure 6100 includes an operation 6102 to interpret a resource utilization requirement for a task system, an operation 6104 to interpret a number of external data sources, an operation 6106 to operate an expert system to predict a forward market price for a resource in response to the resource utilization requirement and the number of external data sources, and an operation 6108 to execute a cryptocurrency transaction on a resource market in response to the predicted forward market price. Operations of the procedure 6100, and further for operations of any systems or procedures throughout the present disclosure that determine a forward market price for a resource may additionally or alternatively include determining a forward market price for a number of forward market time frames, including a predicted price for a spot market at one or more future time values. Operations of the procedure 6100, and further for operations of any system or procedures throughout the present disclosure that perform one or more transactions or operational adjustments in response to a forward market price include operations provide for an improved cost of operation of a task system, and/or to provide for an increased capability (e.g., quality, volume, and/or completion time) of a task system. In certain embodiments, procedure 6100 further includes operations such as those described in reference to FIG. 59 to further

consider substitution costs for a first and second resource, and to perform operations **6108** further in response to the substitution costs for the first and second resource.

Referencing FIG. **62**, an example transaction-enabling system **6200** includes a controller **6202** having a transaction detection circuit **6204** structured to interpret a transaction request value **6206**, wherein the transaction request value **6206** includes a transaction description for one of a proposed or an imminent transaction (e.g., as provided by a user, subscriber to the transaction-enabling system **6200**, application or service utilizing the transaction-enabling system **6200**, etc.), and wherein the transaction description includes a cryptocurrency type value (e.g., one or more cryptocurrency types for the transaction) and a transaction amount value. The controller **6202** further includes a transaction locator circuit **6208** that determines a transaction location parameter **6210** in response to the transaction request value **6206**, where the transaction location parameter **6210** includes at least one of a transaction geographic value (e.g., a physical location where the transaction will be executed, which may be adjusted, for example using server selections, third party transactions, or other methods) and/or a transaction jurisdiction value (e.g., a jurisdictional entity where the laws governing the transaction will be applicable, such as a country, state, province, regional jurisdiction, and/or a governing scheme that is inter-national or trans-national; which may be adjusted, for example using server selections, third party transactions, control of the transaction type or category invoking one of a number of jurisdictions applicable to a particular geographical region, etc.), and a transaction execution circuit **6212** structured to provide a transaction implementation command **6214** in response to the transaction location parameter **6210**.

In certain embodiments, the transaction locator circuit **6208** is further structured to determine the transaction location parameter **6210** based on a tax treatment **6216** of the one of the proposed or imminent transaction, for example selecting a geographical and/or jurisdictional location from a number of available locations that will receive a favorable tax treatment for the transaction. Certain considerations to determine a favorable tax treatment include tax laws and incentives in available jurisdictions according to the type of transaction, specific characteristics of the transacting party including where income, sales, and other aspects related to the transaction will be deemed to occur and the resulting effect on the tax treatment of the transaction, and/or the availability of losses or other offsets to income from a transaction that are applicable to entities involved in the transaction according to the available jurisdictions and/or geographic regions available.

An example transaction-enabling system **6200** further includes the transaction locator circuit **6208** further structured to select the one of the transaction geographic value or the transaction jurisdiction value from a number of available geographic values or jurisdiction values that provides an improved tax treatment **6216** relative to a nominal one of the number of available geographic values or jurisdiction values. For example, an improvement to an execution of a transaction may include determining a nominal tax treatment **6216** (e.g., for a simplest jurisdiction such as a location of a buyer, seller, or delivery location of a purchased item or service), and determining that an alternate available geography or jurisdiction is available. Improvement of a tax treatment should be understood broadly, and can include at least one or more of: reducing a taxable amount on a purchase; achieving a target loss value for a transaction to offset a profit value in a particular geography or jurisdiction

for the transaction; achieving a tax paid target for a jurisdiction, such as to meet a minimum tax threshold as measured separately for an entity from the transaction, and/or to pay taxes to meet a public perception goal or tax payment target; and/or paying taxes in a first category (e.g., sales tax) relative to a second category (e.g., a short term capital gain). Accordingly, in certain embodiments, operations of the transaction locator circuit **6208** may operate to continuously improve or optimize tax treatment of transactions, but may additionally or alternatively find an improved tax treatment location without continuing to optimize a tax treatment for a particular transaction—for example to improve the speed of execution of transactions. In certain embodiments, operations of the transaction locator circuit **6208** may operate to improve a tax treatment of a transaction until a threshold tax treatment value is reached—for example an improvement amount from a nominal transaction location, and/or a target tax treatment amount for the transaction.

An example transaction-enabling system **6200** further includes the transaction locator circuit **6208** further structured to determining the transaction location parameter **6210** in response to a tax treatment **6216** of at least one of the cryptocurrency type value (e.g., where the type of cryptocurrency may affect the tax treatment of the transaction—e.g., according to a country of origin of the currency, a favored currency such as one issued by a government or municipality, a sanctioned currency that may receive unfavorable tax treatment, etc.), or a type of the one of the proposed or imminent transaction (e.g., where the transaction type may lead to variable tax treatment depending upon the jurisdiction, such as a purchase of a clean energy vehicle in a country having an incentive for such a purchase, etc.). A proposed or imminent transaction, as set forth herein, can include a requested transaction (e.g., the requesting entity is attempting to make a transaction) and/or a speculative transaction (e.g., a requesting entity is trying to determine what transaction outcomes are available for the transaction location, tax consequences, and/or total cost of the transaction).

An example transaction locator circuit **6208** operates an expert system **6220** (and/or an AI or machine learning component) configured to use machine learning to continuously improve the determination of the transaction location parameter **6210** relative to a tax treatment **6216** of transactions processed by the controller. Operations to continuously improve the determination of the transaction location parameter **6210** may be performed over a number of transactions, including transactions relating to the same entity or relating to a number of entities, where a given transaction utilizes a transaction location parameter **6210** that is made according to the state of the expert system **6220** at the time of the transaction request value **6206**, and/or wherein the expert system **6220** improves an outcome for the particular transaction such as an improved tax treatment **6216** relative to a nominal, a tax treatment **6216** that is greater than a threshold tax treatment value, and/or a transaction location parameter **6210** that is determined after an optimization period expires (e.g., a time value threshold, which may be fixed or variable) and/or after the expert system **6220** meets convergence criteria (e.g., further improvements appear to be lower than a threshold amount, etc.) while the transaction request value **6206** is pending. An example expert system **6220** is configured to aggregate regulatory information **6222** for cryptocurrency transactions from a number of jurisdictions, and to continuously improve the determination of the transaction location parameter **6210** based on the aggregated regulatory information **6222**. The aggregated regulatory information

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6222 may be updated, refreshed, and/or added to over time. An example expert system 6220 utilizes machine learning to continuously improve the determination of the transaction location parameter 6210 relative to secondary jurisdiction costs related to the cryptocurrency transaction. For example, various regulatory schemes may affect compliance, reporting requirements, privacy considerations (e.g., for data), penalty schemes for particular transaction types, export control or sanctions-related transaction considerations, and/or the desirability or undesirability to execute transactions that occur across jurisdictional boundaries (e.g., invoking customs, treaties, international legal considerations, and/or intra-country considerations such as state or provisional law versus federal or national law). An example expert system 6220 utilizes machine learning to continuously improve the transaction speed for cryptocurrency transactions. An example expert system 6220 utilizes machine learning to continuously improve a favorability of contractual terms related to the cryptocurrency transaction (e.g., meeting purchasing targets in a region, meeting transaction type targets in a region, meeting transaction cost obligations, etc.). An example expert system 6220 utilizes machine learning to continuously improve a compliance of cryptocurrency transactions within the aggregated regulatory information 6222. An example transaction-enabling system 6200 further includes a transaction engine 6218 that is responsive to the transaction implementation command 6214, for example to execute the cryptocurrency transaction in accordance with the transaction location parameter 6210 and/or according to a cryptocurrency type provided as a part of the transaction implementation command 6214.

Referencing FIG. 63, an example procedure 6300 includes an operation 6302 to interpret a cryptocurrency transaction request value, where the transaction request value includes a transaction description for one of a proposed or an imminent transaction, and where the transaction description includes a cryptocurrency type value and a transaction amount value. The example procedure 6300 includes an operation 6304 to determine a transaction location parameter. The example procedure 6300 further includes an operation 6308 to provide a transaction location parameter in response to the transaction request value, where the transaction location parameter includes at least one of a transaction geographic value or a transaction jurisdiction value. In certain embodiments, the example procedure 6300 further includes an operation to provide a transaction implementation command in response to the transaction location parameter. In certain embodiments, an example procedure 6300 includes an operation 6306 to determine a tax treatment and/or a regulatory treatment of the transaction, and to provide 6308 the transaction location parameter in response to the tax and/or regulatory treatment of the transaction.

Referencing FIG. 64, an example procedure 6400 includes an operation 6402 to interpret a cryptocurrency transaction request value, where the transaction request value includes a transaction description for one of a proposed or an imminent transaction, and where the transaction description includes a cryptocurrency type value and a transaction amount value. The example procedure 6400 includes an operation 6404 to determine a transaction location parameter, where the transaction location parameter includes at least one of a transaction geographic value or a transaction jurisdiction value, and an operation 6408 to execute a transaction in response to the transaction location parameter. In certain embodiments, the procedure 6400 further includes an operation 6406 to determine a tax and/or regulatory treatment of the transaction, and to adjust the

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transaction location parameter in response to the tax and/or regulatory treatment of the transaction before the operation 6408 to execute the transaction.

Referencing FIG. 65, an example system 6500 includes a controller 6202 having a smart wrapper 6502 that interprets a transaction request value 6206 from a user, wherein the transaction request value includes a transaction description for an incoming transaction, and wherein the transaction description includes a transaction amount value and at least one of a cryptocurrency type value and a transaction location value. The controller 6202 includes a transaction locator circuit 6208 that determines a transaction location parameter 6210 in response to the transaction request value 6206 and further in response to a plurality of tax treatment 6216 values corresponding to a plurality of transaction location parameters 6210, where the transaction location parameter 6210 includes at least one of a transaction geographic value or a transaction jurisdiction value, and wherein the smart wrapper 6502 further directs an execution of the incoming transaction in response to the transaction location parameter 6210—for example by providing a transaction implementation command 6214 to a transaction engine 6218. In certain embodiments, the smart wrapper 6502 can provide for enhanced speed of transaction execution—for example by storing transaction location parameter 6210 values according to transaction types, amounts, and/or requesting entity characteristics, and thereby provide transaction implementation commands 6214 more quickly than the expert system 6220 provides updated transaction location parameters 6210, while still providing the benefit of the expert system 6220 operations over time as transaction location parameter 6210 values are updated. Additionally or alternatively, operations of the smart wrapper 6502 provide for a consistent and/or selected user interface to the user for accepting transaction request values 6206, and/or to the transaction engine 6218 for sending transaction implementation commands 6214, and allowing for changes in the controller 6202 to remain invisible to the user and/or transaction engine 6218.

Referencing FIG. 66, an example procedure 6600 includes an operation 6602 to interpret a cryptocurrency transaction request value from a user, wherein the transaction request value includes a transaction description for an incoming transaction, and wherein the transaction description includes a transaction amount value and at least one of a cryptocurrency type value and a transaction location value, an operation 6604 to determine a transaction location parameter, an operation 6606 to determine a number of tax treatment values corresponding to a number of available transaction locations, and an operation 6608 to select an available transaction location as the transaction location parameter in response to the number of tax treatment values. The procedure 6600 further includes an operation 6610 to execute a cryptocurrency transaction in response to the selected available transaction location parameter. In certain embodiments, operation 6608 includes selecting an available transaction location as a location having a favorable and/or improved tax treatment, for example relative to a nominal or default transaction location. Example and non-limiting transaction location values for operation 6608 include: a location of a purchaser of the transaction; a location of a seller of the transaction; a location of a delivery of a product or service of the transaction; a location of a supplier of a product or service of the transaction; a residence location of one of the purchaser, seller, or supplier of the transaction; and/or a legally available location for the transaction.

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Referencing FIG. 67, an example transaction-enabling system 6700 includes a controller 6202 having a transaction detection circuit 6204 structured to interpret a plurality of transaction request values 6206, wherein each transaction request value 6206 includes a transaction description for one of a proposed or an imminent transaction, and wherein the transaction description includes a cryptocurrency type value and a transaction amount value. The example controller 6202 further includes a transaction support circuit 6708 structured to interpret a support resource description 6710 including at least one supporting resource for the plurality of transactions, a support utilization circuit 6712 structured to operate an expert system 6220, wherein the expert system 6220 is configured to use machine learning to continuously improve at least one execution parameter 6714 for the plurality of transactions relative to the support resource description 6710, and a transaction execution circuit 6212 structured to command execution of the plurality of transactions (e.g., as transaction implementation commands 6214) in response to the improved at least one execution parameter. In the example of FIG. 67, a support resource is depicted as the transaction engine 6218, although a support resource may be any resource utilized to support the transaction, including at least a mining server, a transaction verification server, an accounting circuit or tax calculation circuit associated with an entity that is providing the transaction request value 6206, and the like. In certain embodiments, the support resource description 6710 includes an energy price, and/or may further include an energy resource consumption to support the transaction or a group of transactions. An example energy price description includes a forward price prediction, a spot price, and/or a forward market price for the energy source. An example energy price description includes one or more energy source selections available to power the execution of the transaction or a group of transactions. An example support resource description 6710 includes a state of charge for an energy source available to power the execution of the transaction or a group of transactions, and/or a charge cost cycle description for an energy storage source available to power the execution of the transaction or a group of transactions. An example system 6700 includes an energy storage source having a battery as a supporting device, and where the expert system 6220 is further configured to use machine learning to improve one or more of: a battery energy transfer efficiency value, a battery life value, and a battery lifetime utilization cost value.

Referencing FIG. 68, an example procedure 6800 includes an operation 6802 to interpret a number of transaction request values, where each transaction request value includes a transaction description for one of a proposed or an imminent transaction, and wherein the transaction description includes a cryptocurrency type value and a transaction amount value. The example procedure 6800 further includes an operation 6804 to interpret a support resource description including at least one supporting resource for the number of transactions, and an operation 6806 to operate an expert system, where the expert system is configured to use machine learning to continuously improve at least one execution parameter for the number of transactions relative to the support resource description. The example procedure 6800 further includes an operation 6808 to command execution of the number of transactions in response to the improved execution parameter(s).

Referencing FIG. 69, an example transaction-enabling system 6900 includes a controller 6902 having: an attention market access circuit 6904 structured to interpret a number

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of attention-related resources 6906 available on an attention related resource market 6907, an intelligent agent circuit 6908 that determines an attention-related resource acquisition value 6910 based on a cost parameter 6916 of at least one of the number of attention-related resources 6906, and an attention acquisition circuit 6912 structured to solicit an attention-related resource 6906 (e.g., by providing an attention-related resource solicitation 6914 to the attention-related resource market 6907) in response to the attention-related resource acquisition value 6910. In certain embodiments, an attention-related resource 6906 may be available for purchase or sale on the attention-related resource market 6907, and/or in certain embodiments an attention-related resource 6906 may be provided by and/or accessible to the controller 6902 for sale on the attention-related resource market 6907.

An example system 6900 includes where the attention acquisition circuit 6912 is further structured to solicit the attention-related resource 6906 by performing an operation such as: purchasing the attention-related resource 6906 from the attention related resource market 6907; selling the attention-related resource 6906 to the attention related resource market 6907; making an offer to sell the attention-related resource 6906 to a second intelligent agent (not shown); and/or making an offer to purchase the attention-related resource from the second intelligent agent. An example system 6900 includes where the number of attention-related resources 6906 include resources such as: an advertising placement; a search listing; a keyword listing; a banner advertisement; a video advertisement; an embedded video advertisement; a panel activity participation; a survey activity participation; a trial activity participation; and/or a pilot activity placement or participation.

An example system 6900 includes one or more of: where the attention related resource market 6907 includes a spot market for at least one of the number of attention-related resources 6906; where the cost parameter 6916 of at least one of the number of attention-related resources 6906 includes a future predicted cost of the at least one of the number of attention-related resources, and where the intelligent agent circuit 6908 is further structured to determine the attention-related resource acquisition value 6910 in response to a comparison of a first cost on the spot market with the cost parameter 6916; where the attention related resource market 6907 includes a forward market for at least one of the number of attention-related resources 6906, and where the cost parameter 6916 of the at least one of the number of attention-related resources 6906 includes a predicted future cost; and/or where the cost parameter 6916 of at least one of the number of attention-related resources 6906 includes a future predicted cost of the at least one of the number of attention-related resources 6906, and where the intelligent agent circuit 6908 is further structured to determine the attention-related resource acquisition value 6910 in response to a comparison of a first cost on the forward market with the cost parameter 6916. An example system includes the intelligent agent circuit 6908 further structured to determine the attention-related resource acquisition value 6910 in response to the cost parameter 6916 of the at least one of the number of attention-related resources having a value that is outside of an expected cost range for the at least one of the number of attention-related resources 6906. An example system includes the intelligent agent circuit 6908 is further structured to determine the attention-related resource acquisition value 6910 in response to a function of: the cost parameter 6916 of the at least one of the number of attention-related resources, and/or an effective-

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ness parameter **6918** of the at least one of the number of attention-related resources **6906**. In certain further embodiments, an example controller **6902** further includes an external data circuit **6920** structured to interpret a social media data source **6922**, and where the intelligent agent circuit **6908** is further structured to determine, in response to the social media data source **6922**, at least one of a future predicted cost of the at least one of the number of attention-related resources **6906**, and to utilize the future predicted cost as the cost parameter **6916** and/or the effectiveness parameter **6918** of the at least one of the number of attention-related resources **6906**.

Referencing FIG. **71**, an example system **7100** includes a fleet of machines, where each machine includes a task system having a core task and further at least one of a compute task or a network task (e.g., task machines **7102**). The system **7100** includes a controller **6902** having: an attention market access circuit **6904** structured to interpret a number of attention-related resources **6906** available on an attention related resource market **6907**; an intelligent agent circuit **6908** structured to determine a number of attention-related resource acquisition value **6910** based on a cost parameter **6916** of at least one of the number of attention-related resources **6906**, and further based on the core task for the corresponding machine of the fleet of machines **7102**. The example system **7100** further includes an attention purchase aggregating circuit **7104** structured to determine an aggregate attention-related resource purchase value **7106** in response to the number of attention-related resource acquisition values **6910** from the intelligent agent circuit corresponding to each machine of the fleet of the machines **7102**; and an attention acquisition circuit **7108** structured to purchase an attention-related resource **6906** in response to the aggregate attention-related resource purchase value **7106**.

An example system **7100** includes where the attention purchase aggregating circuit **7104** is positioned at a location selected from the locations consisting of: at least partially distributed on a number of the controllers corresponding to machines of the fleet of machines **7102**; on a selected controller corresponding to one of the machines of the fleet of machines **7102**; and on a system controller **6902** communicatively coupled to the number of the controllers corresponding to machines of the fleet of machines **7102** (e.g., consistent with the depiction of FIG. **71**). An example system **7100** includes where the attention acquisition circuit **7108** is positioned at a location selected from the locations consisting of: at least partially distributed on a number of the controllers corresponding to machines of the fleet of machines **7102**; on a selected controller corresponding to one of the machines of the fleet of machines **7102**; and on a system controller **6902** communicatively coupled to the number of the controllers corresponding to machines of the fleet of machines **7102** (e.g., consistent with the depiction of FIG. **71**).

Referencing FIG. **70**, an example procedure **7000** includes an operation **7002** to interpret a number of attention-related resources available on an attention market, an operation **7006** to determine an attention-related resource acquisition value based on a cost parameter of at least one of the number of attention-related resources, and an operation **7008** to solicit an attention-related resource in response to the attention-related resource acquisition value. In certain embodiments, the example procedure **7000** further includes an operation **7004** to determine a cost parameter and/or an effectiveness parameter for one or more of the attention-related resources, and to determine the attention-related

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resource acquisition value further in response to the cost parameter and/or the effectiveness parameter.

An example procedure **7000** further includes the operation **7008** to perform the soliciting the attention-related resource by performing an operation such as: purchasing the attention-related resource from the attention market; selling the attention-related resource to the attention market; making an offer to sell the attention-related resource to a second intelligent agent; and/or making an offer to purchase the attention-related resource to the second intelligent agent. An example procedure **7000** further includes where the cost parameter of at least one of the number of attention-related resources includes a future predicted cost of the at least one of the number of attention-related resources, where the procedure **7000** further includes determining the attention-related resource acquisition value in response to a comparison of a first cost on a spot market with the cost parameter. An example procedure **7000** further includes an operation to interpret a social media data source and an operation to determine, in response to the social media data source: a future predicted cost of the at least one of the number of attention-related resources, and to utilize the future predicted cost as the cost parameter; and/or an effectiveness parameter of the at least one of the number of attention-related resources. An example procedure **7000** further includes where the operation **7006** to determine the attention-related resource acquisition value is further based on the at least one of the future predicted cost or the effectiveness parameter determined in response to the social media data source.

Referencing FIG. **72**, an example procedure **7200** includes an operation **7202** to interpret a number of attention-related resources available on an attention market, an operation **7204** to determine an attention-related resource acquisition value for each machine of a fleet of machines based on a cost parameter of at least one of the number of attention-related resources, and further based on a core task for each of a corresponding machine of the fleet of machines. The example procedure **7200** further includes an operation **7208** to determine an aggregate attention-related resource purchase value in response to the number of attention-related resource acquisition values corresponding to each machine of the fleet of the machines, and an operation **7210** to solicit an attention-related resource in response to the aggregate attention-related resource purchase value.

Certain further aspects of an example procedure **7200** are described following, any one or more of which may be present in certain embodiments. An example procedure **7200** includes where the cost parameter of at least one of the number of attention-related resources includes a future predicted cost of the at least one of the number of attention-related resources, and where the procedure further includes an operation to determine each attention-related resource acquisition value in response to a comparison of a first cost on a spot market for attention-related resources with the cost parameter. An example procedure **7200** further includes an operation **7206** to interpret a social media data source to determine, in response to the social media data source, an adjustment for the cost parameter and/or the effectiveness parameter, and to utilize the adjusted cost parameter and/or effectiveness parameter to determine the aggregate attention-related resource purchase value.

Referencing FIG. **73**, an example transaction-enabling system **7300** includes a production facility **7302** including a core task, where the core task includes a production task. The system **7300** further includes a controller **7304** includ-

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ing a facility description circuit **7306** structured to interpret a plurality of historical facility parameter values **7308** and a corresponding plurality of historical facility outcome values **7310**, and a facility prediction circuit **7312** structured to operate an adaptive learning system **7314**, wherein the adaptive learning system **7314** is configured to train a facility production predictor **7316** in response to the plurality of historical facility parameter values **7308** and the corresponding plurality of historical facility outcome values **7310**. The example facility description circuit **7306** is further structured to interpret a plurality of present state facility parameter values **7318**; and wherein the facility prediction circuit **7312** is further structured to operate the adaptive learning system **7314** to predict a present state facility outcome value **7320** in response to the number of present state facility parameter values **7318**. An example and non-limiting facility production predictor **7316** includes a facility model providing for a prediction of a facility outcome value (e.g., a timing, volume, and/or quality parameter) in response to parameters of the facility. In certain embodiments, the adaptive learning system **7314** utilizes machine learning, an expert system, and/or AI operations to determine input values that are related to the outcome values, to add or remove input values that are found to be more consistently or less consistently predictive of facility outcomes, and/or to add relationships (e.g., equations, functions, cut-off values, thresholds, etc.) between the input values and the outcome values. The adaptive learning system **7314** allows for improvement of the predicted outcome values over time, and further allows for the connection of parameters as predictive of the outcome values that may not ordinarily be included within a standard model that might be built—for example in a particular facility, the date of delivery of materials may be relevant to predicting outcomes for the production facility **7302**, where a standard model may focus solely on production parameters due to constraints in the thinking and experience of personnel that build and tune the model. Additionally, standard models are expensive and difficult to adapt to changing conditions at the production facility **7302**, such as aging of equipment, changing of personnel, and/or changing of production processes, volumes, and products.

In certain embodiments, the present state facility outcome value **7320** includes a facility production outcome, such as a volume, time of completion, and/or a quality parameter. Any other aspect of the production facility **7302** may additionally or alternatively be a present state facility outcome value **7320**, including without limitation values such as downtime predictions, equipment or process fault or failure predictions, overtime predictions, waste material generation predictions, and the like. In certain embodiments, the present state facility outcome value **7320** includes a facility production outcome probability distribution, such as a confidence interval, high and low range targets, and/or a mean and standard deviation (or other statistical description, such as a curve, or a non-normal distribution) for an outcome value. In certain embodiments, the present state facility outcome value **7320** may further include information such as: which present state facility parameter values **7318** have the greatest effect on the predicted outcome values **7320**, which present state facility parameter values **7318** have driven recent change in the predicted present state facility outcome values **7320**, and/or which input sources drive the greatest uncertainty within a range of present state facility outcome values **7320**. An example present state facility outcome value **7320** includes at least one value such as: a production volume description of the production task;

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a production quality description of the production task; a facility resource utilization description; an input resource utilization description; and a production timing description of the production task. Example and non-limiting production volume descriptions include a total production volume, a production volume relative to a target value, and/or a specific production volume such as the volume produced per unit of resource input, personnel time utilized, production tooling life utilization, or the like. Example and non-limiting quality descriptions include: a description of products having acceptable quality (e.g., fit for purpose); a description of a scrap or waste rate in products; a distribution of a product parameter such as a test value, tolerance, and/or measurement; and/or a description of a qualitative or categorical distribution of the products. Example and non-limiting facility resource utilization descriptions include: a description of energy consumption for the production facility; a description of personnel utilization of the facility; a description of consumption of a secondary resource for the facility (e.g., recycling, waste production, parking utilization, and/or consumption or production of energy credits); a description of production tooling or other facility asset consumption; trends in any of the foregoing; and/or changes in any of the foregoing. Example and non-limiting input resource utilization descriptions include: a description of a raw resource or input product utilization; a description of capital investment for the production facility; and/or a description of operating costs for the production facility. The examples herein are not limiting to any other aspect of the present disclosure, and are provided only for illustration.

An example facility description circuit **7306** further interprets historical external data from at least one external data source **7322** (e.g., a prior existing production facility and/or an offset production facility), where the adaptive learning system **7314** is further configured to train the facility production predictor **7316** in response to the historical external data. Example and non-limiting external data sources include: a social media data source; a behavioral data source; a spot market price for an energy source; and/or a forward market price for an energy source. An example facility description circuit **7306** further interprets present external data from at least one external data source, where the adaptive learning system **7314** is further configured to predict the present state facility outcome value **7320** further in response to the present external data.

Referencing FIG. **74**, an example procedure **7400** includes an operation **7402** to interpret a plurality of historical facility parameter values and a corresponding plurality of historical facility outcome values; an operation **7408** to operate an adaptive learning system, thereby training a facility production predictor in response to the plurality of historical facility parameter values and the corresponding plurality of historical facility outcome values; an operation **7410** to interpret a number of present state facility parameter values; and an operation **7412** to operate the adaptive learning system to predict a present state facility outcome value in response to the plurality of present state facility parameter values.

An example procedure **7400** further includes an operation **7406** to interpret historical external data from at least one external data source, and to operate the adaptive learning system to further train the facility production predictor in response to the historical external data. An example procedure **7400** includes the operation **7406** to interpret present external data from the at least one data source, and operating

the adaptive learning system to predict the present state facility outcome value further in response to the present external data.

Referencing FIG. 75, an example transaction-enabling system **7500** includes a production facility **7302** including a core task, and a controller **7304**, including: a facility description circuit **7306** structured to interpret a plurality of historical facility parameter values **7308** and a corresponding plurality of historical facility outcome values **7310**; a facility prediction circuit **7312** structured to operate an adaptive learning system **7314**, wherein the adaptive learning system is configured to train a facility resource allocation circuit **7502** in response to the plurality of historical facility parameter values **7308** and the corresponding plurality of historical facility outcome values **7310**; wherein the facility description circuit **7306** is further structured to interpret a plurality of present state facility parameter values **7318**; and where the trained facility resource allocation circuit **7502** is further structured to adjust, in response to the plurality of present state facility parameter values **7318**, a plurality of facility resource values (e.g., by providing adjustments to the facility resource values **7504**).

Example and non-limiting facility resource values include: a provisioning and an allocation of facility energy resources; and/or a provisioning and an allocation of facility compute resources. An example trained facility resource allocation circuit **7502** further adjusts the plurality of facility resource values by producing and/or selecting a favorable facility resource utilization profile from among a set of available facility resource configuration profiles **7506**. An example trained facility resource allocation circuit **7502** is further structured to adjust the plurality of facility resource values **7504** by one of producing or selecting a favorable facility resource output selection from among a set of available facility resource output values. An example trained facility resource allocation circuit **7502** is further structured to adjust the plurality of facility resource values **7504** by producing and/or selecting a favorable facility resource input profile from among a set of available facility resource input profiles. An example trained facility resource allocation circuit **7502** is further structured to adjust the plurality of facility resource values **7504** by producing or selecting a favorable facility resource configuration profile **7506** from among a set of available facility resource configuration profiles.

An example facility description circuit **7306** is further structured to interpret historical external data from at least one external data source **7322**, and wherein the adaptive learning system **7314** is further configured to train the facility resource allocation circuit **7502** in response to the historical external data. Example and non-limiting external data source(s) include at least one data source such as: a social media data source; a behavioral data source; a spot market price for an energy source; and/or a forward market price for an energy source. An example facility description circuit **7306** further interprets present external data from the at least one external data source **7322**, and wherein the trained facility resource allocation circuit **7502** is further structured to adjust the plurality of facility resource values **7504** in response to the present external data.

Referencing FIG. 76, an example procedure **7600** includes an operation **7602** to interpret a plurality of historical facility parameter values and a corresponding plurality of historical facility outcome values; an operation **7606** to operate an adaptive learning system, thereby training a facility resource allocation circuit in response to the plurality of historical facility parameter values and the

corresponding plurality of historical facility outcome values; and an operation **7608** to interpret a plurality of present state facility parameter values. The example procedure **7600** further includes an operation **7610** to adjust, in response to the plurality of present state facility parameter values, a plurality of facility resource values. In certain embodiments, for example where the operations **7610** to adjust the facility resource values by selecting among available profiles for the facility, an example procedure **7600** further includes operations to update the set of available facility resource profiles, for example by removing an ineffective, unused, or lightly used facility resource profile, and/or by adding a new facility resource profile to the set of available facility resource profiles. An example procedure **7600** further includes an operation **7604** to interpret historical external data and/or present external data, where the operation **7606** further includes training the facility resource allocation circuit in response to the historical external data and/or present external data.

Referencing FIG. 77, an example transaction-enabling system **7700** includes a production facility **7302** including a core task; a controller **7304**, including: a facility description circuit **7306** structured to interpret a plurality of historical facility parameter values **7308** and a corresponding plurality of historical facility outcome values **7310**; a facility prediction circuit **7312** structured to operate an adaptive learning system **7314**, wherein the adaptive learning system **7314** is configured to train a facility artificial intelligence (AI) configuration circuit **7702** in response to the plurality of historical facility parameter values **7308** and the corresponding plurality of historical facility outcome values **7310**; where the facility description circuit **7306** is further structured to interpret a plurality of present state facility parameter values **7318**. The example system **7700** includes where the trained facility AI configuration circuit **7702** is further structured to adjust, in response to the plurality of present state facility parameter values, a configuration of a facility AI component **7704** to produce a favorable facility output value. A favorable facility output value includes, without limitation, an improvement and/or optimization of any facility output described in the present disclosure, including at least a production related output and/or a resource utilization output.

An example trained facility AI configuration circuit **7702** further adjusts the configuration of the facility AI component by one of producing or selecting a favorable facility AI component configuration profile **7706** from among a set of available facility AI component configuration profiles. An example favorable facility output value includes, without limitation, an output value such as: a production volume description of the core task; a production quality description of the core task; a facility resource utilization description; an input resource utilization description; and/or a production timing description of the core task. An example facility description circuit **7306** is further structured to interpret historical external data from at least one external data source **7322**, and wherein the adaptive learning system **7314** is further configured to train the facility AI configuration circuit **7702** in response to the historical external data. An example system **7700** includes where the external data source includes at least one data source such as: a social media data source; a behavioral data source; a spot market price for an energy source; and/or a forward market price for an energy source. An example facility description circuit **7306** further interprets present external data from the at least one external data source **7322**, and wherein the trained facility AI configuration circuit **7702** is further structured to

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adjust the configuration of the facility AI component **7704** in response to the present external data.

Referencing FIG. **78**, an example procedure **7800** includes an operation **7802** to interpret a plurality of historical facility parameter values and a corresponding plurality of historical facility outcome values; an operation **7806** to operate an adaptive learning system, thereby training a facility artificial intelligence (AI) configuration circuit in response to the plurality of historical facility parameter values and the corresponding plurality of historical facility outcome values; an operation **7808** to interpret a plurality of present state facility parameter values; and an operation **7810** to operate the trained facility AI configuration circuit to adjust, in response to the plurality of present state facility parameter values, a configuration of a facility AI component to produce a favorable facility output value. An example procedure **7800** further includes an operation **7804** to interpret historical external data and/or present external data from at least one data source, wherein the operation **7806** to operate the trained facility AI configuration circuit is configured the AI facility component is further in response to the historical external data and/or the present external data.

Further referencing FIG. **73**, an example system includes wherein the core task includes a customer relevant output; and the trained facility production predictor **7316** is configured to determine a customer contact indicator **7324** in response to the plurality of present state facility parameter values **7318**; and wherein the controller **7304** further includes a customer notification circuit **7326** structured to provide a notification **7328** to a customer in response to the customer contact indicator **7324**. In certain embodiments, the customer includes one of a current customer and/or a prospective customer. In certain embodiments, determining the customer contact indicator includes performing at least one operation such as: determining whether the customer relevant output will meet a volume request from the customer; determining whether the customer relevant output will meet a quality request from the customer; determining whether the customer relevant output will meet a timing request from the customer; and/or determining whether the customer relevant output will meet an optional request from the customer.

Referencing FIG. **79**, an example procedure **7900** an operation **7902** to interpret a number of historical facility parameter values and a corresponding plurality of historical facility outcome values; an operations **7906** to operate an adaptive learning system, thereby training a facility production predictor in response to the plurality of historical facility parameter values and the corresponding plurality of historical facility outcome values; an operation **7908** to interpret a number of present state facility parameter values; an operation **7910** to operate the trained facility production predictor to determine a customer contact indicator in response to the plurality of present state facility parameter values; and an operation **7912** to provide a notification to a customer in response to the customer contact indicator. An example procedure **7900** further includes an operation **7904** to interpret historical external data and/or present external data from at least one external data source, wherein the operation **7910** to operate the trained facility AI configuration circuit includes further configure the AI facility component in response to the historical external data and/or the present external data.

Referencing FIG. **80**, an example transaction-enabling system **8000** includes a facility **8002** having an energy source and/or an energy utilization requirement, and a compute resource and/or compute task. An example facility

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8002 is an energy and compute facility—for example a server, computing cluster, or the like, that may be paired with an energy source, and/or have an energy utilization requirement for operations of the facility **8002**. The example system **8000** includes a controller **7304** having a facility description circuit **7306** structured to interpret detected conditions **8004**, wherein the detected conditions **8004** include at least one condition such as: an input resource for the facility; a facility resource; an output parameter for the facility; and/or an external condition related to an output of the facility **8002**. The example controller **7304** further includes a facility configuration circuit **8010** structured to operate an adaptive learning system **7314**, wherein the adaptive learning system **7314** is configured to adjust a facility configuration **8006** based on the detected conditions **8004**. An example adaptive learning system **7314** includes at least one of a machine learning system and/or an artificial intelligence (AI) system. An example adaptive learning system **7314** adjusts the facility configuration **8006** by performing a purchase and/or sale transaction on a market **8008**, such as: performing a purchase or sale transaction on one of an energy spot market or an energy forward market; performing a purchase or sale transaction on one of a compute resource spot market or a compute resource forward market; and performing a purchase or sale transaction on one of an energy credit spot market or an energy credit forward market. An example facility **8002** further includes a networking task, and wherein the adaptive learning system **7314** adjusts the facility configuration **8006** by performing a purchase and/or sale transaction on a market **8008**, such as: performing a purchase or sale transaction on one of a network bandwidth spot market or a network bandwidth forward market; and/or performing a purchase or sale transaction on one of a spectrum spot market or a spectrum forward market. An example adaptive learning system **7314** adjusts the facility configuration **8006** by adjusting at least one task of the facility **8002** to reduce the energy utilization requirement of the facility.

Referencing FIG. **81**, an example procedure **8100** includes an operation **8102** to interpret detected conditions relative to a facility, where the detected conditions include at least one condition such as: an input resource for the facility; a facility resource; an output parameter for the facility; and/or an external condition related to an output of the facility. In certain embodiments, operation **8102** includes interpreting detected conditions relating to a set of input resources for the facility. In certain embodiments, the detected conditions relate to at least one resource of the facility. In certain embodiments, the detected conditions relate to an output parameter for the facility. In certain embodiments, the detected conditions relate to a utilization parameter for an output of the facility. The example procedure **8100** further includes an operation **8104** to determine one of a spot market or forward market price for a facility resource. The example procedure **8100** further includes an operation **8106** to operate an adaptive learning system to adjust a facility configuration based on the detected conditions. An example procedure **8100** further includes an operation **8108** to adjust the facility configuration by executing a transaction on a spot market and/or a forward market for a facility resource. In certain embodiments, the facility resource includes one or more of: an energy resource, a compute resource, an energy credit resource, and/or a network bandwidth resource. In certain embodiments, the facility resource includes a spectrum resource.

Referencing FIG. **82**, an example transaction-enabling system **8200** includes a facility **8002** having at least one of

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an energy source or an energy utilization requirement. An example facility **8002** is an energy and compute facility. The example transaction-enabling system **8200** includes a controller **7304** having a facility description circuit **7306** structured to interpret detected conditions **8004**, wherein the detected conditions relate to a set of input resources for the facility **8002**. The example controller **7304** further includes a facility configuration circuit **8010** structured to operate an adaptive learning system **7314**, wherein the adaptive learning system is configured to adjust a facility configuration **8006** based on the detected conditions. An example transaction-enabling system **8200** further includes the adaptive learning system **7314** having at least one of a machine learning system and/or an AI system. An example adaptive learning system **7314** adjusts the facility configuration **8006** by performing a purchase and/or a sale transaction on a market **8008**, such as a spot market and/or a forward market for a facility resource. Example and non-limiting facility resources include: an energy resource; a compute resource; an energy credit resource; a network bandwidth resource; and/or a spectrum resource. In certain embodiments, the facility **8002** further includes a networking task. In certain embodiments, the adaptive learning system **7314** further adjusts at least one facility task **8202** to change an input resource requirement for the facility **8002**, and/or adjusts at least one facility configuration **8006** to change the input resource requirement for the facility **8002**. In certain embodiments, the facility **8002** further includes at least one additional facility resource; where the adaptive learning system **7314** adjusts the facility configuration **8006** by adjusting the compute resource, and further adjusting at the at least one additional facility resource. Example and non-limiting additional facility resources include a network resource, a data storage resource, and/or a spectrum resource.

In certain embodiments, the detected conditions **8004** include an output parameter for the facility **8002**. In certain further embodiments, the adaptive learning system **7314** adjusts the facility configuration **8006** and/or the facility tasks **8202** to provide at least one of: an increased facility output volume, an increased facility quality value, and/or an adjusted facility output time value.

In certain embodiments, the detected conditions **8004** include a utilization parameter for an output of the facility **8002**. In certain further embodiments, the adaptive learning system **7314** adjusts the facility configuration **8006** and/or the facility tasks **8202** to adjust at least one task of the facility **8002** to reduce the utilization parameter for output of the facility **8002**.

Referencing FIG. **83**, an example transaction-enabling system **8300** includes an energy and compute facility **8002** including: at least one of a compute task or a compute resource; and at least one of an energy source or an energy utilization requirement. The example transaction-enabling system **8300** further includes a controller **7304** having a facility model circuit **8302** that operates a digital twin **8304** for the facility **8002**, a facility description circuit **7306** that interprets a set of parameters **8306** from the digital twin for the facility; and a facility configuration circuit **8010** structured to operate an adaptive learning system **7314**, where the adaptive learning system **7314** is configured to adjust a facility configuration **8006** based on the set of parameters **8306** from the digital twin for the facility **8002**. An example adaptive learning system **7314** includes a machine learning system and/or an AI system. An example adaptive learning system **7314** adjusts the facility configuration **8006** by performing a purchase transaction and/or a sale transaction

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of a resource on a market **8008**. Example and non-limiting markets **8008** include a spot market and/or a forward market for a resource. Example and non-limiting resources include an energy resource, a networking resource, a network bandwidth resource, an energy credit resource, and/or a spectrum resource. An example facility **8002** further includes a networking task.

An example system further includes the facility description circuit **7306** interpreting the detected conditions **8004**, where the detected conditions **8004** include one or more of: an input resource for the facility **8002**, a facility resource, an output parameter for the facility **8002**, and/or an external condition related to an output of the facility. An example facility model circuit **8302** is further structured to update the digital twin **8304** in response to the detected conditions **8004**.

Referencing FIG. **84**, an example procedure **8400** includes an operation **8402** to operate a model that is, or that acts as, a digital twin of a facility. An example facility includes an energy resource and/or an energy utilization requirement, a compute resource and/or a compute task, and/or a network resource and/or a network task. The example procedure **8400** further includes an operation **8404** to interpret a set of parameters from the digital twin for the facility, an operation to interpret present state facility parameters/values **8406** and an operation **8408** to operate an adaptive learning system to adjust a facility configuration and/or a facility task based on the set of parameters for the digital twin, and/or based on facility parameter values as detected conditions for the facility. An example procedure **8400** includes an operation **8410** to update the digital twin based on the facility parameter values. Example and non-limiting facility parameter values include values such as: an input resource for the facility; a facility resource; an output parameter for the facility; and/or an external condition related to an output of the facility.

Referencing FIG. **85**, an example transaction-enabling system **8500** includes a machine **8502** having an associated regenerative energy facility **8504**, the machine **8502** having a requirement for at least one of a compute task, a networking task, and/or an energy consumption task. The example system **8500** further includes a controller **8506** including: an energy requirement circuit **8508** structured to determine an amount of energy **8510** for the machine to service the at least one of the compute task, the networking task, and/or the energy consumption task in response to the requirement for the at least one of the compute task, the networking task, and/or the energy consumption task. The example controller **8506** further includes an energy distribution circuit **8512** structured to adaptively improve an energy delivery of energy **8514** produced by the associated regenerative energy facility **8504** between the at least one of the compute task, the networking task, and/or the energy consumption task.

An example system **8500** includes the energy consumption task comprises as a core task, e.g. of the machine **8502**. An example controller **8506** further includes an energy market circuit **8516** structured to access an energy market **8518**, and wherein the energy distribution circuit **8512** is further structured to adaptively improve the energy delivery of the energy **8514** produced by the associated regenerative energy facility **8504** between the compute task, the networking task, the energy consumption task, and/or a sale (e.g., via a transaction **8520**) of the energy produced by the associated regenerative energy facility **8504** on the energy market **8518**. An example energy market **8518** comprises at least one of a spot market or a forward market. An example controller **8506** further includes where the energy distribu-

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tion circuit **8512** further comprises at least one of a machine learning component, an artificial intelligence component, or a neural network component (e.g., as a continuous improvement component **8522**).

Referencing FIG. **86**, an example procedure **8600** includes an operation **8602** to determine an amount of energy for a machine to service at least one of a compute task, a networking task, and/or an energy consumption task in response to a compute task requirement, a networking task requirement, and/or an energy consumption task requirement. The example procedure **8600** further includes an operation **8604** to adaptively improve an energy delivery between the compute task, the networking task, and/or the energy consumption task, where the improved energy deliver is of energy produced by a regenerative energy facility of the machine. In certain embodiments, the procedure **8600** further includes an operation **8606** to access an energy market, and adaptively improving the energy delivery of the energy produced by the regenerative energy facility between: the compute task, the networking task, the energy consumption task, and a sale of the energy produced by the regenerative energy facility on the energy market.

Referencing FIG. **87**, an example transaction-enabling system **8700** includes a machine **8502** having at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement; and a controller **8506** including a resource requirement circuit **8708** structured to determine an amount of a resource **8710** for the machine to service at least one of the compute task requirement, the networking task requirement, and/or the energy consumption task requirement, and a forward resource market circuit **8716** structured to access a forward resource market **8718**. The example controller **8506** further includes a resource distribution circuit **8712** structured to execute a transaction **8720** of the resource on the forward resource market **8718** in response to the determined amount of the resource **8710**. An example system **8700** includes the resource being any one or more of a compute resource, a spectrum allocation resource, an energy credit resource, an energy resource, a data storage resource, an energy storage resource, and/or a network bandwidth resource. An example system **8700** includes the forward resource market **8718** being a forward market for any one or more of the example resources. An example embodiment includes where the transaction(s) **8720** include buying and/or selling the resource. An example resource distribution circuit **8712** is further structured to adaptively improve one of an output value of the machine **8502** or a cost of operation of the machine **8502** using executed transactions **8720** on the forward resource market **8718**. An example resource distribution circuit **8712** further includes at least one of a machine learning component, an artificial intelligence component, and/or a neural network component (e.g., as a continuous improvement component **8722**).

Referencing FIG. **88**, an example procedure **8800** includes an operation **8802** to determine an amount of a resource for a machine to service at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement of the machine, and an operation **8804** to access a forward resource market. The example procedure **8800** further includes an operation **8806** to execute a transaction of the resource on the forward resource market in response to the determined amount of the resource. In certain embodiments, the procedure **8800** further includes an operation **8808** to adaptively improve an

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output value of the machine and/or a cost of operation of the machine using executed transactions on the forward resource market.

Referencing FIG. **89**, an example transaction-enabling system **8900** includes a fleet of machines **8902** each having at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement. The example system **8900** includes a controller **8506** including a resource requirement circuit **8708** structured to determine an amount of a resource **8710** for each of the machines to service the compute task requirements, the networking task requirements, and/or the energy consumption task requirements for each corresponding machine of the fleet of machines **8902**. The example controller **8506** further includes a forward resource market circuit **8716** structured to access a forward resource market **8718**; and a resource distribution circuit **8712** structured to execute an aggregated transaction **8920** of the resource on the forward resource market **8718** in response to the determined amount of the resource **8710** for each of the machines. An example system **8900** includes the resource being any one or more of a compute resource, a spectrum allocation resource, an energy credit resource, an energy resource, a data storage resource, an energy storage resource, and/or a network bandwidth resource. An example system **8900** includes the forward resource market **8718** being a forward market for any one or more of the example resources. An example embodiment includes where the transaction(s) **8920** include buying and/or selling the resource. An example resource distribution circuit **8712** is further structured to adaptively improve one of an aggregate output value of the fleet of machines or a cost of operation of the fleet of machines using executed aggregated transactions **8920** on the forward resource market **8718**. In certain embodiments, a resource distribution circuit **8712** includes a machine learning component, an artificial intelligence component, and/or a neural network component (e.g., as a continuous improvement circuit **8722**).

Referencing FIG. **90**, an example procedure **9000** includes an operation **9002** to determine an amount of a resource, for each of machine of a fleet of machines, to service at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement for each corresponding machine of the fleet of machines. The example procedure **9000** further includes an operation **9004** to access a forward resource market and an operation **9006** to execute an aggregated transaction of the resource on the forward resource market in response to the determined amount of the resource for each of the machines. In certain embodiments, an example procedure **9000** includes an operation **9008** to adaptively improve an aggregate output value of the fleet of machines and/or a cost of operation of the fleet of machines using executed aggregated transactions on the forward resource market.

Referencing FIG. **91**, an example transaction-enabling system **9100**, the system according to one disclosed non-limiting embodiment of the present disclosure can include a fleet of machines **8902** each having a requirement for at least one of a compute task, a networking task, and/or an energy consumption task. The example system **9100** further includes a controller **8506** having a resource requirement circuit **8708** structured to determine an amount of a resource **8710** for each of the machines to service the requirement for the compute task, the networking task, and/or the energy consumption task for each corresponding machine of the fleet of machines **8902**. The example controller **8506** further includes a resource distribution circuit **9112** structured to

adaptively improve a resource utilization **9102** of the resource for each of the machines between the compute task, the networking task, and/or the energy consumption task for each corresponding machine. Example and non-limiting resource(s) **8710** include a compute resource, a spectrum allocation resource, an energy credit resource, an energy resource, a data storage resource, an energy storage resource, and/or a network bandwidth resource. An example resource distribution circuit **9112** includes a machine learning component, an artificial intelligence component, and/or a neural network component (e.g., continuous improvement component **8722**). In certain embodiments, the controller **8506** includes a resource market circuit **9116** structured to access a market **9118** for one or more resources, and where the resource distribution circuit **9112** improves the resource utilization **9102** by executing one or more transactions **8920** on a market **9118** for the one or more resources to improve the resource utilization **9102**. In certain embodiments, the market **9118** may be spot market and/or a forward market for the one or more resources. In certain embodiments, a transaction **8920** includes a purchase and/or a sale of a resource.

Referencing FIG. **92**, an example procedure **9200** includes an operation **9202** to determine an amount of a resource, for each of machine of a fleet of machines, to service a requirement of at least one of a compute task, a networking task, and/or an energy consumption task for each corresponding machine. The procedure **9200** further includes an operation **9204** to adaptively improve a resource utilization of the resource for each of the machines between the compute task, the networking task, and/or the energy consumption task for each corresponding machine. Example and non-limiting operations **9204** include adaptively improving an output value of the fleet of machines, and/or a cost of operation of the fleet of machines. Example and non-limiting operations **9204** to adaptively improve the resource utilization include adjusting workloads for tasks between machines of the fleet of machines, executing one or more transactions for a resource on a forward market and/or a spot market for a resource, and/or adjusting one or more tasks of a machine of the fleet of machines to adjust a total (or aggregate) resource utilization of the fleet of machines. Example and non-limiting operations **9204** to adaptively improve the resource utilization include reducing a total resource utilization, reducing a resource utilization for a particular type of resource, and/or time-shifting resource utilization from a less efficient (e.g., based on cost, capacity, availability, and/or impact of the resource utilization) time frame to a more efficient time frame for the resource utilization. Example and non-limiting operations **9204** include improving the resource utilization by adjusting a consumed resource of one type to a consumed resource of another type by the machine(s), for example by substituting a second resource for a first resource, and/or by substituting a second task for a first task, thereby substituting a second resource for a first resource. Example and non-limiting operations **9204** include executing a transaction related to a second resource in response to a utilization of a first resource, for example where the second resource can be substituted for the first resource, and/or where the second resource provides an economic offset for the first resource.

Referencing FIG. **93**, an example transaction-enabling system **9300** includes a machine **8502** having at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement. The example system **9300** further includes a controller **8506** having a resource requirement circuit **8708** structured to

determine an amount of a resource **8710** for the machine to service at least one of the compute task requirement, the networking task requirement, and/or the energy consumption task requirement. The example controller **8506** further includes a resource market circuit **9116** structured to access a resource market **9118**, and a resource distribution circuit **8712** structured to execute a transaction **8720** of the resource on the resource market **9118** in response to the determined amount of the resource **8710**. In certain embodiments, the resource distribution circuit **8712** is further structured to adaptively improve an output value of the machine, a cost of operation of the machine, and/or a resource utilization of the machine using executed transactions **8720** on the resource market **9118**. Example and non-limiting resources include an energy resource, an energy credit resource, and/or a spectrum allocation resource. An example resource market **9118** includes a spot market and/or a forward market for one or more resources. Example and non-limiting operations of the resource distribution circuit **8712** to adaptively improve the output value, cost of operation, and/or resource utilization of the machine **8502** include: increasing an output volume, quantity, and/or quality; improving an output delivery time (e.g., an earlier delivery and/or a synchronized delivery with some other process of the system); reducing a total resource utilization; reducing a resource utilization for a particular type of resource; reducing a resource consumed per output element produced of the machine; and/or time-shifting resource consumption from a less efficient (e.g., based on cost, capacity, availability, and/or impact of the resource utilization) time frame to a more efficient time frame for the resource consumption. Example and non-limiting operations of the resource distribution circuit **8712** include adjusting a consumed resource of one type to a consumed resource of another type by the machine(s), for example by substituting a second resource for a first resource, and/or by substituting a second task for a first task, thereby substituting a second resource for a first resource. Example and non-limiting operations of the resource distribution circuit **8712** include executing a transaction **8720** related to a second resource in response to a utilization of a first resource, for example where the second resource can be substituted for the first resource, and/or where the second resource provides an economic offset for the first resource. An example resource distribution circuit **8712** includes a machine learning component, an artificial intelligence component, and/or a neural network component (e.g., as a continuous improvement component **8722**).

Referencing FIG. **94**, an example procedure **9400** includes an operation **9402** to determine an amount of a resource for a machine to service at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement; an operation **9404** to access a resource market; and an operation **9406** to execute a transaction on the resource market in response to the amount of the resource. In certain embodiments, an example procedure **9400** further includes an operation **9408** to adaptively improve an output value, a cost of operation, and/or a resource utilization of the machine utilizing the executed transactions on the resource market.

Referencing FIG. **95**, an example transaction-enabling system **9500** includes a fleet of machines **8902** each having at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement. The example system **9500** further includes a controller **8506** having a resource requirement circuit **8708** structured to determine an amount of a resource **8710** for each of the machines of the fleet of machines **8902** to service

at least one of the compute task requirement, the networking task requirement, and/or the energy consumption task requirement for each corresponding machine. The example system **9500** further includes a resource market circuit **9116** structured to access a resource market **9118**, and a resource distribution circuit **9112** structured to execute an aggregated transaction **8920** of the resource on the resource market **9118** in response to the determined amount of the resource **8710** for each of the machines. The example resource market **9118** may include a spot market and/or a forward market for one or more of the resources. The example system **9500** includes the resource distribution circuit **9112** executing the aggregated transaction(s) **8920** to improve a resource performance **9502** of the fleet of machines **8902**, where the resource performance **9502** includes one or more of: a total consumption of one or more resources, a cost of operation of the fleet of machines **8902**, and/or an aggregate output value of the fleet of machines **8902** (e.g., including an output volume, quantity, and/or quality for a given amount of resources utilized, and/or an output volume, quantity, and/or quality per amount of resources utilized). An example resource distribution circuit **9112** is further structured to adaptively improve the resource performance **9502** of the fleet of machines **8902**, and may further include a machine learning component, an artificial intelligence component, and/or a neural network component (e.g., as a continuous improvement component **8722**).

Referencing FIG. **96**, an example procedure **9600** includes an operation **9602** to determine an amount of a resource, for each of machine of a fleet of machines, to service at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement for each corresponding machine. The example procedure **9600** further includes an operation **9604** to access a resource market, and an operation **9606** to execute a transaction on the resource market in response to an aggregated resource amount for the task requirements of each machine of the fleet of machines. In certain embodiments, the procedure **9600** further includes an operation **9608** to adaptively improve an output value, a cost of operation, a resource utilization, and/or a resource performance of the fleet of machines, using executed transactions on the market.

Referencing FIG. **97**, an example transaction-enabling system **9700** includes a machine **8502** having at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement. The example system **9700** further includes a controller **8506** having a resource requirement circuit **8708** structured to determine an amount of a resource for the machine **8502** to service at least one of the compute task requirement, the networking task requirement, and/or the energy consumption task requirement. The example controller **8506** further includes a social media data circuit **9702** structured to interpret data from a plurality of social media data sources **9704**; a forward resource market circuit **8716** structured to access a forward resource market **8718**; a market forecasting circuit **9706** structured to predict a forward market price **9708** of the resource on the forward resource market **8718** in response to the plurality of social media data sources **9704**; and a resource distribution circuit **8712** structured to execute a transaction **8720** of the resource on the forward resource market **8718** in response to the determined amount of the resource **8710** and the predicted forward market price **9708** of the resource. Example and non-limiting resources include a spectrum allocation resource, an energy resource, and/or an energy credit resource. Example and non-limiting trans-

actions **8720** include buying and/or selling one or more of the resources. An example market forecasting circuit **9706** adaptively improves the forward market price **9708** prediction, for example utilizing a machine learning component, an artificial intelligence component, and/or a neural network component (e.g., as a continuous improvement component **9710**). An example resource distribution circuit **8712** is further structured to adaptively improve an operating aspect of the machine **8502**, for example output value of the machine, a cost of operation of the machine, a resource utilization of the machine, and/or a resource performance of the machine using executed transactions **8720** on the forward resource market **8718**. In a further example, the resource distribution circuit **8712** further includes a machine learning component, an artificial intelligence component, and/or a neural network component (e.g., as a continuous improvement component **8722**).

Referencing FIG. **98**, an example procedure **9800** includes an operation **9802** to determine an amount of a resource for a machine to service at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement; an operation **9804** to interpret data from a plurality of social media data sources; an operation **9806** to access a forward resource market; an operation **9808** to predict a forward market price of the resource on the forward resource market in response to the plurality of social media data sources; and an operation **9810** to execute a transaction of the resource on the forward resource market in response to the determined amount of the resource and the predicted forward market price of the resource. An example procedure **9800** further includes an operation **9812** to adaptively improve an output value and/or a cost of operation of the machine using executed transactions on the market. In certain embodiments, the procedure **9800** further includes an operation to adaptively improve the operation **9808** to predict the forward market price of the resource. In certain embodiments, the operation **9812** includes an operation to adaptively improve a resource utilization and/or a resource performance of the machine using the executed transactions on the market.

Referencing FIG. **99**, an example transaction-enabling system **9900** includes a machine **8502** having at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement. The example system **9900** further includes a controller **8506** having a resource requirement circuit **8708** structured to determine an amount of a resource **8710** for the machine **8502** to service at least one of the compute task requirement, the networking task requirement, and/or the energy consumption task requirement, and a resource market circuit **9902** structured to access a resource market **9904**. The example controller **8506** further includes a market testing circuit **9906** structured to execute a first transaction **8720** of the resource on the resource market **9904** in response to the determined amount of the resource **8710**; and an arbitrage execution circuit **9908** structured to execute a second transaction **8720** of the resource on the resource market **9904** in response to the determined amount of the resource and further in response to an outcome **9914** of the execution of the first transaction, wherein the second transaction comprises a larger transaction than the first transaction (e.g., according to the first transaction parameters **9910** defining a lower transaction volume, amount, or cost than the second transaction parameters **9912**). The resource may be any type of resource as set forth throughout the present disclosure. An example arbitrage execution circuit **9908** is further structured to adaptively improve an arbitrage parameter **9916** by

adjusting the first transaction parameter **9910** and/or the second transaction parameter **9912**. Example and non-limiting adjustments to the transaction parameters **9910**, **9912** include adjustments such as: adjusting a relative size of the first transaction and the second transaction; adjusting a resource type of the first transaction and the second transaction; adjusting a selected resource market **9904** of the first transaction and/or the second transaction; adjusting a forward market time-frame and/or a spot market selection of the first transaction parameter **9910** and/or the second transaction parameter **9912**; adjusting a timing between the first transaction and the second transaction; adjusting a time to initiate the first transaction (e.g., a time of day; a calendar date; a relative date to a week beginning, holiday, economic data report, etc.); and/or changing a number of transactions such as a third, fourth, or fifth transactions, including determining transaction parameters **9910**, **9912** for each transaction, a trajectory of volumes and timing for each transaction, and further determining or adjusting parameters for each subsequent transaction based on the transaction outcome of each preceding transaction (or a subset of the preceding transactions, or all of the preceding transactions). In certain embodiments, the arbitrage execution circuit **9908** may include a continuous improvement component, such as a machine learning component, an artificial intelligence component, and/or a neural network component. In certain embodiments, the arbitrage parameter **9916** may include one or more parameters such as: a similarity value in a market response of the first transaction and the second transaction; a confidence value of the first transaction to provide test information for the second transaction; and/or a market effect of the first transaction.

Referencing FIG. **100**, an example procedure **10000** includes an operation **10002** to determine an amount of a resource to service a task requirement of a machine, an operation **10004** to access a resource market, an operation **10006** to execute a first transaction on the resource market, and an operation **10008** to execute a second, larger transaction on the resource market. In certain embodiments, the procedure **10000** further includes an operation **10010** to adaptively improve one of an output value, a cost of operation, a resource utilization, and/or a resource performance of the machine by adjusting parameters of the first and/or second transaction.

Referencing FIG. **101**, an example apparatus **10100** includes a resource requirement circuit **8708** structured to determine an amount of a resource **8710** for a machine to service at least one of a compute task requirement, a networking task requirement, and/or an energy consumption task requirement. The example apparatus **10100** further includes a resource market circuit **9902** structured to access a resource market, for example to establish communications with and/or to communicate transaction parameters (buying, selling, amounts, prices, types of pricing limits, etc.) with the market(s) for the resource(s). The example apparatus **10100** further includes a market testing circuit **9906** structured to execute a first transaction **10102** of the resource on the resource market **9904** in response to the determined amount of the resource **8710**; and an arbitrage execution circuit **9908** structured to execute a second transaction **10106** of the resource on the resource market **9904** in response to the determined amount of the resource and further in response to an outcome **10104** of the execution of the first transaction, wherein the second transaction **10106** comprises a larger transaction than the first transaction **10102** (e.g., according to the first transaction parameters **9910** defining a lower transaction volume, amount, or cost

than the second transaction parameters **9912**). The resource may be any type of resource as set forth throughout the present disclosure. An example arbitrage execution circuit **9908** is further structured to adaptively improve an arbitrage parameter **9916** by adjusting the first transaction parameter **9910** and/or the second transaction parameter **9912**. In certain embodiments, the arbitrage execution circuit **9908** may include and/or communicate with a continuous improvement component **10110**, such as a machine learning component, an AI component, and/or a neural network component. Example and non-limiting adjustments to the transaction parameters **9910**, **9912** include adjustments such as: adjusting a relative size of the first transaction **10102** and the second transaction **10106**; adjusting a resource type of the first transaction **10102** and the second transaction **10106**; adjusting a selected market of the first transaction **10102** and/or the second transaction **10106** (e.g., where more than one market is available); adjusting a forward market time-frame and/or a spot market selection of the first transaction parameter **9910** and/or the second transaction parameter **9912**; adjusting a timing between the first transaction **10102** and the second transaction **10106**; adjusting a time to initiate the first transaction **10102** (e.g., a time of day; a calendar date; a relative date to a week beginning, holiday, economic data report, etc.); and/or changing a number of transactions such as a third, fourth, or fifth transactions (e.g., additional transactions **10108**), including determining parameters **9910**, **9912** for each transaction, a trajectory of volumes and timing for each transaction, and further determining or adjusting parameters for each subsequent transaction based on the transaction outcome of each preceding transaction (or a subset of the preceding transactions, or all of the preceding transactions). In certain embodiments, the arbitrage parameter **9916** may include one or more parameters such as: a similarity value in a market response of the first transaction and the second transaction; a confidence value of the first transaction to provide test information for the second transaction; and/or a market effect of the first transaction.

Referencing FIG. **102**, an example transaction-enabling system **10200** includes a machine **8502** having an associated resource capacity **10202** for a resource, the machine **8502** having a requirement for at least one of a core task, a compute task, an energy storage task, a data storage task, and/or a networking task. The example system **10200** further includes a controller **8506** having a resource requirement circuit **10204** structured to determine an amount of the resource **8710** to service the requirement of the at least one of the core task, the compute task, the energy storage task, the data storage task, and/or the networking task in response to the requirement of the at least one of the core task, the compute task, the energy storage task, the data storage task, and/or the networking task. The example controller **8506** further includes a resource distribution circuit **10206** structured to adaptively improve, in response to the associated resource capacity **10202**, a resource delivery **10208** of the resource between the core task, the compute task, the energy storage task, the data storage task, and/or the networking task. An example associated resource capacity **10202** includes one or more of: a compute capacity for a compute resource; an energy capacity for an energy resource (including at least power, storage, regeneration, torque, and/or credits); a network bandwidth capacity for a networking resource; and/or a data storage capacity for a data storage resource. In certain embodiments, the resource distribution circuit **10206** provides commands to the machine **8502**, and/or provides resource delivery commands **10210** to the associated resource capacity **10202** to implement the deter-

mination and/or improvement of the resource delivery **10208**. In certain embodiments, the controller **8506** determines one or more of a machine output value **10212**, a machine resource utilization **10214**, a machine resource performance **10216**, and/or a machine cost of operation **10218**, and the resource distribution circuit **10206** further adaptively improves the resource delivery **10208** in response to one or more of these. An example resource distribution circuit **10206** includes a continuous improvement circuit **10220**, which may include one or more of a machine learning component, an AI component, and/or a neural network component.

Referencing FIG. **103**, an example procedure **10300** includes an operation **10302** to determine an amount of a resource to service one or more tasks of a machine, an operation **10304** to determine a resource requirement for the machine based on the amount of the resource to service the one or more tasks, and an operation **10306** to adaptively improve, in response to an associated resource capacity of the machine, a resource delivery between the tasks of the machine.

Referencing FIG. **104**, an example transaction-enabling system **10400** includes a fleet of machines **10402** each having an associated resource capacity **10404** for a resource, and each machine of the fleet of machines **10402** further having a requirement for at least one of a core task, a compute task, an energy storage task, a data storage task, and a networking task. The example system **10400** further includes a controller **8506** having a resource requirement circuit **10204** structured to determine an amount of the resource **8710** to service the requirement of the at least one of the core task, the compute task, the energy storage task, the data storage task, and/or the networking task for each machine of the fleet of machines **10402** in response to the requirement of the at least one of the core task, the compute task, the energy storage task, the data storage task, and/or the networking task. The example controller **8506** further includes a resource distribution circuit **10206** structured to adaptively improve, in response to the associated resource capacity **10404**, an aggregated resource delivery **10408** of the resource between the core task, the compute task, the energy storage task, the data storage task, and/or the networking task. An example associated resource capacity **10404** includes one or more of: a compute capacity for a compute resource; an energy capacity for an energy resource (including at least power, storage, regeneration, torque, and/or credits); a network bandwidth capacity for a networking resource; and/or a data storage capacity for a data storage resource. In certain embodiments, the resource distribution circuit **10206** provides commands to the fleet of machines **10402**, and/or provides resource delivery commands **10210** to the associated resource capacity **10404** to implement the determination and/or improvement of the aggregated resource delivery **10408**. In certain embodiments, the controller **8506** determines one or more of a fleet output value **10412**, a fleet resource utilization **10414**, a fleet resource performance **10416**, and/or a fleet cost of operation **10418**, and the resource distribution circuit **10206** further adaptively improves the aggregated resource delivery **10408** in response to one or more of these. An example resource distribution circuit **10206** includes a continuous improvement circuit **10220**, which may include one or more of a machine learning component, an AI component, and/or a neural network component.

An example resource distribution circuit **10206** is further structured to interpret a resource transferability value **10420** between at least two machines of the fleet of machines, and

to adaptively improve the aggregated resource delivery **10408** further in response to the resource transferability value **10420**. Example and non-limiting resource transferability values **10420** include one or more of: an ability to substitute or distribute tasks at least partially between machines of the fleet of machines **10402**; an ability to transfer resources from the associated resource capacities between machines of the fleet of machines **10402**; and/or an ability to substitute a first resource for a second resource within a machine or between machines of the fleet of machines **10402**. A substitution of resources may further include consideration of a rate of resource substitution (e.g., resource consumption per unit time), a capacity of a resource, positive and negative resource flows (e.g., consumption, regeneration, or acquisition of a resource), and/or time frames for the resource substitution (e.g., transferring a resource from a first machine to a second machine at a first time, and transferring from the second machine to the first machine or another machine at a second time).

Referencing FIG. **105**, an example procedure **10500** includes an operation **10502** to determine an aggregated amount of a resource to service a core task, a compute task, an energy storage task, a data storage task, and/or a networking task for each machine of a fleet of machines, and an operation **10504** to determine a resource requirement including at least one of a core task requirement, a compute task requirement, an energy storage task requirement, a data storage task requirement, and/or a networking task requirement for each machine of the fleet of machines. The example procedure **10500** further includes an operation **10506** to adaptively improve, in response to an aggregated associated resource capacity of the fleet of machines, an aggregated resource delivery of the resource between the core task, the compute task, the energy storage task, the data storage task, and/or the networking task for each machine of the fleet of machines.

Referencing FIG. **106**, an example apparatus **10600** includes a controller **8506** having resource requirement circuit **10204** structured to determine an aggregated amount of a resource **8710** to service a core task, a compute task, an energy storage task, a data storage task, and/or a networking task for each machine of a fleet of machines in response to at least one of a core task requirement, a compute task requirement, an energy storage task requirement, a data storage task requirement, and/or a networking task requirement for each machine of the fleet of machines; and a resource distribution circuit **10206** structured to adaptively improve, in response to an aggregated associated resource capacity **10604** of the fleet of machines, an aggregated resource delivery **10408** of the resource between the core task, the compute task, the energy storage task, the data storage task, and/or the networking task for each machine of the fleet of machines. Example associated resource capacities **10604** include one or more of: a compute capacity for a compute resource; an energy capacity for an energy resource (including at least power, storage, regeneration, torque, and/or credits); a network bandwidth capacity for a networking resource; and/or a data storage capacity for a data storage resource. In certain embodiments, the resource distribution circuit **10206** provides commands to the fleet of machines **10402**, and/or provides resource delivery commands **10210** to the associated resource capacity elements to implement the determination and/or improvement of the aggregated resource delivery **10408**. In certain embodiments, the controller **8506** determines one or more of a fleet output value **10412**, a fleet resource utilization **10414**, a fleet resource performance **10416**, and/or a fleet cost of operation **10418**,

and the resource distribution circuit **10206** further adaptively improves the aggregated resource delivery **10408** in response to one or more of these. An example resource distribution circuit **10206** includes a continuous improvement circuit **10220**, which may include one or more of a machine learning component, an AI component, and/or a neural network component.

Referencing FIG. **107**, an example transaction-enabling system **10700** includes a machine **8502** having at least one task requirement, and controller **8506** having a resource requirement circuit **8708** structured to determine an amount of resource(s) **8710** for the machine **8502** to service the task requirement(s). The example controller **8506** further includes a forward resource market circuit **10702** structured to access a forward resource market (resource market(s) **9904**), and a resource market circuit **10706** structured to access a resource market **9904** (e.g., a spot market, or a different forward market). The example controller **8506** further includes a resource distribution circuit **10708** structured to execute a transaction **8720** on the forward resource market and/or the resource market in response to the determined amount of the resource **8710**. An example resource distribution circuit **10708** is further structured to adaptively improve at least one of: an output of the machine, a cost of operation of the machine, a resource utilization of the machine, and/or a resource performance of the machine using the transaction(s) **8720**. In certain embodiments, the resource distribution circuit **10708** includes continuous improvement component, such as a machine learning component, an AI component, and/or a neural network component, to perform the adaptive improvement. In certain embodiments, the resource distribution circuit **10708** may determine a second amount of a second resource for the machine to service the task requirement(s), and may further execute a transaction **8720** of the first resource and/or the second resource to improve an aspect of the machine operation. The second resource may be a substitute resource for the first resource, and/or may be a corresponding resource to the first resource—for example a resource that is expected to make a corresponding or countering price move relative to the first resource. In certain embodiments, the forward resource market is a forward market for a first time scale (e.g., one month out) and the resource market is a forward market for a second time scale (e.g., two months out). In certain embodiments, the forward market is a forward market for the first resource or the second resource, and the resource market is a spot market either the first resource or the second resource (e.g., the same resource as for the forward market, or the other one of the first or second resource), and/or the resource market is a forward market for either the first resource or the second resource (e.g., the same resource as for the forward market but at a different time scale, or the other one of the first or second resource at the same or a different time scale). In certain embodiments, the transaction(s) **8720** may be one of: a sale of a resource, a purchase of a resource, a short sale of a resource, a call option for a resource, a put option for a resource, and/or any of the foregoing with relation to a substitute resource or a correlated resource. An example resource distribution circuit **10708** executes the transaction(s) **8720** using a substitute resource and/or a correlated resource as a substitute for a transaction of the to-be-traded resource. An example resource distribution circuit **10708** executes the transaction (s) **8720** using a substitute resource and/or a correlated resource as an economic offset of the to-be-traded resource. An example resource distribution circuit **10708** executes the

transaction(s) **8720** using a substitute resource and/or a correlated resource in concert with a transaction of the to-be-traded resource.

Referencing FIG. **108**, an example procedure **10800** includes an operation **10802** to determine a resource requirement for task(s) of a machine, and an operation **10804** to access a forward resource market for the resource, a substitute resource, and/or a correlated resource. The example procedure **10800** further includes an operation **10806** to access a resource market for the resource, a substitute resource, and/or a correlated resource. The resource market may be a spot market or a forward market. The example procedure **10800** further includes an operation **10808** to execute a transaction on the forward resource market and/or the resource market in response to the resource requirements. In certain embodiments, the operation **10808** is performed to improve, and/or to adaptively improve, an output value of the machine, a cost of operation of the machine, a resource utilization of the machine, and/or a resource performance of the machine.

Referencing FIG. **109**, an example procedure **10900** includes an operation **10902** to determine a second resource that can substitute for a first resource, and/or that is correlated with the first resource and further includes an operation **10904** that determines that the second resource is a functionally equivalent, operationally acceptable alternative to the first resource or determines the second resource as an economic offset resource. Without limitation to any other aspect of the present disclosure, a correlated resource includes any resource that is expected to make a correlated price move (e.g., either with the first resource or opposite the first resource), and/or any resource that is expected to have correlated availability for purchase with the first resource (e.g., either availability that moves with the first resource or opposite the first resource). The example procedure **10900** further includes an operation **10906** to consider (e.g., to predict and/or adaptively predict using a machine learning, AI, or neural network component) a forward market price of the first resource and/or the second resource, and an operation **10908** to execute a transaction on a market (e.g., a spot market and/or a forward market) of the first resource and/or the second resource.

An example transaction-enabling system includes a machine having a task requirements including a compute task requirement, a networking task requirement, and/or an energy consumption task requirement, and a controller having a number of circuits configured to functionally execute certain operations to execute a transaction of a resource for the machine. The example controller includes a resource requirement circuit structured to determine an amount of a resource for the machine to service the task requirement(s), a forward resource market circuit structured to access a forward resource market, a resource market circuit structured to access a resource market, and a resource distribution circuit structured to execute a transaction of the resource on the resource market and/or the forward resource market in response to the determined amount of the resource.

Certain further aspects of an example system are described following, any one or more of which may be present in certain embodiments. An example system includes where the resource distribution circuit is further structured to adaptively improve at least one of an output of the machine or a resource utilization of the machine, and/or where the resource distribution circuit further includes at least one of a machine learning component, an artificial intelligence component, or a neural network component. An example embodiment includes the resource as one or more

of a compute resource, an energy resource, and/or an energy credit resource. An example system includes where the resource requirement circuit is further structured to determine a second amount of a second resource for the machine to the task requirement(s), and where the resource distribution circuit is further structured to execute a first transaction of the first resource on one of the resource market or the forward resource market, and to execute a second transaction of the second resource on the other one of the resource market or the forward resource market. An example embodiment further includes where the second resource is a substitute resource for the first resource during at least a portion of the operating conditions for the machine. An example system includes where the forward resource market includes a futures market for the resource at a first time scale, and where the resource market includes a spot market for the resource and/or a futures market for the resource at a second time scale. An example system includes the transaction having a transaction type such as: a sale of the resource; a purchase of the resource; a short sale of the resource; a call option for the resource; a put option for the resource; and/or any of the foregoing with regard to at least one of a substitute resource or a correlated resource. An example system includes where the resource distribution circuit is further structured to determine at least one of a substitute resource or a correlated resource, and to further execute at least one transaction of the at least one of the substitute resource or the correlated resource. An example system includes where the resource distribution circuit is further structured to execute the at least one transaction of the at least one of the substitute resource or the correlated resource as a replacement for the transaction of the resource. An example system includes where the resource distribution circuit is further structured to execute the at least one transaction of the at least one of the substitute resource or the correlated resource in concert with the transaction of the resource.

An example procedure includes an operation to determine an amount of a resource for a machine to service at least one task requirement such as a compute task requirement, a networking task requirement, and/or an energy consumption task requirement. The example procedure further includes an operation to access a forward resource market and a resource market, and an operation to execute a transaction of the resource on at least one of the forward resource market or the resource market in response to the determined amount of the resource.

Certain further aspects of an example procedure are described following, any one or more of which may be present in certain embodiments. An example procedure further includes an operation to determine a second amount of a second resource for the machine to service at least one of the task requirement(s), an operation to execute a first transaction of the first resource on one of the resource market or the forward resource market, and an operation to execute a second transaction of the second resource on the other one of the resource market or the forward resource market. An example procedure further includes an operation to determine a substitute resource and/or a correlated resource, and an operation to execute at least one transaction of the substitute resource and/or the correlated resource. An example procedure further includes an operation to execute the transaction of the substitute resource and/or the correlated resource as a replacement for the transaction of the resource. An example procedure further includes an operation to execute the transaction of the substitute resource and/or the correlated resource in concert with the transaction of the resource. An example procedure includes determining

the substitute resource and/or the correlated resource by performing at least one operation such as: determining the correlated resource for the machine as a resource to service alternate tasks that provide acceptable functionality for the machine; determining the correlated resource as a resource that is expected to be correlated with the resource in regard to at least one of a price or an availability; and/or determining the correlated resource as a resource that is expected to have a corresponding price change with the resource, such that a subsequent sale of the correlated resource combined with a spot market purchase of the resource provides for a planned economic outcome.

Detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure, which may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure in virtually any appropriately detailed structure.

The terms “a” or “an,” as used herein, are defined as one or more than one. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open transition).

While only a few embodiments of the present disclosure have been shown and described, it will be obvious to those skilled in the art that many changes and modifications may be made thereunto without departing from the spirit and scope of the present disclosure as described in the following claims. All patent applications and patents, both foreign and domestic, and all other publications referenced herein are incorporated herein in their entireties to the full extent permitted by law.

Certain operations described herein include interpreting, receiving, and/or determining one or more values, parameters, inputs, data, or other information (“receiving data”). Operations to receive data include, without limitation: receiving data via a user input; receiving data over a network of any type; reading a data value from a memory location in communication with the receiving device; utilizing a default value as a received data value; estimating, calculating, or deriving a data value based on other information available to the receiving device; and/or updating any of these in response to a later received data value. In certain embodiments, a data value may be received by a first operation, and later updated by a second operation, as part of the receiving a data value. For example, when communications are down, intermittent, or interrupted, a first receiving operation may be performed, and when communications are restored, an updated receiving operation may be performed.

Certain logical groupings of operations herein, for example methods or procedures of the current disclosure, are provided to illustrate aspects of the present disclosure. Operations described herein are schematically described and/or depicted, and operations may be combined, divided, re-ordered, added, or removed in a manner consistent with the disclosure herein. It is understood that the context of an operational description may require an ordering for one or more operations, and/or an order for one or more operations may be explicitly disclosed, but the order of operations should be understood broadly, where any equivalent grouping of operations to provide an equivalent outcome of operations is specifically contemplated herein. For example, if a value is used in one operational step, the determining of the value may be required before that operational step in

certain contexts (e.g. where the time delay of data for an operation to achieve a certain effect is important), but may not be required before that operation step in other contexts (e.g. where usage of the value from a previous execution cycle of the operations would be sufficient for those purposes). Accordingly, in certain embodiments an order of operations and grouping of operations as described is explicitly contemplated herein, and in certain embodiments re-ordering, subdivision, and/or different grouping of operations is explicitly contemplated herein.

The methods and systems described herein may be deployed in part or in whole through a machine that executes computer software, program codes, and/or instructions on a processor. The present disclosure may be implemented as a method on the machine, as a system or apparatus as part of or in relation to the machine, or as a computer program product embodied in a computer readable medium executing on one or more of the machines. In embodiments, the processor may be part of a server, cloud server, client, network infrastructure, mobile computing platform, stationary computing platform, or other computing platform. A processor may be any kind of computational or processing device capable of executing program instructions, codes, binary instructions and the like. The processor may be or may include a signal processor, digital processor, embedded processor, microprocessor or any variant such as a co-processor (math co-processor, graphic co-processor, communication co-processor and the like) and the like that may directly or indirectly facilitate execution of program code or program instructions stored thereon. In addition, the processor may enable execution of multiple programs, threads, and codes. The threads may be executed simultaneously to enhance the performance of the processor and to facilitate simultaneous operations of the application. By way of implementation, methods, program codes, program instructions and the like described herein may be implemented in one or more thread. The thread may spawn other threads that may have assigned priorities associated with them; the processor may execute these threads based on priority or any other order based on instructions provided in the program code. The processor, or any machine utilizing one, may include non-transitory memory that stores methods, codes, instructions and programs as described herein and elsewhere. The processor may access a non-transitory storage medium through an interface that may store methods, codes, and instructions as described herein and elsewhere. The storage medium associated with the processor for storing methods, programs, codes, program instructions or other type of instructions capable of being executed by the computing or processing device may include but may not be limited to one or more of a CD-ROM, DVD, memory, hard disk, flash drive, RAM, ROM, cache and the like.

A processor may include one or more cores that may enhance speed and performance of a multiprocessor. In embodiments, the process may be a dual core processor, quad core processors, other chip-level multiprocessor and the like that combine two or more independent cores (called a die).

The methods and systems described herein may be deployed in part or in whole through a machine that executes computer software on a server, client, firewall, gateway, hub, router, or other such computer and/or networking hardware. The software program may be associated with a server that may include a file server, print server, domain server, internet server, intranet server, cloud server, and other variants such as secondary server, host server, distributed server and the like. The server may include one or more of

memories, processors, computer readable media, storage media, ports (physical and virtual), communication devices, and interfaces capable of accessing other servers, clients, machines, and devices through a wired or a wireless medium, and the like. The methods, programs, or codes as described herein and elsewhere may be executed by the server. In addition, other devices required for execution of methods as described in this application may be considered as a part of the infrastructure associated with the server.

The server may provide an interface to other devices including, without limitation, clients, other servers, printers, database servers, print servers, file servers, communication servers, distributed servers, social networks, and the like. Additionally, this coupling and/or connection may facilitate remote execution of program across the network. The networking of some or all of these devices may facilitate parallel processing of a program or method at one or more location without deviating from the scope of the disclosure. In addition, any of the devices attached to the server through an interface may include at least one storage medium capable of storing methods, programs, code and/or instructions. A central repository may provide program instructions to be executed on different devices. In this implementation, the remote repository may act as a storage medium for program code, instructions, and programs.

The software program may be associated with a client that may include a file client, print client, domain client, internet client, intranet client and other variants such as secondary client, host client, distributed client and the like. The client may include one or more of memories, processors, computer readable media, storage media, ports (physical and virtual), communication devices, and interfaces capable of accessing other clients, servers, machines, and devices through a wired or a wireless medium, and the like. The methods, programs, or codes as described herein and elsewhere may be executed by the client. In addition, other devices required for execution of methods as described in this application may be considered as a part of the infrastructure associated with the client.

The client may provide an interface to other devices including, without limitation, servers, other clients, printers, database servers, print servers, file servers, communication servers, distributed servers and the like. Additionally, this coupling and/or connection may facilitate remote execution of program across the network. The networking of some or all of these devices may facilitate parallel processing of a program or method at one or more location without deviating from the scope of the disclosure. In addition, any of the devices attached to the client through an interface may include at least one storage medium capable of storing methods, programs, applications, code and/or instructions. A central repository may provide program instructions to be executed on different devices. In this implementation, the remote repository may act as a storage medium for program code, instructions, and programs.

The methods and systems described herein may be deployed in part or in whole through network infrastructures. The network infrastructure may include elements such as computing devices, servers, routers, hubs, firewalls, clients, personal computers, communication devices, routing devices and other active and passive devices, modules and/or components as known in the art. The computing and/or non-computing device(s) associated with the network infrastructure may include, apart from other components, a storage medium such as flash memory, buffer, stack, RAM, ROM and the like. The processes, methods, program codes, instructions described herein and elsewhere may be

executed by one or more of the network infrastructural elements. The methods and systems described herein may be adapted for use with any kind of private, community, or hybrid cloud computing network or cloud computing environment, including those which involve features of software as a service (SaaS), platform as a service (PaaS), and/or infrastructure as a service (IaaS).

The methods, program codes, and instructions described herein and elsewhere may be implemented on a cellular network having multiple cells. The cellular network may either be frequency division multiple access (FDMA) network or code division multiple access (CDMA) network. The cellular network may include mobile devices, cell sites, base stations, repeaters, antennas, towers, and the like. The cell network may be a GSM, GPRS, 3G, EVDO, mesh, or other networks types.

The methods, program codes, and instructions described herein and elsewhere may be implemented on or through mobile devices. The mobile devices may include navigation devices, cell phones, mobile phones, mobile personal digital assistants, laptops, palmtops, netbooks, pagers, electronic books readers, music players and the like. These devices may include, apart from other components, a storage medium such as a flash memory, buffer, RAM, ROM and one or more computing devices. The computing devices associated with mobile devices may be enabled to execute program codes, methods, and instructions stored thereon. Alternatively, the mobile devices may be configured to execute instructions in collaboration with other devices. The mobile devices may communicate with base stations interfaced with servers and configured to execute program codes. The mobile devices may communicate on a peer-to-peer network, mesh network, or other communications network. The program code may be stored on the storage medium associated with the server and executed by a computing device embedded within the server. The base station may include a computing device and a storage medium. The storage device may store program codes and instructions executed by the computing devices associated with the base station.

The computer software, program codes, and/or instructions may be stored and/or accessed on machine readable media that may include: computer components, devices, and recording media that retain digital data used for computing for some interval of time; semiconductor storage known as random access memory (RAM); mass storage typically for more permanent storage, such as optical discs, forms of magnetic storage like hard disks, tapes, drums, cards and other types; processor registers, cache memory, volatile memory, non-volatile memory; optical storage such as CD, DVD; removable media such as flash memory (e.g., USB sticks or keys), floppy disks, magnetic tape, paper tape, punch cards, standalone RAM disks, Zip drives, removable mass storage, off-line, and the like; other computer memory such as dynamic memory, static memory, read/write storage, mutable storage, read only, random access, sequential access, location addressable, file addressable, content addressable, network attached storage, storage area network, bar codes, magnetic ink, and the like.

The methods and systems described herein may transform physical and/or intangible items from one state to another. The methods and systems described herein may also transform data representing physical and/or intangible items from one state to another.

The elements described and depicted herein, including in flow charts and block diagrams throughout the figures, imply logical boundaries between the elements. However,

according to software or hardware engineering practices, the depicted elements and the functions thereof may be implemented on machines through computer executable media having a processor capable of executing program instructions stored thereon as a monolithic software structure, as standalone software modules, or as modules that employ external routines, code, services, and so forth, or any combination of these, and all such implementations may be within the scope of the present disclosure. Examples of such machines may include, but may not be limited to, personal digital assistants, laptops, personal computers, mobile phones, other handheld computing devices, medical equipment, wired or wireless communication devices, transducers, chips, calculators, satellites, tablet PCs, electronic books, gadgets, electronic devices, devices having artificial intelligence, computing devices, networking equipment, servers, routers and the like. Furthermore, the elements depicted in the flow chart and block diagrams or any other logical component may be implemented on a machine capable of executing program instructions. Thus, while the foregoing drawings and descriptions set forth functional aspects of the disclosed systems, no particular arrangement of software for implementing these functional aspects should be inferred from these descriptions unless explicitly stated or otherwise clear from the context. Similarly, it will be appreciated that the various steps identified and described above may be varied, and that the order of steps may be adapted to particular applications of the techniques disclosed herein. All such variations and modifications are intended to fall within the scope of this disclosure. As such, the depiction and/or description of an order for various steps should not be understood to require a particular order of execution for those steps, unless required by a particular application, or explicitly stated or otherwise clear from the context.

The methods and/or processes described above, and steps associated therewith, may be realized in hardware, software or any combination of hardware and software suitable for a particular application. The hardware may include a general-purpose computer and/or dedicated computing device or specific computing device or particular aspect or component of a specific computing device. The processes may be realized in one or more microprocessors, microcontrollers, embedded microcontrollers, programmable digital signal processors or other programmable device, along with internal and/or external memory. The processes may also, or instead, be embodied in an application specific integrated circuit, a programmable gate array, programmable array logic, or any other device or combination of devices that may be configured to process electronic signals. It will further be appreciated that one or more of the processes may be realized as a computer executable code capable of being executed on a machine-readable medium.

The computer executable code may be created using a structured programming language such as C, an object oriented programming language such as C++, or any other high-level or low-level programming language (including assembly languages, hardware description languages, and database programming languages and technologies) that may be stored, compiled or interpreted to run on one of the above devices, as well as heterogeneous combinations of processors, processor architectures, or combinations of different hardware and software, or any other machine capable of executing program instructions.

Thus, in one aspect, methods described above and combinations thereof may be embodied in computer executable code that, when executing on one or more computing devices, performs the steps thereof. In another aspect, the

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methods may be embodied in systems that perform the steps thereof, and may be distributed across devices in a number of ways, or all of the functionality may be integrated into a dedicated, standalone device or other hardware. In another aspect, the means for performing the steps associated with the processes described above may include any of the hardware and/or software described above. All such permutations and combinations are intended to fall within the scope of the present disclosure.

While the disclosure has been disclosed in connection with the preferred embodiments shown and described in detail, various modifications and improvements thereon will become readily apparent to those skilled in the art. Accordingly, the spirit and scope of the present disclosure is not to be limited by the foregoing examples, but is to be understood in the broadest sense allowable by law.

The use of the terms “a” and “an” and “the” and similar referents in the context of describing the disclosure (especially in the context of the following claims) is to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein may be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

While the foregoing written description enables one skilled to make and use what is considered presently to be the best mode thereof, those skilled in the art will understand and appreciate the existence of variations, combinations, and equivalents of the specific embodiment, method, and examples herein. The disclosure should therefore not be limited by the above described embodiment, method, and examples, but by all embodiments and methods within the scope and spirit of the disclosure.

Any element in a claim that does not explicitly state “means for” performing a specified function, or “step for” performing a specified function, is not to be interpreted as a “means” or “step” clause as specified in 35 U.S.C. § 112(f). In particular, any use of “step of” in the claims is not intended to invoke the provision of 35 U.S.C. § 112(f). The term “set” as used herein refers to a group having one or more members.

Persons skilled in the art may appreciate that numerous design configurations may be possible to enjoy the functional benefits of the inventive systems. Thus, given the wide variety of configurations and arrangements of embodiments of the present invention the scope of the invention is reflected by the breadth of the claims below rather than narrowed by the embodiments described above.

What is claimed is:

1. A computer-implemented method of maintaining a distributed ledger by a plurality of distributed ledger nodes, the method comprising:

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generating, by the plurality of distributed ledger nodes, the distributed ledger based on a plurality of distributed ledger transactions;

receiving, by the plurality of distributed ledger nodes, a plurality of transaction requests, each transaction request specifying licensing terms including 1) an identification of a respective copyrighted data element, 2) a description of a scope of utilization of the respective copyrighted data element, and 3) a description of sub-licensing rights for the respective copyrighted data element;

embedding, by the plurality of distributed ledger nodes, the licensing terms for each of the plurality of transaction requests into an aggregate intellectual property (IP) stack within a smart contract stored on the distributed ledger, wherein the smart contract is configured to: access the distributed ledger comprising the licensing terms,

provide provable access to each of the respective copyrighted data elements in the aggregate IP stack, add new copyrighted data elements to the aggregate IP stack in response to IP addition requests, adjust a royalty distribution in response to changes in the aggregate IP stack, and

automatically distribute royalties associated with each of the respective copyrighted data elements to a plurality of respective owners associated with the respective copyrighted data elements;

automatically executing, by the plurality of distributed ledger nodes, a licensing operation on the distributed ledger in response to a request to license a particular copyrighted data element from a first party, wherein the licensing operation provides access to the particular copyrighted data element and commits the first party to the licensing terms;

automatically executing, by the plurality of distributed ledger nodes, a sub-licensing operation on the distributed ledger in response to a request to sub-license the particular copyrighted data element, wherein the sub-licensing operation provides access to the particular copyrighted data element, commits a second party to the licensing terms, and stores a licensing chain indicating the licensing operation and the sub-licensing operation on the distributed ledger; and

automatically executing, by the plurality of distributed ledger nodes, an apportioning operation that distributes royalty amounts associated with the particular copyrighted data element to the plurality of respective owners in accordance with the licensing chain stored on the distributed ledger and the royalty distribution in the smart contract.

2. The method of claim 1, further comprising continuously improving the apportioning operation by:

receiving, by the plurality of distributed ledger nodes, updated royalty distribution proportions from an external machine learning system that maintains a training data set for a machine learning component, trains the machine learning component, and outputs the updated royalty distribution proportions using the trained machine learning component;

automatically adjusting, by the plurality of distributed ledger nodes, the royalty distribution in the smart contract based on the received updated royalty distribution proportions;

automatically executing, by the plurality of distributed ledger nodes, subsequent apportioning operations using the adjusted royalty distribution; and

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transmitting, by the plurality of distributed ledger nodes, outcomes of the subsequent apportioning operations to the external machine learning system in response to requests for updating the training data set.

3. The method of claim 2, wherein the machine learning component outputs the updated royalty distribution proportions based on a fractional contribution of each respective owner to the respective copyrighted data element.

4. The method of claim 3, wherein the respective copyrighted data element comprises source code, wherein the fractional contribution indicates a value of contributions by each respective owner to the source code.

5. The method of claim 1, wherein the smart contract is further configured to store the licensing operation and the sub-licensing operation.

6. The method of claim 1, wherein the smart contract is further configured to store a dependent licensing operation.

7. The method of claim 1, wherein the smart contract is further configured to verify ownership of the respective copyrighted data elements.

8. The method of claim 1, wherein the smart contract is further configured to verify licensing of the particular copyrighted data element based on the licensing chain stored on the distributed ledger.

9. The method of claim 1, wherein the smart contract is further configured to collect royalty payments associated with the particular copyrighted data element in accordance with the licensing chain stored on the distributed ledger.

10. The method of claim 1, wherein the description of the scope of utilization of the respective copyrighted data element specifies one or more of a time frame for using the respective copyrighted data element, a field of use for the respective copyrighted data element, or a volume of use of the respective copyrighted data element.

11. A system that maintains a distributed ledger, the system comprising:

a plurality of distributed ledger nodes, wherein each distributed ledger node of the plurality of distributed ledger nodes comprises one or more physical processors and memory storing instructions that configure the corresponding distributed ledger node to perform steps, wherein the plurality of distributed ledger nodes are configured to perform steps comprising:

generating the distributed ledger based on a plurality of distributed ledger transactions;

receiving a plurality of transaction requests, each transaction request specifying licensing terms including 1) an identification of a respective copyrighted data element, 2) a description of a scope of utilization of the respective copyrighted data element, and 3) a description of sub-licensing rights for the respective copyrighted data element;

embedding the licensing terms for each of the plurality of transaction requests into an aggregate intellectual property (IP) stack within a smart contract stored on the distributed ledger, wherein the smart contract is configured to:

access the distributed ledger comprising the licensing terms,

provide provable access to each of the respective copyrighted data elements in the aggregate IP stack,

add new copyrighted data elements to the aggregate IP stack in response to IP addition requests,

adjust a royalty distribution in response to changes in the aggregate IP stack, and

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automatically distribute royalties associated with each of the respective copyrighted data elements to a plurality of respective owners associated with the respective copyrighted data elements;

automatically executing a licensing operation on the distributed ledger in response to a request to license a particular copyrighted data element from a first party, wherein the licensing operation provides access to the particular copyrighted data element and commits the first party to the licensing terms;

automatically executing a sub-licensing operation on the distributed ledger in response to a request to sub-license the particular copyrighted data element, wherein the sub-licensing operation provides access to the particular copyrighted data element, commits a second party to the licensing terms, and stores a licensing chain indicating the licensing operation and the sub-licensing operation on the distributed ledger; and

automatically executing an apportioning operation that distributes royalty amounts associated with the particular copyrighted data element to the plurality of respective owners in accordance with the licensing chain stored on the distributed ledger and the royalty distribution in the smart contract.

12. The system of claim 11, further comprising an external machine learning system that maintains a training data set for a machine learning component, trains the machine learning component, and outputs updated royalty distribution proportions using the trained machine learning component, wherein the plurality of distributed ledger nodes are further configured to continuously improve the apportioning operation by performing steps comprising:

receiving the updated royalty distribution proportions from the external machine learning system;

automatically adjusting the royalty distribution in the smart contract based on the received updated royalty distribution proportions;

automatically executing subsequent apportioning operations using the adjusted royalty distribution; and transmitting outcomes of the subsequent apportioning operations to the external machine learning system in response to requests for updating the training data set.

13. The system of claim 12, wherein the machine learning component outputs the updated royalty distribution proportions based on a fractional contribution of each respective owner to the respective copyrighted data element.

14. The system of claim 13, wherein the respective copyrighted data element comprises source code, wherein the fractional contribution indicates a value of contributions by each respective owner to the source code.

15. The system of claim 11, wherein the smart contract is further configured to store the licensing operation and the sub-licensing operation.

16. The system of claim 11, wherein the smart contract is further configured to store a dependent licensing operation.

17. The system of claim 11, wherein the smart contract is further configured to verify ownership of the respective copyrighted data elements.

18. The system of claim 11, wherein the smart contract is further configured to verify licensing of the particular copyrighted data element based on the licensing chain stored on the distributed ledger.

19. The system of claim 11, wherein the smart contract is further configured to collect royalty payments associated with the particular copyrighted data element in accordance with the licensing chain stored on the distributed ledger.

20. The system of claim 11, wherein the description of the scope of utilization of the respective copyrighted data element specifies one or more of a time frame for using the respective copyrighted data element, a field of use for the respective copyrighted data element, or a volume of use of the respective copyrighted data element. 5

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